

Almost-Sure Radial Laws for Nested Random Polynomials and Erdős Problem #522

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Abstract

Let $(a_k)_{k \geq 0}$ be independent Rademacher random variables and let $f_n(z) = \sum_{k=0}^n a_k z^k$. Erdős asked in 1961 whether, almost surely, these random Littlewood polynomials have $n/2 + o(n)$ zeros in the closed unit disk, counted with multiplicity, as $n \rightarrow \infty$. Here, we answer this question, Erdős Problem #522, in the affirmative. In fact, we prove that, almost surely, $\frac{1}{n} \#\{z : f_n(z) = 0, |z| \leq 1 + x/n\}$ converges, at the natural scale $1/n$ around the unit circle, uniformly on compact subsets of $\mathbb{R}$ to $\frac{1}{2}(1 + \coth x - 1/x)$, interpreted as $1/2$ when $x = 0$. The same unit-disk and radial laws hold when the coefficients are independent and identically distributed with the Steinhaus distribution, the standard real Gaussian, the standard complex Gaussian, or any nondegenerate distribution which is bounded and centrally symmetric. For Rademacher coefficients we also prove the stronger almost sure bound $\#\{z : f_n(z) = 0, |z| \leq 1\} = n/2 + O(n/\log n)$. Since different degrees share their coefficients, we first prove the result along a subsequence of degrees, and then fill in the other degrees with Rouché's theorem, once we show that few zeros near the unit circle have a small derivative. All main results have been formally verified in Lean.

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1 Introduction

We consider random polynomials whose coefficients are taken from the same sequence for every degree. Throughout, zeros are counted with multiplicity. Our main result is the following radial law.

Theorem 1.1 (Radial law). *Let $(\xi_k)_{k \geq 0}$ be independent and identically distributed, with the Rademacher, Steinhaus, standard real Gaussian or standard complex Gaussian distribution, or with a nondegenerate bounded centrally symmetric distribution on $\mathbb{R}$ or $\mathbb{C}$. Put $f_n(z) = \sum_{k=0}^n \xi_k z^k$, let $\nu_n(r)$ be the number of zeros of f_n in $\{|z| \leq r\}$, counted with multiplicity, with $\nu_n = 0$ when $f_n \equiv 0$, and put*

$$\Phi(x) = \frac{\int_0^1 t e^{2xt} dt}{\int_0^1 e^{2xt} dt}.$$

On one event of probability one, for every $A > 0$,

$$\lim_{n \rightarrow \infty} \sup_{|x| \leq A} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| = 0.$$

Explicitly,

$$\Phi(0) = \frac{1}{2}, \quad \Phi(x) = \frac{e^{2x}}{e^{2x} - 1} - \frac{1}{2x} \quad (x \neq 0).$$

We prove Theorem 1.1 for Rademacher coefficients in Section 6, and for the other distributions in Section 7. Here the Rademacher distribution gives ± 1 with probability $\frac{1}{2}$ each, the Steinhaus distribution is uniform on the unit circle, and the standard complex Gaussian distribution has density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$.

In particular, almost surely we have $\nu_n(1 + x/n)/n \rightarrow \Phi(x)$ simultaneously for every $x \in \mathbb{R}$, and the convergence is uniform on each compact interval of x . To obtain the explicit formula, note that the quotient defining Φ is the derivative of $\frac{1}{2} \log \int_0^1 e^{2xt} dt$, and that $\int_0^1 e^{2xt} dt = (e^{2x} - 1)/(2x)$ for $x \neq 0$, so that

$$\Phi(x) = \frac{d}{dx} \frac{1}{2} \log \frac{e^{2x} - 1}{2x} = \frac{e^{2x}}{e^{2x} - 1} - \frac{1}{2x} = \frac{1}{2} \left(1 + \coth x - \frac{1}{x} \right).$$

In particular $\Phi(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $\Phi(x) \rightarrow 1$ as $x \rightarrow \infty$. In Figure 1 we compare the limiting density Φ' with the roots of 77 polynomials with random signs. Taking $x = 0$ in Theorem 1.1, we obtain the following strong law.

Corollary 1.2 (The unweighted strong law). *In the setting of Theorem 1.1, almost surely*

$$\frac{\nu_n(1)}{n} \longrightarrow \frac{1}{2}.$$

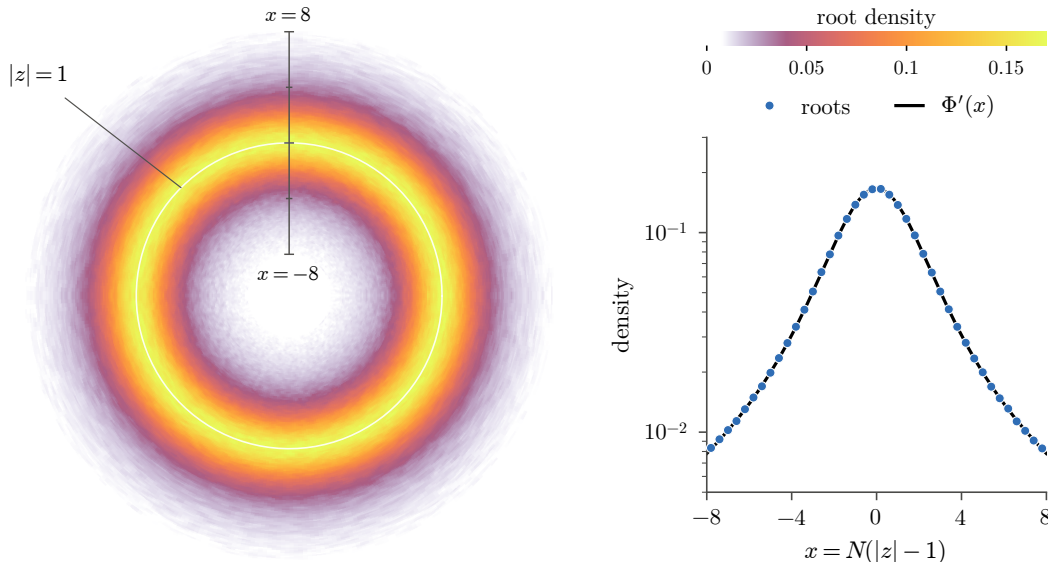
For Rademacher coefficients, in the notation of Erdős problem #522 [3], where $R_n = \nu_n(1)$ is the number of roots in the closed unit disk, Corollary 1.2 states that $R_n/(n/2) \rightarrow 1$ almost surely. For these coefficients we first prove Corollary 1.2 directly in Section 6.1, and then we prove Theorem 1.1 in Section 6.2 by the same construction.

For Rademacher coefficients we also obtain a rate of convergence in the unit disk.

Theorem 1.3 (Logarithmic rate). *In the setting of Theorem 1.1, for Rademacher coefficients,*

$$\limsup_{n \rightarrow \infty} (\log n) \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| \leq 2000 \log 2 \quad \text{almost surely.}$$

In Section 8 we also determine sufficient exponents for a sparse subsequence of degrees in the construction (Proposition 8.1), bound the number of roots in sublevel components that meet circles near the unit circle (Corollary 8.2), and compute the logarithmic variance profile of polynomially weighted polynomials with random signs (Proposition 8.3).



7.7×10^6 roots of 77 polynomials with random signs, $N=10^5$

Figure 1: Density of the zeros of 77 random Littlewood polynomials of degree 10^5 near the unit circle (left), and its angular average with the limiting density Φ' (right).

The question of convergence in probability appears in Hayman's problem collection [7, 8, Problem 4.15]. Yakir [24] proved that the proportion of roots in the unit disk converges to $1/2$ in probability, by combining concentration of an angular logarithmic integral with Jensen's formula. In his convention of n coefficients, with probability tending to one, the number of roots in the unit disk differs from $n/2$ by less than $n^{9/10}$. Do [6] proved that, for nested Kac polynomials with iid real coefficients of mean zero, variance one, and finite $(2 + \varepsilon)$ th moment, the number of real roots in $[-1, 1]$, divided by $\log n$, tends to $1/\pi$ almost surely. That proof uses maximal inequalities to pass from a sparse subsequence of degrees to all degrees. For the complex roots considered here, we compare consecutive degrees by root matching under the appended coefficient tail.

Conventions. Sections 2–6 treat Rademacher coefficients, which we write ϵ_k , and Section 7 treats the other distributions, whose coefficients we write ξ_k . In $\nu_n(r)$ the disk is closed, and the denominator in every limit is the nominal degree n . For a coefficient distribution with an atom at zero, $\nu_n = 0$ whenever f_n is identically zero. The radii $1 + x/n$ with $|x| \leq A$ are positive once $n > A$, and only sufficiently large n are used when $x < 0$. We write $C(K)$ for a positive constant depending only on the parameter K , which may increase from one estimate to the next, and log for

the natural logarithm. Whenever j indexes blocks of degrees, $N = j^8$ is the first degree of the block $j^8 \leq n < (j+1)^8$, except in the proof of Proposition 8.1, where $N = \lfloor j^q \rfloor$. We call these first degrees the sparse subsequence of degrees. We write μ for the normalized arc measure $d\theta/(2\pi)$ on the circle $\mathbb{T}$ of angles θ .

The proof

The main difficulty is to pass from the sparse subsequence of degrees to every degree, which requires control of root movement under every partial appended tail. We first show that most roots lie in a narrow annulus, and then control the roots at which the derivative is small. Figure 2 shows how these steps depend on one another.

We first estimate the angular average of the logarithm of the normalized polynomial (Theorem 3.1). For this we use one-point and two-point Gaussian approximations to the occupation of small disks by that polynomial. Logarithmic integrability controls the contribution near zero. Along the sparse subsequence of degrees, the resulting probability bounds are summable. By Jensen's formula at four radii (Lemmas 2.1 and 2.4, and Corollary 3.3) we get

$$\frac{1}{N} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\} \leq \frac{2 \log 2}{K} + o(1),$$

so that K controls the fraction of roots outside the annulus.

For each root in this annulus with small derivative there is a nearby mesh point at which the polynomial and its derivative are both small. We bound the first two moments of the number of these mesh points through the covariance matrices of the corresponding real random vectors, which have four coordinates for one mesh point and eight for a pair. To bound the number of roots assigned to one mesh point, we use Hanson–Wright concentration for arc energy and Jensen's formula. By Lemma 4.5, outside a summable exceptional event there are at most $N^{31/32}$ annular roots with $|f'_N(\alpha)| < N^{3/2-1/64}$.

We localize each remaining root by a Taylor expansion (Lemma 5.4). A common regular sublevel of $|f_N|$ then yields disjoint domains, each of which contains one root. We compare every degree n of a block $j^8 \leq n < (j+1)^8$ with its first degree $N = j^8$. By a maximal bound on every appended tail (Lemma 4.6), the hypothesis of Rouché's theorem holds on the boundary of each selected domain. The tail amplitude controls the localization radius. The exponent eight ensures that the probability bounds are summable and that the localization radius is smaller than the half-width of the thin band.

By root matching (Theorem 5.1) we have

$$|\nu_n(1) - \nu_N(1)| \leq c_N + U_N + (\deg f_n - \deg f_N),$$

where c_N is the number of selected domains whose closures meet the unit circle and U_N is the number of unselected roots. Through the thin bands of Lemma 2.5, the same logarithmic concentration estimate shows that the normalized count of the roots in the crossing domains tends to zero, since these roots lie in a thinner annulus. The unselected roots contribute at most $2 \log 2/K + o_K(1)$ to the normalized count. For each fixed K the probabilities of the exceptional events are summable over the blocks, so that by the Borel–Cantelli lemma these bounds hold for all large blocks on an event Ω_K of probability one. Finally, we let K tend to infinity on $\bigcap_{K \geq 2} \Omega_K$, which yields the limit along all degrees on the original probability space.

In the probabilistic estimates we use Raič’s convex-set normal approximation [17], the logarithmic integrability of Nazarov–Nishry–Sodin [16], and the Hanson–Wright inequality [19]. We compute the constants and their dependence on the annular width in Appendix A.

The Lean formalization [13] proves the unit-disk and radial laws on the shared coefficient space. Appendix C describes the theorem correspondence.

Organization

In Section 2 we apply Jensen’s formula to radial counts and compute the variance profile, and in Section 3 we prove logarithmic concentration for signs. Section 4 contains the local-count, derivative, and maximal-tail estimates. We prove root matching in Section 5, and then Corollary 1.2 and Theorem 1.1 in Section 6. In Section 7 we treat the other coefficient distributions and prove Theorem 1.1 and Corollary 1.2 for them. The rate, the choice of the sparse subsequence of degrees, nearby circles, and weighted variance profiles follow in Section 8. Section 9 collects open problems, and Section 10 reports numerical observations. In Appendix B we collect the general forms of the concentration, mesh, and diagonalization estimates.

2 Jensen’s formula and the variance profile

We first consider the polynomials $f_N(z) = \sum_{k=0}^N \epsilon_k z^k$ with random signs. Recall that $d\mu(\theta) = d\theta/(2\pi)$, and set

$$\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}, \quad J_N(r) = \int_{\mathbb{T}} \log |f_N(re^{i\theta})| d\mu(\theta).$$

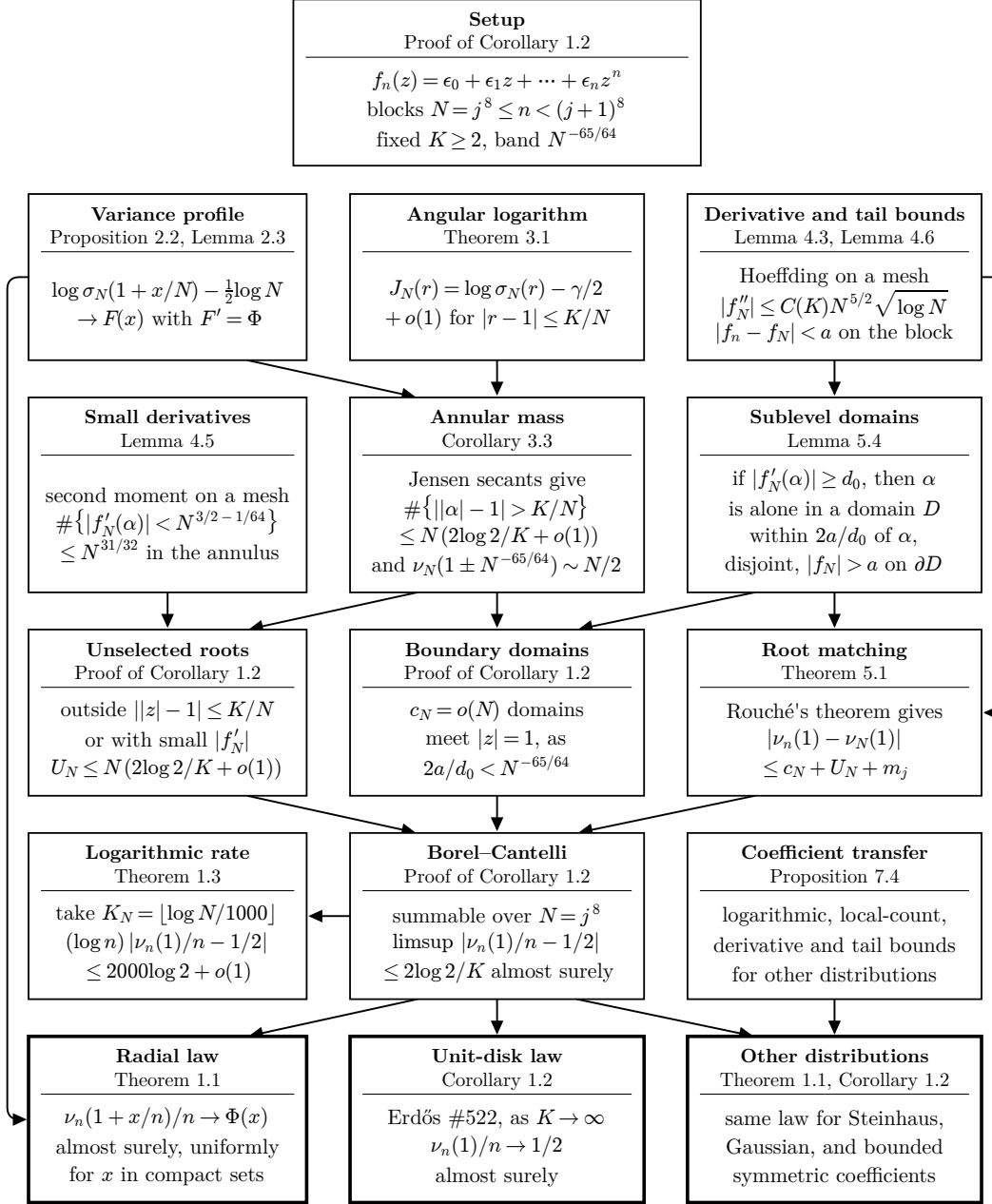


Figure 2: Structure of the proof. Arrows point from inputs to the steps that use them.

The results of this section are deterministic: they hold for every nonzero polynomial f_N of degree at most N , and the randomness of the coefficients enters only through the concentration estimates of Section 3. We write γ for Euler's constant.

Lemma 2.1 (Four-radius bound for radial mass). *Let f_N be a nonzero polynomial of degree at most N , and fix $K > 0$, $N > K$. We set*

$$(x_1, x_2, x_3, x_4) = (-K, -K/2, K/2, K), \quad r_i = 1 + x_i/N.$$

Let $\sigma_N(r) > 0$ be a normalization, and suppose that at each of these four radii

$$(1) \quad |J_N(r_i) - \log \sigma_N(r_i) + \gamma/2| \leq \varepsilon_i, \quad i = 1, 2, 3, 4.$$

Then

$$(2) \quad \frac{\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\}}{N} \leq \frac{\deg f_N}{N} + \frac{\log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} - \frac{\log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)}.$$

In particular, suppose that along a sequence of degrees $\deg f_N/N \rightarrow 1$, that the errors ε_i tend to zero, and that $\log \sigma_N(1 + x/N)$, minus a deterministic centering that may depend on N but not on x , converges to $F(x)$ uniformly on $[-K, K]$. Then the limiting upper bound for $\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\}/N$ is

$$(3) \quad 1 + \frac{2}{K} \{F(-K/2) - F(-K) - F(K) + F(K/2)\}.$$

Here the centering $-\gamma/2$ is $\mathbb{E} \log |G|$, where G has density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$, and it is the same at all four radii. If the four bounds (1) hold outside four exceptional events, then (2) holds outside their union, whose probability is at most the sum of their probabilities. The limiting bound (3) holds for every sample sequence on which the four bounds hold for all sufficiently large degrees.

Proof. We will bound the numbers of roots in $\{|z| < r_1\}$ and in $\{|z| > r_4\}$ in terms of the slopes of two secants of the convex function $t \mapsto J_N(e^t)$, and then replace J_N by $\log \sigma_N$ using (1). We write $f_N(z) = a \prod_{j=1}^{\deg f_N} (z - \alpha_j)$. By Jensen's formula [22, §3.61] for one factor, $\int_{\mathbb{T}} \log |e^t e^{i\theta} - \alpha| d\mu(\theta) = \max(t, \log |\alpha|)$ for $\alpha \neq 0$, and the integral is t for $\alpha = 0$, so that

$$J_N(e^t) = \log |a| + \sum_{j=1}^{\deg f_N} \max(t, \log |\alpha_j|), \quad t \in \mathbb{R},$$

where a term with $\alpha_j = 0$ is read as t . Since the term of α_j has right derivative 1 in t if $|\alpha_j| \leq e^t$, and 0 otherwise, the function $t \mapsto J_N(e^t)$ is convex, and its right

derivative is $\nu_N(e^t)$. Since this function is also piecewise linear, it is the integral of its right derivative, and after the substitution $r = e^s$ we obtain

$$(4) \quad J_N(e^{t'}) - J_N(e^t) = \int_t^{t'} \nu_N(e^s) ds = \int_{e^t}^{e^{t'}} \frac{\nu_N(r)}{r} dr, \quad t < t'.$$

Since $N > K$, we have $0 < r_1 < r_2 < r_3 < r_4$, and since ν_N is nondecreasing, (4) gives

$$\begin{aligned} \nu_N(r_1) &\leq \frac{1}{\log(r_2/r_1)} \int_{r_1}^{r_2} \frac{\nu_N(r)}{r} dr = \frac{J_N(r_2) - J_N(r_1)}{\log(r_2/r_1)}, \\ \frac{J_N(r_4) - J_N(r_3)}{\log(r_4/r_3)} &= \frac{1}{\log(r_4/r_3)} \int_{r_3}^{r_4} \frac{\nu_N(r)}{r} dr \leq \nu_N(r_4). \end{aligned}$$

By (1), in which $\gamma/2$ cancels in differences, we have

$$\begin{aligned} J_N(r_2) - J_N(r_1) &\leq \log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2, \\ J_N(r_4) - J_N(r_3) &\geq \log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4. \end{aligned}$$

Let m be the number of roots with $||\alpha| - 1| > K/N$. There are at most $\nu_N(r_1)$ roots in the inner tail $\{|z| < r_1\}$ and exactly $\deg f_N - \nu_N(r_4)$ roots in the outer tail $\{|z| > r_4\}$, and hence, by the last two displays,

$$\begin{aligned} \frac{m}{N} &\leq \frac{\nu_N(r_1)}{N} + \frac{\deg f_N - \nu_N(r_4)}{N} \leq \frac{\deg f_N}{N} + \frac{J_N(r_2) - J_N(r_1)}{N \log(r_2/r_1)} - \frac{J_N(r_4) - J_N(r_3)}{N \log(r_4/r_3)} \\ &\leq \frac{\deg f_N}{N} + \frac{\log \sigma_N(r_2) - \log \sigma_N(r_1) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} \\ &\quad - \frac{\log \sigma_N(r_4) - \log \sigma_N(r_3) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)}. \end{aligned}$$

This proves (2).

For the limit, let c be the deterministic centering of the hypothesis, which does not depend on x , and let $F_N(x) = \log \sigma_N(1 + x/N) - c$, so that $\log \sigma_N(r_i) = F_N(x_i) + c$. We have

$$\begin{aligned} N \log(r_2/r_1) &= \int_{-K}^{-K/2} \frac{dx}{1 + x/N} \longrightarrow K/2, \\ N \log(r_4/r_3) &= \int_{K/2}^K \frac{dx}{1 + x/N} \longrightarrow K/2. \end{aligned}$$

Since c cancels in the differences, the right-hand side of (2) is the expression in the first line below, and since $\deg f_N/N \rightarrow 1$, $\varepsilon_i \rightarrow 0$ and $F_N(x_i) \rightarrow F(x_i)$, we obtain

$$\frac{\deg f_N}{N} + \frac{F_N(-K/2) - F_N(-K) + \varepsilon_1 + \varepsilon_2}{N \log(r_2/r_1)} - \frac{F_N(K) - F_N(K/2) - \varepsilon_3 - \varepsilon_4}{N \log(r_4/r_3)}$$

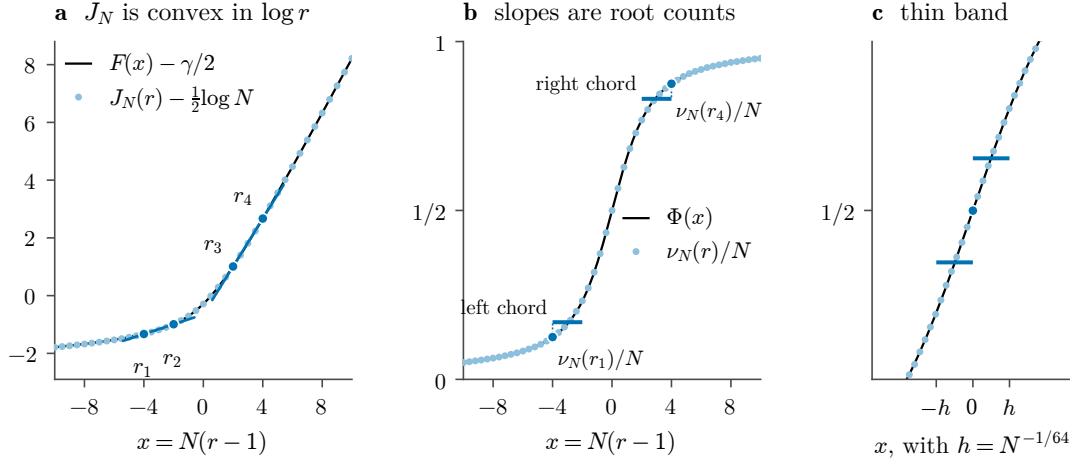


Figure 3: Jensen secants for one sequence at $N = 10^5$. The chord slopes bound the normalized zero counts, which approach Φ .

$$\begin{aligned} &\longrightarrow 1 + \frac{F(-K/2) - F(-K)}{K/2} - \frac{F(K) - F(K/2)}{K/2} \\ &= 1 + \frac{2}{K} \{F(-K/2) - F(-K) - F(K) + F(K/2)\}, \end{aligned}$$

so that the limit superior of m/N is at most (3). $\square$

Figure 3 shows these secants for one sequence at $N = 10^5$.

Proposition 2.2 (The unweighted logarithmic variance profile). *For the normalization $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$ of this section,*

$$\log \sigma_N(1 + x/N) - \frac{1}{2} \log N \longrightarrow F(x) := \frac{1}{2} \log \int_0^1 e^{2xt} dt$$

uniformly on compact intervals of x . The profile satisfies

$$F'(x) = \Phi(x), \quad F(x) = x + F(-x), \quad F'(0) = \frac{1}{2}.$$

The bound in (3) is

$$\frac{2}{K} \log \frac{2}{1 + e^{-K}} \leq \frac{2 \log 2}{K}.$$

Proof. We first compare each term $(1 + x/N)^{2k}$ of $\sigma_N(1 + x/N)^2$ with $e^{2xk/N}$, which turns $\sigma_N(1 + x/N)^2/N$ into a Riemann sum for $\int_0^1 e^{2xt} dt$, and then derive the identities for F and the value of (3) from this integral.

For $|x| \leq K$, $N \geq 2K$ and $0 \leq k \leq N$, expanding the logarithm in its power series and using $|x/N| \leq K/N \leq 1/2$, we obtain

$$|2k \log(1 + x/N) - 2xk/N| = 2k \left| -\frac{1}{2} \left(\frac{x}{N} \right)^2 + \frac{1}{3} \left(\frac{x}{N} \right)^3 - \dots \right|$$

$$\leq k \left(\frac{x}{N} \right)^2 \left(1 + \left| \frac{x}{N} \right| + \left(\frac{x}{N} \right)^2 + \cdots \right) = \frac{k(x/N)^2}{1 - |x/N|} \leq 2k \left(\frac{x}{N} \right)^2 \leq 2K^2/N.$$

Exponentiating, we get

$$(5) \quad e^{-2K^2/N} \leq \frac{(1 + x/N)^{2k}}{e^{2xk/N}} \leq e^{2K^2/N}.$$

Since $(x, t) \mapsto e^{2xt}$ is uniformly continuous on $[-K, K] \times [0, 1]$ and the term $k = 0$ is $1/N$, the Riemann sums $\frac{1}{N} \sum_{k=0}^N e^{2xk/N}$ converge to $\int_0^1 e^{2xt} dt$ uniformly for $|x| \leq K$. Multiplying (5) by $e^{2xk/N}/N$ and summing over $0 \leq k \leq N$, we see that $\sigma_N(1 + x/N)^2/N = \frac{1}{N} \sum_{k=0}^N (1 + x/N)^{2k}$ lies between $e^{-2K^2/N}$ and $e^{2K^2/N}$ times the Riemann sum, so that it has the same limit. Since this limit is at least e^{-2K} and the logarithm is uniformly continuous on $[e^{-2K}/2, \infty)$, we obtain, uniformly for $|x| \leq K$,

$$\log \sigma_N(1 + x/N) - \frac{1}{2} \log N = \frac{1}{2} \log \frac{\sigma_N(1 + x/N)^2}{N} \longrightarrow \frac{1}{2} \log \int_0^1 e^{2xt} dt = F(x).$$

Differentiating the positive integral under the integral sign, we have

$$F'(x) = \frac{1}{2} \frac{\int_0^1 2te^{2xt} dt}{\int_0^1 e^{2xt} dt} = \frac{\int_0^1 te^{2xt} dt}{\int_0^1 e^{2xt} dt} = \Phi(x),$$

and in particular $F'(0) = \int_0^1 t dt = \frac{1}{2}$. The change of variables $t \mapsto 1 - t$ gives the symmetry

$$F(x) = \frac{1}{2} \log \int_0^1 e^{2x(1-t)} dt = \frac{1}{2} \log \left(e^{2x} \int_0^1 e^{-2xt} dt \right) = x + F(-x).$$

By integration we also obtain

$$\begin{aligned} F(-K/2) - F(-K) &= \frac{1}{2} \log \frac{\int_0^1 e^{-Kt} dt}{\int_0^1 e^{-2Kt} dt} = \frac{1}{2} \log \frac{(1 - e^{-K})/K}{(1 - e^{-2K})/(2K)} \\ &= \frac{1}{2} \log \frac{2(1 - e^{-K})}{(1 - e^{-K})(1 + e^{-K})} = \frac{1}{2} \log \frac{2}{1 + e^{-K}}. \end{aligned}$$

Using $F(x) = x + F(-x)$ at $x = K$ and $x = K/2$ in (3), we get

$$\begin{aligned} 1 + \frac{2}{K} \{F(-K/2) - F(-K) - F(K) + F(K/2)\} \\ &= 1 + \frac{2}{K} \left\{ F(-K/2) - F(-K) - (K + F(-K)) + \left(\frac{K}{2} + F(-K/2) \right) \right\} \\ &= \frac{4}{K} \{F(-K/2) - F(-K)\} = \frac{2}{K} \log \frac{2}{1 + e^{-K}} \leq \frac{2 \log 2}{K}. \end{aligned}$$

□

This proof also yields a rate of convergence in the profile limit.

Lemma 2.3 (Rate in the variance profile). *For $K > 0$, $N \geq 2K$ and $|x| \leq K$,*

$$\left| \log \sigma_N(1 + x/N) - \frac{1}{2} \log N - F(x) \right| \leq \frac{K^2 + e^{4K}/2}{N}.$$

Proof. We use (5) to compare $\sigma_N(1 + x/N)^2/N$ with a Riemann sum of e^{2xt} , and the monotonicity of $t \mapsto e^{2xt}$ to compare the Riemann sum with the integral. Let

$$S_N(x) = \frac{1}{N} \sum_{k=0}^N e^{2xk/N}, \quad I(x) = \int_0^1 e^{2xt} dt,$$

so that $F = \frac{1}{2} \log I$. Multiplying (5) by $e^{2xk/N}/N$ and summing over $0 \leq k \leq N$, we get $e^{-2K^2/N} S_N(x) \leq \sigma_N(1 + x/N)^2/N \leq e^{2K^2/N} S_N(x)$, and hence

$$\left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log S_N(x) \right| \leq \frac{2K^2}{N}.$$

The function $g(t) = e^{2xt}$ is nondecreasing on $[0, 1]$ for $x \geq 0$ and nonincreasing for $x \leq 0$, so that $I(x)$ lies between the Riemann sums $\frac{1}{N} \sum_{k=0}^{N-1} g(k/N)$ and $\frac{1}{N} \sum_{k=1}^N g(k/N)$, in the order that gives

$$I(x) \leq \frac{1}{N} \sum_{k=1}^N g(k/N) \leq S_N(x) = \frac{1}{N} \sum_{k=0}^{N-1} g(k/N) + \frac{g(1)}{N} \leq I(x) + \frac{g(1)}{N}, \quad x \geq 0,$$

$$I(x) \leq \frac{1}{N} \sum_{k=0}^{N-1} g(k/N) \leq S_N(x) = \frac{1}{N} \sum_{k=1}^N g(k/N) + \frac{g(0)}{N} \leq I(x) + \frac{g(0)}{N}, \quad x \leq 0.$$

In either case, since $g(0) = 1$ and $g(1) = e^{2x} \leq e^{2K}$,

$$I(x) \leq S_N(x) \leq I(x) + \frac{\max(g(0), g(1))}{N} \leq I(x) + \frac{e^{2K}}{N}.$$

For $|x| \leq K$ we also have $I(x) \geq e^{-2K}$, since $e^{2xt} \geq e^{-2K}$ for $0 \leq t \leq 1$. By the mean value theorem, $|\log a - \log b| \leq |a - b|/\min(a, b)$ for $a, b > 0$, and hence

$$0 \leq \log S_N(x) - \log I(x) \leq \frac{e^{2K}/N}{e^{-2K}} = \frac{e^{4K}}{N}.$$

Combining the two estimates, we obtain

$$\begin{aligned} \left| \log \sigma_N(1 + x/N) - \frac{1}{2} \log N - F(x) \right| &= \frac{1}{2} \left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log I(x) \right| \\ &\leq \frac{1}{2} \left| \log \frac{\sigma_N(1 + x/N)^2}{N} - \log S_N(x) \right| + \frac{1}{2} \left| \log S_N(x) - \log I(x) \right| \\ &\leq \frac{1}{2} \left(\frac{2K^2}{N} + \frac{e^{4K}}{N} \right) = \frac{K^2 + e^{4K}/2}{N}. \end{aligned}$$

□

Figure 4 shows $\Phi(x)$ as the mean of t under the limit, in $t = k/N$, of the tilted coefficient mass $Nr^{2k}/\sigma_N(r)^2$ at $r = 1 + x/N$.

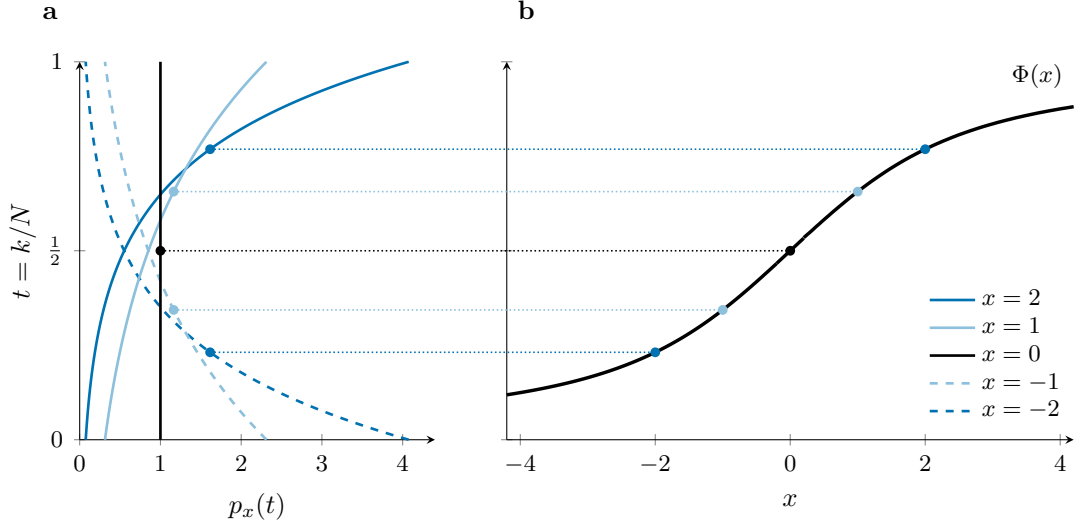


Figure 4: The densities $p_x(t) = 2xe^{2xt}/(e^{2x} - 1)$ (a) and their means $\Phi(x)$ (b).

2.1 Radial mass and thin bands

In the next two lemmas we turn bounds on J_N at a few radii into bounds on root counts, and in Section 3 these bounds are supplied by Theorem 3.1 outside exceptional events with summable probabilities.

Lemma 2.4 (Annular mass). *Let $K \geq 2$ and $N \geq 2K$, let f_N be a nonzero polynomial of degree at most N , and let $r_i = 1 + x_i/N$ with $(x_1, x_2, x_3, x_4) = (-K, -K/2, K/2, K)$. Let $\varepsilon, D \geq 0$ be such that*

$$\begin{aligned} |J_N(r_i) - \log \sigma_N(r_i) + \gamma/2| &\leq \varepsilon, \\ \left| \log \sigma_N(r_i) - \frac{1}{2} \log N - F(x_i) \right| &\leq D/N, \quad i = 1, 2, 3, 4, \end{aligned}$$

and $\varepsilon + D/N \leq K/8$. Then

$$\begin{aligned} \frac{\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\}}{N} &\leq \frac{2}{K} \log \frac{2}{1 + e^{-K}} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right) \\ &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right). \end{aligned}$$

By Lemma 2.3, the second condition holds with $D = K^2 + e^{4K}/2$.

Proof. We apply Lemma 2.1 with $\varepsilon_i = \varepsilon$, replace $\log \sigma_N(r_i)$ by $\frac{1}{2} \log N + F(x_i)$, compare both denominators with $K/2$, and evaluate the leading term by the symmetry $F(x) = x + F(-x)$. Let m be the number of roots with $||\alpha| - 1| > K/N$, and let $\delta = 2\varepsilon + 2D/N$. By the second hypothesis, in which $\frac{1}{2} \log N$ cancels in

differences,

$$\begin{aligned}\log \sigma_N(r_2) - \log \sigma_N(r_1) &\leq F(-K/2) - F(-K) + \frac{2D}{N}, \\ \log \sigma_N(r_4) - \log \sigma_N(r_3) &\geq F(K) - F(K/2) - \frac{2D}{N}.\end{aligned}$$

Since $1/2 \leq 1 + x/N \leq 1$ for $-K \leq x \leq 0$, because $N \geq 2K$, and $1 + x/N \geq 1$ for $x \geq 0$, the denominators satisfy

$$N \log(r_2/r_1) = \int_{-K}^{-K/2} \frac{dx}{1 + x/N} \geq \frac{K}{2}, \quad N \log(r_4/r_3) = \int_{K/2}^K \frac{dx}{1 + x/N} \leq \frac{K}{2}.$$

The numerators $F(-K/2) - F(-K) + \delta$ and $F(K) - F(K/2) - \delta$ in the chain below are nonnegative. Indeed, by Proposition 2.2, $F' = \Phi$, and $\Phi(x)$ is the mean of t under the probability density proportional to e^{2xt} on $[0, 1]$, so that $\Phi \geq 0$. Differentiating this mean under the integral sign, we get

$$\begin{aligned}\Phi'(x) &= \frac{d}{dx} \frac{\int_0^1 t e^{2xt} dt}{\int_0^1 e^{2xt} dt} = \frac{\int_0^1 2t^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} - \frac{\int_0^1 t e^{2xt} dt \int_0^1 2t e^{2xt} dt}{\left(\int_0^1 e^{2xt} dt\right)^2} \\ (6) \quad &= 2 \left(\frac{\int_0^1 t^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} - \left(\frac{\int_0^1 t e^{2xt} dt}{\int_0^1 e^{2xt} dt} \right)^2 \right) = 2 \operatorname{Var}_x(t) \geq 0,\end{aligned}$$

where $\operatorname{Var}_x(t)$ is the variance of t under the same density. Thus Φ is nondecreasing, and $\Phi(0) = F'(0) = 1/2$ by Proposition 2.2, so that, with $F(K) - F(K/2) = \int_{K/2}^K \Phi \geq \frac{K}{2} \Phi(0)$ and the hypothesis $\varepsilon + D/N \leq K/8$,

$$F(-K/2) - F(-K) \geq 0, \quad F(K) - F(K/2) \geq \frac{K}{4} \geq 2 \left(\varepsilon + \frac{D}{N} \right) = \delta.$$

By (2) with $\varepsilon_i = \varepsilon$, which applies since $N \geq 2K > K$, by $\deg f_N \leq N$, and by the above bounds on the differences of $\log \sigma_N$, on the denominators and on the numerators, we have

$$\begin{aligned}\frac{m}{N} &\leq \frac{\deg f_N}{N} + \frac{\log \sigma_N(r_2) - \log \sigma_N(r_1) + 2\varepsilon}{N \log(r_2/r_1)} - \frac{\log \sigma_N(r_4) - \log \sigma_N(r_3) - 2\varepsilon}{N \log(r_4/r_3)} \\ &\leq 1 + \frac{F(-K/2) - F(-K) + \delta}{N \log(r_2/r_1)} - \frac{F(K) - F(K/2) - \delta}{N \log(r_4/r_3)} \\ &\leq 1 + \frac{2}{K} \{F(-K/2) - F(-K) + \delta\} - \frac{2}{K} \{F(K) - F(K/2) - \delta\} \\ &= 1 + \frac{2}{K} \{F(-K/2) - F(-K) - F(K) + F(K/2)\} + \frac{4\delta}{K} \\ &= \frac{4}{K} \{F(-K/2) - F(-K)\} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right) = \frac{2}{K} \log \frac{2}{1 + e^{-K}} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right) \\ &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(\varepsilon + \frac{D}{N} \right),\end{aligned}$$

where the fifth line uses $F(x) = x + F(-x)$ and the evaluation of $F(-K/2) - F(-K)$ in the proof of Proposition 2.2. $\square$

The two secants contribute equally to the leading term $\frac{2}{K} \log \frac{2}{1+e^{-K}} \leq \frac{2 \log 2}{K}$ of Lemma 2.4, each at most $\log 2/K$: the inner secant of Lemma 2.1 controls the roots in $|z| < 1 - K/N$, and the outer secant controls those in $|z| > 1 + K/N$. Indeed, by Lemma 2.3 and $N \log(r_2/r_1) \rightarrow K/2$, the inner secant $(\log \sigma_N(r_2) - \log \sigma_N(r_1))/(N \log(r_2/r_1))$ tends to

$$\frac{2[F(-K/2) - F(-K)]}{K} = \frac{1}{K} \log \frac{2}{1+e^{-K}} \leq \frac{\log 2}{K},$$

by the evaluation of $F(-K/2) - F(-K)$ in the proof of Proposition 2.2. There are at most $N - \nu_N(r_4)$ roots in $|z| > 1 + K/N$, and $\nu_N(r_4)/N$ is at least the outer secant, as in the proof of Lemma 2.1. By the same limits and the symmetry $F(x) = x + F(-x)$, 1 minus the outer secant tends to

$$\begin{aligned} 1 - \frac{2}{K} \{F(K) - F(K/2)\} &= 1 - \frac{2}{K} \left\{ \frac{K}{2} + F(-K) - F(-K/2) \right\} \\ &= \frac{2}{K} \{F(-K/2) - F(-K)\}. \end{aligned}$$

Lemma 2.5 (Thin bands). *Fix $0 < \kappa \leq 1/64$, let $N \geq 4$, and let f_N be a nonzero polynomial of degree at most N . Suppose that for some $\varepsilon \geq 0$,*

$$\left| J_N(1 + qN^{-1-\kappa}) - \log \sigma_N(1 + qN^{-1-\kappa}) + \gamma/2 \right| \leq \varepsilon, \quad q = -2, -1, 0, 1, 2.$$

Then, for $q = -1, 0, 1$,

$$\left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + 4\varepsilon N^\kappa.$$

Proof. We follow the proof of Lemma 2.1, now at five radii spaced by $N^{-1-\kappa}$. At each of the three middle radii, the count ν_N lies between the slopes of two secants of $t \mapsto J_N(e^t)$, and a bound on the second derivative of $\log \sigma_N(e^t)$ shows that the slopes of the corresponding secants of $\log \sigma_N(e^t)$, divided by N , are within $2N^{-\kappa}$ of $1/2$. Let $g(t) = \log \sigma_N(e^t)$. Differentiating the finite sum, we see that

$$\begin{aligned} g'(t) &= \frac{\sum_{k=0}^N k e^{2kt}}{\sum_{k=0}^N e^{2kt}}, \\ g''(t) &= 2 \left\{ \frac{\sum_{k=0}^N k^2 e^{2kt}}{\sum_{k=0}^N e^{2kt}} - \left(\frac{\sum_{k=0}^N k e^{2kt}}{\sum_{k=0}^N e^{2kt}} \right)^2 \right\}. \end{aligned}$$

That is, under the probability weights proportional to e^{2kt} on $\{0, 1, \dots, N\}$, $g'(t)$ is the mean of k and $g''(t)$ is twice its variance $\text{Var}_t(k)$. At $t = 0$ the weights are

uniform, so that $g'(0) = \frac{1}{N+1} \sum_{k=0}^N k = \frac{N}{2}$. For a random variable Y with values in $[m, M]$, since $\mathbb{E}(Y - m)(M - Y) \geq 0$, we have

$$\begin{aligned} \text{Var}(Y) &= (\mathbb{E}Y - m)(M - \mathbb{E}Y) - \mathbb{E}(Y - m)(M - Y) \\ &\leq (\mathbb{E}Y - m)(M - \mathbb{E}Y) \leq \frac{(M - m)^2}{4}. \end{aligned}$$

Applying this bound with $Y = k$, $m = 0$ and $M = N$, we obtain

$$(7) \quad g''(t) = 2 \text{Var}_t(k) \leq 2 \cdot \frac{N^2}{4} = \frac{N^2}{2}.$$

Let $t_q = \log(1 + qN^{-1-\kappa})$ for $q = -2, -1, 0, 1, 2$. Since $N \geq 4$, we have $N^{-1-\kappa} \leq 1/4$, and since $|\log(1 + y)| \leq |y|/(1 - |y|) \leq 2|y|$ for $|y| \leq 1/2$, the logarithmic radii and their gaps satisfy

$$\begin{aligned} |t_q| &\leq 2|q|N^{-1-\kappa} \leq 4N^{-1-\kappa}, \\ t_{q+1} - t_q &= \int_{1+qN^{-1-\kappa}}^{1+(q+1)N^{-1-\kappa}} \frac{du}{u} \geq \frac{N^{-1-\kappa}}{1 + (q+1)N^{-1-\kappa}} \geq \frac{N^{-1-\kappa}}{2}, \quad q = -2, -1, 0, 1. \end{aligned}$$

By the hypothesis at $t_{-2}, \dots, t_2$, in which $\gamma/2$ cancels in differences, by (4) and since $\nu_N(e^t)$ is nondecreasing in t , for $q = -1, 0, 1$ we have

$$\begin{aligned} \frac{g(t_q) - g(t_{q-1}) - 2\varepsilon}{N(t_q - t_{q-1})} &\leq \frac{J_N(e^{t_q}) - J_N(e^{t_{q-1}})}{N(t_q - t_{q-1})} = \frac{1}{N(t_q - t_{q-1})} \int_{t_{q-1}}^{t_q} \nu_N(e^t) dt \\ &\leq \frac{\nu_N(1 + qN^{-1-\kappa})}{N} \leq \frac{1}{N(t_{q+1} - t_q)} \int_{t_q}^{t_{q+1}} \nu_N(e^t) dt \\ &= \frac{J_N(e^{t_{q+1}}) - J_N(e^{t_q})}{N(t_{q+1} - t_q)} \leq \frac{g(t_{q+1}) - g(t_q) + 2\varepsilon}{N(t_{q+1} - t_q)}. \end{aligned}$$

Let $[a, b]$ be $[t_{q-1}, t_q]$ or $[t_q, t_{q+1}]$. Since $g'(0) = N/2$, and $|g'(t) - g'(0)| \leq \frac{N^2}{2}|t|$ by (7) and the mean value theorem, with $|t| \leq 4N^{-1-\kappa}$ on $[a, b]$, we have

$$\begin{aligned} \left| \frac{g(b) - g(a)}{N(b - a)} - \frac{1}{2} \right| &= \frac{1}{N(b - a)} \left| \int_a^b (g'(t) - g'(0)) dt \right| \\ &\leq \frac{1}{N(b - a)} \int_a^b |g'(t) - g'(0)| dt \leq \frac{1}{N(b - a)} \int_a^b \frac{N^2}{2} |t| dt \\ &\leq \frac{N}{2} \cdot 4N^{-1-\kappa} = 2N^{-\kappa}. \end{aligned}$$

Moreover, since $b - a \geq N^{-1-\kappa}/2$, we have $2\varepsilon/(N(b - a)) \leq 4\varepsilon N^\kappa$. By the secant bounds and the last two estimates, we have

$$\begin{aligned} \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} &\leq \left(\frac{g(t_{q+1}) - g(t_q)}{N(t_{q+1} - t_q)} - \frac{1}{2} \right) + \frac{2\varepsilon}{N(t_{q+1} - t_q)} \leq 2N^{-\kappa} + 4\varepsilon N^\kappa, \\ \frac{1}{2} - \frac{\nu_N(1 + qN^{-1-\kappa})}{N} &\leq \left(\frac{1}{2} - \frac{g(t_q) - g(t_{q-1})}{N(t_q - t_{q-1})} \right) + \frac{2\varepsilon}{N(t_q - t_{q-1})} \leq 2N^{-\kappa} + 4\varepsilon N^\kappa. \end{aligned}$$

□

3 Logarithmic concentration for polynomials with random signs

We now estimate $J_N(r) - \log \sigma_N(r)$ for the polynomials with random signs. Let $X_r(\theta) = f_N(re^{i\theta})/\sigma_N(r)$, so that $J_N(r) - \log \sigma_N(r) = \int \log |X_r| d\mu$. Recall that G has density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$. We fix $K \geq 2$ before letting N tend to infinity. Exact bounds and sufficient degree thresholds are collected in Appendix A.

Theorem 3.1 (Concentration of the angular logarithm). *Fix $K \geq 2$. For all sufficiently large N , uniformly for $r \in [1 - K/N, 1 + K/N]$,*

$$(8) \quad \begin{aligned} & \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > C(K)N^{-1/32}(\log N)^7\} \\ & \leq C(K)N^{-3/8} + N^{-12} + C(K)N^{-7/16}(\log N)^{-12}. \end{aligned}$$

Here the constants and the degree threshold depend only on K . In particular, these probabilities are summable along $N = j^8$, since there $N^{-3/8} = j^{-3}$ and $N^{-7/16} = j^{-7/2}$.

Theorem 3.1 provides, at every radius of the annulus $[1 - K/N, 1 + K/N]$, the logarithmic bounds that Lemmas 2.4 and 2.5 require (Corollary 3.3). We prove Theorem 3.1 at the end of Section 3.2. In Section 3.1 we use one- and two-point Gaussian approximations to control the occupation of small disks by X_r , and in Section 3.2 we use the logarithmic integrability of Nazarov–Nishry–Sodin to control the contribution near zero.

Figure 5 illustrates the theorem for one nested sequence.

3.1 One- and two-point bounds

We show that, for radii $r, s \in [1 - K/N, 1 + K/N]$ and for angles θ, ϕ outside small exceptional sets, the probabilities that $X_r(\theta)$ lies in a disk centered at 0 and that $(X_r(\theta), X_s(\phi))$ lies in a product of two such disks differ by $O_K(N^{-1/2})$ from those for one and for two independent copies of G . Summation by parts shows that the covariance matrices are close to $\frac{1}{2}I$, Raič's theorem and an interpolation of densities compare the laws with that of G , and we then deduce bounds for the mean and the variance of the angular occupation of a disk.

By the definition of X_r ,

$$\begin{pmatrix} \Re X_r(\theta) \\ \Im X_r(\theta) \end{pmatrix} = \sum_{k=0}^N \epsilon_k \frac{r^k}{\sigma_N(r)} \begin{pmatrix} \cos(k\theta) \\ \sin(k\theta) \end{pmatrix}.$$

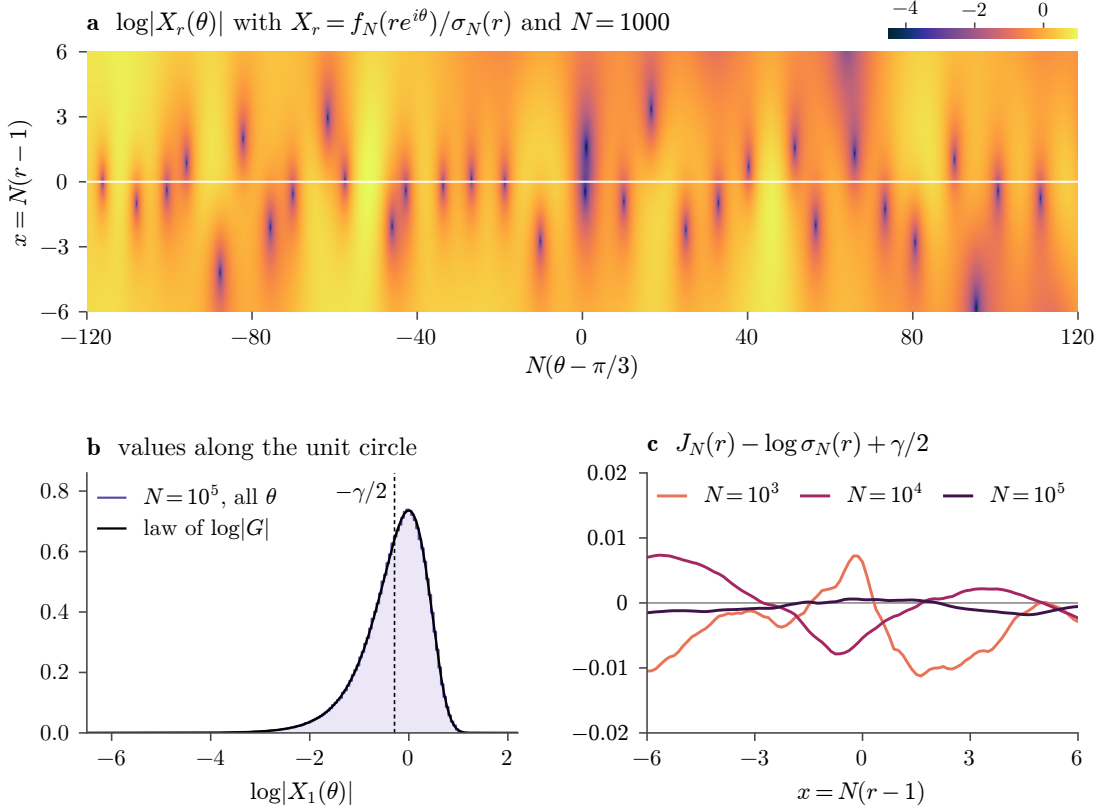


Figure 5: The angular logarithm for one sequence near $e^{i\pi/3}$, with the zeros as dark spots (a), its distribution (b), and its centered mean at three degrees (c).

Since the signs are independent with mean zero and variance one, the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is given by

$$V(r, \theta) = \frac{1}{\sigma_N(r)^2} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos^2(k\theta) & \cos(k\theta) \sin(k\theta) \\ \cos(k\theta) \sin(k\theta) & \sin^2(k\theta) \end{pmatrix}.$$

For two points, the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta), \Re X_s(\phi), \Im X_s(\phi))$ is the 4×4 matrix

$$\begin{pmatrix} V(r, \theta) & V(r, s, \theta, \phi) \\ V(r, s, \theta, \phi)^T & V(s, \phi) \end{pmatrix},$$

where

$$V(r, s, \theta, \phi) = \frac{1}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k \begin{pmatrix} \cos(k\theta) \cos(k\phi) & \cos(k\theta) \sin(k\phi) \\ \sin(k\theta) \cos(k\phi) & \sin(k\theta) \sin(k\phi) \end{pmatrix},$$

so that $V(r, \theta) = V(r, r, \theta, \theta)$. By the product-to-sum identities with $x, y \in \{k\theta, k\phi\}$, and since $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$, we get

$$V(r, \theta) = \frac{1}{2} I_2 + \frac{1}{2\sigma_N(r)^2} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos(2k\theta) & \sin(2k\theta) \\ \sin(2k\theta) & -\cos(2k\theta) \end{pmatrix},$$

$$V(r, s, \theta, \phi) = \frac{1}{2\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k \times \begin{pmatrix} \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)) & \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)) \\ \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)) \end{pmatrix}.$$

The oscillatory sums in these blocks, in the angles 2θ , 2ϕ , $\theta - \phi$ and $\theta + \phi$, are small unless one of these angles is close to $2\pi\mathbb{Z}$, and we remove the angles where this happens.

For one point, we remove the set $\text{dist}(\theta, \pi\mathbb{Z}) < N^{-1/2}$, two arcs of length $2N^{-1/2}$ centered at 0 and π , of normalized measure $4N^{-1/2}/(2\pi) = 2N^{-1/2}/\pi \leq 2N^{-1/2}$. Every retained angle satisfies $\text{dist}(2\theta, 2\pi\mathbb{Z}) = 2\text{dist}(\theta, \pi\mathbb{Z}) \geq 2N^{-1/2}$. For two points, we remove this set in each coordinate, together with the sets $\text{dist}(\theta - \phi, 2\pi\mathbb{Z}) < N^{-1/2}$ and $\text{dist}(\theta + \phi, 2\pi\mathbb{Z}) < N^{-1/2}$. For fixed ϕ each of the last two defines an arc of θ of normalized length $N^{-1/2}/\pi$, so that in the angular square the union of all the removed sets has measure at most $2N^{-1/2} + 2N^{-1/2} + 2N^{-1/2}/\pi \leq 10N^{-1/2}$.

The entries of these matrices are real and imaginary parts of weighted exponential sums, which we bound at the retained angles by summation by parts. Let $r, s \in [1 - K/N, 1 + K/N]$, $N \geq 2K$ and $\beta \in \{0, 1, 2\}$, and for $\text{dist}(\psi, 2\pi\mathbb{Z}) > 0$ set

$$b_k = (rs)^k (k/N)^\beta, \quad S_\ell(\psi) = \sum_{k=0}^{\ell} e^{ik\psi},$$

where $(k/N)^0 = 1$, including $k = 0$, and $S_{-1}(\psi) = 0$. Summing the geometric series, and since $|1 - e^{i\psi}| = 2|\sin(\psi/2)| \geq 2\text{dist}(\psi, 2\pi\mathbb{Z})/\pi$, we get

$$S_\ell(\psi) = \frac{1 - e^{i(\ell+1)\psi}}{1 - e^{i\psi}}, \quad |S_\ell(\psi)| \leq \frac{2}{|1 - e^{i\psi}|} \leq \frac{\pi}{\text{dist}(\psi, 2\pi\mathbb{Z})}.$$

Summation by parts yields

$$\sum_{k=0}^N b_k e^{ik\psi} = \sum_{k=0}^N b_k (S_k(\psi) - S_{k-1}(\psi)) = b_N S_N(\psi) + \sum_{k=0}^{N-1} (b_k - b_{k+1}) S_k(\psi).$$

For $\beta = 0$ the weights are geometric. For $\beta > 0$ we have $b_0 = 0$ and $b_{k+1}/b_k = (rs)(1 + 1/k)^\beta$ for $k \geq 1$, and these ratios are nonincreasing, which implies that the weights first increase and then decrease, with either part possibly empty. Let $b_* = \max_{k \leq N} b_k$. Since $b_k \leq (rs)^k \leq \max(1, rs)^N$ with $rs \leq (1 + K/N)^2$, and since the variation of weights that first increase and then decrease is the rise from b_0 to b_* plus the fall from b_* to b_N , we have

$$b_* \leq (1 + K/N)^{2N} \leq e^{2K},$$

$$\begin{aligned} \sum_{k=0}^{N-1} |b_{k+1} - b_k| &= 2b_* - b_0 - b_N, \\ b_N + \sum_{k=0}^{N-1} |b_{k+1} - b_k| &\leq b_0 + b_N + 2b_* \leq 4e^{2K}. \end{aligned}$$

Consequently, since $4\pi \leq 10e^{2K}$ for $K \geq 2$,

$$\begin{aligned} (9) \quad \left| N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik\psi} \right| &\leq \frac{\pi}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})} \left(b_N + \sum_{k=0}^{N-1} |b_{k+1} - b_k| \right) \\ &\leq \frac{4\pi e^{2K}}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})} \leq \frac{10e^{4K}}{N \operatorname{dist}(\psi, 2\pi\mathbb{Z})}. \end{aligned}$$

We keep the larger constant $10e^{4K}$, from which the later bounds are computed.

If $N \geq 2K$ and $r \in [1 - K/N, 1 + K/N]$, then, since $K/N \leq 1/2$ and $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$, each term of $\sigma_N(r)^2$ satisfies $e^{-4K} \leq (1 - K/N)^{2N} \leq r^{2k} \leq (1 + K/N)^{2N} \leq e^{2K}$, $0 \leq k \leq N$. Summing these $N + 1$ terms, we get the radial bounds

$$(10) \quad Ne^{-4K} \leq (N+1)e^{-4K} \leq \sigma_N(r)^2 \leq (N+1)e^{2K} \leq 2Ne^{2K}.$$

At the retained angles, every entry of $V(r, \theta) - \frac{1}{2}I_2$ is $N/(2\sigma_N(r)^2)$ times the real or the imaginary part of the sum in (9) with $s = r$, $\beta = 0$ and $\psi = 2\theta$, and every entry of $V(r, s, \theta, \phi)$ is $N/(2\sigma_N(r)\sigma_N(s))$ times the real or the imaginary part of that sum with $\beta = 0$ and $\psi = \theta - \phi$, plus or minus that of the same sum with $\psi = \theta + \phi$. By (9), the lower bound in (10) and the retained distances, these entries are at most, in absolute value,

$$\begin{aligned} (11) \quad \frac{N}{2\sigma_N(r)^2} \left| \frac{1}{N} \sum_{k=0}^N r^{2k} e^{2ik\theta} \right| &\leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot 2N^{-1/2}} \leq 10e^{8K} N^{-1/2}, \\ \frac{N}{2\sigma_N(r)\sigma_N(s)} \sum_{\pm} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta \pm \phi)} \right| &\leq \frac{e^{4K}}{2} \cdot 2 \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 10e^{8K} N^{-1/2}. \end{aligned}$$

The first bound also holds for the entries of $V(s, \phi) - \frac{1}{2}I_2$. Since the operator norm of a $d \times d$ matrix is at most d times its largest entry in absolute value, it follows that, at the retained angles, $V(r, \theta) - \frac{1}{2}I_2$ and the 4×4 covariance matrix minus $\frac{1}{2}I_4$ have operator norm at most $40e^{8K} N^{-1/2}$. In particular, for all sufficiently large N , depending only on K , every eigenvalue lies in $[1/4, 3/4]$.

Raič's theorem [17, Theorem 1.1] applies to independent centered real vectors with total covariance I_d , and it states that, for a measurable convex set D ,

$$\left| \mathbb{P}\left(\sum_k Y_k \in D\right) - \mathbb{P}(G_d \in D) \right| \leq (42d^{1/4} + 16) \sum_k \mathbb{E}\|Y_k\|^3,$$

where G_d is standard Gaussian in $\mathbb{R}^d$. More generally, for a covariance matrix Σ we write G_Σ for a centered Gaussian vector with covariance Σ , so that $G_d = G_{I_d}$. Distances and entropies between such vectors are those of their laws. For two values, the deterministic summands and their covariance are

$$u_k = \begin{pmatrix} r^k \cos(k\theta)/\sigma_N(r) \\ r^k \sin(k\theta)/\sigma_N(r) \\ s^k \cos(k\phi)/\sigma_N(s) \\ s^k \sin(k\phi)/\sigma_N(s) \end{pmatrix}, \quad \Sigma = \sum_{k=0}^N u_k u_k^T,$$

so that $\sum_{k=0}^N \epsilon_k u_k$ is the column vector $(\Re X_r(\theta), \Im X_r(\theta), \Re X_s(\phi), \Im X_s(\phi))^T$ and Σ is its covariance matrix, the 4×4 matrix above. Let $Y_k = \epsilon_k \Sigma^{-1/2} u_k$. Since the signs are independent and centered, with $\mathbb{E} \epsilon_k^2 = 1$, and $\|Y_k\| \leq \|\Sigma^{-1/2}\| \|u_k\| \leq 2\|u_k\|$, because every eigenvalue of Σ is at least $1/4$, we have, by $r^{2k}, s^{2k} \leq e^{2K}$ and the lower bound in (10),

$$\begin{aligned} \sum_{k=0}^N \mathbb{E}[Y_k Y_k^T] &= \Sigma^{-1/2} \left(\sum_{k=0}^N u_k u_k^T \right) \Sigma^{-1/2} = I_4, \\ \|u_k\|^2 &= \frac{r^{2k}}{\sigma_N(r)^2} + \frac{s^{2k}}{\sigma_N(s)^2} \leq \frac{2e^{6K}}{N}, \\ \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 &\leq (N+1) \left(\frac{2\sqrt{2}e^{3K}}{\sqrt{N}} \right)^3 \leq 32\sqrt{2}e^{9K}N^{-1/2} < 46e^{9K}N^{-1/2}. \end{aligned}$$

For one point, we use only the first two coordinates, with $\Sigma = V(r, \theta)$, and we obtain the smaller bound

$$\|u_k\|^2 = \frac{r^{2k}}{\sigma_N(r)^2} \leq \frac{e^{6K}}{N}, \quad \sum_{k=0}^N \mathbb{E}\|Y_k\|^3 \leq (N+1) \left(\frac{2e^{3K}}{\sqrt{N}} \right)^3 \leq 16e^{9K}N^{-1/2}.$$

Disks and products of disks, and their images under whitening, are measurable and convex.

To compare the resulting Gaussian with the law of $G_{I_d/2}$, let $\Delta = \Sigma - \frac{1}{2}I_d$, where $d = 2$ or 4 , so that $\|\Delta\| \leq 40e^{8K}N^{-1/2}$ by the operator norm bound above. Let p_λ be the centered Gaussian density with covariance $\Sigma_\lambda = \frac{1}{2}I_d + \lambda\Delta$, $0 \leq \lambda \leq 1$. Then every Σ_λ has eigenvalues in $[1/4, 3/4]$, since $\|\lambda\Delta\| \leq \|\Delta\| \leq 1/4$ for all sufficiently large N . Differentiating the density and integrating, we obtain

$$\begin{aligned} \partial_\lambda p_\lambda(x) &= \frac{p_\lambda(x)}{2} \left\{ x^T \Sigma_\lambda^{-1} \Delta \Sigma_\lambda^{-1} x - \text{tr}(\Sigma_\lambda^{-1} \Delta) \right\}, \\ \int_{\mathbb{R}^d} |\partial_\lambda p_\lambda(x)| dx &= \frac{1}{2} \mathbb{E} \left| G_d^T \Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2} G_d - \text{tr}(\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}) \right| \\ &\leq d \|\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}\| \leq 4d \|\Delta\|, \end{aligned}$$

$$(12) \quad \text{TV}(G_\Sigma, G_{I_d/2}) \leq \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\partial_\lambda p_\lambda(x)| \, dx \, d\lambda \leq 2d \|\Delta\| \leq 320e^{8K} N^{-1/2}.$$

The first line follows from $\partial_\lambda \log \det \Sigma_\lambda = \text{tr}(\Sigma_\lambda^{-1} \Delta)$ and $\partial_\lambda \Sigma_\lambda^{-1} = -\Sigma_\lambda^{-1} \Delta \Sigma_\lambda^{-1}$, and the second from the change of variables $x \mapsto \Sigma_\lambda^{-1/2} x$, which carries the law with density p_λ to that of G_d . In an orthonormal eigenbasis of the symmetric matrix $\Sigma_\lambda^{-1/2} \Delta \Sigma_\lambda^{-1/2}$, whose d eigenvalues are at most its norm in absolute value, the squared coordinates of G_d have mean one, so that the quadratic form and the trace have mean absolute value at most d times the norm. The last line follows from $p_1 - p_0 = \int_0^1 \partial_\lambda p_\lambda \, d\lambda$, since the total variation distance is half the L^1 distance between the densities.

Let $F_G(t) = \mathbb{P}(|G| \leq t) = 1 - e^{-t^2}$. For $t, t' \geq 0$, let D be the product of disks $\{x \in \mathbb{R}^4 : x_1^2 + x_2^2 \leq t^2, x_3^2 + x_4^2 \leq t'^2\}$ when $d = 4$, and the disk $\{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq t^2\}$ when $d = 2$. Since $\sum_k \epsilon_k u_k = \Sigma^{1/2} \sum_k Y_k$ and $\Sigma^{1/2} G_d$ has the law of G_Σ , we have $\mathbb{P}(\sum_k \epsilon_k u_k \in D) = \mathbb{P}(\sum_k Y_k \in \Sigma^{-1/2} D)$ and $\mathbb{P}(G_\Sigma \in D) = \mathbb{P}(G_d \in \Sigma^{-1/2} D)$, so that, by Raič's theorem with the third-moment bounds above and by (12), for retained angles and pairs,

$$(13) \quad \begin{aligned} & \left| \mathbb{P}(\sum_k \epsilon_k u_k \in D) - \mathbb{P}(G_{I_d/2} \in D) \right| \\ & \leq \left| \mathbb{P}(\sum_k Y_k \in \Sigma^{-1/2} D) - \mathbb{P}(G_d \in \Sigma^{-1/2} D) \right| + \left| \mathbb{P}(G_\Sigma \in D) - \mathbb{P}(G_{I_d/2} \in D) \right| \\ & \leq (42d^{1/4} + 16) \cdot 46e^{9K} N^{-1/2} + 320e^{8K} N^{-1/2} \leq C(K) N^{-1/2}. \end{aligned}$$

For $d = 4$ the left-hand side is $|\mathbb{P}(|X_r(\theta)| \leq t, |X_s(\phi)| \leq t') - F_G(t)F_G(t')|$, since the two coordinate pairs of $G_{I_4/2}$ are independent copies of G , and for $d = 2$ it is $|\mathbb{P}(|X_r(\theta)| \leq t) - F_G(t)|$.

Let $A_t = \mu\{\theta : |X_r(\theta)| \leq t\}$ be the angular occupation. By Tonelli's theorem,

$$\begin{aligned} \mathbb{E}A_t &= \int \mathbb{P}(|X_r(\theta)| \leq t) \, d\mu(\theta), \\ \mathbb{E}A_t^2 &= \iint \mathbb{P}(|X_r(\theta)| \leq t, |X_r(\phi)| \leq t) \, d\mu(\theta) \, d\mu(\phi). \end{aligned}$$

On the removed sets, of measures at most $2N^{-1/2}$ and $10N^{-1/2}$, the integrands differ from $F_G(t)$ and $F_G(t)^2$ by at most one, and on the retained angles and pairs we use (13), with $s = r$ and $t' = t$ for the pairs. Hence, uniformly for $t \geq 0$, since $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| &\leq 2N^{-1/2} + C(K)N^{-1/2} \leq C(K)N^{-1/2}, \\ |\mathbb{E}A_t^2 - F_G(t)^2| &\leq 10N^{-1/2} + C(K)N^{-1/2} \leq C(K)N^{-1/2}, \\ \text{Var}(A_t) &= \mathbb{E}A_t^2 - F_G(t)^2 - (\mathbb{E}A_t - F_G(t))(\mathbb{E}A_t + F_G(t)) \end{aligned}$$

$$(14) \quad \leq |\mathbb{E}A_t^2 - F_G(t)^2| + |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \leq C(K)N^{-1/2}.$$

The two-point argument also permits different radii and different disk thresholds.

3.2 Logarithmic integrability and angular energy

The normalized Fourier coefficients satisfy $\sum_{k=0}^N |r^k/\sigma_N(r)|^2 = 1$. By Corollary 1.2 of Nazarov–Nishry–Sodin [16], which uses normalization in $L^2(\Omega \times \mathbb{T})$, we have

$$(15) \quad \mathbb{E} \int |\log |X_r||^p d\mu \leq (C_* p)^{6p}, \quad p \geq 1,$$

where $C_* \geq 1$ is an absolute constant, which does not depend on the normalized coefficients. Here zeros on the circle give only integrable logarithmic singularities. By Parseval's identity, the angular energy satisfies, for every choice of signs,

$$(16) \quad \int |X_r|^2 d\mu = \frac{1}{\sigma_N(r)^2} \sum_{k=0}^N \epsilon_k^2 r^{2k} = 1.$$

We clip the logarithm at $\pm T$, control the mean and the variance of the clipped integral by the occupation bounds, and control the two tails by the logarithmic moments and the angular energy outside exceptional events. Table 1 lists the three quantities that depend on the clipping level T and what each requires at $T = c \log N$. The first and the last quantities tend to zero for every $c > 0$, so

Quantity	Bound	At $T = c \log N$	Requirement
Gaussian mass of $\{ z \leq e^{-T}\}$	$F_G(e^{-T}) \leq e^{-2T}$	N^{-2c}	tends to zero
Chebyshev bound for $A_{e^{-T}}$	$C(K)N^{-1/2}e^{4T}$	$C(K)N^{4c-1/2}$	summable in j
Lower tail ($e^{-2T} \geq N^{-1/2}$)	order $(\log N)^7 e^{-T}$	$(\log N)^7 N^{-c}$	tends to zero

Table 1: The quantities in the clipping argument of Lemma 3.2 that depend on the clipping level $T = c \log N$, for signs, where $H = C(K)$. The third column holds for the last row when $c \leq 1/4$. At $N = j^8$ the second is $C(K)j^{32c-4}$, which is summable when $32c - 4 < -1$, that is, $c < 3/32$.

that any c in $(0, 3/32)$ would do, and we take $c = 1/32$, which makes the lower tail $(\log N)^7 N^{-1/32}$, the error term of Theorem 3.1. Then $T = (\log N)/32$, so that $e^{-2T} = N^{-1/16} \geq N^{-1/2}$, $e^{-T} = N^{-1/32}$ and $N^{-1/2}e^{4T} = N^{-3/8}$, which is j^{-3} at $N = j^8$. We state the argument for coefficients that need not be signs, since Section 7 applies it to other coefficient distributions.

Theorem B.1, which we apply below, uses two facts about the standard complex Gaussian variable G . For every $T \geq 0$, since $F_G(e^{-T}) \leq e^{-2T}$ and $\mathbb{P}(|G| > e^T) \leq$

$$e^{-2\tau} \mathbb{E}|G|^2 = e^{-2\tau},$$

$$(17) \quad \int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau \leq \frac{1}{2}e^{-2T}, \quad \int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau \leq \frac{1}{2}e^{-2T}.$$

Since $|G|^2$ has the exponential distribution with mean one, the classical integral for Euler's constant gives

$$(18) \quad \mathbb{E} \log |G| = \frac{1}{2} \mathbb{E} \log(|G|^2) = \frac{1}{2} \int_0^\infty e^{-y} \log y dy = -\frac{\gamma}{2}.$$

Lemma 3.2 (Clipping). *Let $N \geq e$ and $r > 0$, and let X_r and A_t be defined as above for a random polynomial f_N whose coefficients need not be signs. Let $H \geq 0$, $C > 0$ and $B_E > 0$ be constants and E an event such that, for all $t \geq 0$ and $p \geq 1$,*

$$|\mathbb{E}A_t - F_G(t)| \leq HN^{-1/2}, \quad \text{Var}(A_t) \leq HN^{-1/2},$$

$$\int_E \int |\log |X_r||^p d\mu d\mathbb{P} \leq (Cp)^{6p},$$

and $\int |X_r|^2 d\mu \leq B_E$ on E . Let $T = (\log N)/32$ and

$$\begin{aligned} \varepsilon &= N^{-1/32}(\log N)^7 + 2HTN^{-1/2} \\ &\quad + (4eC)^6 \sqrt{2+H} N^{-1/32}(\log N)^7 + (B_E/2 + 1)N^{-1/16} \\ &\leq (2 + 2H + (4eC)^6 \sqrt{2+H} + B_E/2) N^{-1/32}(\log N)^7. \end{aligned}$$

Then

$$\mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon\} \leq HN^{-3/8} + N^{-12} + \frac{H}{256} N^{-7/16} (\log N)^{-12} + \mathbb{P}(E^c).$$

Proof. This is Theorem B.1 in the variant form given at the end of its statement, with $X = X_r$, $d_1 = v = HN^{-1/2}$, $\beta = 6$, $E_H = E$, $T = (\log N)/32$, $u = N^{-1/32}(\log N)^7$, $b = N^{-1/16}$, $q = \lceil \log N \rceil$ and $\lambda = (2e \log N/q)^6 \log N$. Its hypotheses hold, since the occupation bounds of the lemma are the bounds on $\mathbb{E}A_t$ and $\text{Var}(A_t)$ of that form, the moment bound of the lemma is (69) with $E_H = E$ and $\beta = 6$, and $\int |X_r|^2 d\mu \leq B_E$ on E . For $N \geq e$ we have $\log N \leq q \leq \log N + 1 \leq 2 \log N$, so that $\lambda \geq e^6 \log N > 1$,

$$\lambda(2Cq)^6 = \left(\frac{2e \log N}{q}\right)^6 (\log N)(2Cq)^6 = (4eC)^6 (\log N)^7$$

and

$$\lambda^{-2q} = \left[\left(\frac{q}{2e \log N}\right)^{12} (\log N)^{-2} \right]^q \leq [e^{-12} (\log N)^{-2}]^q \leq N^{-12},$$

where the last two steps use $q \leq 2 \log N$, $\log N \geq 1$ and $q \geq \log N$. By the first identity, the Markov event for U in the proof of Theorem B.1 is $E \cap \{U >$

$(4eC)^{12}(\log N)^{14}\}$, where $U = \int |\log |X_r||^2 d\mu$. Since $e^{-2T} = N^{-1/16}$ and $N^{-1/2} \leq N^{-1/16}$,

$$\sqrt{e^{-2T} + d_1 + b} = \sqrt{2N^{-1/16} + HN^{-1/2}} \leq \sqrt{2 + H} N^{-1/32},$$

so that the error $\eta_{\log}$ of (72) is at most ε . The first three terms of (73) are $v/b^2 = HN^{-1/2}N^{1/8} = HN^{-3/8}$, $\lambda^{-2q} \leq N^{-12}$ and $4T^2v/u^2 = 4HT^2N^{-7/16}(\log N)^{-14} = (H/256)N^{-7/16}(\log N)^{-12}$, since $4T^2 = (\log N)^2/256$, and $p_E + p_H$ is replaced by $\mathbb{P}(E^c)$. Since $J_N(r) - \log \sigma_N(r) = \int \log |X_r| d\mu$ and $\mathbb{E} \log |G| = -\gamma/2$ by (18), this is the bound of the lemma. The bound on ε in the statement uses $TN^{-1/2} \leq N^{-1/32}(\log N)^7$ and $N^{-1/16} \leq N^{-1/32}(\log N)^7$ for $N \geq e$. $\square$

Proof of Theorem 3.1. By (14), (15) and (16), Lemma 3.2 applies with H the constant $C(K)$ of (14), $C = C_*$, $E = \Omega$ and $B_E = 1$. Since C_* is an absolute constant, its error ε is at most $C(K)N^{-1/32}(\log N)^7$, and, since $\mathbb{P}(E^c) = 0$, its probability bound is at most the right-hand side of (8). All constants above depend only on K , and all bounds hold uniformly for $r \in [1 - K/N, 1 + K/N]$. $\square$

3.3 Radial mass and thin bands with high probability

Corollary 3.3 (Annular mass and thin bands). *Fix $K \geq 2$ and $0 < \kappa \leq 1/64$, and take N sufficiently large. Outside four events, each of probability at most the right-hand side of (8),*

$$(19) \quad \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} \leq \frac{2 \log 2}{K} + C(K)N^{-1/32}(\log N)^7.$$

Outside five further events of the same kind, for $q = -1, 0, 1$,

$$(20) \quad \left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + C(K)N^{\kappa-1/32}(\log N)^7.$$

By Theorem 3.1, the sum of the probabilities of these nine exceptional events is summable at $N = j^8$.

Proof. We apply the deterministic Lemmas 2.4 and 2.5 outside the exceptional events of Theorem 3.1, at the radii that these lemmas use. Their polynomial hypothesis holds, since $f_N(0) = \epsilon_0 \neq 0$.

The four radii $r_i = 1 + x_i/N$ of Lemma 2.4 lie in $[1 - K/N, 1 + K/N]$, and hence, by Theorem 3.1, the first hypothesis of Lemma 2.4 holds outside the four logarithmic exceptional events, with $\varepsilon = C(K)N^{-1/32}(\log N)^7$. By Lemma 2.3, which applies since $|x_i| \leq K$ and $N \geq 2K$, the second hypothesis holds with

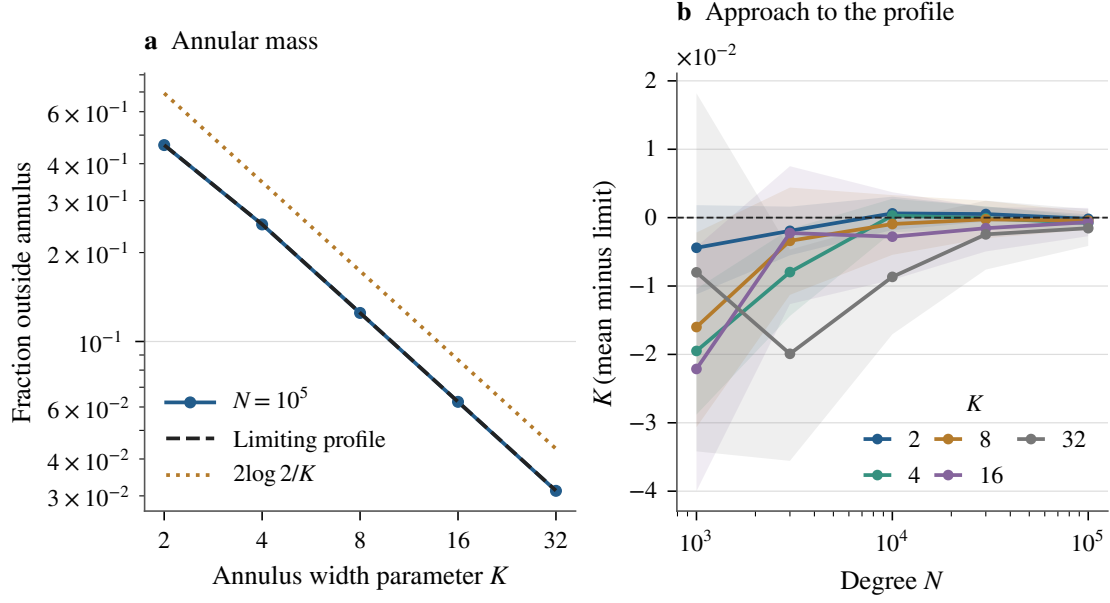


Figure 6: The fraction of zeros outside the annulus of width K/N .

$D = K^2 + e^{4K}/2$, and $\varepsilon + D/N \leq K/8$ for all sufficiently large N . Therefore Lemma 2.4 and $1/N \leq N^{-1/32}(\log N)^7$ for $N \geq 3$ yield

$$\begin{aligned} \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(C(K) N^{-1/32} (\log N)^7 + \frac{K^2 + e^{4K}/2}{N} \right) \\ &\leq \frac{2 \log 2}{K} + \frac{8}{K} \left(C(K) + K^2 + \frac{e^{4K}}{2} \right) N^{-1/32} (\log N)^7 \\ &\leq \frac{2 \log 2}{K} + C(K) N^{-1/32} (\log N)^7, \end{aligned}$$

which is (19).

The five radii $1 + qN^{-1-\kappa}$, $q = -2, \dots, 2$, satisfy $|qN^{-1-\kappa}| \leq 2N^{-1-\kappa} \leq K/N$, since $N^{-\kappa} \leq 1$ and $K \geq 2$, so that they lie in the same annulus. Outside the five logarithmic exceptional events at these radii, the hypothesis of Lemma 2.5 holds with $\varepsilon = C(K)N^{-1/32}(\log N)^7$, and hence the lemma, which applies for $N \geq 4$, bounds the left-hand side of (20) by $2N^{-\kappa} + 4C(K)N^{-1/32}(\log N)^7N^\kappa = 2N^{-\kappa} + 4C(K)N^{\kappa-1/32}(\log N)^7$. Both terms tend to zero, since $\kappa \leq 1/64 < 1/32$. $\square$

Figure 6 compares the fraction of roots outside the annulus with $2 \log 2/K$, the part of (19) that does not tend to zero. Panel (a) also shows the limit $1 - \{\Phi(K) - \Phi(-K)\}$ of the fraction, and panel (b) shows K times its difference from this limit, with pointwise 95% bootstrap bands, both over 77 sequences.

4 Local zeros and small derivatives

4.1 Local zero counts

We will use the Hanson–Wright inequality of Rudelson–Vershynin [19, Theorem 1.1]: if ξ is a random vector with independent centered real coordinates whose ψ_2 norm is at most Λ , M is a real matrix and $t > 0$, then

$$\mathbb{P}\{|\xi^T M \xi - \mathbb{E} \xi^T M \xi| > t\} \leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{t^2}{\Lambda^4 \|M\|_{\text{HS}}^2}, \frac{t}{\Lambda^2 \|M\|} \right\} \right],$$

where $c_{\text{HW}} > 0$ is an absolute constant. Rudelson and Vershynin use the moment convention for ψ_2 , and signs satisfy both it and the equivalent exponential-moment convention with $\Lambda = 2$, since $p^{-1/2}(\mathbb{E}|\epsilon_k|^p)^{1/p} = p^{-1/2} \leq 2$ for $p \geq 1$ and $\mathbb{E} \exp(\epsilon_k^2/2^2) = e^{1/4} \leq 2$. Hence, for independent uniform signs $\epsilon = (\epsilon_0, \dots, \epsilon_N)$, a real $(N+1) \times (N+1)$ matrix M and $t > 0$,

$$(21) \quad \mathbb{P}\{|\epsilon^T M \epsilon - \mathbb{E} \epsilon^T M \epsilon| > t\} \leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{t^2}{16 \|M\|_{\text{HS}}^2}, \frac{t}{4 \|M\|} \right\} \right].$$

We first bound the local zero counts when the coefficients are independent signs times bounded amplitudes, and then we specialize this bound to signs.

Lemma 4.1 (Local zero counts for bounded amplitudes). *Fix $K \geq 2$, $B \geq 1$ and $\frac{1}{2} \leq \eta \leq 1$, and put*

$$A(B, \eta) = \max \left(1, \frac{1024\pi B^2}{c_{\text{HW}} \eta} \right), \quad t_N(B, \eta) = \frac{A(B, \eta) \log N}{N},$$

$$C_1(K, B, \eta) = \frac{4A(B, \eta) + 4K + 4 + \log B}{\log 2}.$$

Let $N \geq e$ with $t_N(B, \eta) \leq 1$. Let $\epsilon_0, \dots, \epsilon_N$ be independent uniform signs, and let $a_0, \dots, a_N$ be real or complex random variables, independent of the signs, with $|a_k| \leq B$ for every k . Let $\mathcal{E}$ be an event determined by $a_0, \dots, a_N$ on which

$$\sum_{k=0}^N |a_k|^2 \geq \eta(N+1),$$

and put $P_N(z) = \sum_{k=0}^N \epsilon_k a_k z^k$. Then the probability that $\mathcal{E}$ occurs and some disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains more than $C_1(K, B, \eta) \log N$ zeros of P_N is at most N^{-3} .

Proof. The idea is to show that the mean of $|P_N|^2$ over each arc of a fine angular grid is at least half its conditional expectation, and then to apply Jensen's formula at a point of the arc at which $|P_N|$ is large.

We condition on $a_0, \dots, a_N$ and assume that $\mathcal{E}$ occurs, so that the signs remain independent and uniform. We take $J = \lceil 2\pi/t_N(B, \eta) \rceil$ equally spaced angular centers, with the closed arc of radian length $2t_N(B, \eta)$ around each center. Since $t_N(B, \eta) \leq 1 < \pi$, every such arc I has $\mu(I) = t_N(B, \eta)/\pi$, and since $A(B, \eta) \geq 1$ and $\log N \geq 1$,

$$J \leq \frac{2\pi}{t_N(B, \eta)} + 1 = \frac{2\pi N}{A(B, \eta) \log N} + 1 \leq 2\pi N + 1 \leq 8N.$$

For each such arc I , the conditional real Gram matrix is defined by

$$(A_I)_{jk} = \int_I \Re(a_j \bar{a}_k e^{i(j-k)\theta}) d\mu(\theta), \quad 0 \leq j, k \leq N.$$

For every real vector x , since the x_k are real and by Parseval's identity on the whole circle, we have

$$\begin{aligned} 0 \leq x^T A_I x &= \int_I \left| \sum_{k=0}^N x_k a_k e^{ik\theta} \right|^2 d\mu(\theta) \leq \int \left| \sum_{k=0}^N x_k a_k e^{ik\theta} \right|^2 d\mu(\theta) \\ &= \sum_{k=0}^N x_k^2 |a_k|^2 \leq B^2 \|x\|^2. \end{aligned}$$

It follows that the eigenvalues of the real symmetric matrix A_I lie in $[0, B^2]$, and since they are nonnegative, we have

$$\begin{aligned} \|A_I\| &\leq B^2, \quad \|A_I\|_{\text{HS}}^2 \leq \|A_I\| \operatorname{tr} A_I \leq B^2 \operatorname{tr} A_I, \\ \operatorname{tr} A_I &= \mu(I) \sum_{k=0}^N |a_k|^2 \geq \eta(N+1)\mu(I) = \frac{\eta(N+1)t_N(B, \eta)}{\pi}, \end{aligned}$$

where the trace identity follows from $(A_I)_{kk} = |a_k|^2 \mu(I)$, and the inequality after it from $\mathcal{E}$. Taking $x = \epsilon$ in the identity above and using $\mathbb{E}\epsilon_j \epsilon_k = \mathbf{1}\{j = k\}$, we obtain

$$Q_I := \int_I |P_N(e^{i\theta})|^2 d\mu(\theta) = \epsilon^T A_I \epsilon, \quad \mathbb{E}_\epsilon Q_I = \operatorname{tr} A_I.$$

If $Q_I < \frac{1}{2} \operatorname{tr} A_I$, then $|Q_I - \mathbb{E}_\epsilon Q_I| > \frac{1}{2} \operatorname{tr} A_I$. By (21) with $M = A_I$ and $t = \frac{1}{2} \operatorname{tr} A_I$, and the bounds above, we have

$$\begin{aligned} \mathbb{P}_\epsilon \{Q_I < \tfrac{1}{2} \operatorname{tr} A_I\} &\leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{(\operatorname{tr} A_I)^2}{64 \|A_I\|_{\text{HS}}^2}, \frac{\operatorname{tr} A_I}{8 \|A_I\|} \right\} \right] \\ &\leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{\operatorname{tr} A_I}{64 B^2}, \frac{\operatorname{tr} A_I}{8 B^2} \right\} \right] = 2 \exp \left[-\frac{c_{\text{HW}} \operatorname{tr} A_I}{64 B^2} \right] \end{aligned}$$

$$\leq 2 \exp \left[-\frac{c_{\text{HW}} \eta (N+1) A(B, \eta) \log N}{64\pi B^2 N} \right] \leq 2e^{-16 \log N} = 2N^{-16},$$

where the last inequality holds since $A(B, \eta) \geq 1024\pi B^2 / (c_{\text{HW}} \eta)$. By the union bound over the at most $8N$ arcs, we get

$$\mathbb{P}_\epsilon \{Q_I < \tfrac{1}{2} \text{tr } A_I \text{ for some arc } I\} \leq (8N)(2N^{-16}) = 16N^{-15} \leq N^{-3}.$$

Taking the expectation over $a_0, \dots, a_N$ on $\mathcal{E}$, we see that the probability that $\mathcal{E}$ occurs and some arc has $Q_I < \tfrac{1}{2} \text{tr } A_I$ is at most N^{-3} .

It remains to bound the number of zeros when $\mathcal{E}$ occurs and every arc has $Q_I \geq \tfrac{1}{2} \text{tr } A_I$. In this case, averaging over each closed arc and using the lower bound on $\text{tr } A_I$, we get

$$\max_{\theta \in I} |P_N(e^{i\theta})|^2 \geq \frac{Q_I}{\mu(I)} \geq \frac{\text{tr } A_I}{2\mu(I)} \geq \frac{\eta(N+1)}{2}.$$

By continuity, there exists $z_1 = e^{i\theta_1}$ on the arc such that $|P_N(z_1)|^2 \geq \eta(N+1)/2$. Now let $||z_0| - 1| \leq 1/N$. Then the angular center nearest to $z_0/|z_0|$ is at angular distance at most $\pi/J \leq t_N(B, \eta)/2$ from it, and such a point z_1 on its arc is at angular distance at most $t_N(B, \eta)$ from the center, so that $|z_0/|z_0| - z_1| \leq 3t_N(B, \eta)/2$, since a chord is at most its arc. Let $R = (2K+1)/N + 2t_N(B, \eta)$. For $|z - z_0| \leq 2K/N$, since $|z_0 - z_0/|z_0|| = ||z_0| - 1| \leq 1/N$, we have

$$|z - z_1| \leq |z - z_0| + |z_0 - z_0/|z_0|| + |z_0/|z_0| - z_1| \leq \frac{2K}{N} + \frac{1}{N} + \frac{3t_N(B, \eta)}{2} < R,$$

so that the target disk lies strictly within $\{|z - z_1| < R\}$. For $|z - z_1| \leq 2R$ we have $|z| \leq 1 + 2R$, and hence

$$|P_N(z)| \leq \sum_{k=0}^N |a_k| |z|^k \leq B(N+1)(1+2R)^N \leq B(N+1)e^{2NR}.$$

Let $n(s)$ denote the number of zeros of P_N in $|z - z_1| \leq s$. Since $P_N(z_1) \neq 0$, Jensen's formula applies on the disk $|z - z_1| \leq 2R$, and using $2NR = 4K + 2 + 4A(B, \eta) \log N$ we get

$$\begin{aligned} \#\{\alpha : |\alpha - z_1| < R\} &\leq n(R) \leq \frac{1}{\log 2} \int_R^{2R} \frac{n(s)}{s} ds \\ &\leq \frac{\max_{|z - z_1| = 2R} \log |P_N(z)| - \log |P_N(z_1)|}{\log 2} \\ &\leq \frac{\log(B(N+1)e^{2NR}) - \frac{1}{2} \log(\eta(N+1)/2)}{\log 2} \end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{1}{2} \log(N+1) + \log B + \frac{1}{2} \log(2/\eta) + 2NR}{\log 2} \\
&= \frac{\frac{1}{2} \log(N+1) + \log B + \frac{1}{2} \log(2/\eta) + 4K + 2 + 4A(B, \eta) \log N}{\log 2} \\
&\leq \frac{\log N + (4K + 3 + \log B) \log N + 4A(B, \eta) \log N}{\log 2} = C_1(K, B, \eta) \log N.
\end{aligned}$$

The second inequality is the zero-count step of [22, §8.21], the third is Jensen's formula [22, §3.61] for $P_N(z_1 + z)$, and the last uses $\frac{1}{2} \log(N+1) \leq \log N$ for $N \geq 2$, $\frac{1}{2} \log(2/\eta) \leq \frac{1}{2} \log 4 = \log 2$ for $\eta \geq \frac{1}{2}$, and $\log B + \log 2 + 4K + 2 \leq (4K + 3 + \log B) \log N$, which holds for $\log N \geq 1$ since $\log B \geq 0$ and $\log 2 < 1$. Since the containment is strict, this count includes the zeros on the boundary of the target disk, and since the angular cover is finite and does not depend on z_0 , the bound holds for every admissible center z_0 at once. $\square$

Lemma 4.2 (Local zero counts). *Fix $K \geq 2$ and take N sufficiently large. Except on an event of probability at most N^{-3} , every disk*

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C(K) \log N$ zeros of f_N .

The local-count constant is computed in Section A.4.

Proof. We apply Lemma 4.1 with $a_k = 1$ for every k , $B = 1$, $\eta = 1$ and $\mathcal{E} = \Omega$, which we may do since $\sum_{k=0}^N |a_k|^2 = N + 1$. Then $P_N = f_N$, and since $\log 1 = 0$, the constants of that lemma are

$$\begin{aligned}
A(1, 1) &= \max(1, 1024\pi/c_{\text{HW}}) = A, & t_N(1, 1) &= \frac{A \log N}{N} = t_N, \\
C_1(K, 1, 1) &= \frac{4A + 4K + 4}{\log 2} = C_1(K),
\end{aligned}$$

which are the constants of Section A.4. The conditions $N \geq e$ and $t_N \leq 1$ hold for all sufficiently large N , since $t_N \rightarrow 0$. This proves the lemma with $C(K) = C_1(K)$. $\square$

4.2 Derivative detection and small-ball estimates

In this subsection we fix $K \geq 2$, let $L = N^{1/64}$, and consider the annulus $||z| - 1| \leq K/N$.

Lemma 4.3 (Second derivative). *For every $N \geq \max\{2, (K+1)/3\}$, outside an event of probability at most N^{-10} ,*

$$(22) \quad \sup_{|z| \leq 1 + (K+1)/N} |f_N''(z)| \leq C(K) N^{5/2} \sqrt{\log N},$$

and one can take $C(K) = 100e^{K+4}$.

Proof. We first bound f_N'' at finitely many points of the circle $|z| = 1 + (K+1)/N$ by Hoeffding's inequality, then pass to the whole circle using a deterministic bound on f_N''' , and finally to the disk by the maximum principle. Bounds of this order for the supremum of a random trigonometric polynomial go back to Salem and Zygmund [12, Chapter 6].

We choose $\lceil N^3 \rceil \leq 2N^3$ equally spaced points on the boundary circle, and at each sample z we take $b = 100e^{K+4}N^{5/2}\sqrt{\log N}$. Since $|z| = 1 + (K+1)/N$ and $0 \leq 2k-4 \leq 2N$ for $2 \leq k \leq N$, we have $|z|^{2k-4} \leq (1 + (K+1)/N)^{2N} \leq e^{2K+2}$, and hence

$$\sum_{k=2}^N k^2(k-1)^2|z|^{2k-4} \leq e^{2K+2} \sum_{k=2}^N k^4 \leq 2e^{2K+2}N^5.$$

By Hoeffding's inequality [9, Theorem 2], applied to both tails, $\mathbb{P}\{|\sum_k \epsilon_k c_k| \geq t\} \leq 2\exp[-t^2/(2\sum_k c_k^2)]$ for real numbers c_k and $t > 0$. Applying this real bound to the real and imaginary parts $\Re f_N''(z) = \sum_{k=2}^N \epsilon_k k(k-1)\Re(z^{k-2})$ and $\Im f_N''(z) = \sum_{k=2}^N \epsilon_k k(k-1)\Im(z^{k-2})$, whose squared coefficients are at most $k^2(k-1)^2|z|^{2k-4}$, we get

$$\begin{aligned} \mathbb{P}\{|f_N''(z)| \geq b/2\} &\leq \mathbb{P}\left\{|\Re f_N''(z)| \geq \frac{b}{2\sqrt{2}}\right\} + \mathbb{P}\left\{|\Im f_N''(z)| \geq \frac{b}{2\sqrt{2}}\right\} \\ &\leq 4\exp\left[-\frac{(b/2)^2}{4\sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}}\right] \leq 4\exp\left[-\frac{b^2}{32e^{2K+2}N^5}\right] \\ &= 4\exp\left[-\frac{10000e^6}{32}\log N\right] \leq 4\exp[-100\log N] = 4N^{-100}, \end{aligned}$$

where the first step holds since $|f_N''(z)| \geq b/2$ implies that one of the two parts is at least $b/(2\sqrt{2})$ in absolute value.

Since $N \geq (K+1)/3$, the boundary circle has radius $1 + (K+1)/N \leq 4$ and length at most 8π , so that each point of it lies within arc length $4\pi/N^3$ of a sample. Moreover, we have the deterministic bound

$$\begin{aligned} \sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| &\leq \sup_{|z| \leq 1+(K+1)/N} \sum_{k=3}^N k(k-1)(k-2)|z|^{k-3} \\ &\leq N^3 \cdot N \cdot \left(1 + \frac{K+1}{N}\right)^N \leq e^{K+1}N^4. \end{aligned}$$

Integrating f_N''' along the arc from a point of the circle to its nearest sample, we see from this bound that the interpolation error is at most $4\pi e^{K+1}N$, which is less than $b/2$ for every $N \geq 2$, that is,

$$\frac{4\pi}{N^3} \sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| \leq \frac{4\pi}{N^3} e^{K+1}N^4 = 4\pi e^{K+1}N$$

$$< 50e^{K+4}N^{5/2}\sqrt{\log N} = \frac{b}{2},$$

since $4\pi < 50e^3\sqrt{\log 2}$ and $N^{3/2}\sqrt{\log N} \geq \sqrt{\log 2}$. By the union bound over the samples, the probability that $|f_N''(z)| \geq b/2$ for some sample z is at most $(2N^3)(4N^{-100}) = 8N^{-97} \leq N^{-10}$. Outside this event, every point ζ of the circle of radius $1 + (K+1)/N$ and its nearest sample z satisfy $|f_N''(\zeta)| \leq |f_N''(z)| + 4\pi e^{K+1}N$, so that $|f_N''| < b/2 + b/2 = b$ on the circle $|z| = 1 + (K+1)/N$, and hence, by the maximum principle, on the disk $|z| \leq 1 + (K+1)/N$. This proves (22) with the constant $100e^{K+4}$. $\square$

We will use the following inequalities for the block diagonal comparison of the paired jet covariance in the proof of Lemma 4.5. Devroye, Mehrabian and Reddad [5] give upper and lower bounds that match up to constant factors for the total variation distance between Gaussians with equal means, and we need only an upper bound.

Lemma 4.4 (An entropy comparison). *Let p, q be probability densities, with p absolutely continuous with respect to q , and set*

$$D(p\|q) = \int p \log(p/q), \quad \text{TV}(p, q) = \frac{1}{2} \int |p - q|.$$

Then

$$\text{TV}(p, q) \leq \sqrt{D(p\|q)/2}.$$

For positive-definite covariance matrices Σ, Σ_0 in dimension d , let G_Σ and G_{Σ_0} be centered Gaussian vectors with these covariances, and set

$$R = \Sigma_0^{-1/2} \Sigma \Sigma_0^{-1/2}.$$

Then

$$D(G_\Sigma\|G_{\Sigma_0}) = \frac{1}{2} \{\text{tr } R - d - \log \det R\}.$$

If the eigenvalues of R are $1 + \eta_1, \dots, 1 + \eta_d$ and $|\eta_j| \leq \delta \leq 1/2$, then

$$D(G_\Sigma\|G_{\Sigma_0}) \leq \frac{1}{2} \sum_{j=1}^d \eta_j^2 \leq \frac{d}{2} \delta^2,$$

$$\text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \frac{\sqrt{d}}{2} \delta.$$

Proof. The first inequality is Pinsker's inequality [23, Lemma 2.5(i)]. The formula for $D(G_\Sigma\|G_{\Sigma_0})$ is [18, eq. (A.23)] with zero means and with Σ, Σ_0 in place of the covariances Σ_0, Σ_1 there, as $\text{tr}(\Sigma_0^{-1}\Sigma) = \text{tr } R$ and $\log \det(\Sigma_0\Sigma^{-1}) = -\log \det R$.

Since the eigenvalues of R are $1 + \eta_1, \dots, 1 + \eta_d$, we have $\text{tr } R = \sum_j (1 + \eta_j)$ and $\log \det R = \sum_j \log(1 + \eta_j)$, so that

$$\text{tr } R - d - \log \det R = \sum_{j=1}^d \{\eta_j - \log(1 + \eta_j)\}.$$

If $|\eta_j| \leq \delta \leq 1/2$, then $0 \leq \eta_j - \log(1 + \eta_j) \leq \eta_j^2$, since $y - \log(1 + y) = \int_0^y t/(1+t) dt$, $t/(1+t)$ has the sign of t for $t > -1$, and $|t/(1+t)| \leq 2|t|$ for $|t| \leq 1/2$. It follows that

$$D(G_\Sigma \| G_{\Sigma_0}) = \frac{1}{2} \sum_{j=1}^d \{\eta_j - \log(1 + \eta_j)\} \leq \frac{1}{2} \sum_{j=1}^d \eta_j^2 \leq \frac{d}{2} \delta^2,$$

and hence, by the first inequality,

$$\text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \sqrt{D(G_\Sigma \| G_{\Sigma_0})/2} \leq \sqrt{\frac{d}{4} \delta^2} = \frac{\sqrt{d}}{2} \delta.$$

□

Lemma 4.5 (Few small derivatives). *Let $B_{K,N}$ count the roots α of f_N with $|\alpha| - 1| \leq K/N$ and $|f'_N(\alpha)| < N^{3/2}/L$. For all sufficiently large N ,*

$$\mathbb{P}\{B_{K,N} > N^{31/32}\} \leq N^{-3} + N^{-10} + C(K)N^{-3/8}(\log N)^4.$$

Proof. The idea is to detect each annular root with a small derivative, outside two thin sectors around the real axis, at a nearby point of a fixed mesh at which both f_N and zf'_N are small. We bound the mean and the variance of the number of such mesh points by one- and two-point Gaussian approximations, and then the number of roots by Chebyshev's inequality and the local zero counts, which also bound the roots in the two sectors directly.

By the Taylor bounds below, at a mesh point z within h of a root, $|f_N(z)|/\sqrt{N}$ is at most a constant times $Nh/L + N^2h^2\sqrt{\log N}$, and for the mean count below both terms must be at most a constant times $L^{-2}(\log N)^{-1/2}$, which holds when h is at most a constant times $1/(NL\sqrt{\log N})$. Since a smaller h would only increase the number of mesh points, we choose a deterministic mesh with spacing

$$h = \frac{c}{NL\sqrt{\log N}},$$

where $c > 0$ is sufficiently small and depends only on K . The Taylor bounds below require $100e^{K+4}c \leq 1/2$, which also implies that $c \leq 1$. The construction in Section A.6 takes $\lceil 2K/(Nh) \rceil + 1$ rows, at equally spaced radii from $1 - K/N$ to $1 + K/N$, and on each row the same $\lceil 8\pi/h \rceil$ equally spaced angles, so that

adjacent rows are at most h apart, and adjacent angles at most $h/4$ apart. Since $L\sqrt{\log N}/c \geq 1$ for $c \leq 1$ and $N \geq e$, the number of mesh points satisfies

$$\begin{aligned} \left(\left\lceil \frac{2K}{Nh} \right\rceil + 1\right) \left\lceil \frac{8\pi}{h} \right\rceil &\leq \left(\frac{2K}{Nh} + 2\right) \left(\frac{8\pi}{h} + 1\right) \\ &= \left(\frac{2KL\sqrt{\log N}}{c} + 2\right) \left(\frac{8\pi NL\sqrt{\log N}}{c} + 1\right) \\ &\leq \frac{(2K+2)L\sqrt{\log N}}{c} \cdot \frac{(8\pi+1)NL\sqrt{\log N}}{c} \leq \frac{53(K+1)}{c^2} NL^2 \log N, \end{aligned}$$

and since c depends only on K , the mesh has at most $C(K)NL^2 \log N$ points.

We retain the mesh points $re^{i\theta}$ such that $\text{dist}(\theta, \pi\mathbb{Z}) \geq N^{-1/2}$, or equivalently $\text{dist}(2\theta, 2\pi\mathbb{Z}) \geq 2N^{-1/2}$. These are the angles retained in Section 3.1, and we use them in the covariance bounds below. Let α be a root with $||\alpha| - 1| \leq K/N$ and $\text{dist}(\arg \alpha, \pi\mathbb{Z}) \geq 2N^{-1/2}$. We assign to it the mesh point z on the row nearest to $|\alpha|$ whose angle is nearest to $\arg \alpha$. Then $||\alpha| - |z|| \leq h/2$, the angles of α and z differ by at most $h/8$ modulo 2π , and $|z| \leq 1 + K/N \leq 2$ for $N \geq K$, so that, since a chord is at most its arc,

$$|\alpha - z| \leq ||\alpha| - |z|| + |z| |e^{i\arg \alpha} - e^{i\arg z}| \leq \frac{h}{2} + 2 \cdot \frac{h}{8} < h.$$

Moreover, since $h \leq 1/N \leq N^{-1/2}$ for $c \leq 1$ and $N \geq e$, we have

$$\text{dist}(\arg z, \pi\mathbb{Z}) \geq \text{dist}(\arg \alpha, \pi\mathbb{Z}) - \frac{h}{8} \geq 2N^{-1/2} - N^{-1/2} = N^{-1/2}.$$

Thus every annular root α with $\text{dist}(\arg \alpha, \pi\mathbb{Z}) \geq 2N^{-1/2}$ has a retained mesh point within h .

Now let α be such a root with $|f'_N(\alpha)| < N^{3/2}/L$, and let z be its assigned mesh point. Since $f_N(\alpha) = 0$, by Taylor's theorem with integral remainder at α we have

$$\begin{aligned} \frac{|f_N(z)|}{\sqrt{N}} &= \frac{1}{\sqrt{N}} \left| f'_N(\alpha)(z - \alpha) + (z - \alpha)^2 \int_0^1 (1-t) f''_N(\alpha + t(z - \alpha)) dt \right| \\ &\leq \frac{|f'_N(\alpha)|h}{\sqrt{N}} + \frac{\sup |f''_N| h^2}{2\sqrt{N}} \leq \frac{Nh}{L} + \frac{C(K)}{2} N^2 h^2 \sqrt{\log N} \\ &= \frac{c}{L^2 \sqrt{\log N}} + \frac{C(K)c^2}{2L^2 \sqrt{\log N}} \leq \frac{C(K)}{L^2 \sqrt{\log N}}, \end{aligned}$$

where we used $|z - \alpha| \leq h$, $|f'_N(\alpha)| < N^{3/2}/L$ and, on the event of Lemma 4.3, (22) with $C(K) = 100e^{K+4}$, and the last step holds with the same $C(K)$, since $c + C(K)c^2/2 = c(1 + C(K)c/2) \leq 5c/4 \leq C(K)$ by $C(K)c \leq 1/2$ and $c \leq 1$. Since $f'_N(z) = f'_N(\alpha) + (z - \alpha) \int_0^1 f''_N(\alpha + t(z - \alpha)) dt$ and $|z| \leq 2$, we also have

$$\frac{|zf'_N(z)|}{N^{3/2}} \leq 2 \left(\frac{|f'_N(\alpha)|}{N^{3/2}} + \frac{\sup |f''_N| h}{N^{3/2}} \right) \leq \frac{2(1 + C(K)c)}{L} \leq \frac{3}{L},$$

since $C(K)c \leq 1/2$ and, by (22), $\sup |f_N''| h/N^{3/2} \leq C(K)N^{5/2}\sqrt{\log N} h/N^{3/2} = C(K)c/L$. In both chains the supremum of the second derivative is taken over the disk $|\zeta| \leq 1 + (K+1)/N$, which contains the segment from α to z , since α and z lie in the convex disk $|\zeta| \leq 1 + K/N$. Thus the root produces the test

$$(23) \quad |f_N(z)|/\sqrt{N} \leq u = \frac{C(K)}{L^2\sqrt{\log N}}, \quad |zf_N'(z)|/N^{3/2} \leq v = 3/L.$$

If a root α is assigned to a mesh point z , then, since $|z - z/|z|| = ||z| - 1| \leq K/N$ and $h \leq 1/N$,

$$|\alpha - z/|z|| \leq |\alpha - z| + |z - z/|z|| \leq h + K/N < 2(K+2)/N.$$

Hence, by Lemma 4.2 with $K+2$ in place of K , applied to the disk $\{\zeta : |\zeta - z/|z|| \leq 2(K+2)/N\}$, whose center lies on the unit circle, at most $C_1(K+2)\log N$ roots are assigned to each cell.

We consider

$$W(z) = (W_0(z), W_1(z)) = \left(\frac{f_N(z)}{\sqrt{N}}, \frac{zf_N'(z)}{N^{3/2}} \right),$$

as a real vector with coordinates $\Re W_0(z), \Im W_0(z), \Re W_1(z), \Im W_1(z)$. For $z = re^{i\theta}$, we have $W_0(z) = \sum_{k=0}^N \epsilon_k r^k e^{ik\theta} / \sqrt{N}$ and $W_1(z) = \sum_{k=0}^N \epsilon_k (k/N) r^k e^{ik\theta} / \sqrt{N}$, and since the signs are independent with mean zero and variance one, the real covariance is

$$A_z = \frac{1}{N} \sum_{k=0}^N r^{2k} \times \begin{pmatrix} \cos^2(k\theta) & \cos(k\theta)\sin(k\theta) & \frac{k}{N}\cos^2(k\theta) & \frac{k}{N}\cos(k\theta)\sin(k\theta) \\ \cos(k\theta)\sin(k\theta) & \sin^2(k\theta) & \frac{k}{N}\cos(k\theta)\sin(k\theta) & \frac{k}{N}\sin^2(k\theta) \\ \frac{k}{N}\cos^2(k\theta) & \frac{k}{N}\cos(k\theta)\sin(k\theta) & \frac{k^2}{N^2}\cos^2(k\theta) & \frac{k^2}{N^2}\cos(k\theta)\sin(k\theta) \\ \frac{k}{N}\cos(k\theta)\sin(k\theta) & \frac{k}{N}\sin^2(k\theta) & \frac{k^2}{N^2}\cos(k\theta)\sin(k\theta) & \frac{k^2}{N^2}\sin^2(k\theta) \end{pmatrix}.$$

Using

$$\cos^2(k\theta) = \frac{1 + \cos(2k\theta)}{2}, \quad \sin^2(k\theta) = \frac{1 - \cos(2k\theta)}{2}, \quad \cos(k\theta)\sin(k\theta) = \frac{\sin(2k\theta)}{2},$$

we write $A_z = A_z^{(0)} + E_z$, where

$$A_z^{(0)} = \frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} 1 & 0 & k/N & 0 \\ 0 & 1 & 0 & k/N \\ k/N & 0 & (k/N)^2 & 0 \\ 0 & k/N & 0 & (k/N)^2 \end{pmatrix}$$

and

$$E_z = \frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} \cos(2k\theta) & \sin(2k\theta) & \frac{k}{N} \cos(2k\theta) & \frac{k}{N} \sin(2k\theta) \\ \sin(2k\theta) & -\cos(2k\theta) & \frac{k}{N} \sin(2k\theta) & -\frac{k}{N} \cos(2k\theta) \\ \frac{k}{N} \cos(2k\theta) & \frac{k}{N} \sin(2k\theta) & \frac{k^2}{N^2} \cos(2k\theta) & \frac{k^2}{N^2} \sin(2k\theta) \\ \frac{k}{N} \sin(2k\theta) & -\frac{k}{N} \cos(2k\theta) & \frac{k^2}{N^2} \sin(2k\theta) & -\frac{k^2}{N^2} \cos(2k\theta) \end{pmatrix}.$$

Every entry of E_z is half the real or imaginary part of an oscillatory sum in (9), with $r = s$ and $\psi = 2\theta$, namely, up to sign, of $N^{-1} \sum_{k=0}^N (r \cdot r)^k (k/N)^\beta e^{ik \cdot 2\theta}$ with $\beta \in \{0, 1, 2\}$. For retained angles we have $\text{dist}(2\theta, 2\pi\mathbb{Z}) \geq 2N^{-1/2}$, and the norm of a 4×4 matrix is at most four times its largest entry in absolute value. Thus, by (9), which applies since $r \in [1 - K/N, 1 + K/N]$ and $N \geq 2K$, we have

$$\begin{aligned} \|E_z\| &\leq 4 \cdot \frac{1}{2} \cdot \frac{10e^{4K}}{N \text{dist}(2\theta, 2\pi\mathbb{Z})} \leq \frac{20e^{4K}}{N \cdot 2N^{-1/2}} \\ &= 10e^{4K} N^{-1/2} \leq 20e^{4K} N^{-1/2} \leq e^{-4K}/320, \end{aligned}$$

where we keep the cruder constant $20e^{4K}$, from which the threshold is computed, and the last step holds if and only if $N^{1/2} \geq 20 \cdot 320e^{8K} = 6400e^{8K}$, that is, at the appendix threshold $N \geq (6400e^{8K})^2$.

To bound $A_z^{(0)}$ from below, we list the coordinates in the order $\Re W_0(z)$, $\Re W_1(z)$, $\Im W_0(z)$, $\Im W_1(z)$. In this order $A_z^{(0)}$ is block diagonal with two equal blocks

$$\frac{1}{2N} \sum_{k=0}^N r^{2k} \begin{pmatrix} 1 & k/N \\ k/N & (k/N)^2 \end{pmatrix},$$

and reordering the coordinates does not change the eigenvalues. Each matrix in this sum is positive semidefinite, since it is the product of the column $(1, k/N)^T$ with its transpose. The radial weights are at least e^{-4K} , since $r^{2k} \geq \min(1, r)^{2N} \geq (1 - K/N)^{2N} \geq e^{-4K}$ for $0 \leq k \leq N$, by $r \geq 1 - K/N$, $K/N \leq 1/2$ and $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$. It follows that each block is at least $e^{-4K}/2$ times the unweighted matrix

$$\begin{aligned} M_N &= \frac{1}{N} \sum_{k=0}^N \begin{pmatrix} 1 & k/N \\ k/N & (k/N)^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{N} \sum_{k=0}^N 1 & \frac{1}{N^2} \sum_{k=0}^N k \\ \frac{1}{N^2} \sum_{k=0}^N k & \frac{1}{N^3} \sum_{k=0}^N k^2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{N+1}{2N} & \frac{N+1}{2N} \\ \frac{N+1}{2N} & \frac{(N+1)(2N+1)}{6N^2} \end{pmatrix} \end{aligned}$$

in the positive semidefinite order, and hence $\lambda_{\min}(A_z^{(0)}) \geq (e^{-4K}/2)\lambda_{\min}(M_N)$. By a direct calculation, we have

$$\det M_N = \frac{N+1}{N} \cdot \frac{(N+1)(2N+1)}{6N^2} - \left(\frac{N+1}{2N}\right)^2 = \frac{(N+1)^2 \{2(2N+1) - 3N\}}{12N^3}$$

$$\begin{aligned}
&= \frac{(N+1)^2(N+2)}{12N^3}, \\
\text{tr } M_N &= \frac{N+1}{N} + \frac{(N+1)(2N+1)}{6N^2} = \frac{(N+1)(8N+1)}{6N^2}.
\end{aligned}$$

Since the smaller eigenvalue of the positive semidefinite matrix M_N is $\det M_N$ divided by the larger one, and the larger one is at most $\text{tr } M_N$, we get

$$\lambda_{\min}(M_N) \geq \frac{\det M_N}{\text{tr } M_N} = \frac{(N+1)(N+2)}{2N(8N+1)} \geq \frac{1}{16}.$$

By Weyl's inequality, at the appendix threshold $N \geq (6400e^{8K})^2$ we have

$$\begin{aligned}
\lambda_{\min}(A_z) &\geq \lambda_{\min}(A_z^{(0)}) - \|E_z\| \geq \frac{e^{-4K}}{2} \lambda_{\min}(M_N) - \|E_z\| \\
(24) \quad &\geq \frac{e^{-4K}}{32} - \frac{e^{-4K}}{320} \geq \frac{e^{-4K}}{80} = \lambda,
\end{aligned}$$

where the last line defines λ .

For two points $z = re^{i\theta}$ and $w = se^{i\phi}$, we consider the real vector $(W(z), W(w))$ in $\mathbb{R}^8$, with the four coordinates of $W(z)$ followed by those of $W(w)$, and denote its covariance matrix by

$$\Sigma = \begin{pmatrix} A_z & C_{z,w} \\ C_{z,w}^T & A_w \end{pmatrix},$$

where A_z, A_w are the four-dimensional covariance matrices above. By the independence of the signs, the cross block is

$$\begin{aligned}
C_{z,w} &= \frac{1}{N} \sum_{k=0}^N (rs)^k \\
&\times \begin{pmatrix} \cos(k\theta) \cos(k\phi) & \cos(k\theta) \sin(k\phi) & \frac{k}{N} \cos(k\theta) \cos(k\phi) & \frac{k}{N} \cos(k\theta) \sin(k\phi) \\ \sin(k\theta) \cos(k\phi) & \sin(k\theta) \sin(k\phi) & \frac{k}{N} \sin(k\theta) \cos(k\phi) & \frac{k}{N} \sin(k\theta) \sin(k\phi) \\ \frac{k}{N} \cos(k\theta) \cos(k\phi) & \frac{k}{N} \cos(k\theta) \sin(k\phi) & \frac{k^2}{N^2} \cos(k\theta) \cos(k\phi) & \frac{k^2}{N^2} \cos(k\theta) \sin(k\phi) \\ \frac{k}{N} \sin(k\theta) \cos(k\phi) & \frac{k}{N} \sin(k\theta) \sin(k\phi) & \frac{k^2}{N^2} \sin(k\theta) \cos(k\phi) & \frac{k^2}{N^2} \sin(k\theta) \sin(k\phi) \end{pmatrix}.
\end{aligned}$$

Each of its four 2×2 blocks is $(k/N)^\beta$, $\beta \in \{0, 1, 2\}$, times the matrix of the products $\cos(k\theta) \cos(k\phi)$, $\cos(k\theta) \sin(k\phi)$, $\sin(k\theta) \cos(k\phi)$ and $\sin(k\theta) \sin(k\phi)$. Using the trigonometric identities

$$\begin{aligned}
2 \cos(k\theta) \cos(k\phi) &= \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)), \\
2 \cos(k\theta) \sin(k\phi) &= \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)), \\
2 \sin(k\theta) \cos(k\phi) &= \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)), \\
2 \sin(k\theta) \sin(k\phi) &= \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)),
\end{aligned}$$

we see that this matrix of products is $\frac{1}{2}\Theta_k$, where

$$\Theta_k = \begin{pmatrix} \cos(k(\theta - \phi)) + \cos(k(\theta + \phi)) & \sin(k(\theta + \phi)) - \sin(k(\theta - \phi)) \\ \sin(k(\theta + \phi)) + \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) - \cos(k(\theta + \phi)) \end{pmatrix},$$

and hence

$$C_{z,w} = \frac{1}{2N} \sum_{k=0}^N (rs)^k \begin{pmatrix} \Theta_k & \frac{k}{N} \Theta_k \\ \frac{k}{N} \Theta_k & \frac{k^2}{N^2} \Theta_k \end{pmatrix}.$$

For retained pairs, both $\text{dist}(\theta - \phi, 2\pi\mathbb{Z})$ and $\text{dist}(\theta + \phi, 2\pi\mathbb{Z})$ are at least $N^{-1/2}$. Every entry of $C_{z,w}$ is half a sum or a difference of the real or imaginary parts of $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta-\phi)}$ and $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta+\phi)}$, with $\beta \in \{0, 1, 2\}$. By (9) with $\psi = \theta \mp \phi$, each of these two sums is at most $10e^{4K}/(N \cdot N^{-1/2}) = 10e^{4K}N^{-1/2}$ in absolute value, and hence so is the largest entry of $C_{z,w}$. In particular, for

$$\Sigma_0 = \begin{pmatrix} A_z & 0 \\ 0 & A_w \end{pmatrix}, \quad \Sigma - \Sigma_0 = \begin{pmatrix} 0 & C_{z,w} \\ C_{z,w}^T & 0 \end{pmatrix},$$

we have

$$\|\Sigma - \Sigma_0\| = \|C_{z,w}\| \leq 4 \cdot 10e^{4K}N^{-1/2} = 40e^{4K}N^{-1/2},$$

where the first step holds since the eigenvalues of the symmetric matrix $\Sigma - \Sigma_0$, which maps (x, y) to $(C_{z,w}y, C_{z,w}^Tx)$ for $x, y \in \mathbb{R}^4$, are plus and minus the singular values of $C_{z,w}$. Since Σ_0 is block diagonal and z and w are retained, $\lambda_{\min}(\Sigma_0) = \min\{\lambda_{\min}(A_z), \lambda_{\min}(A_w)\} \geq \lambda$. By Weyl's inequality, we have

$$\lambda_{\min}(\Sigma) \geq \lambda_{\min}(\Sigma_0) - \|\Sigma - \Sigma_0\| \geq \lambda - 40e^{4K}N^{-1/2} \geq \lambda/2$$

for $N \geq (6400e^{8K})^2$, since the last step holds if and only if $40e^{4K}N^{-1/2} \leq \lambda/2 = e^{-4K}/160$, that is, $N^{1/2} \geq 40 \cdot 160e^{8K} = 6400e^{8K}$.

To apply Raič's theorem, we write the paired jet as a sum of independent vectors and whiten it by its covariance. We set

$$b_k(z) = \frac{r^k}{\sqrt{N}} \begin{pmatrix} \cos(k\theta) \\ \sin(k\theta) \\ \frac{k}{N} \cos(k\theta) \\ \frac{k}{N} \sin(k\theta) \end{pmatrix}, \quad b_k = \begin{pmatrix} b_k(z) \\ b_k(w) \end{pmatrix}, \quad Y_k = \epsilon_k \Sigma^{-1/2} b_k.$$

Then $(W(z), W(w)) = \sum_{k=0}^N \epsilon_k b_k$ and $\mathbb{E}Y_k = 0$, and the covariances $\text{Cov}(Y_k)$, $0 \leq k \leq N$, sum to the identity, since $\text{Cov}(Y_k) = \Sigma^{-1/2} b_k b_k^T \Sigma^{-1/2}$ and $\sum_{k=0}^N b_k b_k^T = \Sigma$. Since $r^{2k}, s^{2k} \leq (1 + K/N)^{2N} \leq e^{2K}$ for $0 \leq k \leq N$ and $\|\Sigma^{-1/2}\| = \lambda_{\min}(\Sigma)^{-1/2} \leq (2/\lambda)^{1/2}$, we have

$$\|b_k\|^2 = \frac{(r^{2k} + s^{2k})(1 + (k/N)^2)}{N} \leq \frac{4e^{2K}}{N},$$

$$\begin{aligned}
\sum_{k=0}^N \mathbb{E} \|Y_k\|^3 &\leq \sum_{k=0}^N \|\Sigma^{-1/2}\|^3 \|b_k\|^3 \leq (N+1)(2/\lambda)^{3/2} \left(\frac{2e^K}{\sqrt{N}} \right)^3 \\
&\leq 16e^{3K} (2/\lambda)^{3/2} N^{-1/2}.
\end{aligned}$$

Let $U \subset \mathbb{R}^4$ be the product of the disks $x_1^2 + x_2^2 \leq u^2$ and $x_3^2 + x_4^2 \leq v^2$, so that the test (23) at z is the event $W(z) \in U$, and the tests at z and w together are the event $(W(z), W(w)) \in U \times U$. The products of disks that define the tests are convex, and so are their images under whitening. Since $\Sigma^{-1/2}(W(z), W(w)) = \sum_k Y_k$ and $\Sigma^{-1/2}G_\Sigma$ has the law of G_8 , it follows from Raič's theorem, as stated in Section 3.1, with $d = 8$ and the convex set $\Sigma^{-1/2}(U \times U)$, that

$$\begin{aligned}
&|\mathbb{P}\{(W(z), W(w)) \in U \times U\} - \mathbb{P}\{G_\Sigma \in U \times U\}| \\
&= \left| \mathbb{P}\left\{ \sum_{k=0}^N Y_k \in \Sigma^{-1/2}(U \times U) \right\} - \mathbb{P}\{G_8 \in \Sigma^{-1/2}(U \times U)\} \right| \\
&\leq (42 \cdot 8^{1/4} + 16) \sum_{k=0}^N \mathbb{E} \|Y_k\|^3 \\
&\leq (87 \cdot 16) e^{3K} (2/\lambda)^{3/2} N^{-1/2} \leq 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2}.
\end{aligned}$$

Similarly, the four-dimensional marginal, whose summands have smaller norms and whose covariance has least eigenvalue at least λ , satisfies the same bound. Indeed, $W(z) = \sum_k \epsilon_k b_k(z)$ has covariance A_z , $\|b_k(z)\|^2 = r^{2k}(1 + (k/N)^2)/N \leq 2e^{2K}/N$ and $\|A_z^{-1/2}\| \leq \lambda^{-1/2}$. Therefore, by Raič's theorem with $d = 4$, applied to the vectors $\epsilon_k A_z^{-1/2} b_k(z)$ and the convex set $A_z^{-1/2}U$, we have

$$\begin{aligned}
|\mathbb{P}\{W(z) \in U\} - \mathbb{P}\{G_{A_z} \in U\}| &\leq (42 \cdot 4^{1/4} + 16) \sum_{k=0}^N \|A_z^{-1/2}\|^3 \|b_k(z)\|^3 \\
&\leq 76(N+1) \lambda^{-3/2} \left(\frac{2e^{2K}}{N} \right)^{3/2} = 76(N+1) e^{3K} (2/\lambda)^{3/2} N^{-3/2} \\
&\leq 152 e^{3K} (2/\lambda)^{3/2} N^{-1/2} \leq 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2}.
\end{aligned}$$

The density of G_{A_z} is at most $(2\pi)^{-2}(\det A_z)^{-1/2}$, and $(\det A_z)^{-1/2} \leq \lambda^{-2}$ since every eigenvalue of A_z is at least λ . Moreover, the volume of U is $\pi u^2 \cdot \pi v^2$. Hence we obtain the one-point bound

$$\begin{aligned}
\mathbb{P}\{W(z) \in U\} &\leq \mathbb{P}\{G_{A_z} \in U\} + 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2} \\
&\leq (2\pi)^{-2} (\det A_z)^{-1/2} \cdot \pi u^2 \cdot \pi v^2 + 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2} \\
&\leq (2\pi)^{-2} \lambda^{-2} \cdot \pi^2 u^2 v^2 + 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2} \\
&= \frac{u^2 v^2}{4\lambda^2} + 2000 e^{3K} (2/\lambda)^{3/2} N^{-1/2}
\end{aligned}$$

$$\begin{aligned}
&= \frac{9C(K)^2}{4\lambda^2} \cdot \frac{L^{-6}}{\log N} + 2000e^{3K}(2/\lambda)^{3/2}N^{-1/2} \\
(25) \quad &\leq C(K)L^{-6}/\log N + C(K)N^{-1/2}.
\end{aligned}$$

Here the first step is the four-dimensional Raič bound, and in the fifth we used $u^2v^2 = 9C(K)^2L^{-6}/\log N$, by (23). The last step holds since $\lambda = e^{-4K}/80$ depends only on K .

For a separated pair, we compare its Gaussian law with the product of the actual marginal Gaussian laws by relative entropy, which, by Lemma 4.4, depends on Σ and Σ_0 only through the eigenvalues of the matrix R below, and one norm bound on $\Sigma_0^{-1/2}(\Sigma - \Sigma_0)\Sigma_0^{-1/2}$ places these near one. We set

$$R = \Sigma_0^{-1/2}\Sigma\Sigma_0^{-1/2}, \quad \delta = \frac{40e^{4K}}{\lambda\sqrt{N}}.$$

The inverse square root acts separately on the two marginal blocks. The matrix R minus the identity is $\Sigma_0^{-1/2}(\Sigma - \Sigma_0)\Sigma_0^{-1/2}$, and, since $\lambda_{\min}(\Sigma_0) \geq \lambda$ and $40e^{4K}N^{-1/2} \leq \lambda/2$ for $N \geq (6400e^{8K})^2$,

$$\begin{aligned}
\|\Sigma_0^{-1/2}(\Sigma - \Sigma_0)\Sigma_0^{-1/2}\| &\leq \|\Sigma_0^{-1/2}\|^2 \|\Sigma - \Sigma_0\| = \|\Sigma_0^{-1}\| \|\Sigma - \Sigma_0\| \\
&\leq \lambda^{-1} \cdot 40e^{4K}N^{-1/2} = \delta \leq \frac{1}{2}.
\end{aligned}$$

Since R minus the identity is symmetric with norm at most $\delta \leq \frac{1}{2}$, every eigenvalue of R lies in $[1 - \delta, 1 + \delta]$. Hence Lemma 4.4 applies in dimension $d = 8$, where $(\sqrt{d}/2)\delta = \sqrt{2}\delta$, and gives

$$(26) \quad \text{TV}(G_\Sigma, G_{\Sigma_0}) \leq \sqrt{2}\delta \leq \frac{80e^{4K}}{\lambda\sqrt{N}} \leq C(K)N^{-1/2}.$$

The Gaussian vector G_{Σ_0} is a pair of independent vectors G_{A_z} and G_{A_w} , and the two marginal covariances can differ.

Let I_z be the indicator of the two inequalities in (23), that is, of the event $W(z) \in U$, let $p_z = \mathbb{E}I_z$, $g_z = \mathbb{P}\{G_{A_z} \in U\}$ and $g_{z,w} = \mathbb{P}\{G_\Sigma \in U \times U\}$. Combining the pair approximation with the two marginal approximations, we get

$$\begin{aligned}
|\text{Cov}(I_z, I_w)| &= |\mathbb{E}(I_z I_w) - p_z p_w| \\
&\leq |\mathbb{E}(I_z I_w) - g_{z,w}| + |g_{z,w} - g_z g_w| + |g_z g_w - p_z p_w| \\
&\leq C(K)N^{-1/2} + 80e^{4K}/(\lambda\sqrt{N}) + |g_z - p_z| + |g_w - p_w| \\
&\leq C(K)N^{-1/2}.
\end{aligned}$$

In the third line, the first term is bounded by the eight-dimensional Raič bound, since $\mathbb{E}(I_z I_w) = \mathbb{P}\{(W(z), W(w)) \in U \times U\}$, and the second by (26), since $g_z g_w =$

$\mathbb{P}\{G_{\Sigma_0} \in U \times U\}$ by the independence of G_{A_z} and G_{A_w} . For the third we used $|g_z g_w - p_z p_w| \leq |g_z - p_z| |g_w + p_z| |g_w - p_w| \leq |g_z - p_z| + |g_w - p_w|$, and then the four-dimensional Raič bound.

Let $z_1, \dots, z_m$ be the retained mesh points, and let $Z = \sum_{\ell=1}^m I_{z_\ell}$ be the number of successful tests. For each retained mesh point $re^{i\theta}$, at most $10mN^{-1/2}$ partners lie in the removed pair bands. Indeed, a mesh point $se^{i\phi}$ lies in a removed pair band of $re^{i\theta}$ when $\text{dist}(\theta - \phi, 2\pi\mathbb{Z}) < N^{-1/2}$ or $\text{dist}(\theta + \phi, 2\pi\mathbb{Z}) < N^{-1/2}$, which on each row confines ϕ to two arcs of length $2N^{-1/2}$, centered at θ and at $-\theta$, of total length at most $4N^{-1/2}$. The $\lceil 8\pi/h \rceil$ angles of a row are $2\pi/\lceil 8\pi/h \rceil$ apart, so that these arcs contain at most $2(N^{-1/2}\lceil 8\pi/h \rceil/\pi + 1) \leq N^{-1/2}\lceil 8\pi/h \rceil$ of them, since $\lceil 8\pi/h \rceil \geq 8\pi/h \geq 8\pi N$ and $(1 - 2/\pi)8\pi N^{1/2} > 2$. The removed real sectors $\text{dist}(\phi, \pi\mathbb{Z}) < N^{-1/2}$ are also two arcs of length $2N^{-1/2}$, so that every row keeps at least $(1 - N^{-1/2})\lceil 8\pi/h \rceil \geq \frac{1}{2}\lceil 8\pi/h \rceil$ retained angles for $N \geq 4$, and the whole mesh has at most $2m$ points. Summing over the rows, at most $N^{-1/2} \cdot 2m \leq 10mN^{-1/2}$ partners of $re^{i\theta}$ lie in the removed pair bands. We bound their covariances by one, and the others by $C(K)N^{-1/2}$, and sum over all ordered pairs, including the diagonal, which lies in the band $\text{dist}(\theta - \phi, 2\pi\mathbb{Z}) < N^{-1/2}$. By (25) and $m \leq C(K)NL^2 \log N$, we get

$$\begin{aligned} \mathbb{E}Z &= \sum_{\ell=1}^m \mathbb{P}(I_{z_\ell} = 1) \leq m \left\{ \frac{C(K)L^{-6}}{\log N} + C(K)N^{-1/2} \right\} \\ &\leq C(K)NL^2 \log N \left\{ \frac{L^{-6}}{\log N} + N^{-1/2} \right\} = C(K)\{NL^{-4} + N^{1/2}L^2 \log N\}, \\ \text{Var } Z &= \sum_{\ell,j=1}^m \text{Cov}(I_{z_\ell}, I_{z_j}) \leq 10m^2N^{-1/2} + C(K)m^2N^{-1/2} \\ &\leq C(K)(NL^2 \log N)^2 N^{-1/2} = C(K)N^{3/2}L^4(\log N)^2. \end{aligned}$$

It remains to count the roots in the sectors $\text{dist}(\arg z, \pi\mathbb{Z}) < 2N^{-1/2}$, $||z| - 1| \leq K/N$. We place centers on the unit circle, with angular gaps at most $1/N$, on the two intervals of total length $8N^{-1/2}$. Then every point of the sectors is within $(K+1)/N$ of a center, since its modulus differs from one by at most K/N and its angle differs from that of the nearest center by at most $1/(2N)$. Including endpoints and roundings, this requires at most $40\sqrt{N}$ centers, since each interval, of length $4N^{-1/2}$, needs at most $\lceil 4N^{1/2} \rceil + 1 \leq 4\sqrt{N} + 2$ of them. Hence the disks of radius $2(K+2)/N > (K+1)/N$ around these centers cover the sectors, and on the common local-count event of Lemma 4.2, with $K+2$ in place of K , each of them contains at most $C_1(K+2) \log N$ roots, so that there are at most $40C_1(K+2)\sqrt{N} \log N$ roots in their union. This cover can also be used for random

endpoint coefficients, that is, when the constant and the leading coefficients of the polynomial are random and may be small, as for the coefficient distributions of Section 7, since it uses only the local zero count and no lower bound on these coefficients.

Recall that $B_{K,N}$ counts the annular roots with derivative smaller than $N^{3/2}/L$. By the assignment and the sector cover, we have

$$B_{K,N} \leq C_1(K+2)(\log N)Z + 40C_1(K+2)\sqrt{N} \log N.$$

Indeed, on the local-count event of Lemma 4.2, with $K+2$ in place of K , and the derivative event of Lemma 4.3, a counted root in the sectors is one of the roots of the sector cover, and every other counted root is assigned to a retained mesh point z_ℓ , which has $I_{z_\ell} = 1$ by (23) and receives at most $C_1(K+2) \log N$ roots. Let $Q = N^{31/32}/(2C_1(K+2) \log N)$. Substituting $L = N^{1/64}$ into the bounds on the mean and the variance, we obtain

$$\begin{aligned} \mathbb{E}Z &\leq C(K)\{N^{1-4/64} + N^{1/2+2/64} \log N\} = C(K)\{N^{15/16} + N^{17/32} \log N\}, \\ \frac{\mathbb{E}Z}{Q} &\leq C(K)\{N^{-1/32} \log N + N^{-7/16}(\log N)^2\}, \\ \text{Var } Z &\leq C(K)N^{3/2+4/64}(\log N)^2 = C(K)N^{25/16}(\log N)^2, \end{aligned}$$

where the second line uses $1/Q = 2C_1(K+2)(\log N)N^{-31/32}$. Since the right-hand side of the second line and $C(K)\sqrt{N} \log N/N^{31/32} = C(K)N^{-15/32} \log N$ tend to 0 as $N \rightarrow \infty$, it follows that $\mathbb{E}Z \leq Q/2$ and that the sector contribution is at most $N^{31/32}/2$ for all sufficiently large N . Therefore, on the local-count event of Lemma 4.2 and the derivative event of Lemma 4.3, $B_{K,N} > N^{31/32}$ implies that $Z > Q$, since then $C_1(K+2)(\log N)Z > N^{31/32}/2$. By Chebyshev's inequality at deviation $Q/2$, and since $\mathbb{E}Z \leq Q/2$, we have

$$\begin{aligned} \mathbb{P}\{Z > Q\} &\leq \mathbb{P}\{|Z - \mathbb{E}Z| \geq Q/2\} \leq \frac{4 \text{Var } Z}{Q^2} \\ &\leq C(K) \frac{N^{25/16}(\log N)^2}{N^{31/16}(\log N)^{-2}} = C(K)N^{-3/8}(\log N)^4. \end{aligned}$$

Adding the probabilities of the exceptional events of Lemmas 4.2 and 4.3, we get

$$(27) \quad \mathbb{P}\{B_{K,N} > N^{31/32}\} \leq N^{-3} + N^{-10} + C(K)N^{-3/8}(\log N)^4,$$

whose three terms are the local-count exception of Lemma 4.2, the derivative exception of Lemma 4.3, and the mesh-deviation exception. The inequalities that ensure the above two thresholds are given explicitly in Section A.6. $\square$

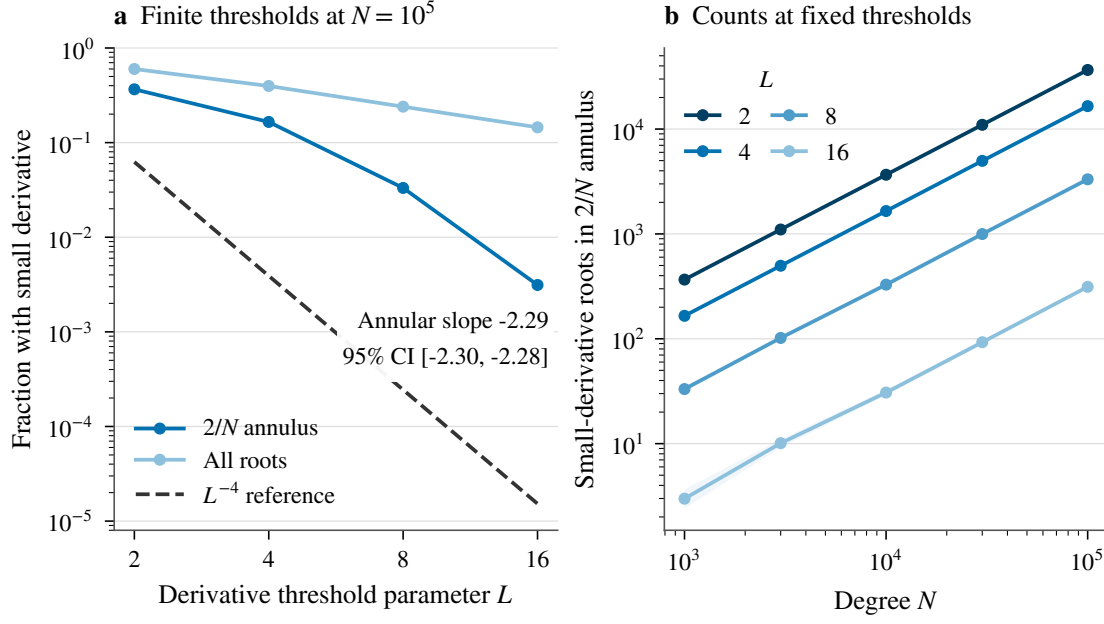


Figure 7: Zeros with small derivative, their fraction (a) and number (b) over 77 sequences.

Figure 7 shows the fraction of annular roots with a small derivative for fixed L , whereas the lemma takes $L = N^{1/64}$. Panel (a) compares the number of such zeros at $N = 10^5$, in the annulus and among all zeros, divided by N , with the reference L^{-4} , which is the power of L in the leading term of the bound $\mathbb{E}Z \leq C(K)\{NL^{-4} + N^{1/2}L^2 \log N\}$ above.

4.3 Bounds for appended tails

Lemma 4.6 (Appended tails). *Let $K \geq 2$, $N \geq \max\{4, K\}$ and $1 \leq m_{\max} \leq N$. Set*

$$\rho = 1 + K/N, \quad g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k z^k, \quad 0 \leq m \leq m_{\max},$$

and put

$$a = 15e^{2K} \sqrt{m_{\max} \log N}.$$

Outside an event of probability at most N^{-10} ,

$$\max_{m \leq m_{\max}} \sup_{|z| \leq \rho} |g_m(z)| < a.$$

Proof. As in the proof of Lemma 4.3 for the second derivative, we combine Hoeffding's inequality at the samples of the circle $|z| = \rho$, a deterministic bound on g'_m

between the samples, and the maximum principle, now with the constants of the tails.

Note that the empty tail is identically zero. We choose $\lceil N^3 \rceil \leq 2N^3$ equally spaced samples on the boundary circle. For every prefix, since $\rho \geq 1$ and $k \leq N + m \leq 2N$, we have $\sum_{k=N+1}^{N+m} \rho^{2k} \leq m\rho^{4N} \leq m_{\max}e^{4K}$. For a sample z and $1 \leq m \leq m_{\max}$, we apply Hoeffding's inequality to $\Re g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k \Re(z^k)$ and $\Im g_m(z) = \sum_{k=N+1}^{N+m} \epsilon_k \Im(z^k)$, whose squared coefficients are at most $|z|^{2k} = \rho^{2k}$. As in the proof of Lemma 4.3, and since $(2a/3)^2 = 100e^{4K}m_{\max} \log N$, we get

$$\begin{aligned} \mathbb{P}\{|g_m(z)| \geq 2a/3\} &\leq \mathbb{P}\left\{|\Re g_m(z)| \geq \frac{2a}{3\sqrt{2}}\right\} + \mathbb{P}\left\{|\Im g_m(z)| \geq \frac{2a}{3\sqrt{2}}\right\} \\ &\leq 4 \exp\left[-\frac{(2a/3)^2}{4 \sum_{k=N+1}^{N+m} \rho^{2k}}\right] \leq 4 \exp\left[-\frac{(2a/3)^2}{4m_{\max}e^{4K}}\right] \\ &= 4 \exp[-25 \log N] = 4N^{-25}. \end{aligned}$$

Since there are at most $2N^3$ samples and at most $m_{\max} \leq N$ nonempty prefixes, there are at most $2N^4$ such pairs, and by the union bound the probability that $|g_m(z)| \geq 2a/3$ for one of them is at most $(2N^4)(4N^{-25}) = 8N^{-21} \leq N^{-10}$. For the derivative we have the deterministic bound

$$\sup_{|z| \leq \rho} |g'_m(z)| \leq \sum_{k=N+1}^{N+m} k\rho^{k-1} \leq m \cdot 2N \cdot \rho^{2N} \leq 2Nm_{\max}e^{2K}.$$

Since $N \geq K$, the circle $|z| = \rho$ has radius at most 2 and length at most 4π , so that every point of it lies within arc length $2\pi/N^3$ of a sample. Therefore, since $m_{\max} \leq \sqrt{m_{\max}}\sqrt{N}$ and $4\pi < 40 \leq 5N^{3/2}\sqrt{\log N}$ for $N \geq 4$, the error of nearest-sample interpolation is at most

$$\frac{2\pi}{N^3} \cdot 2Nm_{\max}e^{2K} = \frac{4\pi m_{\max}e^{2K}}{N^2} \leq 4\pi e^{2K} \sqrt{m_{\max}} N^{-3/2} < 5e^{2K} \sqrt{m_{\max} \log N} = a/3.$$

Outside this event, every point ζ of the circle of radius ρ and its nearest sample z therefore satisfy, for $1 \leq m \leq m_{\max}$,

$$|g_m(\zeta)| \leq |g_m(z)| + \frac{4\pi m_{\max}e^{2K}}{N^2} < \frac{2a}{3} + \frac{a}{3} = a.$$

By the maximum principle, the boundary bound extends to the disk. Hence $\max_{m \leq m_{\max}} \sup_{|z| \leq \rho} |g_m(z)| < a$ outside an event of probability at most N^{-10} , and this one event controls every appended tail g_m at once. $\square$

5 Root matching under perturbations

For a nonzero polynomial P , denote by $\nu_P(r)$ the number of roots of P in the closed disk $\{|z| \leq r\}$, counted with multiplicity.

Theorem 5.1 (Root matching). *Let $P, Q = P + G$ be nonzero complex polynomials. Let $D_1, \dots, D_s$ be pairwise disjoint open disks, each containing exactly one root of P counted with multiplicity. Suppose for a fixed $a > 0$ that*

$$|P(z)| \geq a > |G(z)| \quad (z \in \partial D_j, \ 1 \leq j \leq s).$$

Denote by $c(r)$ the number of these disks whose closures meet $\{|z| = r\}$. Then

$$(28) \quad |\nu_Q(r) - \nu_P(r)| \leq c(r) + (\deg P - s) + \max(0, \deg Q - \deg P).$$

The same assertion holds for pairwise disjoint bounded open domains whose boundaries are finite unions of smooth Jordan curves, with exactly one root in each domain and the same boundary domination.

Proof. The idea is to find exactly one root of Q in each selected domain by Rouché's theorem, and to count the other roots crudely, using the fact that for each polynomial the number of them in $\{|z| \leq r\}$ lies between zero and their total number.

We use Rouché's theorem in the form of [1, Chapter 4, Section 5.2, p. 153]. Let γ be a cycle that is homologous to zero in a region Ω and has winding number $n(\gamma, z) \in \{0, 1\}$ at every point z off γ , and let f, g be analytic in Ω with $|f - g| < |f|$ on γ . Then f and g have equally many zeros, counted with multiplicity, in $\{z : n(\gamma, z) = 1\}$. We take $\Omega = \mathbb{C}$, in which every cycle is homologous to zero, $f = P$, $g = Q$, and $\gamma = \partial D_j$, with its outer curve oriented counterclockwise and its inner curves, if any, clockwise. Since D_j is the set of points inside its outer curve and outside its inner curves, and a smooth Jordan curve winds once about each point inside it and zero times about each point outside it, with the sign of its orientation, we have $n(\partial D_j, z) = 1$ for $z \in D_j$ and $n(\partial D_j, z) = 0$ for $z \notin \overline{D_j}$. On ∂D_j we have $|f - g| = |G| < a \leq |P| = |f|$ by hypothesis. Hence D_j contains as many roots of Q as of P , that is, exactly one, counted with multiplicity.

Since the selected domains are pairwise disjoint and each of them contains exactly one root of P and exactly one root of Q , exactly s roots of each polynomial lie in these domains. We call the other roots unmatched. Their numbers are $\deg P - s$ and $\deg Q - s = \deg P - s + \deg Q - \deg P$. If the closure of a selected domain misses $\{|z| = r\}$, then, since the domain is connected and the complement of the circle is the union of the disjoint open sets $\{|z| < r\}$ and $\{|z| > r\}$, it lies entirely inside or entirely outside that circle. Its root of P is therefore counted by $\nu_P(r)$ if and only if its root of Q is counted by $\nu_Q(r)$, so that this domain does not change the count. On the other hand, each crossing domain, that is, one whose closure meets the circle, changes the count by at most one, since each of its two

roots adds 0 or 1 to its count. Hence the total change from selected roots is at most $c(r)$.

The numbers of unmatched roots inside the circle belong to $[0, \deg P - s]$ and $[0, \deg Q - s]$, and their difference therefore has absolute value at most

$$\max(\deg P - s, \deg Q - s) = \deg P - s + \max(0, \deg Q - \deg P),$$

by the above expression for $\deg Q - s$. Since $\nu_Q(r) - \nu_P(r)$ is the change from selected roots plus the difference of the unmatched counts, we obtain (28) by the triangle inequality. Roots on the target circle are included in the inside count, since the disk in $\nu_P(r)$ and $\nu_Q(r)$ is closed. Each selected root is simple, since its total multiplicity is one. $\square$

Remark 5.2. If $\deg Q \geq \deg P$, the last term in (28) is the degree difference. For arbitrary degrees the positive part is necessary. Indeed, take $P(z) = z^2$, $Q(z) = 1$, no disks, and $0 < r < 1$. With the signed degree difference we would have

$$2 \not\leq 2 + (0 - 2).$$

In this example we have $s = c(r) = 0$, $\nu_P(r) = 2$, since the double root 0 lies in $\{|z| \leq r\}$, and $\nu_Q(r) = 0$, so that $|\nu_Q(r) - \nu_P(r)| = 2$, $\deg P - s = 2$ and $\deg Q - \deg P = 0 - 2$.

In the same way we can compare the root counts of P and Q in two different disks.

Corollary 5.3 (Matching across two radii). *Let $P, Q = P + G$ and the domains $D_1, \dots, D_s$ satisfy the hypotheses of Theorem 5.1, in either of its forms. Let $r, r' > 0$, and denote by $c(r, r')$ the number of these domains whose closures meet the closed annulus $\{\min(r, r') \leq |z| \leq \max(r, r')\}$. Then*

$$|\nu_Q(r') - \nu_P(r)| \leq c(r, r') + (\deg P - s) + \max(0, \deg Q - \deg P).$$

Proof. Each domain D_j contains exactly one root of P , by hypothesis, and exactly one root of Q , by Rouché's theorem on ∂D_j , applied as in the proof of Theorem 5.1, where $|G| < a \leq |P|$. That step does not involve the circle, so that only the counting changes. As before, we call the remaining $\deg P - s$ roots of P and $\deg Q - s$ roots of Q unmatched. Let A be the closed annulus between the two circles. A domain D_j whose closure misses A lies in $\{|z| < \min(r, r')\}$ or in $\{|z| > \max(r, r')\}$, since it is connected and the complement of A is the union of these two disjoint open sets. In the first case its root of P is counted by $\nu_P(r)$ and its root of Q by $\nu_Q(r')$, and in the second case neither is counted, so that the domain contributes the same

amount to $\nu_Q(r')$ and to $\nu_P(r)$. On the other hand, each of the $c(r, r')$ remaining domains changes the difference by at most one, since each of its two roots adds 0 or 1 to its count. The numbers of unmatched roots counted by $\nu_P(r)$ and by $\nu_Q(r')$ belong to $[0, \deg P - s]$ and $[0, \deg Q - s]$, so that, as in the proof of Theorem 5.1, their difference has absolute value at most $\deg P - s + \max(0, \deg Q - \deg P)$. Since $\nu_Q(r') - \nu_P(r)$ is the sum of the two contributions, the corollary follows from the triangle inequality by adding their bounds. For $r = r'$ the annulus is the circle $\{|z| = r\}$, so that $c(r, r) = c(r)$, and the bound reduces to (28). $\square$

Lemma 5.4 (Disjoint sublevel domains near regular roots). *Let $a > 0$ and let α be a root of a nonconstant polynomial P , with $|P'(\alpha)| \geq d_0 > 0$. Suppose $|P''| \leq M_2$ throughout the closed disk of radius $2a/d_0$ around α , and $4M_2a \leq d_0^2$. Then*

$$|P| \geq 3a/2 \quad \text{on } |z - \alpha| = 2a/d_0,$$

that disk contains exactly one root, and the component of $\{|P| < a\}$ containing α lies inside it. For any finite family of such roots with a common a , one can choose a common level $b \in (a, 3a/2)$ for which the corresponding components of $\{|P| < b\}$ are pairwise disjoint bounded smooth domains, each with exactly one root. They satisfy the hypotheses of Theorem 5.1 whenever $|G| < a$ on their closures.

Proof. The idea is to compare P on the circle $|z - \alpha| = 2a/d_0$ with its linear part $P'(\alpha)(z - \alpha)$ at α , using Taylor's theorem for the lower bound on the circle and Rouché's theorem for the single root in the disk. For a family of roots we then choose a regular level b between a and $3a/2$, and the lower bound keeps each selected component of $\{|P| < b\}$ inside the disk of its root.

Let $|z - \alpha| \leq 2a/d_0$. Then

$$\begin{aligned} |P(z) - P'(\alpha)(z - \alpha)| &= |P(z) - P(\alpha) - P'(\alpha)(z - \alpha)| \\ &= |z - \alpha|^2 \left| \int_0^1 (1 - u) P''(\alpha + u(z - \alpha)) du \right| \\ &\leq M_2 |z - \alpha|^2 \int_0^1 (1 - u) du = \frac{M_2}{2} |z - \alpha|^2, \end{aligned}$$

where the second equality is Taylor's theorem with integral remainder for the function $u \mapsto P(\alpha + u(z - \alpha))$ on $[0, 1]$, and the inequality holds since $|P''| \leq M_2$ on the segment from α to z , which lies in the closed disk of radius $2a/d_0$ around α . For $|z - \alpha| = 2a/d_0$, we therefore have

$$|P(z)| \geq |P'(\alpha)(z - \alpha)| - |P(z) - P'(\alpha)(z - \alpha)| \geq |P'(\alpha)| |z - \alpha| - \frac{M_2}{2} |z - \alpha|^2$$

$$\geq d_0 \cdot \frac{2a}{d_0} - \frac{M_2(2a/d_0)^2}{2} = 2a - \frac{2M_2a^2}{d_0^2} \geq 2a - \frac{a}{2} = \frac{3a}{2},$$

where the third inequality uses $|P'(\alpha)| \geq d_0$, and the last is the hypothesis $4M_2a \leq d_0^2$ multiplied by $a/(2d_0^2)$.

On the same circle, as in the chain above, we also have

$$\begin{aligned} |P(z) - P'(\alpha)(z - \alpha)| &\leq \frac{M_2}{2}|z - \alpha|^2 = \frac{M_2(2a/d_0)^2}{2} = \frac{2M_2a^2}{d_0^2} \leq \frac{a}{2} \\ &< 2a = d_0 \cdot \frac{2a}{d_0} \leq |P'(\alpha)||z - \alpha| = |P'(\alpha)(z - \alpha)|. \end{aligned}$$

Thus $f = P'(\alpha)(z - \alpha)$ and $g = P$ satisfy $|f - g| < |f|$ on the circle. Rouché's theorem, in the form used in the proof of Theorem 5.1, with $\Omega = \mathbb{C}$ and γ the counterclockwise circle, which winds once about each point of the open disk and zero times about each point outside the closed disk, therefore shows that P has as many roots in the disk $|z - \alpha| < 2a/d_0$ as the linear polynomial $P'(\alpha)(z - \alpha)$, counted with multiplicity. This linear polynomial is nonzero, since $|P'(\alpha)| \geq d_0 > 0$, and its only root α lies in the disk. Hence the open disk contains exactly one root, and so does the closed disk, since $|P| \geq 3a/2 > 0$ on the circle. Since $|P| \geq 3a/2 > a$ on the boundary circle, the component of $\{|P| < a\}$ containing α does not meet this circle, and it therefore lies inside the disk, since any connected set meeting both the interior and the exterior of the disk meets its boundary.

For a finite family of roots, it suffices that the common level b satisfies $b < 3a/2$, so that the lower bound $|P| \geq 3a/2$ on the circles keeps the components inside the disks, and $b > a$, so that $|P| = b > a > |G|$ on their boundaries. It must also avoid the values of $|P|$ at the zeros of P' , so that the level $\{|P| = b\}$ is regular. We therefore choose

$$b \in (a, 3a/2) \setminus \{|P(w)| : P'(w) = 0\}.$$

The excluded set is finite, since P is nonconstant and P' is a nonzero polynomial with finitely many roots, and hence there exists such a b . This level is regular, since by the Cauchy–Riemann equations the gradient of $|P|^2$, written as a complex number, is $2P\overline{P'}$, and on $\{|P| = b\}$ we have $P \neq 0$ since $b > 0$, and $P' \neq 0$ by the choice of b . The set $\{|P| = b\}$ is also compact, since it is closed, as $|P|$ is continuous, and bounded, as $|P(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$ for the nonconstant polynomial P . By the implicit function theorem, it is a compact smooth one-dimensional manifold, and hence a finite union of smooth circles.

Let α be a designated root, that is, a root of the family. Then the component of $\{|P| < b\}$ containing α is connected and does not meet the circle $|z - \alpha| = 2a/d_0$, where $|P| \geq 3a/2 > b$, so that it lies inside this circle. Since the disk contains

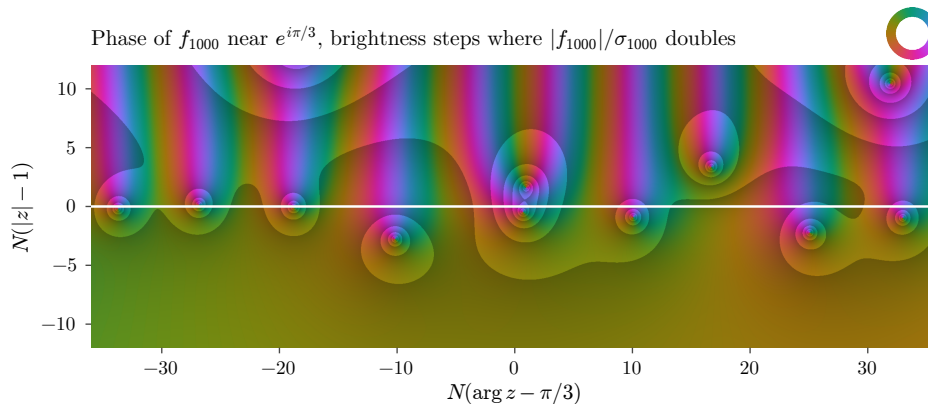


Figure 8: Phase portrait of f_{1000} near $e^{i\pi/3}$. All hues meet at each zero, and the level curves around isolated zeros bound disjoint components.

exactly one root of P , the component contains exactly its designated root. In particular, two designated roots lie in distinct components, and distinct components of $\{|P| < b\}$ are disjoint. Each component is open, as a component of an open set, and bounded, since it lies in a disk. Its boundary lies in $\{|P| = b\}$, since a boundary point z satisfies $|P(z)| \leq b$ by continuity, and if $|P(z)| < b$, then a small disk about z lies in $\{|P| < b\}$ and meets the component, hence lies in it, which is impossible. Near each point of the regular level $\{|P| = b\}$, the implicit function theorem shows that $\{|P| < b\}$ is one side of the level curve. If the point lies on the boundary of the component, then this side meets the component and, being connected, lies in it, so that the nearby points of the curve lie on the boundary as well. Consequently, the boundary of the component meets each of the smooth circles above in an open and closed subset, and it is a union of some of these circles. These components are therefore disjoint bounded smooth domains, with $|P| = b > a > |G|$ on their boundaries, by the choice of b and the hypothesis on G , and each of them contains exactly one root of P , counted with multiplicity, so that they satisfy the hypotheses of Theorem 5.1. This holds even when their localization disks overlap, since their disjointness comes from the roots they contain, not from the disks. $\square$

Figure 8 shows, for one polynomial with random signs, the level curves of $|f_N|/\sigma_N(|z|)$ at the powers of 2 around its zeros.

6 Proof of the radial law

In Section 6.1 we first prove the unit-disk law, Corollary 1.2. We then prove Theorem 1.1 in Section 6.2, by the same construction at the moving radii $1 + x/n$.

6.1 The unit circle

Proof of Corollary 1.2. The idea is to compare every degree n in a block $j^8 \leq n \leq (j+1)^8$ with the first degree $N = j^8$ of the block, by root matching (Theorem 5.1). A root of f_N is matched unless it lies outside the annulus $||z| - 1| \leq K/N$ or has a small derivative there, and a matched root can cross the unit circle only if its domain meets that circle.

Step 1: parameters. Fix an integer $K \geq 2$. We compare the degrees $N = j^8 \leq n \leq (j+1)^8 = N + m_j$. In this proof the annulus is $||z| - 1| \leq K/N$, and, as in Section 4.2, $L = N^{1/64}$. For the appended tails, let ρ and a be as in Lemma 4.6, with $K+2$ in place of K and $m_{\max} = m_j$:

$$\rho = 1 + \frac{K+2}{N}, \quad a = 15e^{2(K+2)}\sqrt{m_j \log N}.$$

This lemma bounds every partial tail by a on $|z| \leq \rho$. We write M_2 for the bound (22) of Lemma 4.3 on $|f_N''|$ over $|z| \leq 1 + (K+1)/N$, and d_0 for the derivative level of Lemma 4.5, below which that lemma counts the annular roots in $B_{K,N}$:

$$M_2 = 100e^{K+4}N^{5/2}\sqrt{\log N}, \quad d_0 = N^{3/2}/L.$$

These parameters must satisfy the conditions of Table 2. All of them hold for N beyond a threshold that depends only on K , since the block exponent 8 makes m_j of order $N^{7/8}$, and hence a of order $N^{7/16}\sqrt{\log N}$, while $d_0 = N^{3/2-1/64}$.

Condition	Needed for	Holds by
$m_j \leq N$	Lemma 4.6	(29) and $N^{1/8} = j \geq 9$
$N \geq K+2$	Lemma 4.6	large j
$2a/d_0 \leq 1/N$	Step 2: the disk of radius $2a/d_0$ about an annular root lies in $ z \leq 1 + (K+1)/N$, where $ f_N'' \leq M_2$	(32)
$4M_2a \leq d_0^2$	Step 2: the hypothesis of Lemma 5.4	(33)
$2a/d_0 < N^{-65/64}$	Step 3: the half-width $N^{-65/64}$ of the thin band	(35)

Table 2: The conditions on the parameters of Step 1 in Section 6.1, which hold for all large j .

With blocks j^β in place of j^8 , m_j would be of order $N^{1-1/\beta}$, and $2a/d_0$ and $4M_2a/d_0^2$ of orders $N^{(1-1/\beta)/2-95/64}$ and $N^{(1-1/\beta)/2-15/32}$, up to logarithmic factors, so that Steps 2 and 3 would need $(1-1/\beta)/2 < 15/32$, that is, $\beta < 16$. On the

other hand, the bounds of order $N^{-3/8}$ on the probabilities of the exceptional events in Step 5 are summable along $N = j^\beta$ only if $3\beta/8 > 1$, that is, $\beta > 8/3$. The exponent 8 lies in this range. More precisely, for $j \geq 60$, we have

$$\begin{aligned}
 m_j &= (j+1)^8 - j^8 \leq 8(j+1)^7 = 8\left(1 + \frac{1}{j}\right)^7 N^{7/8} \\
 (29) \quad &\leq 8\left(\frac{61}{60}\right)^7 N^{7/8} \leq 9N^{7/8}, \\
 (30) \quad a &\leq 45e^{2(K+2)} N^{7/16} \sqrt{\log N}, \\
 \frac{2a}{d_0} &= \frac{2a}{N^{3/2-1/64}} \leq 90e^{2(K+2)} N^{7/16+1/64-3/2} \sqrt{\log N} \\
 (31) \quad &= 90e^{2(K+2)} N^{-67/64} \sqrt{\log N}, \\
 (32) \quad \frac{2Na}{d_0} &\leq 90e^{2(K+2)} N^{-3/64} \sqrt{\log N} \longrightarrow 0, \\
 \frac{4M_2a}{d_0^2} &= \frac{400e^{K+4} N^{5/2} \sqrt{\log N} a}{N^{3-1/32}} \leq 180 \cdot 100e^{K+4} e^{2(K+2)} N^{7/16+1/32-1/2} \log N \\
 (33) \quad &= 180 \cdot 100e^{K+4} e^{2(K+2)} N^{-1/32} \log N \longrightarrow 0.
 \end{aligned}$$

In (29) we use the mean value theorem for the eighth power on $[j, j+1]$, $j^7 = N^{7/8}$, $j \geq 60$ and $8(61/60)^7 < 9$, and (30) follows from it. In the last three bounds we substitute (30) and $d_0 = N^{3/2-1/64}$.

Step 2: selected domains. We call an annular root α , that is, a root of f_N with $||\alpha| - 1| \leq K/N$, good if $|f'_N(\alpha)| \geq N^{3/2}/L$. If $|z - \alpha| \leq 2a/d_0$, then $|z| \leq |\alpha| + 2a/d_0 \leq 1 + K/N + 1/N = 1 + (K+1)/N < \rho$, since $2a/d_0 \leq 1/N$ by Step 1. Hence, outside the exceptional event of Lemma 4.3, we have $|f''_N| \leq M_2$ throughout this disk, and Lemma 5.4, applied with $P = f_N$, $|f'_N(\alpha)| \geq d_0$ and $4M_2a \leq d_0^2$, shows that the disk contains exactly one zero and that $|f_N| \geq 3a/2$ on its boundary.

Let $b \in (a, 3a/2)$ be a common regular level. By Lemma 5.4, applied to the family of good annular roots with the common amplitude a , the selected components of $\{|f_N| < b\}$ are disjoint smooth bounded domains. Each contains one zero and lies in its localization disk, since it is connected, contains its root α and does not meet the circle $|z - \alpha| = 2a/d_0$, where $|f_N| \geq 3a/2 > b$. The level may depend on f_N , but the estimates hold uniformly for every regular level in this interval. For a measurable choice, we take the first rational in $(a, 3a/2)$, in a fixed enumeration, outside the finite set of critical-value moduli.

Step 3: unselected roots and crossings. Let U_N denote the number of unselected roots. Since every good annular root selects a domain that contains no other root, the unselected roots are the roots with $||\alpha| - 1| > K/N$ and the annular roots that

are not good, which $B_{K,N}$ counts. Consequently,

$$\begin{aligned}
 U_N &\leq \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| > K/N\} + B_{K,N}, \\
 \frac{U_N}{N} &\leq \frac{2\log 2}{K} + C(K)N^{-1/32}(\log N)^7 + N^{-1/32} \\
 (34) \qquad &= \frac{2\log 2}{K} + O_K(N^{-1/32}(\log N)^7),
 \end{aligned}$$

where the first count is bounded by (19) of Corollary 3.3, and $B_{K,N} \leq N^{31/32}$ by Lemma 4.5, outside the exceptional events of these two results. Let c_N be the number of selected domains whose closures meet the unit circle. The root of such a domain lies within $2a/d_0$ of the unit circle, since the closure lies within $2a/d_0$ of the root and $||\alpha| - 1| \leq |\alpha - z|$ for a point z of the closure on the unit circle. We count these roots with the thin-band bound (20) of Corollary 3.3, which controls ν_N at the radii $1 + qN^{-1-\kappa}$ for $0 < \kappa \leq 1/64$. We take the largest value $\kappa = 1/64$, so that the two error terms $2N^{-\kappa}$ and $C(K)N^{\kappa-1/32}(\log N)^7$ of (20) are both $N^{-1/64}$, up to constants and the factor $(\log N)^7$, and the band half-width $N^{-1-\kappa} = N^{-65/64}$ is larger than $2a/d_0$ for large N , since $65/64 < 67/64$. More precisely, by (31) we have

$$\begin{aligned}
 (35) \qquad N^{65/64} \frac{2a}{d_0} &\leq 90e^{2(K+2)}N^{-67/64+1+1/64}\sqrt{\log N} \\
 &= 90e^{2(K+2)}N^{-1/32}\sqrt{\log N} \longrightarrow 0,
 \end{aligned}$$

and, once $2a/d_0 < N^{-65/64}$,

$$\begin{aligned}
 (36) \qquad c_N &\leq \#\{\alpha : f_N(\alpha) = 0, \ 1 - N^{-65/64} < |\alpha| < 1 + N^{-65/64}\} \\
 &= \#\{\alpha : f_N(\alpha) = 0, \ |\alpha| < 1 + N^{-65/64}\} - \nu_N(1 - N^{-65/64}) \\
 &\leq \nu_N(1 + N^{-65/64}) - \nu_N(1 - N^{-65/64}),
 \end{aligned}$$

where the first inequality holds since each such root satisfies $||\alpha| - 1| \leq 2a/d_0 < N^{-65/64}$ and distinct domains have distinct roots, and the equality since the inner inequality is strict. We apply (20) with $\kappa = 1/64$ and with 2 in place of K , which suffices since the five radii $1 + qN^{-65/64}$, $q = -2, \dots, 2$, lie within $2N^{-65/64} \leq 2/N$ of the unit circle. For all sufficiently large N , outside five events, one at each of these radii, we obtain

$$(37) \qquad \left| \frac{\nu_N(1 + qN^{-65/64})}{N} - \frac{1}{2} \right| \leq 2N^{-1/64} + C(2)N^{-1/64}(\log N)^7, \qquad q = -1, 0, 1.$$

Dividing (36) by N and centering at $\frac{1}{2}$, we obtain, whenever $2a/d_0 < N^{-65/64}$,

$$\begin{aligned}
 (38) \quad \frac{c_N}{N} &\leq \frac{\nu_N(1 + N^{-65/64}) - \nu_N(1 - N^{-65/64})}{N} \\
 &= \left(\frac{\nu_N(1 + N^{-65/64})}{N} - \frac{1}{2} \right) - \left(\frac{\nu_N(1 - N^{-65/64})}{N} - \frac{1}{2} \right) \\
 &\leq \left| \frac{\nu_N(1 + N^{-65/64})}{N} - \frac{1}{2} \right| + \left| \frac{\nu_N(1 - N^{-65/64})}{N} - \frac{1}{2} \right|.
 \end{aligned}$$

By the cases $q = \pm 1$ of (37), it follows that

$$\frac{c_N}{N} \leq 4N^{-1/64} + 2C(2)N^{-1/64}(\log N)^7 = O(N^{-1/64}(\log N)^7) = o(1).$$

By Theorem 3.1, each of the five events has probability at most the right-hand side of (8) at $K = 2$, which is summable along $N = j^8$, as noted after that theorem. By the Borel–Cantelli lemma, almost surely only finitely many of these events occur along $N = j^8$, and hence the case $q = 0$ of (37) holds for all large j . Since its right-hand side tends to zero, $\nu_N(1)/N \rightarrow 1/2$ almost surely along $N = j^8$.

Step 4: comparison within a block. For $N < n \leq N + m_j$, we write $f_n = f_N + g_{n-N}$, where g_m is as in Lemma 4.6, with $K + 2$ in place of K . Since that lemma bounds all the partial tails at once, outside its tail event we have $|g_{n-N}| < a$ on $|z| \leq \rho$ simultaneously for all these n , and $|z| \leq \rho$ contains the closures of the selected domains by Step 2. By Lemma 5.4, the selected domains then satisfy the hypotheses of Theorem 5.1 with $P = f_N$, $G = g_{n-N}$ and $Q = f_n$. Moreover, $\deg f_n = n$ and $\deg f_N = N$, since $\epsilon_n, \epsilon_N \neq 0$. Each of the s selected domains contains exactly one root of f_N , and the other roots of f_N are the unselected ones, so that $\deg f_N - s = N - s = U_N$. By Theorem 5.1, with degrees N and n and with $r = 1$, where $c(1) = c_N$, we have

$$\begin{aligned}
 \max_{N < n \leq N + m_j} |\nu_n(1) - \nu_N(1)| &\leq \max_{N < n \leq N + m_j} \left\{ c_N + (\deg f_N - s) \right. \\
 &\quad \left. + \max(0, \deg f_n - \deg f_N) \right\} \\
 &= \max_{N < n \leq N + m_j} \left\{ c_N + U_N + (n - N) \right\} = c_N + U_N + m_j.
 \end{aligned}$$

Since $m_j/N \rightarrow 0$, the bound on c_N/N in Step 3, (34) and (29) give, outside the exceptional events of Steps 1–4,

$$\begin{aligned}
 \frac{1}{N} \max_{N < n \leq N + m_j} |\nu_n(1) - \nu_N(1)| &\leq \frac{c_N}{N} + \frac{U_N}{N} + \frac{m_j}{N} \leq O(N^{-1/64}(\log N)^7) \\
 &\quad + \frac{2 \log 2}{K} + O_K(N^{-1/32}(\log N)^7) + 9N^{-1/8}
 \end{aligned}$$

$$= \frac{2 \log 2}{K} + o_K(1).$$

For every degree n in this block, the matching bound above, which holds trivially for $n = N$, together with $N \leq n$ and $n - N \leq m_j$, yields

$$\begin{aligned} \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &= \left| \frac{N}{n} \left(\frac{\nu_N(1)}{N} - \frac{1}{2} \right) + \frac{\nu_n(1) - \nu_N(1)}{n} - \frac{n - N}{2n} \right| \\ &\leq \frac{N}{n} \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N + U_N + m_j}{n} + \frac{n - N}{2n} \\ (39) \quad &\leq \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N + U_N}{N} + \frac{3m_j}{2N}. \end{aligned}$$

Step 5: exceptional events, and $K \rightarrow \infty$. Let $E_{K,j}$ denote the common exceptional event in block j , the union of the events of Table 3, where $\Pi_K(N)$ is the right-hand side of (8). For all large j ,

$$(40) \quad \mathbb{P}(E_{K,j}) \leq 4\Pi_K(N) + 5\Pi_2(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4.$$

In this bound the shared local-count and derivative events are counted once each. We count the maximal-tail event once for the whole block, since outside it the tail bound of Step 4 holds for all the degrees of the block simultaneously. With $N = j^8$ we have $\Pi_K(j^8) = C(K)j^{-3} + j^{-96} + C(K)j^{-7/2}(8 \log j)^{-12}$, since $N^{-3/8} = j^{-3}$, $N^{-12} = j^{-96}$, $N^{-7/16} = j^{-7/2}$ and $\log N = 8 \log j$, and (40) reads

$$\mathbb{P}(E_{K,j}) \leq 4\Pi_K(j^8) + 5\Pi_2(j^8) + j^{-24} + 2j^{-80} + C(K)j^{-3}(8 \log j)^4.$$

Hence, for each fixed K , every term of (40) is summable over j , since the exponents of j are -3 , -96 , $-7/2$, -24 and -80 , and a power of $\log j$ does not affect the convergence. In finite form, (40) is (64) with $R = 2$, the width of the five thin-band events, and the root-count bounds of Step 3 are (62) and (63).

Let $\Omega_K = (\limsup_j E_{K,j})^c$. Since $\limsup_j E_{K,j} \subset \bigcup_{j \geq J} E_{K,j}$ for every J , the summability above gives $\mathbb{P}(\Omega_K^c) \leq \lim_{J \rightarrow \infty} \sum_{j \geq J} \mathbb{P}(E_{K,j}) = 0$, and since there are countably many integers $K \geq 2$, $\mathbb{P}(\bigcap_{K=2}^{\infty} \Omega_K) \geq 1 - \sum_{K=2}^{\infty} \mathbb{P}(\Omega_K^c) = 1$. On this intersection, for each $K \geq 2$, all large j and every n with $N \leq n \leq N + m_j$, the bounds (39), the case $q = 0$ of (37), the bound on c_N/N in Step 3, (34) and (29) imply that

$$\begin{aligned} \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N}{N} + \frac{U_N}{N} + \frac{3m_j}{2N} \\ &\leq 2N^{-1/64} + C(2)N^{-1/64}(\log N)^7 + 4N^{-1/64} + 2C(2)N^{-1/64}(\log N)^7 \\ &\quad + \frac{2 \log 2}{K} + C(K)N^{-1/32}(\log N)^7 + N^{-1/32} + \frac{27}{2}N^{-1/8}. \end{aligned}$$

Event	Source	Used in	Probability
Four radial events	(19)	Step 3, U_N	$\Pi_K(N)$ each
Five thin-band events	(20), width 2	Step 3, $c_N, \nu_N(1)$	$\Pi_2(N)$ each
Local-count event	Lemma 4.2	Step 3, $B_{K,N}$	N^{-3}
Derivative event	Lemma 4.3	Steps 1–3	N^{-10}
Mesh-deviation event	Lemma 4.5	Step 3, $B_{K,N}$	$C(K)N^{-3/8}(\log N)^4$
Maximal-tail event	Lemma 4.6	Step 4	N^{-10}

Table 3: The exceptional events whose union is $E_{K,j}$ in Step 5 of Section 6.1. The derivative event is the event that (22) fails, and the local-count, derivative and mesh-deviation events are those of Lemma 4.5.

Since every large n lies in such a block, and each term on the right other than $2 \log 2/K$ tends to zero as $j \rightarrow \infty$, we conclude that

$$0 \leq \limsup_{n \rightarrow \infty} \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| \leq \inf_{K \geq 2} \frac{2 \log 2}{K} = 0.$$

□

Remark 6.1 (A deterministic block envelope). There are deterministic numbers $\varepsilon_j \rightarrow 0$ and events E_j , defined on the original probability space, such that $\sum_j \mathbb{P}(E_j) < \infty$ and, off E_j ,

$$\max_{j^8 \leq n < (j+1)^8} \frac{|\nu_n(1) - \nu_{j^8}(1)|}{j^8} \leq \varepsilon_j.$$

Proof. We apply Lemma B.7 with the exponent 8, $n_j = j^8$ and $X_n = \nu_n(1)$, for which its conclusion is the assertion of the remark. It requires, for each positive integer k , numbers a_k , $b_k(j)$ and $p_k(j)$ such that, for every j ,

$$\mathbb{P} \left\{ \max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} > a_k + b_k(j) \right\} \leq p_k(j),$$

where $a_k \rightarrow 0$ as $k \rightarrow \infty$, and where $b_k(j) \rightarrow 0$ as $j \rightarrow \infty$ and $\sum_j p_k(j) < \infty$ for each fixed k .

Let k be a positive integer, and put $K = k + 1$, so that $K \geq 2$, as the proof of Corollary 1.2 requires. Since every threshold in that proof depends only on K , there is a deterministic $j_0(K) \geq 60$ such that for every $j \geq j_0(K)$ the bound (40) holds and, outside the event $E_{K,j}$ of that proof, with $N = j^8$, the matching bound of Step 4, (34), the bound on c_N/N in Step 3 with $C(K) \geq 4 + 2C(2)$, and (29) give

$$\max_{N < n \leq N + m_j} |\nu_n(1) - \nu_N(1)| \leq c_N + U_N + m_j,$$

$$\begin{aligned}\frac{U_N}{N} &\leq \frac{2 \log 2}{K} + C(K)N^{-1/32}(\log N)^7 + N^{-1/32}, \\ \frac{c_N}{N} &\leq C(K)N^{-1/64}(\log N)^7, \\ \frac{m_j}{N} &\leq 9N^{-1/8}.\end{aligned}$$

We set $a_k = 2 \log 2 / K$,

$$b_k(j) = C(K)N^{-1/32}(\log N)^7 + N^{-1/32} + C(K)N^{-1/64}(\log N)^7 + 9N^{-1/8},$$

and let $p_k(j)$ be the bound (40) for $\mathbb{P}(E_{K,j})$ when $j \geq j_0(K)$, and $p_k(j) = 1$ when $j < j_0(K)$. Since the block $n_j \leq n < n_{j+1}$ is contained in $N \leq n \leq N + m_j$, and the increment vanishes at $n = N$, the four bounds above give, for $j \geq j_0(K)$ and outside $E_{K,j}$,

$$\begin{aligned}\max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} &\leq \frac{1}{N} \max_{N < n \leq N + m_j} |\nu_n(1) - \nu_N(1)| \\ &\leq \frac{c_N}{N} + \frac{U_N}{N} + \frac{m_j}{N} \leq a_k + b_k(j).\end{aligned}$$

Hence the probability in the above requirement is at most $\mathbb{P}(E_{K,j}) \leq p_k(j)$ when $j \geq j_0(K)$, and at most $1 = p_k(j)$ when $j < j_0(K)$. Moreover, $a_k \rightarrow 0$ as $k \rightarrow \infty$, and, for fixed k , $b_k(j) \rightarrow 0$ as $j \rightarrow \infty$ and $\sum_j p_k(j) < \infty$, since only finitely many of its terms equal one and the others are summable by Step 5 of that proof. By Lemma B.7, there are therefore deterministic $\varepsilon_j \rightarrow 0$ and events E_j with $\sum_j \mathbb{P}(E_j) < \infty$, off which the normalized block increment is at most ε_j . We may take E_j to be the event on which the normalized block increment exceeds ε_j , since it is contained in the event given by the lemma. (In the proof of Lemma B.7 the given events are this event for $j \geq J_1$ and Ω for $j < J_1$, where J_1 is the first of the integers J_k chosen there, and $\sum_{j \geq J_1} \mathbb{P}(E_j) \leq 1$.) All the events here are defined through the random signs on the original probability space. $\square$

With $n_j = j^8$ and $X_n = \nu_n(1)$, the numbers ε_j and the events E_j of Remark 6.1 are those required by Theorem B.6.

Figure 9 shows one block, from degree 1000 to degree 1421, of the sequence with seed 522000.

6.2 Moving radii

Proof of Theorem 1.1. The idea is to repeat the construction of Section 6.1 with the unit circle replaced by the circle of radius $1 + x/N$, for one x at a time, comparing the degrees in a block by Corollary 5.3, and then to pass to all x by monotonicity.

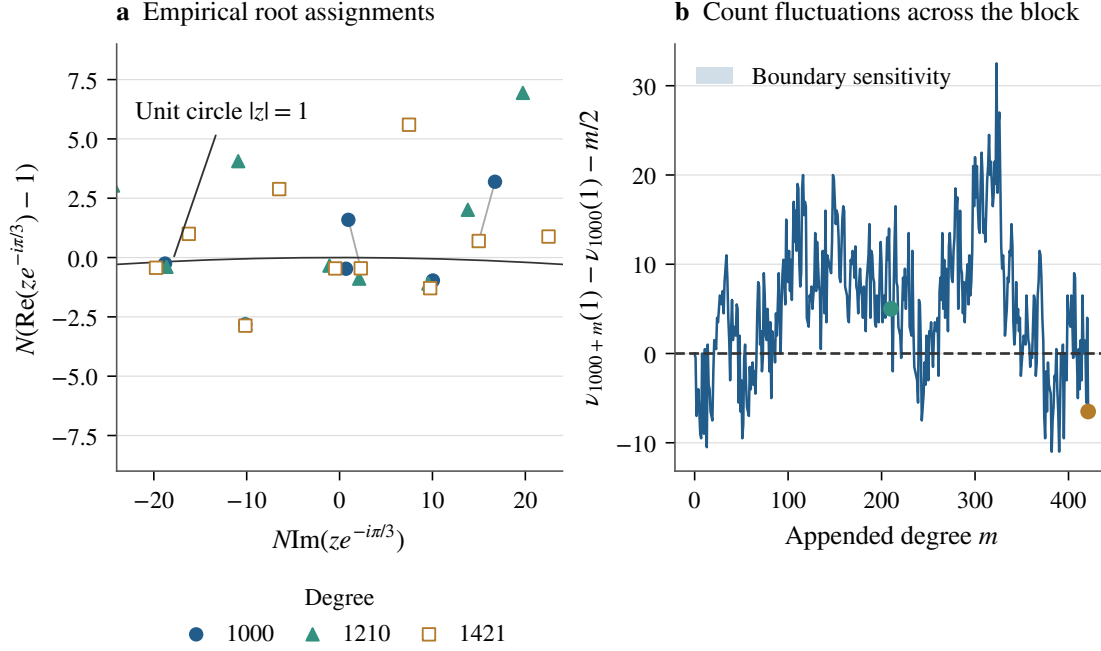


Figure 9: Zeros of one sequence at degrees 1000, 1210 and 1421 near $e^{i\pi/3}$, joined by an uncertified empirical assignment (a), and the change in the count across the block (b).

The profile in Proposition 2.2 satisfies

$$F' = \Phi, \quad 0 \leq \Phi \leq 1, \quad 0 \leq \Phi' = 2 \operatorname{Var}_x(t) \leq \frac{1}{2}.$$

Here $F' = \Phi$ is part of Proposition 2.2, and $0 \leq \Phi \leq 1$ since, by its definition in Theorem 1.1, $\Phi(x)$ is the mean of t under the probability density proportional to e^{2xt} on $[0, 1]$. By (6) in the proof of Lemma 2.4, $\Phi'(x) = 2 \operatorname{Var}_x(t)$, where $\operatorname{Var}_x(t)$ is the variance of t under the same density, and since the variance is at most the mean square deviation from $\frac{1}{2}$,

$$\operatorname{Var}_x(t) \leq \frac{\int_0^1 (t - \frac{1}{2})^2 e^{2xt} dt}{\int_0^1 e^{2xt} dt} \leq \frac{1}{4}.$$

Fix x and an integer $R \geq |x| + 3$. We use the logarithmic bounds at the radii $1 + (x + qh)/N$, $|q| \leq 2$, for an offset $h \in (0, 1]$ that must meet the three requirements of Table 4, for the annular width K fixed below. The value $h = N^{-1/64}$ meets them, since the quantities in the last column of that table tend to zero. For the first degree $N = j^8$ of a block, we therefore set

$$h = N^{-1/64}, \quad r_N = 1 + x/N.$$

Requirement	Reason	At $h = N^{-1/64}$
$h \rightarrow 0$	the secants of F over steps of length h approximate $\Phi(x)$ with an error of order h	$N^{-1/64}$
$N^{-1/32}(\log N)^7/h \rightarrow 0$	the secants divide the logarithmic error $N^{-1/32}(\log N)^7$ of Theorem 3.1 by h	$N^{-1/64}(\log N)^7$
h/N exceeds a bound of order $N^{-67/64}\sqrt{\log N} + N^{-9/8}$	the band of half-width h/N about $ z = 1 + x/N$ contains the root of every selected domain that meets the annulus between $ z = 1 + x/N$ and $ z = 1 + x/n$, for every degree n of the block	$N^{-1/32}\sqrt{\log N}$ and $N^{-7/64}$

Table 4: The requirements on the offset h in the proof of Theorem 1.1, for fixed K and x . The last column gives h , the quotient in the first column, and the two terms of the bound divided by h/N .

We apply Theorem 3.1, with R in place of K , at the five radii $r_q = 1 + y_q/N$, where $y_q = x + qh$ and $q = -2, -1, 0, 1, 2$. Since $h \leq 1$, we have $|y_q| \leq |x| + 2h < R$, so that every r_q lies in $[1 - R/N, 1 + R/N]$. For all sufficiently large N , outside five exceptional events, one at each radius r_q , this theorem gives $|J_N(r_q) - \log \sigma_N(r_q) + \gamma/2| \leq C(R)N^{-1/32}(\log N)^7$, and, for $N \geq 2R$, Lemma 2.3 with R in place of K gives $|\log \sigma_N(r_q) - \frac{1}{2} \log N - F(y_q)| \leq (R^2 + e^{4R}/2)/N$. Adding the two bounds, and using $1/N \leq N^{-1/32}(\log N)^7$ for $N \geq 3$, we get, outside the five exceptional events,

$$\begin{aligned}
 \left| J_N(r_q) - \frac{1}{2} \log N + \frac{\gamma}{2} - F(y_q) \right| &\leq C(R)N^{-1/32}(\log N)^7 + \frac{R^2 + e^{4R}/2}{N} \\
 (41) \qquad \qquad \qquad &\leq \eta_N := (C(R) + R^2 + e^{4R}/2)N^{-1/32}(\log N)^7.
 \end{aligned}$$

For $q = -1, 0, 1$ we apply Lemma B.3 at y_q , with R in place of K , the centering $\frac{1}{2} \log N$, the profile F of Proposition 2.2, $M_F = 1/2$ and $B_F = 1$, which hold by the bounds on $\Phi = F'$ above. Its logarithmic error is $\eta_{\log} = C(R)N^{-1/32}(\log N)^7$, by Theorem 3.1 with $\mathbb{E} \log |G| = -\gamma/2$, and its profile error is $\varepsilon = (R^2 + e^{4R}/2)/N$, by Lemma 2.3 on $[-R, R]$ for $N \geq 2R$, so that $\eta_{\log} + \varepsilon \leq \eta_N$ as in (41). The lemma applies: outside the five exceptional events $J_N(r_q)$ is finite, so that f_N is nonzero; $|y_q| + h \leq |x| + 2h < R < N$, the last since $N \geq 2R$; and its three radii $1 + (y_q - h)/N$, r_q and $1 + (y_q + h)/N$ are among the five radii $r_{q'}$. Outside the five

exceptional events it gives

$$\left| \frac{\nu_N(r_q)}{N} - \Phi(y_q) \right| \leq \frac{h}{4} + \frac{R}{N} + \frac{2\eta_N(1+R/N)}{h}.$$

Since $|\Phi(y_q) - \Phi(x)| \leq |q|h/2$ by $0 \leq \Phi' \leq 1/2$, since $|q|h/2 + h/4 \leq 2h$ for $|q| \leq 1$ and $1 + R/N \leq 2$, and since $r_N + qh/N = r_q$, it follows that, for $q = -1, 0, 1$,

$$\begin{aligned} \left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| &\leq \frac{|q|h}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2\eta_N(1+R/N)}{h} \\ &\leq \varepsilon_N := 2h + \frac{R}{N} + \frac{4\eta_N}{h} \\ &= 2N^{-1/64} + \frac{R}{N} + 4(C(R) + R^2 + e^{4R}/2)N^{-1/64}(\log N)^7 \\ (42) \quad &= O_R(N^{-1/64}(\log N)^7). \end{aligned}$$

Explicit constants for this estimate are given in (65). Moreover, the probabilities of the five logarithmic exceptional events are summable along $N = j^8$, and $\varepsilon_N \rightarrow 0$. Subtracting the two endpoint counts and using (42) with $q = \pm 1$, we get

$$\begin{aligned} &\#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - r_N| < h/N\} \\ &= \#\{\alpha : f_N(\alpha) = 0, \ |\alpha| < r_N + h/N\} - \nu_N(r_N - h/N) \\ &\leq \nu_N(r_N + h/N) - \nu_N(r_N - h/N) \\ &= N \left(\frac{\nu_N(r_N + h/N)}{N} - \Phi(x) \right) + N \left(\Phi(x) - \frac{\nu_N(r_N - h/N)}{N} \right) \\ (43) \quad &\leq 2\varepsilon_N N, \end{aligned}$$

where the first equality holds since the inner inequality is strict.

Fix an integer annular width $K \geq R$, and let a, ρ, d_0 , the common regular level $b \in (a, 3a/2)$ and the selected domains be those of Steps 1 and 2 of Section 6.1 at this width. By Step 2, each selected domain contains exactly one root α of f_N , and its closure lies in the closed disk of radius $2a/d_0$ about α , where $2a/d_0$ is bounded by (31). For $N \leq n \leq N + m_j$, with $m_j = (j+1)^8 - j^8$ as in Step 1 of Section 6.1, the target radius moves by $|(1+x/n) - (1+x/N)| = |x|(n-N)/(nN) \leq |x|m_j/N^2$, and $m_j \leq 9N^{7/8}$ for $j \geq 60$ by (29). Since $N/h = N^{65/64}$, these bounds and (31) yield

$$\begin{aligned} \frac{N}{h} \left(\frac{2a}{d_0} + \frac{|x|m_j}{N^2} \right) &\leq 90e^{2(K+2)}N^{-67/64+65/64}\sqrt{\log N} + 9|x|N^{7/8-2+65/64} \\ &= 90e^{2(K+2)}N^{-1/32}\sqrt{\log N} + 9|x|N^{-7/64} \longrightarrow 0. \end{aligned}$$

Hence $2a/d_0 + |x|m_j/N^2 < h/N$ for all N large enough, depending on K and x . Let D be a selected domain whose closure meets the closed annulus between the circles of radii r_N and $1 + x/n$, let α be its root of f_N , and let z be a point of this annulus in the closure of D . Then $|z - \alpha| \leq 2a/d_0$, and $|z|$ lies between r_N and $1 + x/n$, so that

$$\begin{aligned} \left| |\alpha| - r_N \right| &\leq \left| |\alpha| - |z| \right| + \left| |z| - r_N \right| \leq |\alpha - z| + \left| |z| - r_N \right| \\ &\leq \frac{2a}{d_0} + \left| \left(1 + \frac{x}{n}\right) - r_N \right| \leq \frac{2a}{d_0} + \frac{|x|m_j}{N^2} < \frac{h}{N}. \end{aligned}$$

Hence the root of f_N in such a domain lies in the strict band of (43). This is the case whenever two matched roots lie on different sides of their respective target circles, since a connected domain whose closure misses the closed annulus lies inside both circles or outside both (proof of Corollary 5.3), the complement of the closed annulus being the union of the disjoint open sets inside the smaller circle and outside the larger one. Since distinct domains have distinct roots, by (43) the number $c(r_N, 1 + x/n)$ of these domains is at most $2\varepsilon_N N$.

As in Step 4 of Section 6.1, outside the tail event of Lemma 4.6, with $K + 2$ in place of K and $m_{\max} = m_j$, the appended tail $f_n - f_N$ has modulus less than a on $|z| \leq \rho$, which contains the selected domains, while $|f_N| > a$ on their boundaries. Since this lemma bounds all the partial tails simultaneously, a single maximal-tail event controls the entire block and all target radii. The polynomials f_N and f_n have degrees exactly N and n , since their leading coefficients are ± 1 , and, with s the number of selected domains and U_N the number of unselected roots, $\deg f_N - s = N - s = U_N$, as in Step 4. By Corollary 5.3, with $P = f_N$, $Q = f_n$, $r = r_N$ and $r' = 1 + x/n$, we have

$$\begin{aligned} &\max_{N \leq n \leq N+m_j} |\nu_n(1 + x/n) - \nu_N(r_N)| \\ &\leq \max_{N \leq n \leq N+m_j} \left\{ c(r_N, 1 + x/n) + (\deg f_N - s) + \max(0, \deg f_n - \deg f_N) \right\} \\ &= \max_{N \leq n \leq N+m_j} \left\{ c(r_N, 1 + x/n) + U_N + (n - N) \right\} \leq 2\varepsilon_N N + U_N + m_j, \end{aligned}$$

and U_N/N is bounded by (34) at this width K .

Let $E_{K,R,x,j}$ be the union of the events of Table 3 at the width K , with the five thin-band events replaced by the five logarithmic events of Theorem 3.1 at the width R and the radii r_q . With $\Pi_K(N)$ as in Step 5 of Section 6.1, and $\Pi_R(N)$ the same with R in place of K , for all large j ,

$$(44) \quad \mathbb{P}(E_{K,R,x,j}) \leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4.$$

This is (40) with $5\Pi_2(N)$ replaced by $5\Pi_R(N)$, and its finite form is (64). At $N = j^8$ its terms are those of Step 5 of Section 6.1 with Π_R in place of Π_2 , so that, for each fixed K, R, x , it is summable over j .

For $N \leq n \leq N + m_j$ we split $\nu_n(1 + x/n)/n - \Phi(x)$ into three terms, and by (42) with $q = 0$, the matching bound above, $N \leq n \leq N + m_j$ and $0 \leq \Phi \leq 1$, we have

$$\begin{aligned} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| &= \left| \frac{N}{n} \left(\frac{\nu_N(r_N)}{N} - \Phi(x) \right) \right. \\ &\quad \left. + \frac{\nu_n(1 + x/n) - \nu_N(r_N)}{n} - \frac{n - N}{n} \Phi(x) \right| \\ &\leq \frac{N}{n} \left| \frac{\nu_N(r_N)}{N} - \Phi(x) \right| + \frac{|\nu_n(1 + x/n) - \nu_N(r_N)|}{n} \\ &\quad + \frac{n - N}{n} \Phi(x) \leq \varepsilon_N + \frac{2\varepsilon_N N + U_N + m_j}{N} + \frac{m_j}{N} \\ &= 3\varepsilon_N + \frac{U_N}{N} + \frac{2m_j}{N}. \end{aligned}$$

Let $\Omega_{K,R,x} = (\limsup_j E_{K,R,x,j})^c$. As in Step 5 of Section 6.1, the summability of (44) gives $\mathbb{P}(\Omega_{K,R,x}^c) \leq \lim_{J \rightarrow \infty} \sum_{j \geq J} \mathbb{P}(E_{K,R,x,j}) = 0$. On $\Omega_{K,R,x}$ the preceding bounds hold for all large j , and hence, by (34) and $m_j \leq 9N^{7/8}$, for all large j and $N \leq n \leq N + m_j$ we get

$$\left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq 3\varepsilon_N + \frac{2 \log 2}{K} + C(K)N^{-1/32}(\log N)^7 + N^{-1/32} + 18N^{-1/8}.$$

Every large n lies in such a block, and each term on the right other than $2 \log 2/K$ tends to zero as $j \rightarrow \infty$. Hence, on the intersection Ω_x of the events $\Omega_{K,R,x}$ over integers $K \geq R$, which has probability one as a countable intersection of events of probability one,

$$0 \leq \limsup_{n \rightarrow \infty} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq \frac{2 \log 2}{K} \rightarrow 0 \quad (K \rightarrow \infty).$$

As a countable intersection, the rational event $\bigcap_{x \in \mathbb{Q}} \Omega_x$ also has probability one.

For any real x , choose rationals $x_- < x < x_+$. By monotonicity we have $\nu_n(1 + x_-/n)/n \leq \nu_n(1 + x/n)/n \leq \nu_n(1 + x_+/n)/n$, and taking limits on the rational event we get

$$\Phi(x_-) \leq \liminf_{n \rightarrow \infty} \frac{\nu_n(1 + x/n)}{n} \leq \limsup_{n \rightarrow \infty} \frac{\nu_n(1 + x/n)}{n} \leq \Phi(x_+).$$

Letting x_- increase to x and x_+ decrease to x through rationals, the continuity of Φ shows that the limit inferior and the limit superior are both $\Phi(x)$. To prove

the uniform convergence, fix a compact interval I and $\delta > 0$, write $\widehat{\Phi}_n(x) = \nu_n(1 + x/n)/n$, and choose rational numbers $x_0 < x_1 < \dots < x_M$ with $I \subset (x_0, x_M)$ and $x_{i+1} - x_i < \delta$. For $x \in [x_i, x_{i+1}] \cap I$ we have

$$\widehat{\Phi}_n(x_i) - \Phi(x_{i+1}) \leq \widehat{\Phi}_n(x) - \Phi(x) \leq \widehat{\Phi}_n(x_{i+1}) - \Phi(x_i),$$

since $\widehat{\Phi}_n$ is nondecreasing, as ν_n is, and Φ is nondecreasing, as $\Phi' \geq 0$. Taking the supremum over $x \in I$, we get

$$\begin{aligned} \sup_{x \in I} |\widehat{\Phi}_n(x) - \Phi(x)| &\leq \max_{0 \leq i \leq M} |\widehat{\Phi}_n(x_i) - \Phi(x_i)| + \max_{0 \leq i < M} \{\Phi(x_{i+1}) - \Phi(x_i)\} \\ &\leq \max_{0 \leq i \leq M} |\widehat{\Phi}_n(x_i) - \Phi(x_i)| + \frac{\delta}{2}, \end{aligned}$$

where the second inequality follows from $|\Phi'| \leq 1/2$ and $x_{i+1} - x_i < \delta$ by the mean value theorem. On the rational event the maximum tends to zero, and we then let δ tend to zero. Finally, we intersect over intervals with integer endpoints. All these intersections take place on the given probability space. Since the rational event does not depend on I , the uniform convergence holds on it for every compact interval I at once, and in particular for $I = [-A, A]$ for every $A > 0$, which is the assertion of Theorem 1.1. $\square$

7 Other coefficient distributions

We now take one sequence of iid coefficients with one of the distributions below, use the same sequence at every degree, and write

$$f_N(z) = \sum_{k=0}^N \xi_k z^k, \quad \sigma_N(r)^2 = \mathbb{E}|\xi_0|^2 \sum_{k=0}^N r^{2k},$$

and, as in Section 3, $X_r(\theta) = f_N(re^{i\theta})/\sigma_N(r)$. In the three classes for which we prove a root law, the coefficients have mean zero and positive second moment. We define the logarithmic integral J_N and the closed-disk zero count as before.

We first bound the local zero counts. Steinhaus and bounded centrally symmetric coefficients have the law of independent signs times bounded amplitudes, so that Lemma 4.1 applies to them, while Gaussian amplitudes are unbounded, and we treat them separately in Lemma 7.3.

Lemma 7.1 (Local zero counts for Steinhaus coefficients). *Let the ξ_k be independent and uniform on the unit circle. Fix $K \geq 2$, and put*

$$A = \max(1, 1024\pi/c_{\text{HW}}), \quad t_N = \frac{A \log N}{N}, \quad C_1(K) = \frac{4A + 4K + 4}{\log 2}.$$

Let $N \geq e$ with $t_N \leq 1$. Except on an event of probability at most N^{-3} , every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_1(K) \log N$ zeros of f_N .

Proof. The idea is to write each coefficient as a sign times a point of a fixed semicircle and to apply Lemma 4.1. The event in the lemma depends only on the law of $(\xi_0, \dots, \xi_N)$. Let the Z_k be uniform on a fixed semicircle, independent of each other and of the signs. Since $-e^{i\varphi} = e^{i(\varphi+\pi)}$, the products $\epsilon_k Z_k$ are independent and uniform on the unit circle, so we may take $\xi_k = \epsilon_k Z_k$. Since $|Z_k| = 1$ and $\sum_{k=0}^N |Z_k|^2 = N + 1$, we may apply Lemma 4.1 with $a_k = Z_k$, $B = 1$, $\eta = 1$ and $\mathcal{E} = \Omega$. Then $P_N = f_N$, and, since $\log 1 = 0$, the constants of that lemma are

$$\begin{aligned} A(1, 1) &= \max(1, 1024\pi/c_{\text{HW}}) = A, & t_N(1, 1) &= t_N, \\ C_1(K, 1, 1) &= \frac{4A + 4K + 4}{\log 2} = C_1(K). \end{aligned}$$

This proves the lemma. □

Lemma 7.2 (Local zero counts for bounded coefficients). *Let the ξ_k be independent with a fixed nondegenerate distribution on $\mathbb{R}$ or $\mathbb{C}$ that is centrally symmetric, that is, $\text{Law}(\xi_0) = \text{Law}(-\xi_0)$, and normalize*

$$\mathbb{E}|\xi_0|^2 = 1, \quad |\xi_0| \leq B, \quad B \geq 1.$$

Fix $K \geq 2$, and put

$$\begin{aligned} A_B &= \max(1, 2048\pi B^2/c_{\text{HW}}), & t_{N,B} &= \frac{A_B \log N}{N}, \\ C_{1,B}(K) &= \frac{4A_B + 4K + 4 + \log B}{\log 2}. \end{aligned}$$

Let $N \geq e$ with $t_{N,B} \leq 1$. Except on an event of probability at most $\exp[-(N + 1)/(2B^4)] + N^{-3}$, every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_{1,B}(K) \log N$ zeros of f_N .

Proof. The idea is to write each coefficient as a sign times an independent copy of itself, to bound the probability that the amplitudes have small total energy, and to apply Lemma 4.1 with $\eta = \frac{1}{2}$. The event in the lemma depends only on the law of $(\xi_0, \dots, \xi_N)$. Let $a_0, \dots, a_N$ be independent with the coefficient distribution, and

independent of the signs. By symmetry, $\text{Law}(\epsilon_k a_k) = \frac{1}{2} \text{Law}(a_k) + \frac{1}{2} \text{Law}(-a_k) = \text{Law}(\xi_k)$, and the products are independent, so that we may take $\xi_k = \epsilon_k a_k$. Let $\mathcal{E} = \{\sum_{k=0}^N |a_k|^2 \geq (N+1)/2\}$. Since $|a_k|^2 \in [0, B^2]$ and $\mathbb{E}|a_k|^2 = 1$, it follows from Hoeffding's inequality [9, Theorem 2] for the $N+1$ variables $|a_k|^2$ at the deviation $(N+1)/2$ that

$$\begin{aligned} \mathbb{P}(\mathcal{E}^c) &= \mathbb{P}\left\{\sum_{k=0}^N (|a_k|^2 - 1) < -\frac{N+1}{2}\right\} \\ &\leq \exp\left[-\frac{2((N+1)/2)^2}{(N+1)B^4}\right] = \exp\left[-\frac{N+1}{2B^4}\right]. \end{aligned}$$

We apply Lemma 4.1 with these a_k , this B , $\eta = \frac{1}{2}$ and this event $\mathcal{E}$. Then $P_N = f_N$, and the constants of that lemma are

$$\begin{aligned} A(B, \tfrac{1}{2}) &= \max\left(1, \frac{1024\pi B^2}{c_{\text{HW}}/2}\right) = \max\left(1, \frac{2048\pi B^2}{c_{\text{HW}}}\right) = A_B, \\ t_N(B, \tfrac{1}{2}) &= \frac{A_B \log N}{N} = t_{N,B}, \\ C_1(K, B, \tfrac{1}{2}) &= \frac{4A_B + 4K + 4 + \log B}{\log 2} = C_{1,B}(K). \end{aligned}$$

It follows that the probability that $\mathcal{E}$ occurs and some disk contains more than $C_{1,B}(K) \log N$ zeros is at most N^{-3} . Adding $\mathbb{P}(\mathcal{E}^c)$ proves the lemma. $\square$

Lemma 7.3 (Local zero counts for Gaussian coefficients). *Let $\xi_k = g_k$, where the g_k are independent standard real Gaussians, or let $\xi_k = (g_k + ih_k)/\sqrt{2}$, where all g_k, h_k are independent standard real Gaussians. Fix $K \geq 2$, let t_N and $C_1(K)$ be as in Lemma 7.1, and put*

$$C_{1,G}(K) = C_1(K) + \frac{2}{\log 2}.$$

Let $N \geq e$ with $t_N \leq 1$. Except on an event of probability at most $N^{-3} + e^{-3(N+1)/32}$, every disk

$$\{z : |z - z_0| \leq 2K/N\}, \quad ||z_0| - 1| \leq 1/N,$$

contains at most $C_{1,G}(K) \log N$ zeros of f_N .

Proof. Since the Gaussian amplitudes are unbounded, Lemma 4.1 does not apply, but its proof still works once $|f_N|$ is bounded outside an event of exponentially small probability. Let J , the arcs I and the real Gram matrix A_I be those of the proof of Lemma 4.1 with $a_k = 1$, $B = 1$ and $\eta = 1$, so that $J \leq 8N$, $\mu(I) = t_N/\pi$, the eigenvalues of A_I lie in $[0, 1]$, $\text{tr } A_I = (N+1)t_N/\pi$, $\|A_I\| \leq 1$ and $\|A_I\|_{\text{HS}}^2 \leq \text{tr } A_I$. Let $Q_I = \int_I |f_N(e^{i\theta})|^2 d\mu(\theta)$.

Real coefficients. With $g = (g_0, \dots, g_N)$, we have $Q_I = g^T A_I g$ and $\mathbb{E}Q_I = \text{tr } A_I$, since $\mathbb{E}g_j g_k = \mathbf{1}\{j = k\}$. Standard real Gaussian coordinates have ψ_2 norm at most two in the moment convention of Section 4.1. Indeed, $y^{p/2} e^{-y/4} \leq (2p/e)^{p/2}$ for $y \geq 0$, and $\mathbb{E}e^{g_k^2/4} = (1 - \frac{1}{2})^{-1/2} = \sqrt{2}$, so that, writing $|g_k|^p = (g_k^2)^{p/2} e^{-g_k^2/4} \cdot e^{g_k^2/4}$, for $p \geq 1$ we have

$$p^{-1/2}(\mathbb{E}|g_k|^p)^{1/p} \leq p^{-1/2}(\sqrt{2}(2p/e)^{p/2})^{1/p} = 2^{1/(2p)}(2/e)^{1/2} \leq 2/\sqrt{e} < 2.$$

Therefore (21) holds with g in place of ϵ , and the Hanson–Wright chain in the proof of Lemma 4.1 at $B = \eta = 1$ gives $\mathbb{P}\{Q_I < \frac{1}{2} \text{tr } A_I\} \leq 2N^{-16}$.

Complex coefficients. For an arc I , the complex Gram matrix is given by

$$M_{jk} = \int_I e^{i(k-j)\theta} d\mu(\theta), \quad 0 \leq j, k \leq N,$$

so that, expanding the square,

$$Q_I = \int_I \left| \sum_{k=0}^N \xi_k e^{ik\theta} \right|^2 d\mu(\theta) = \sum_{j,k} \bar{\xi}_j M_{jk} \xi_k.$$

For a complex vector x , the same expansion and Parseval's identity show that $\sum_{j,k} \bar{x}_j M_{jk} x_k = \int_I |\sum_k x_k e^{ik\theta}|^2 d\mu$ lies between 0 and $\int |\sum_k x_k e^{ik\theta}|^2 d\mu = \|x\|^2$, so that the eigenvalues of M lie in $[0, 1]$, and $\text{tr } M = (N+1)\mu(I)$. We write the arc energy as a quadratic form in the real vector $(g, h) = (g_0, \dots, g_N, h_0, \dots, h_N)$ with the matrix

$$\widetilde{M} = \frac{1}{2} \begin{pmatrix} \Re M & -\Im M \\ \Im M & \Re M \end{pmatrix}.$$

Indeed, since $\xi_k = (g_k + ih_k)/\sqrt{2}$, Q_I is real and $M = \Re M + i\Im M$, taking real parts in $Q_I = \frac{1}{2}(g - ih)^T M (g + ih)$ we get

$$Q_I = \frac{1}{2} \{g^T (\Re M) g + h^T (\Re M) h + h^T (\Im M) g - g^T (\Im M) h\} = (g, h)^T \widetilde{M} (g, h).$$

Let λ_ℓ be the eigenvalues of M . If $Mv = \lambda_\ell v$, then taking real and imaginary parts shows that $(\Re v, \Im v)$ and $(-\Im v, \Re v)$ are eigenvectors of $\widetilde{M}$ with eigenvalue $\lambda_\ell/2$, and for an orthonormal eigenbasis of M these $2N+2$ real vectors are orthonormal. Hence each $\lambda_\ell/2$ occurs twice as an eigenvalue of $\widetilde{M}$, so that, since $\lambda_\ell^2 \leq \lambda_\ell$ for $\lambda_\ell \in [0, 1]$, we have $\widetilde{M} \geq 0$, $\text{tr } \widetilde{M} = \text{tr } M$, $\|\widetilde{M}\| \leq \frac{1}{2}$ and $\|\widetilde{M}\|_{\text{HS}}^2 = \frac{1}{2} \sum_\ell \lambda_\ell^2 \leq \text{tr } M/2$. In particular $\mathbb{E}Q_I = \text{tr } \widetilde{M} = \text{tr } M$, since the $2N+2$ coordinates of (g, h) are independent standard Gaussians. As in the real case, the g_k and h_k have moment ψ_2 norm at most two, so that the Hanson–Wright inequality at deviation $\text{tr } M/2$ and the bounds on $\|\widetilde{M}\|_{\text{HS}}^2$ and $\|\widetilde{M}\|$ give, for each arc,

$$\mathbb{P}\{Q_I < \text{tr } M/2\} \leq 2 \exp \left[-c_{\text{HW}} \min \left\{ \frac{(\text{tr } M/2)^2}{16\|\widetilde{M}\|_{\text{HS}}^2}, \frac{\text{tr } M/2}{4\|\widetilde{M}\|} \right\} \right]$$

$$\begin{aligned}
&\leq 2 \exp[-c_{\text{HW}} \min\{\text{tr } M/32, \text{tr } M/4\}] = 2e^{-c_{\text{HW}} \text{tr } M/32} \\
&\leq 2e^{-c_{\text{HW}} \text{tr } M/64} = 2 \exp\left[-\frac{c_{\text{HW}}(N+1)t_N}{64\pi}\right] \leq 2N^{-16},
\end{aligned}$$

since $\text{tr } M = (N+1)\mu(I) = (N+1)t_N/\pi$, as in the real case.

Both cases. In both cases $\mathbb{E}Q_I = (N+1)\mu(I)$, and, by the union bound over at most $8N$ arcs, $\mathbb{P}\{Q_I < \frac{1}{2}\mathbb{E}Q_I \text{ for some arc } I\} \leq (8N)(2N^{-16}) = 16N^{-15} \leq N^{-3}$. On the complement of this event, averaging over each closed arc gives $\max_{\theta \in I} |f_N(e^{i\theta})|^2 \geq Q_I/\mu(I) \geq (N+1)/2$, and by continuity there exists $z_1 = e^{i\theta_1}$ on the arc such that $|f_N(z_1)|^2 \geq (N+1)/2$. Given z_0 , we take such a z_1 on the arc of the angular center nearest to $z_0/|z_0|$. As in the proof of Lemma 4.1, the target disk then lies strictly within $\{|z - z_1| < R\}$, where $R = (2K+1)/N + 2t_N$.

For the energy, let $S = \sum_{k=0}^N |\xi_k|^2$, which is $\sum_{k=0}^N g_k^2$ in the real case. For $0 \leq y \leq 1/8$ and real coefficients $\xi_k = g_k$, we have $\mathbb{E}e^{yg_k^2} = (1-2y)^{-1/2}$, and hence

$$\log \mathbb{E}e^{y(g_k^2-1)} = -y - \frac{1}{2} \log(1-2y) = \sum_{j \geq 2} \frac{2^{j-1}y^j}{j} \leq \frac{y^2}{1-2y} \leq 2y^2,$$

since $2^{j-1}/j \leq 2^{j-2}$ for $j \geq 2$, $\sum_{j \geq 2} 2^{j-2}y^j = y^2/(1-2y)$ and $1-2y \geq 3/4$. For complex coefficients, $|\xi_k|^2$ has the exponential distribution with mean one, so that $\mathbb{E}e^{y|\xi_k|^2} = (1-y)^{-1}$ and

$$\log \mathbb{E}e^{y(|\xi_k|^2-1)} = -y - \log(1-y) = \sum_{j \geq 2} \frac{y^j}{j} \leq \frac{y^2}{2(1-y)} \leq 2y^2,$$

since $1/j \leq 1/2$ for $j \geq 2$ and $1-y \geq 7/8$. In both cases, by independence, the exponential Markov inequality at $y = 1/8$ and the bound $2y^2 = 2/64$ for each of the $N+1$ factors, we have

$$\begin{aligned}
\mathbb{P}\left\{\sum_{k=0}^N |\xi_k|^2 > 2(N+1)\right\} &\leq e^{-2(N+1)/8} \mathbb{E}e^{S/8} = e^{-(N+1)/8} \mathbb{E}e^{(S-(N+1))/8} \\
&= e^{-(N+1)/8} \prod_{k=0}^N \mathbb{E}e^{(|\xi_k|^2-1)/8} \\
&\leq \exp\left[-\frac{N+1}{8} + \frac{2(N+1)}{64}\right] = e^{-3(N+1)/32}.
\end{aligned}$$

On the complement of this event, the Cauchy–Schwarz inequality bounds the polynomial in the local Jensen disk by

$$\begin{aligned}
|f_N(z)| &\leq \left(\sum_{k=0}^N |\xi_k|^2\right)^{1/2} \left(\sum_{k=0}^N |z|^{2k}\right)^{1/2} \\
&\leq \sqrt{2(N+1)}\sqrt{N+1}e^{2NR} = \sqrt{2}(N+1)e^{2NR},
\end{aligned}$$

for $|z - z_1| \leq 2R$, where $|z| \leq 1 + 2R \leq e^{2R}$, so that $\sum_{k=0}^N |z|^{2k} \leq (N+1)e^{4NR}$. By the zero-count step [22, §8.21] and Jensen's formula [22, §3.61] at the nonzero center value, as in the proof of Lemma 4.1, and by the two bounds on $|f_N|$ above, we have

$$\begin{aligned} \#\{\alpha : |\alpha - z_1| < R\} &\leq \frac{\log(\sqrt{2}(N+1)e^{2NR}) - \frac{1}{2}\log((N+1)/2)}{\log 2} \\ &\leq C_1(K) \log N + \frac{1}{2} \leq C_{1,G}(K) \log N. \end{aligned}$$

The middle step is the last display of that proof at $B = \eta = 1$, whose maximum bound $(N+1)e^{2NR}$ lacks the factor $\sqrt{2}$, which adds $\log \sqrt{2}/\log 2 = \frac{1}{2}$, and the last step holds since $\frac{1}{2} \leq (2/\log 2) \log N$ for $N \geq 2$. $\square$

In the radial estimates below, $K \geq 2$ is fixed, N is sufficiently large, and $r \in [1 - K/N, 1 + K/N]$.

Proposition 7.4 (Transfer to other coefficient distributions). *Let the coefficient distribution be Steinhaus, standard real Gaussian, standard complex Gaussian, or a fixed nondegenerate bounded centrally symmetric distribution on $\mathbb{R}$ or $\mathbb{C}$. For the bounded distribution, normalize*

$$\mathbb{E}|\xi_0|^2 = 1, \quad |\xi_0| \leq B, \quad B \geq 1.$$

The occupation and jet estimates of Sections 3 and 4, and the local-count, derivative and maximal-tail estimates, hold with the model constants and exceptional probabilities in Section A.11. These probabilities are summable along $N = j^8$.

Put $V = \int |X_r|^2 d\mu$ and, for a bounded distribution, put

$$C_B = 2\{C_* + \max(\frac{1}{2}\log 2, \log B) + 1\}.$$

For $p \geq 1$, the table gives an upper bound for

$$\int_E \int |\log |X_r||^p d\mu d\mathbb{P}.$$

The bounds in the first two rows also hold without restriction to E .

Coefficient distribution	Logarithmic	
	moment	Energy on E
Steinhaus	$\leq (C_*p)^{6p}$	$E = \Omega, V = 1$
Standard real or complex Gaussian	$\leq (16p)^p$	$E = \{V \leq 2\}$
Bounded centrally symmetric	$\leq (C_Bp)^{6p}$	$E = \{V \geq 1/2\}, V \leq B^2$

The corresponding energy exceptions satisfy

$$\mathbb{P}(E^c) \leq \begin{cases} 0, & \text{Steinhaus,} \\ \exp[-3N/(32e^{6K})], & \text{Gaussian,} \\ \exp[-N/(2B^4e^{6K})], & \text{bounded centrally symmetric.} \end{cases}$$

In particular, for all sufficiently large N ,

$$\begin{aligned} & \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > C(K)N^{-1/32}(\log N)^7\} \\ & \leq C(K)N^{-3/8} + N^{-12} + C(K)N^{-7/16}(\log N)^{-12} + \mathbb{P}(E^c). \end{aligned}$$

Here the constant $C(K)$ and the degree cutoff may depend on the coefficient distribution.

Proof. The idea is to check, for each distribution, the inputs of Sections 3 and 4, and to redo every computation in which an input changes, applying Lemma 3.2 for the logarithmic concentration. Table 5 collects the inputs that change. The one- and two-point bounds of Section 3.1, and the jet bounds in the proof of Lemma 4.5, use the coefficients only through the independence and centering of the summands, their covariance matrices and their third moments, and the proof of Theorem 3.1 uses them only through (14), (15) and (16). In the proofs of Lemmas 4.3 and 4.6, the coefficients enter through a tail bound at each sample and a deterministic bound on a derivative. Since $\mathbb{E}|\xi_0|^2 = 1$, the normalization $\sigma_N(r)$ and the radial bounds (10) are those of the sign case.

Steinhaus coefficients. Let ξ_k be independent uniform points on the unit circle, and write $\xi_k = e^{i\omega_k}$, with independent ω_k uniform on $[0, 2\pi)$. Then $\mathbb{E}\xi_k = 0$, $|\xi_k|^2 = 1$ and $\mathbb{E}|\xi_k|^3 = 1$. To show that the one-point covariance is exactly circular, and that (9) controls the two-point covariance kernel, fix angles θ, ϕ and an index k . Since $2\omega_k$ is uniform modulo 2π , the variables $\cos(2\omega_k + k(\theta + \phi))$ and $\sin(2\omega_k + k(\theta + \phi))$ have mean zero. Therefore, by the product-to-sum identities, we have

$$\begin{aligned} \mathbb{E}[\cos(\omega_k + k\theta) \cos(\omega_k + k\phi)] &= \frac{1}{2}\mathbb{E}[\cos(k(\theta - \phi)) + \cos(2\omega_k + k(\theta + \phi))] \\ &= \frac{1}{2}\cos(k(\theta - \phi)), \\ \mathbb{E}[\sin(\omega_k + k\theta) \sin(\omega_k + k\phi)] &= \frac{1}{2}\mathbb{E}[\cos(k(\theta - \phi)) - \cos(2\omega_k + k(\theta + \phi))] \\ &= \frac{1}{2}\cos(k(\theta - \phi)), \\ \mathbb{E}[\sin(\omega_k + k\theta) \cos(\omega_k + k\phi)] &= \frac{1}{2}\mathbb{E}[\sin(k(\theta - \phi)) + \sin(2\omega_k + k(\theta + \phi))] \\ &= \frac{1}{2}\sin(k(\theta - \phi)), \\ \mathbb{E}[\cos(\omega_k + k\theta) \sin(\omega_k + k\phi)] &= \frac{1}{2}\mathbb{E}[\sin(2\omega_k + k(\theta + \phi)) - \sin(k(\theta - \phi))] \end{aligned}$$

Quantity	Steinhaus	Gaussian	Bounded	Source
Local count	$C_1(K+2)$	$C_{1,G}(K+2)$	$C_{1,B}(K+2)$	Lemmas 7.1–7.3
Exception beyond N^{-3}	none	$e^{-3(N+1)/32}$	$e^{-(N+1)/(2B^4)}$	Lemmas 7.1–7.3
f_N'' and tail bounds	signs	signs	B times signs	Lemmas 4.3, 4.6
Envelope exception	none	$2(2N+1)e^{-N^2/2}$	none	Lemmas 4.3, 4.6
Occupation, mean	signs	$2 + 320e^{8K}$	$2 + c_B$	(14)
Occupation, variance	signs	$14 + 960e^{8K}$	$14 + 3c_B$	(14)
$\mathbb{P}\{B_{K,N} > N^{31/32}\}$	(27)	(49)	(50)	Lemma 4.5

Table 5: The inputs of the proof of Proposition 7.4 that depend on the coefficient distribution, with the sign-case result that each row transfers. The Gaussian column covers the real and the complex case, “signs” means the value of the sign case, the occupation constants multiply $N^{-1/2}$, and the bounds on f_N'' and on the tails fail with probability at most N^{-10} each, for Gaussian coefficients on the envelope event $\max_{k \leq 2N} |\xi_k| \leq N$.

$$= -\frac{1}{2} \sin(k(\theta - \phi)).$$

Since $\Re(\xi_k e^{ik\theta}) = \cos(\omega_k + k\theta)$ and $\Im(\xi_k e^{ik\theta}) = \sin(\omega_k + k\theta)$, the four lines are

$$(45) \quad \begin{aligned} \mathbb{E}[\Re(\xi_k e^{ik\theta}) \Re(\xi_k e^{ik\phi})] &= \mathbb{E}[\Im(\xi_k e^{ik\theta}) \Im(\xi_k e^{ik\phi})] = \frac{1}{2} \cos(k(\theta - \phi)), \\ \mathbb{E}[\Im(\xi_k e^{ik\theta}) \Re(\xi_k e^{ik\phi})] &= -\mathbb{E}[\Re(\xi_k e^{ik\theta}) \Im(\xi_k e^{ik\phi})] = \frac{1}{2} \sin(k(\theta - \phi)). \end{aligned}$$

Let $z = re^{i\theta}$ and $w = se^{i\phi}$, with $r, s \in [1 - K/N, 1 + K/N]$. Since the ξ_k are independent and centered, multiplying (45) by $(rs)^k / (\sigma_N(r)\sigma_N(s))$ and summing over k shows, since $\sum_k r^{2k} = \sigma_N(r)^2$, that the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is $\frac{1}{2} I_2$, and that the cross covariance of $(\Re X_r(\theta), \Im X_r(\theta))$ and $(\Re X_s(\phi), \Im X_s(\phi))$ is

$$\frac{1}{2\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k \begin{pmatrix} \cos(k(\theta - \phi)) & -\sin(k(\theta - \phi)) \\ \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) \end{pmatrix}.$$

Each entry of this matrix is, up to sign, $N/(2\sigma_N(r)\sigma_N(s))$ times the real or the imaginary part of the sum in (9) with $\beta = 0$ and $\psi = \theta - \phi$. For retained pairs, by (9) and the lower bound in (10) we have

$$\frac{N}{2\sigma_N(r)\sigma_N(s)} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta - \phi)} \right| \leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 5e^{8K} N^{-1/2} \leq 10e^{8K} N^{-1/2}.$$

Hence the covariance matrices of Section 3.1 satisfy the entry bounds (11), with the one-point error equal to zero, and the eigenvalue bound and the total variation

bound (12) of that section, which use the covariance only through these entry bounds, hold with the same constants.

We apply Raič's theorem to independent centered real vectors obtained from each complex coefficient. The k th summand of the real vector with the coordinates $\Re X_r(\theta)$, $\Im X_r(\theta)$, $\Re X_s(\phi)$ and $\Im X_s(\phi)$ is

$$\begin{pmatrix} r^k \cos(\omega_k + k\theta)/\sigma_N(r) \\ r^k \sin(\omega_k + k\theta)/\sigma_N(r) \\ s^k \cos(\omega_k + k\phi)/\sigma_N(s) \\ s^k \sin(\omega_k + k\phi)/\sigma_N(s) \end{pmatrix}.$$

These summands are independent and centered, their covariance matrices sum to that of the vector, and the squared norm of the k th summand is $r^{2k}/\sigma_N(r)^2 + s^{2k}/\sigma_N(s)^2$, as for signs. After the whitening of Section 3.1, the sums of their third moments are therefore at most $46e^{9K}N^{-1/2}$ and, for one point, at most $16e^{9K}N^{-1/2}$, as computed there, so that the normal-approximation errors have the same bounds as for signs. It follows that (13) and the occupation bounds (14) hold for Steinhaus coefficients, with the constants of the sign case.

For the jets in the proof of Lemma 4.5, we have $W_l(z) = N^{-1/2} \sum_{k=0}^N \xi_k(k/N)^l z^k$ for $l \in \{0, 1\}$. Multiplying (45) by $(k/N)^{l+m}(rs)^k/N$ and summing over k shows that the cross covariance of $(\Re W_l(z), \Im W_l(z))$ and $(\Re W_m(w), \Im W_m(w))$ is

$$\frac{1}{2N} \sum_{k=0}^N (rs)^k \left(\frac{k}{N}\right)^{l+m} \begin{pmatrix} \cos(k(\theta - \phi)) & -\sin(k(\theta - \phi)) \\ \sin(k(\theta - \phi)) & \cos(k(\theta - \phi)) \end{pmatrix}, \quad l, m \in \{0, 1\}.$$

For $w = z$ this shows that the covariance matrix of $W(z)$ is the nonoscillatory matrix of that proof, with no oscillatory part. Its least eigenvalue is therefore at least $e^{-4K}/32$, which is larger than the bound $e^{-4K}/80$ of (24) used there. For a retained pair, every entry of the cross covariance of $W(z)$ and $W(w)$ is, up to sign, half the real or the imaginary part of $N^{-1} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta - \phi)}$, with $\beta = l + m \in \{0, 1, 2\}$, which by (9) is at most $5e^{4K}N^{-1/2} \leq 10e^{4K}N^{-1/2}$ in absolute value, the bound used there. Moreover, the summands of $(W(z), W(w))$ are independent and centered, and the squared norm of the k th summand is $(r^{2k} + s^{2k})(1 + (k/N)^2)/N$, as for signs. Therefore the Raič, small-ball and entropy steps of that proof hold with the same constants.

For complex numbers c_k , the variables $\Re(\xi_k c_k)$ are independent and centered, with $|\Re(\xi_k c_k)| \leq |c_k|$, so that, by Hoeffding's inequality [9, Theorem 2] for both tails, for every level $u > 0$,

$$\mathbb{P}\left\{\left|\Re \sum_k \xi_k c_k\right| \geq u\right\} \leq 2 \exp\left[-\frac{u^2}{2 \sum_k |c_k|^2}\right],$$

and the same bound holds for the imaginary part, since also $|\Im(\xi_k c_k)| \leq |c_k|$. This is the bound used in the proofs of Lemmas 4.3 and 4.6, whose deterministic bounds use only $|\xi_k| \leq 1$. Hence these lemmas hold for Steinhaus coefficients, with the same constants and exceptional probabilities.

Write $\xi_k = \epsilon_k Z_k$ as in the proof of Lemma 7.1, and denote expectations over the signs and over the Z_k by $\mathbb{E}_\epsilon$ and $\mathbb{E}_Z$. Conditional on all Z_k , let $b_k = Z_k r^k / \sigma_N(r)$, so that $X_r(\theta) = \sum_k \epsilon_k b_k e^{ik\theta}$ and, since $|Z_k| = 1$, $\sum_{k=0}^N |b_k|^2 = \sigma_N(r)^{-2} \sum_{k=0}^N |Z_k|^2 r^{2k} = 1$. Hence (15) for the normalized coefficients b_k , whose constant C_* does not depend on them, gives $\mathbb{E}_\epsilon \int |\log |\sum_{k=0}^N \epsilon_k b_k e^{ik\theta}||^p d\mu \leq (C_* p)^{6p}$, and, by Fubini's theorem, $\mathbb{E} \int |\log |X_r||^p d\mu = \mathbb{E}_Z [\mathbb{E}_\epsilon \int |\log |X_r||^p d\mu] \leq (C_* p)^{6p}$.

For the local count we use Lemma 7.1 (Table 5). In the proof of Lemma 4.5, the coefficients enter only through this local count, Lemma 4.3, the covariances and the summand norms, which hold here with the same constants, so that (27) holds for Steinhaus coefficients. By Parseval's identity and $|\xi_k| = 1$, $\int |X_r|^2 d\mu = \sigma_N(r)^{-2} \sum_k |\xi_k|^2 r^{2k} = 1$ for every realization, which is (16). With (14), (15) and (16), the proof of Theorem 3.1 applies word for word, and we obtain (8) with the same constants, which is the concentration bound of the proposition, with $E = \Omega$ and $\mathbb{P}(E^c) = 0$.

Real Gaussian coefficients. Let the coefficients be independent standard real Gaussians g_k . The real vector $(\Re X_r(\theta), \Im X_r(\theta))$ and the real vector with the coordinates $\Re X_r(\theta), \Im X_r(\theta), \Re X_s(\phi)$ and $\Im X_s(\phi)$ are the sums over k of g_k times the deterministic summands of Section 3.1, and $\mathbb{E} g_k = 0$ and $\mathbb{E} g_j g_k = \mathbf{1}\{j = k\}$, as for signs, so that these vectors have the laws of G_Σ for the covariance matrices Σ of that section. Hence the normal approximation is exact, and only the total variation bound (12) of that section remains. For retained angles and pairs, $t, t' \geq 0$ and the product of disks D of Section 3.1, we get

$$(46) \quad \begin{aligned} \left| \mathbb{P}\{|X_r(\theta)| \leq t, |X_s(\phi)| \leq t'\} - F_G(t)F_G(t') \right| &= \left| \mathbb{P}(G_\Sigma \in D) - \mathbb{P}(G_{I_4/2} \in D) \right| \\ &\leq \text{TV}(G_\Sigma, G_{I_4/2}) \leq 320e^{8K} N^{-1/2}. \end{aligned}$$

Here the equality holds since the two coordinate pairs of $G_{I_4/2}$ are independent copies of G , and the last step is (12), which uses Σ only through the bound $40e^{8K} N^{-1/2}$ on the norm of Σ minus half the identity. With the disk of radius t and the one-point covariance, the same three steps show that $|\mathbb{P}\{|X_r(\theta)| \leq t\} - F_G(t)| \leq 320e^{8K} N^{-1/2}$. As in Section 3.1, the removed angles and pairs have measures at most $2N^{-1/2}$ and $10N^{-1/2}$, and there the integrands differ from $F_G(t)$ and $F_G(t)^2$ by at most one.

Hence, by Tonelli's theorem, as in (14), uniformly for $t \geq 0$,

$$\begin{aligned}
|\mathbb{E}A_t - F_G(t)| &\leq (2 + 320e^{8K})N^{-1/2}, \\
|\mathbb{E}A_t^2 - F_G(t)^2| &\leq (10 + 320e^{8K})N^{-1/2}, \\
\text{Var}(A_t) &= \mathbb{E}A_t^2 - F_G(t)^2 - (\mathbb{E}A_t - F_G(t))(\mathbb{E}A_t + F_G(t)) \\
&\leq |\mathbb{E}A_t^2 - F_G(t)^2| + |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \\
&\leq (10 + 320e^{8K})N^{-1/2} + 2(2 + 320e^{8K})N^{-1/2} \\
&= (14 + 960e^{8K})N^{-1/2},
\end{aligned}$$

where the last line uses the first two and $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$. These are the occupation bounds (14) for real Gaussian coefficients.

The vector $(W(z), W(w))$ is the sum over k of g_k times the deterministic summands in the proof of Lemma 4.5, so that it has the law of $G_{\Sigma_{z,w}}$, where $\Sigma_{z,w}$ is its covariance matrix (the matrix Σ of that proof), and $W(z)$ has the law of G_{Σ_z} , where Σ_z is the covariance matrix of $W(z)$ (the matrix A_z of that proof). Hence the covariance bounds of that proof hold, and its Raič steps are exact. Let $D' \subset \mathbb{R}^4$ be the product of the two disks of radii $C(K)N^{-1/32}/\sqrt{\log N}$ and $3N^{-1/64}$ that defines the test (23), where $N^{1/64}$ is the parameter L of that proof. Bounding the density of G_{Σ_z} and the volume of D' as in that proof, with $(\det \Sigma_z)^{-1/2} \leq (e^{-4K}/80)^{-2} = 6400e^{8K}$ by (24), we get

$$\begin{aligned}
\mathbb{P}\{W(z) \in D'\} &= \mathbb{P}\{G_{\Sigma_z} \in D'\} \leq (2\pi)^{-2}(\det \Sigma_z)^{-1/2} \cdot \pi^2 \frac{C(K)^2 N^{-1/16}}{\log N} \cdot 9N^{-1/32} \\
&\leq 14400e^{8K} C(K)^2 N^{-3/32} / \log N \leq C(K) N^{-3/32} / \log N.
\end{aligned}$$

Since $N^{-3/32} = (N^{1/64})^{-6}$, this is (25) without its term $C(K)N^{-1/2}$. For separated pairs, with the block diagonal matrix Σ_0 of that proof, we have

$$\begin{aligned}
&\left| \text{Cov}(\mathbf{1}\{W(z) \in D'\}, \mathbf{1}\{W(w) \in D'\}) \right| \\
&= \left| \mathbb{P}\{G_{\Sigma_{z,w}} \in D' \times D'\} - \mathbb{P}\{G_{\Sigma_z} \in D'\} \mathbb{P}\{G_{\Sigma_w} \in D'\} \right| \\
&\leq \text{TV}(G_{\Sigma_{z,w}}, G_{\Sigma_0}) \leq 6400e^{8K} N^{-1/2},
\end{aligned}$$

where the first inequality holds since the product of the last two probabilities is $\mathbb{P}\{G_{\Sigma_0} \in D' \times D'\}$, by the independence of the two blocks of G_{Σ_0} , and the last is (26) with the eigenvalue bound $e^{-4K}/80$ of (24).

At every angle $\mathbb{E}|X_r(\theta)|^2 = \sigma_N(r)^{-2} \sum_k r^{2k} = 1$, so that the two real covariance eigenvalues are nonnegative and sum to one, and the larger one is at least $1/2$. Since $|X_r|$ is at least the absolute value of its projection onto an eigenvector with variance $v \geq 1/2$, and the Gaussian density is at most $1/\sqrt{2\pi v}$, we have

$\mathbb{P}\{|X_r| \leq u\} \leq 2u/\sqrt{2\pi v} \leq 2u$ for $u \geq 0$. With $u = e^{-\tau}$, layer-cake integration yields the negative logarithmic moments

$$\begin{aligned} \mathbb{E}(\log^- |X_r|)^p &= p \int_0^\infty \tau^{p-1} \mathbb{P}\{|X_r| < e^{-\tau}\} d\tau \\ &\leq 2p \int_0^\infty \tau^{p-1} e^{-\tau} d\tau = 2\Gamma(p+1), \quad p \geq 1. \end{aligned}$$

For the positive part, write $R = \sqrt{2}|G|$. Then $\sqrt{2}\Re G$ and $\sqrt{2}\Im G$ are independent standard Gaussians, and R is the norm of the vector they form. In a covariance eigenbasis, the coordinates of $(\Re X_r, \Im X_r)$ are independent centered Gaussians with variances at most one, so we can couple X_r so that $|X_r| \leq R$. Since $\log^+ x \leq x$, and $R^2/2$ has the exponential distribution with mean one, we have $\mathbb{E}(\log^+ |X_r|)^p \leq \mathbb{E}R^p = 2^{p/2}\Gamma(1+p/2)$. For $a > 0$, since $\max_{y \geq 0} y^a e^{-y/2} = (2a/e)^a$ and $\int_0^\infty e^{-y/2} dy = 2$, we have $\Gamma(a+1) = \int_0^\infty y^a e^{-y/2} e^{-y/2} dy \leq 2(2a/e)^a$. Taking $a = p$ and $a = p/2$, and using $4 \leq (4e)^p$ and $2 \leq (32ep)^{p/2}$ for $p \geq 1$, we obtain $2\Gamma(p+1) \leq 4(2p/e)^p \leq (8p)^p$ and $2^{p/2}\Gamma(1+p/2) \leq 2(2p/e)^{p/2} \leq (8p)^p$. Adding the two logarithmic parts and integrating over angle yields

$$(47) \quad \mathbb{E} \int |\log |X_r||^p d\mu \leq 2(8p)^p \leq (16p)^p, \quad p \geq 1.$$

For the angular energy we set $V = \sum_{k=0}^N w_k g_k^2$, with $w_k = r^{2k}/\sigma_N(r)^2$, which is $\int |X_r|^2 d\mu$ by Parseval's identity. For the weights, $\sum_k w_k = 1$ and $w_{\max} = \max_k w_k \leq e^{6K}/N$, by $r^{2k} \leq e^{2K}$ and the lower bound in (10). By the bound on $\log \mathbb{E}e^{y(g_k^2-1)}$ for $0 \leq y \leq 1/8$ in the proof of Lemma 7.3, with $y = \lambda w_k$, and by independence and $\mathbb{E}e^{\lambda w_k g_k^2} = (1 - 2\lambda w_k)^{-1/2}$, we have, for $0 \leq \lambda \leq (8w_{\max})^{-1}$,

$$\log \mathbb{E}e^{\lambda(V-1)} = \sum_k \{-\lambda w_k - \tfrac{1}{2} \log(1 - 2\lambda w_k)\} \leq 2\lambda^2 \sum_k w_k^2.$$

In the exponential Markov inequality we choose $\lambda = (8w_{\max})^{-1}$, the largest admissible value, which is the best one, since the exponent $-\lambda + 2\lambda^2 w_{\max}$ below has derivative $-1 + 4\lambda w_{\max} < 0$ for $0 \leq \lambda < (4w_{\max})^{-1}$, and so decreases on the admissible range. Then, since $\sum_k w_k^2 \leq w_{\max} \sum_k w_k = w_{\max}$,

$$\begin{aligned} (48) \quad \mathbb{P}(V > 2) &\leq e^{-\lambda} \mathbb{E}e^{\lambda(V-1)} \leq \exp\left(-\lambda + 2\lambda^2 \sum_k w_k^2\right) \leq \exp(-\lambda + 2\lambda^2 w_{\max}) \\ &= \exp[-3/(32w_{\max})] \leq \exp[-3N/(32e^{6K})]. \end{aligned}$$

Laurent and Massart [15] give sharper tail bounds for such weighted sums of squared Gaussians, but this bound suffices here.

By (47), $\mathbb{E} \int |\log |X_r||^p d\mu \leq (16p)^p \leq (16p)^{6p}$ for $p \geq 1$, and hence this bound also holds for the integral over any event. With this moment bound and the

occupation bounds above, whose constants satisfy $2 + 320e^{8K} < 14 + 960e^{8K}$, Lemma 3.2 applies on the energy event $E = \{V \leq 2\}$, with

$$H = 14 + 960e^{8K}, \quad C = 16, \quad B_E = 2,$$

so that $4eC = 64e$ and $B_E/2 + 1 = 2$. Its error is at most $C(K)N^{-1/32}(\log N)^7$, and its probability bound, with $\mathbb{P}(E^c) = \mathbb{P}(V > 2)$ bounded in (48), is the concentration bound of the proposition.

For the local count we use the real case of Lemma 7.3 (Table 5). For real numbers c_k , the sum $Y = \sum_k c_k g_k$ is a centered real Gaussian of variance $v = \sum_k c_k^2$. For $v > 0$ and a level $u > 0$, by the exponential Markov inequality at $\lambda = u/v$ we have

$$\mathbb{P}\{Y \geq u\} \leq e^{-\lambda u} \mathbb{E}e^{\lambda Y} = \exp(-\lambda u + \lambda^2 v/2) = e^{-u^2/(2v)},$$

and hence $\mathbb{P}\{|Y| \geq u\} \leq 2e^{-u^2/(2v)}$, since $-Y$ has the law of Y . This is the form of Hoeffding's inequality used in the proofs of Lemmas 4.3 and 4.6. The real and imaginary parts of $f_N''(z)$ and of a tail $\sum_{k=N+1}^n g_k z^k$ of Lemma 4.6, with $N < n \leq N + m_{\max}$, are such sums, with variances at most $\sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}$ and $\sum_{k=N+1}^n |z|^{2k}$. Applying the above bound to the real and the imaginary part at the level divided by $\sqrt{2}$, at each sample z we have

$$\begin{aligned} \mathbb{P}\{|f_N''(z)| \geq 50e^{K+4}N^{5/2}\sqrt{\log N}\} \\ \leq 4 \exp \left[-\frac{(50e^{K+4}N^{5/2}\sqrt{\log N})^2}{4 \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}} \right] \leq 4N^{-100}, \end{aligned}$$

and, for $N < n \leq N + m_{\max}$ and $\rho = 1 + K/N$,

$$\begin{aligned} \mathbb{P}\left\{ \left| \sum_{k=N+1}^n g_k z^k \right| \geq 10e^{2K}\sqrt{m_{\max} \log N} \right\} \\ \leq 4 \exp \left[-\frac{(10e^{2K}\sqrt{m_{\max} \log N})^2}{4 \sum_{k=N+1}^n \rho^{2k}} \right] \leq 4N^{-25}, \end{aligned}$$

where the levels are half the threshold $100e^{K+4}N^{5/2}\sqrt{\log N}$ of Lemma 4.3 and two thirds of the amplitude $15e^{2K}\sqrt{m_{\max} \log N}$ of Lemma 4.6, and the last steps are those of the sign case. For interpolation, we consider the event $\max_{k \leq 2N} |g_k| \leq N$, whose complement has probability at most $2(2N+1)e^{-N^2/2}$, by the above bound with $v = 1$ and $u = N$ and a union bound over the $2N+1$ coefficients. On this event, the deterministic bounds in the proofs of Lemmas 4.3 and 4.6, which used $|\epsilon_k| = 1$, gain the factor N :

$$\sup_{|z| \leq 1+(K+1)/N} |f_N'''(z)| \leq N \cdot e^{K+1}N^4 = e^{K+1}N^5,$$

$$\sup_{|z| \leq \rho} \left| \sum_{k=N+1}^n k g_k z^{k-1} \right| \leq N \cdot 2N m_{\max} e^{2K} = 2N^2 m_{\max} e^{2K}.$$

Each interpolation error is the arc-length spacing of the samples times such a supremum, so the spacing must shrink by the factor N to keep the interpolation errors of the sign case. We therefore use $\lceil N^4 \rceil \leq 2N^4$ equally spaced boundary samples in both proofs, in place of $\lceil N^3 \rceil$. Then every point of the circle $|z| = 1 + (K+1)/N$, of length at most 8π , lies within arc length $4\pi/N^4$ of a sample, and every point of the circle $|z| = \rho$, of length at most 4π , within $2\pi/N^4$. Therefore the interpolation errors are

$$\frac{4\pi}{N^4} \cdot e^{K+1} N^5 = 4\pi e^{K+1} N, \quad \frac{2\pi}{N^4} \cdot 2N^2 m_{\max} e^{2K} = \frac{4\pi m_{\max} e^{2K}}{N^2},$$

which are those of the sign case. There are at most $2N^4$ samples, and at most $2N^4 \cdot N$ pairs of a sample and one of the at most N nonempty tails, so that, by the union bound, the two unions have probabilities at most $(2N^4)(4N^{-100}) = 8N^{-96} \leq N^{-10}$ and $(2N^4 \cdot N)(4N^{-25}) = 8N^{-20} \leq N^{-10}$. We conclude that, on the event $\max_{k \leq 2N} |g_k| \leq N$, the bounds of Lemmas 4.3 and 4.6 hold for real Gaussian coefficients with the same constants, each outside an event of probability at most N^{-10} .

In the proof of Lemma 4.5, the cell assignment and the sector cover now use the local count of Lemma 7.3, with $K+2$ in place of K . From them we obtain

$$B_{K,N} \leq C_{1,G}(K+2)(\log N) \cdot \#\{\text{successful tests}\} + 40C_{1,G}(K+2)\sqrt{N} \log N,$$

where the successful tests are the retained mesh points at which (23) holds, and we take $Q = N^{31/32}/(2C_{1,G}(K+2) \log N)$ there. With the jet bounds above, the mean and variance bounds and the Chebyshev chain of that proof hold with its constants $C(K)$, which now also depend on $C_{1,G}(K+2)$. Adding the exceptional probabilities of the local count, of the derivative bound and of the event $\max_{k \leq 2N} |g_k| \leq N$, we get

$$(49) \quad \mathbb{P}\{B_{K,N} > N^{31/32}\} \\ \leq N^{-3} + e^{-3(N+1)/32} + N^{-10} + 2(2N+1)e^{-N^2/2} + C(K)N^{-3/8}(\log N)^4.$$

Since Gaussian coefficients are nonzero almost surely, every prefix has degree exactly N , and $f_N(0) = g_0 \neq 0$.

Standard complex Gaussian coefficients. The conservative bounds of the real case also hold for $\xi_k = (g_k + ih_k)/\sqrt{2}$, where all g_k, h_k are independent standard

real Gaussians. We have $\mathbb{E}|\xi_k|^2 = 1$, and $|\xi_k|^2$ has the exponential distribution with mean one. Since $\Re(\xi_k e^{ik\theta}) = (g_k \cos(k\theta) - h_k \sin(k\theta))/\sqrt{2}$ and $\Im(\xi_k e^{ik\theta}) = (g_k \sin(k\theta) + h_k \cos(k\theta))/\sqrt{2}$, and since $\mathbb{E}g_k^2 = \mathbb{E}h_k^2 = 1$ and $\mathbb{E}g_k h_k = 0$, we get

$$\begin{aligned}\mathbb{E}[\Re(\xi_k e^{ik\theta})\Re(\xi_k e^{ik\phi})] &= \frac{1}{2}\{\cos(k\theta)\cos(k\phi) + \sin(k\theta)\sin(k\phi)\} = \frac{1}{2}\cos(k(\theta - \phi)), \\ \mathbb{E}[\Im(\xi_k e^{ik\theta})\Im(\xi_k e^{ik\phi})] &= \frac{1}{2}\{\sin(k\theta)\sin(k\phi) + \cos(k\theta)\cos(k\phi)\} = \frac{1}{2}\cos(k(\theta - \phi)), \\ \mathbb{E}[\Im(\xi_k e^{ik\theta})\Re(\xi_k e^{ik\phi})] &= \frac{1}{2}\{\sin(k\theta)\cos(k\phi) - \cos(k\theta)\sin(k\phi)\} = \frac{1}{2}\sin(k(\theta - \phi)), \\ \mathbb{E}[\Re(\xi_k e^{ik\theta})\Im(\xi_k e^{ik\phi})] &= \frac{1}{2}\{\cos(k\theta)\sin(k\phi) - \sin(k\theta)\cos(k\phi)\} = -\frac{1}{2}\sin(k(\theta - \phi)),\end{aligned}$$

which is (45). Since the covariance matrices of the real vectors of Section 3.1 and of the proof of Lemma 4.5 are computed from (45) and the independence of the coefficients alone, they are those of the Steinhaus case, and they satisfy the entry and eigenvalue bounds shown there. These vectors are linear images of $(g_0, \dots, g_N, h_0, \dots, h_N)$, and hence centered Gaussian vectors, so that the normal approximation is again exact. Since the chain (46) and the occupation bounds after it use Σ only through the norm bound $40e^{8K}N^{-1/2}$, they hold word for word. In the real Gaussian case, the jet bounds use only the least eigenvalue $e^{-4K}/80$ of (24) and the total variation bound (26) in the proof of Lemma 4.5, and so they also hold word for word.

For nonnegative weights with $\sum_k w_k = 1$, let $w_{\max} = \max_k w_k$ as before. For $0 \leq \lambda w_k \leq 1/8$, by independence, $\mathbb{E}e^{\lambda w_k |\xi_k|^2} = (1 - \lambda w_k)^{-1}$, $\sum_k w_k = 1$ and the bound on $\log \mathbb{E}e^{y(|\xi_k|^2 - 1)}$ in the proof of Lemma 7.3 with $y = \lambda w_k$, we have

$$\begin{aligned}\log \mathbb{E} \exp \left[\lambda \left(\sum_k w_k |\xi_k|^2 - 1 \right) \right] &= \sum_k \{-\lambda w_k - \log(1 - \lambda w_k)\} \\ &\leq 2\lambda^2 \sum_k w_k^2 \leq 2\lambda^2 w_{\max}.\end{aligned}$$

With $\lambda = (8w_{\max})^{-1}$, the largest admissible value, as in the real case, Markov's inequality gives

$$\begin{aligned}\mathbb{P} \left\{ \sum_k w_k |\xi_k|^2 > 2 \right\} &\leq e^{-\lambda} \mathbb{E} \exp \left[\lambda \left(\sum_k w_k |\xi_k|^2 - 1 \right) \right] \\ &\leq \exp(-\lambda + 2\lambda^2 w_{\max}) = \exp[-3/(32w_{\max})].\end{aligned}$$

For radial weights, $w_{\max} \leq e^{6K}/N$. Since $\sum_k w_k |\xi_k|^2 = V$ by Parseval's identity, this yields the bound on the probability of the energy exception in (48). For uniform weights, we get the local-count exception $e^{-3(N+1)/32}$ of Lemma 7.3.

For d coefficients, since $\mathbb{P}\{|\xi_k| > N\} = e^{-N^2}$, the common envelope satisfies $\mathbb{P}\{\max_{k < d} |\xi_k| > N\} \leq de^{-N^2} \leq 2de^{-N^2/2}$. A linear combination $Y = \sum_k c_k \xi_k$

is circular Gaussian with $v = \sum_k |c_k|^2$, so that $Y = 0$ almost surely if $v = 0$, and, if $v > 0$, $\mathbb{P}\{|Y| > u\} = e^{-u^2/v} \leq 4e^{-u^2/(4v)}$ for $u \geq 0$, since $|Y|^2/v$ has the exponential distribution with mean one. With $d = 2N + 1$, the envelope event is $\max_{k \leq 2N} |\xi_k| \leq N$, and its complement has the probability bound $2(2N + 1)e^{-N^2/2}$ of the real case. The sample events are $|Y| \geq u$ with $Y = f_N''(z)$, for which $v = \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}$, and with Y a tail $\sum_{k=N+1}^n \xi_k z^k$, for which $v = \sum_{k=N+1}^n |z|^{2k}$. At the two levels of the real case, $4e^{-u^2/(4v)}$ is the bound in the first line of the corresponding displays there, so that the sample bounds are again $4N^{-100}$ and $4N^{-25}$. The numbers of samples and of tails are also those of the real case, and hence the same sample thresholds and union estimates apply. On the envelope event, $|\xi_k| \leq N$ for $k \leq 2N$, which is all that the deterministic bounds on f_N''' and on the derivatives of the tails use, so that the N^4 grid yields the same interpolation bounds and the same derivative and tail amplitudes.

For the local count we use the complex case of Lemma 7.3, whose constant and exceptional probability are those of the real case (Table 5), and so are the derivative bound on the envelope event and the jet bounds. It follows that (49) also holds for these coefficients. For the logarithmic moments, X_r is a standard complex Gaussian, that is, circular with $\mathbb{E}|X_r|^2 = 1$, so that $|X_r|^2$ has the exponential distribution with mean one. Hence, for $u \geq 0$ and $p \geq 1$, we have $\mathbb{P}\{|X_r| \leq u\} = 1 - e^{-u^2} \leq \min(u^2, 1) \leq 2u$, and, with $u = e^{-\tau}$ as in the real case, $\log^+ x \leq x$ and $\mathbb{E}|X_r|^p = \mathbb{E}(|X_r|^2)^{p/2}$,

$$\mathbb{E}(\log^- |X_r|)^p = p \int_0^\infty \tau^{p-1} \mathbb{P}\{|X_r| < e^{-\tau}\} d\tau \leq 2\Gamma(p+1) \leq (8p)^p,$$

$$\mathbb{E}(\log^+ |X_r|)^p \leq \mathbb{E}|X_r|^p = \Gamma(1 + p/2) \leq 2^{p/2}\Gamma(1 + p/2) \leq (8p)^p,$$

where the last inequalities are those of the real Gaussian case. Adding the two parts and integrating over angle, we obtain $\mathbb{E} \int |\log |X_r||^p d\mu \leq 2(8p)^p \leq (16p)^p$. Since the occupation bounds, the moment bound $(16p)^p$ and the bound (48) on $\mathbb{P}(V > 2)$ hold here with the constants of the real case, Lemma 3.2 applies with the parameters of the real case and gives the same exceptional probabilities.

Centrally symmetric bounded coefficients. Recall that a real or complex coefficient distribution is centrally symmetric if $\text{Law}(\xi) = \text{Law}(-\xi)$, and nondegenerate if it is not a point mass. A centrally symmetric point mass sits at zero, so that a centrally symmetric distribution is nondegenerate exactly when $\mathbb{P}(|\xi| > 0) > 0$, which for a bounded distribution is equivalent to $\mathbb{E}|\xi|^2 > 0$. We first bound the covariances and the third moments, then the derivatives and tails, and then the logarithmic estimates, which use the occupation bounds. For complex coefficients, symmetry implies that $\mathbb{E}\xi = 0$. Let $\zeta = \mathbb{E}\xi^2$, so that $|\zeta| = |\mathbb{E}\xi^2| \leq \mathbb{E}|\xi|^2 = 1$. For $z = re^{i\theta}$

and $w = se^{i\phi}$, the ordinary and unconjugated covariance kernels are

$$\begin{aligned}\mathcal{K}(z, w) &= \mathbb{E}[X_r(\theta)\overline{X_s(\phi)}] = \frac{1}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k e^{ik(\theta-\phi)}, \\ \mathcal{P}(z, w) &= \mathbb{E}[X_r(\theta)X_s(\phi)] = \frac{\zeta}{\sigma_N(r)\sigma_N(s)} \sum_{k=0}^N (rs)^k e^{ik(\theta+\phi)}.\end{aligned}$$

In real coordinates their cross covariance is

$$\frac{1}{2} \begin{pmatrix} \Re(\mathcal{K}(z, w) + \mathcal{P}(z, w)) & \Im(\mathcal{P}(z, w) - \mathcal{K}(z, w)) \\ \Im(\mathcal{P}(z, w) + \mathcal{K}(z, w)) & \Re(\mathcal{K}(z, w) - \mathcal{P}(z, w)) \end{pmatrix}.$$

Indeed, with $X_r(\theta) = x_1 + ix_2$ and $X_s(\phi) = y_1 + iy_2$, where x_1, x_2, y_1, y_2 are real, expanding the products gives

$$\begin{aligned}\mathcal{K}(z, w) &= \mathbb{E}[x_1y_1 + x_2y_2] + i\mathbb{E}[x_2y_1 - x_1y_2], \\ \mathcal{P}(z, w) &= \mathbb{E}[x_1y_1 - x_2y_2] + i\mathbb{E}[x_1y_2 + x_2y_1],\end{aligned}$$

and adding and subtracting the real parts, and the imaginary parts, gives the entries of the matrix above. For $w = z$ we have $\mathcal{K}(z, z) = 1$, since $\sum_k r^{2k} = \sigma_N(r)^2$, so that the covariance matrix of $(\Re X_r(\theta), \Im X_r(\theta))$ is

$$\frac{1}{2}I_2 + \frac{1}{2} \begin{pmatrix} \Re\mathcal{P}(z, z) & \Im\mathcal{P}(z, z) \\ \Im\mathcal{P}(z, z) & -\Re\mathcal{P}(z, z) \end{pmatrix}, \quad \mathcal{P}(z, z) = \frac{\zeta}{\sigma_N(r)^2} \sum_{k=0}^N r^{2k} e^{2ik\theta}.$$

Hence every entry of this matrix minus $\frac{1}{2}I_2$ is at most $\frac{1}{2}|\mathcal{P}(z, z)|$ in absolute value, and every entry of the cross covariance is at most $\frac{1}{2}(|\mathcal{K}(z, w)| + |\mathcal{P}(z, w)|)$. For retained angles and pairs, by $|\zeta| \leq 1$, (9) and the lower bound in (10) we have

$$\begin{aligned}\frac{1}{2}|\mathcal{P}(z, z)| &\leq \frac{N}{2\sigma_N(r)^2} \left| \frac{1}{N} \sum_{k=0}^N r^{2k} e^{2ik\theta} \right| \\ &\leq \frac{e^{4K}}{2} \cdot \frac{10e^{4K}}{N \cdot 2N^{-1/2}} \leq 10e^{8K}N^{-1/2}, \\ \frac{1}{2}(|\mathcal{K}(z, w)| + |\mathcal{P}(z, w)|) &\leq \frac{N}{2\sigma_N(r)\sigma_N(s)} \sum_{\pm} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k e^{ik(\theta \pm \phi)} \right| \\ &\leq \frac{e^{4K}}{2} \cdot 2 \cdot \frac{10e^{4K}}{N \cdot N^{-1/2}} = 10e^{8K}N^{-1/2}.\end{aligned}$$

These are the entry bounds (11). Since $|\zeta| \leq 1$, the Abel bounds for the oscillatory terms are unchanged, including the case $|\zeta| = 1$, when the coefficient distribution is supported on a line. For the derivative coordinates, write, as in the proof of

Lemma 4.5, $W_0(z) = f_N(z)/\sqrt{N}$ and $W_1(z) = zf'_N(z)/N^{3/2}$. For $l, m \in \{0, 1\}$, by independence and centering we have

$$\begin{aligned}\mathbb{E}[W_l(z)\overline{W_m(w)}] &= \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^{l+m} e^{ik(\theta-\phi)}, \\ \mathbb{E}[W_l(z)W_m(w)] &= \frac{\zeta}{N} \sum_{k=0}^N (rs)^k (k/N)^{l+m} e^{ik(\theta+\phi)}.\end{aligned}$$

Each real covariance block has the same form as the above 2×2 matrix, with these two sums in the places of $\mathcal{K}(z, w)$ and $\mathcal{P}(z, w)$. The nonoscillatory moment matrix still has the normalization $\mathbb{E}|\xi|^2 = 1$. It follows that, for $w = z$, the first sum is real, and that the covariance matrix Σ_z of $W(z)$ is the nonoscillatory matrix of the proof of Lemma 4.5 plus a matrix each entry of which is, up to sign, half the real or the imaginary part of $\zeta N^{-1} \sum_{k=0}^N r^{2k} (k/N)^\beta e^{2ik\theta}$, with $\beta \in \{0, 1, 2\}$. Since $|\zeta| \leq 1$, these entries satisfy the bounds of that proof, so that this matrix has norm at most $20e^{4K}N^{-1/2}$ and the least eigenvalue of Σ_z is at least $e^{-4K}/80$, as in (24). For a retained pair, (9) and $|\zeta| \leq 1$ bound every entry of the cross covariance of $W(z)$ and $W(w)$ in absolute value by

$$\begin{aligned}\frac{1}{2} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta-\phi)} \right| + \frac{|\zeta|}{2} \left| \frac{1}{N} \sum_{k=0}^N (rs)^k (k/N)^\beta e^{ik(\theta+\phi)} \right| \\ \leq \frac{1}{2} \cdot 10e^{4K}N^{-1/2} + \frac{1}{2} \cdot 10e^{4K}N^{-1/2} = 10e^{4K}N^{-1/2},\end{aligned}$$

which is the bound of that proof. We conclude that the same angular exclusions and covariance lower bounds apply.

Raič's theorem yields occupation and mesh estimates with constants multiplied by the finite third moment. Indeed, the k th summand of the real vector with the coordinates $\Re X_r(\theta)$, $\Im X_r(\theta)$, $\Re X_s(\phi)$ and $\Im X_s(\phi)$ is the image of ξ_k under a real linear map. These summands are independent and centered, their covariance matrices sum to that of the vector, and the norm of the k th summand is $|\xi_k|$ times the value for signs, since $|\xi_k e^{ik\theta}| = |\xi_k|$. By the entry bounds, the eigenvalues of the covariance matrix Σ are at least $1/4$, as in Section 3.1, so that $\|\Sigma^{-1/2}\| \leq 2$. Therefore the whitened k th summand has norm at most $2\sqrt{2}|\xi_k|e^{3K}N^{-1/2}$, and the third-moment sum in Raič's theorem is at most

$$\begin{aligned}\sum_{k=0}^N \mathbb{E} \left(2\sqrt{2}|\xi_k|e^{3K}N^{-1/2} \right)^3 &= (N+1) \mathbb{E}|\xi_0|^3 16\sqrt{2}e^{9K}N^{-3/2} \\ &\leq 32\sqrt{2} \mathbb{E}|\xi_0|^3 e^{9K}N^{-1/2} \leq 46B^3 e^{9K}N^{-1/2}.\end{aligned}$$

For one point, the whitened k th summand has norm at most $2|\xi_k|e^{3K}N^{-1/2}$, since the squared norm of the summand before whitening is then $|\xi_k|^2 r^{2k}/\sigma_N(r)^2 \leq$

$|\xi_k|^2 e^{6K}/N$, and

$$\sum_{k=0}^N \mathbb{E} \left(2|\xi_k| e^{3K} N^{-1/2} \right)^3 = (N+1) \mathbb{E} |\xi_0|^3 8e^{9K} N^{-3/2} \leq 16B^3 e^{9K} N^{-1/2}.$$

Hence, for retained angles and pairs, by the triangle inequality, Raič's theorem and the total variation bound we have, as in (13),

$$\begin{aligned} & \left| \mathbb{P}\{|X_r(\theta)| \leq t, |X_s(\phi)| \leq t'\} - F_G(t)F_G(t') \right| \\ & \leq (42 \cdot 4^{1/4} + 16) \cdot 46B^3 e^{9K} N^{-1/2} + 320e^{8K} N^{-1/2} \leq C(K, B)N^{-1/2}, \end{aligned}$$

and the same bound holds for $|\mathbb{P}\{|X_r(\theta)| \leq t\} - F_G(t)|$, since the one-point bound is $(42 \cdot 2^{1/4} + 16) \cdot 16B^3 e^{9K} N^{-1/2} + 320e^{8K} N^{-1/2}$, which is smaller. Here and below, $C(K, B)$ denotes a constant depending only on K and B , which may increase from one estimate to the next. Let $c_B = (42 \cdot 4^{1/4} + 16) \cdot 46B^3 e^{9K} + 320e^{8K}$, so that the above two-point bound, and the smaller one-point bound, are at most $c_B N^{-1/2}$. The derivation of (14) from (13), written out above for real Gaussian coefficients, with the removed angles and pairs contributing $2N^{-1/2}$ and $10N^{-1/2}$, gives, uniformly for $t \geq 0$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| & \leq (2 + c_B)N^{-1/2}, \\ |\mathbb{E}A_t^2 - F_G(t)^2| & \leq (10 + c_B)N^{-1/2}, \\ \text{Var}(A_t) & \leq |\mathbb{E}A_t^2 - F_G(t)^2| + 2|\mathbb{E}A_t - F_G(t)| \\ & \leq (10 + c_B)N^{-1/2} + 2(2 + c_B)N^{-1/2} = (14 + 3c_B)N^{-1/2}. \end{aligned}$$

In the proof of Lemma 4.5, the k th summand of $(W(z), W(w))$ has $|\xi_k|$ times the norm of the k th summand for signs, and $\|\Sigma_{z,w}^{-1/2}\| \leq (160e^{4K})^{1/2}$, since the least eigenvalue of the covariance matrix $\Sigma_{z,w}$ of $(W(z), W(w))$ is at least $e^{-4K}/160$, as there. It follows that its third-moment sum is at most

$$\mathbb{E}|\xi_0|^3 \cdot 16e^{3K}(160e^{4K})^{3/2}N^{-1/2} \leq 16B^3 e^{3K}(160e^{4K})^{3/2}N^{-1/2},$$

and that its Raič bound $2000e^{3K}(160e^{4K})^{3/2}N^{-1/2}$ is multiplied by at most B^3 .

For the derivative and tail estimates, we apply Hoeffding's inequality, as in the Steinhaus case, to $\Re(\xi_k c_k)$ and $\Im(\xi_k c_k)$, now with $|\Re(\xi_k c_k)|, |\Im(\xi_k c_k)| \leq B|c_k|$. In the proof of Lemma 4.3, at each sample and at B times the level used there, we obtain

$$\begin{aligned} & \mathbb{P}\{|f_N''(z)| \geq 50Be^{K+4}N^{5/2}\sqrt{\log N}\} \\ & \leq 4 \exp \left[-\frac{(50Be^{K+4}N^{5/2}\sqrt{\log N})^2}{4B^2 \sum_{k=2}^N k^2(k-1)^2 |z|^{2k-4}} \right] \end{aligned}$$

$$= 4 \exp \left[-\frac{(50e^{K+4}N^{5/2}\sqrt{\log N})^2}{4 \sum_{k=2}^N k^2(k-1)^2|z|^{2k-4}} \right] \leq 4N^{-100},$$

where the last step is that of the sign case. As in the sign case, the union over the at most $2N^3$ samples has probability at most $8N^{-97} \leq N^{-10}$. Since $|\xi_k| \leq B$, the deterministic bound on f_N''' gains the factor B , so that the interpolation error is at most $4\pi Be^{K+1}N < 50Be^{K+4}N^{5/2}\sqrt{\log N}$. Hence $\sup_{|z| \leq 1+(K+1)/N} |f_N''(z)| \leq 100Be^{K+4}N^{5/2}\sqrt{\log N}$ outside an event of probability at most N^{-10} . In the proof of Lemma 4.6, at each sample, for $N < n \leq N + m_{\max}$ and at B times the level used there, we have

$$\begin{aligned} & \mathbb{P} \left\{ \left| \sum_{k=N+1}^n \xi_k z^k \right| \geq 10Be^{2K} \sqrt{m_{\max} \log N} \right\} \\ & \leq 4 \exp \left[-\frac{(10Be^{2K} \sqrt{m_{\max} \log N})^2}{4B^2 \sum_{k=N+1}^n \rho^{2k}} \right] \\ & = 4 \exp \left[-\frac{(10e^{2K} \sqrt{m_{\max} \log N})^2}{4 \sum_{k=N+1}^n \rho^{2k}} \right] \leq 4N^{-25}. \end{aligned}$$

As in the sign case, the union over the at most $2N^4$ pairs of a sample and a nonempty tail has probability at most $8N^{-21} \leq N^{-10}$. Similarly, the deterministic bound on the derivatives of the tails gains the factor B , so that the interpolation error is at most $4\pi Bm_{\max}e^{2K}/N^2 < 5Be^{2K}\sqrt{m_{\max} \log N}$. Hence $\max_{N < n \leq N+m_{\max}} \sup_{|z| \leq \rho} \left| \sum_{k=N+1}^n \xi_k z^k \right| < 15Be^{2K}\sqrt{m_{\max} \log N}$ outside an event of probability at most N^{-10} . The derivative and tail constants depend on B as in Section A.11.

With these bounds, the proof of Lemma 4.5 applies with constants depending on K and B . Its second-derivative bound is now $M_B N^{5/2} \sqrt{\log N}$, with $M_B = 100Be^{K+4}$, and its Taylor bounds require that the mesh constant of that proof, here $\eta \leq 1$, satisfy $M_B \eta \leq 1/2$, so that the spacing is $h = \eta/(NL\sqrt{\log N})$ as there. For an annular root α with $|f_N'(\alpha)| < N^{3/2}/L$, outside the two real sectors, and its assigned mesh point z , Taylor's theorem at α , as in that proof, with $|z - \alpha| \leq h$, $|z| \leq 2$ and the bound $M_B N^{5/2} \sqrt{\log N}$ on $|f_N''|$, gives

$$\begin{aligned} \frac{|f_N(z)|}{\sqrt{N}} & \leq \frac{Nh}{L} + \frac{M_B}{2} N^2 h^2 \sqrt{\log N} = \frac{\eta + M_B \eta^2/2}{L^2 \sqrt{\log N}} \leq \frac{M_B}{L^2 \sqrt{\log N}}, \\ \frac{|zf_N'(z)|}{N^{3/2}} & \leq \frac{2(1 + M_B \eta)}{L} \leq \frac{3}{L}, \end{aligned}$$

since $Nh = \eta/(L\sqrt{\log N})$, $\eta + M_B \eta^2/2 \leq 5\eta/4 \leq M_B$ and $M_B \eta \leq 1/2$. It follows that a root produces the test (23) with $u_B = M_B/(L^2 \sqrt{\log N})$ in place of its value threshold and the same derivative threshold $3/L$. By the mesh count of that proof,

there are at most $53(K+1)\eta^{-2}NL^2\log N = C(K,B)N^{33/32}\log N$ points, since $L^2 = N^{1/32}$. With the eigenvalue bounds of that proof and its two Raič bounds multiplied by at most B^3 , as shown above, the chain (25) becomes

$$\begin{aligned}\mathbb{P}\{\text{the test holds at } z\} &\leq \frac{u_B^2(3/L)^2}{4(e^{-4K}/80)^2} + 2000B^3e^{3K}(160e^{4K})^{3/2}N^{-1/2} \\ &= \frac{9 \cdot 80^2 e^{8K} M_B^2}{4} \cdot \frac{L^{-6}}{\log N} + 2000B^3e^{3K}(160e^{4K})^{3/2}N^{-1/2} \\ &\leq C(K,B)L^{-6}/\log N + C(K,B)N^{-1/2},\end{aligned}$$

where the first step collects the four-dimensional Raič bound and the density and volume bounds of that chain, with $2/(e^{-4K}/80) = 160e^{4K}$. For a separated pair, the three-term bound for the covariance of the two test indicators in that proof holds with its two Raič terms multiplied by at most B^3 and with the total variation bound (26) unchanged, so that this covariance is at most $C(K,B)N^{-1/2}$. Let $m_B \leq C(K,B)NL^2\log N$ be the number of retained mesh points and S the number of successful tests. Then, by the mean and variance calculations of that proof, we have

$$\begin{aligned}\mathbb{E}S &\leq m_B \left\{ \frac{C(K,B)L^{-6}}{\log N} + C(K,B)N^{-1/2} \right\} \leq C(K,B)\{NL^{-4} + N^{1/2}L^2\log N\}, \\ \text{Var } S &\leq 10m_B^2N^{-1/2} + C(K,B)m_B^2N^{-1/2} \leq C(K,B)N^{3/2}L^4(\log N)^2,\end{aligned}$$

where the variance bound counts the covariances of the at most $10m_B^2N^{-1/2}$ ordered pairs in the removed bands as one and the others as $C(K,B)N^{-1/2}$. For the local count we use Lemma 7.2, with $K+2$ in place of K , so that

$$B_{K,N} \leq C_{1,B}(K+2)(\log N) \cdot \#\{\text{successful tests}\} + 40C_{1,B}(K+2)\sqrt{N}\log N,$$

and we take $Q = N^{31/32}/(2C_{1,B}(K+2)\log N)$. With $L = N^{1/64}$, the substitution of that proof yields

$$\begin{aligned}\frac{\mathbb{E}S}{Q} &\leq C(K,B)\{N^{-1/32}\log N + N^{-7/16}(\log N)^2\}, \\ \frac{40C_{1,B}(K+2)\sqrt{N}\log N}{N^{31/32}} &= 40C_{1,B}(K+2)N^{-15/32}\log N,\end{aligned}$$

and both tend to zero, so that $\mathbb{E}S \leq Q/2$ and the sector contribution is at most $N^{31/32}/2$ for all sufficiently large N , depending on K and B . On the local-count and derivative events, $B_{K,N} > N^{31/32}$ then forces $S > Q$, and by Chebyshev's inequality at deviation $Q/2$ we have

$$\mathbb{P}\{S > Q\} \leq \frac{4 \text{Var } S}{Q^2} \leq C(K,B) \frac{N^{25/16}(\log N)^2}{N^{31/16}(\log N)^{-2}} = C(K,B)N^{-3/8}(\log N)^4,$$

since $\mathbb{E}S \leq Q/2$. Adding the exceptional probabilities of Lemma 7.2 and of the above derivative bound, we obtain

$$(50) \quad \mathbb{P}\{B_{K,N} > N^{31/32}\} \leq N^{-3} + \exp[-(N+1)/(2B^4)] + N^{-10} + C(K, B)N^{-3/8}(\log N)^4.$$

Write $\xi_k = \epsilon_k a_k$ as in the proof of Lemma 7.2, with independent signs and independent, possibly complex, amplitudes a_k with the coefficient distribution, independent of the signs. Conditional on the amplitudes, with the weights $w_k = r^{2k}/\sigma_N(r)^2$ of the Gaussian case, by Parseval's identity and $|\xi_k| = |a_k|$ we have $V = \sum_{k=0}^N w_k |a_k|^2$ and $\sum_{k=0}^N w_k = 1$. Since $0 \leq |a_k|^2 \leq B^2$, we have $V \leq B^2$ and $\mathbb{E}V = \sum_k w_k \mathbb{E}|a_k|^2 = 1$. By Hoeffding's inequality for the independent variables $w_k |a_k|^2$, with values in intervals of lengths $w_k B^2$, and since $\sum_k w_k^2 \leq w_{\max} \sum_k w_k = w_{\max} \leq e^{6K}/N$, we have

$$(51) \quad \begin{aligned} \mathbb{P}(V < 1/2) &\leq \mathbb{P}\{V - \mathbb{E}V \leq -1/2\} \leq \exp\left[-\frac{2(1/2)^2}{\sum_k (w_k B^2)^2}\right] \\ &= \exp\left[-\frac{1}{2B^4 \sum_k w_k^2}\right] \leq \exp[-1/(2B^4 w_{\max})] \leq \exp[-N/(2B^4 e^{6K})]. \end{aligned}$$

On the event $E = \{V \geq 1/2\}$ of the statement, let $b_k = r^k a_k / (\sigma_N(r) \sqrt{V})$. Then $\sum_{k=0}^N |b_k|^2 = 1$, $X_r(\theta) = \sqrt{V} \sum_{k=0}^N \epsilon_k b_k e^{ik\theta}$, and $|\log \sqrt{V}| \leq \max(\frac{1}{2} \log 2, \log B)$, since $1/2 \leq V \leq B^2$ on E . Denote expectations over signs and amplitudes by $\mathbb{E}_\epsilon$ and $\mathbb{E}_a$. By NNS, $\mathbb{E}_\epsilon \int |\log |X_r/\sqrt{V}||^p d\mu \leq (C_* p)^{6p}$ for $p \geq 1$. Hence, by Fubini's theorem, since E depends only on the amplitudes, and by $|\log |X_r|| \leq |\log |X_r/\sqrt{V}|| + |\log \sqrt{V}|$ and $(x+y)^p \leq 2^{p-1}(x^p + y^p)$ for $x, y \geq 0$, we have

$$\begin{aligned} \int_E \int |\log |X_r||^p d\mu d\mathbb{P} &= \mathbb{E}_a \left[1_E \mathbb{E}_\epsilon \int |\log |X_r||^p d\mu \right] \\ &\leq 2^{p-1} \{ (C_* p)^{6p} + \max(\tfrac{1}{2} \log 2, \log B)^p \} \\ &\leq 2^p \left[(C_* + \max(\tfrac{1}{2} \log 2, \log B) + 1)p \right]^{6p} \\ &\leq [2(C_* + \max(\tfrac{1}{2} \log 2, \log B) + 1)p]^{6p}, \end{aligned}$$

where the third step holds since the base $(C_* + \max(\frac{1}{2} \log 2, \log B) + 1)p$ is at least 1, at least $C_* p$ and at least $\max(\frac{1}{2} \log 2, \log B)$.

By the definition of C_B , the last bound is $(C_B p)^{6p}$, and $V \leq B^2$ on E . With the occupation bounds above, whose constants satisfy $2 + c_B < 14 + 3c_B$, Lemma 3.2 therefore applies on E , with

$$H = 14 + 3c_B, \quad C = C_B, \quad B_E = B^2.$$

Its error is at most $C(K, B)N^{-1/32}(\log N)^7$, and its probability bound, with $\mathbb{P}(E^c) \leq \exp[-N/(2B^4e^{6K})]$ by (51), is the concentration bound of the proposition. The explicit bound is computed in Section A.11.

Summability. At $N = j^8$ the polynomial terms of the exceptional probabilities are summable, by the last column of Table 14. The remaining terms are

$$e^{-3(N+1)/32}, \quad e^{-3N/(32e^{6K})}, \quad 2(2N+1)e^{-N^2/2}, \quad e^{-(N+1)/(2B^4)}, \quad e^{-N/(2B^4e^{6K})}.$$

All but the third are at most $e^{-\delta N}$ for a constant $\delta > 0$ depending only on K and B . Since $e^{-y} \leq 2y^{-2}$ for $y > 0$, we have, for every $\delta > 0$,

$$(52) \quad \begin{aligned} \sum_{j \geq 1} e^{-\delta j^8} &\leq \frac{2}{\delta^2} \sum_{j \geq 1} j^{-16} < \infty, \\ \sum_{j \geq 1} 2(2j^8 + 1)e^{-j^{16}/2} &\leq \sum_{j \geq 1} 6j^8 \cdot 8j^{-32} = 48 \sum_{j \geq 1} j^{-24} < \infty, \end{aligned}$$

where in the second chain we used $2(2j^8 + 1) \leq 6j^8$ and $e^{-j^{16}/2} \leq 2(j^{16}/2)^{-2} = 8j^{-32}$. Hence the exceptional probabilities are summable along $N = j^8$, which completes the proof. $\square$

Proof of Corollary 1.2 for the other distributions. Multiplying every coefficient by a nonzero constant does not change the zeros of f_n , and a nondegenerate centrally symmetric distribution has positive second moment, since it is not concentrated at zero. We may therefore assume $\mathbb{E}|\xi_0|^2 = 1$ for a bounded distribution, as in Proposition 7.4, and this holds for the other three distributions. We follow the proof for Rademacher coefficients in Section 6.1 step by step, with the estimates of Proposition 7.4 instead of those of Sections 3 and 4, and we write out every step in which a constant, a degree or an exceptional probability changes. By Proposition 7.4, the occupation, logarithmic, energy, annular mesh, local-count and maximal-tail estimates hold, the logarithmic estimate on the energy event of that proposition when the coefficient distribution requires it, and the additional exceptional probabilities are exponential in N or explicitly summable. Since $\mathbb{E}|\xi_0|^2 = 1$, the normalization $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$ is that of the sign case, so that, as in Proposition 2.2, the variance profile is

$$F(x) = \frac{1}{2} \log \int_0^1 e^{2xt} dt, \quad F'(0) = \frac{1}{2},$$

and Lemma 2.3 applies unchanged. Together with the matching argument of Section 6.1, these estimates yield the conclusion for nonzero coefficients. In what follows, let $\Pi_K(N)$ be the right-hand side of the concentration bound of Proposition 7.4 without its term $\mathbb{P}(E^c)$, with the constants of the distribution, and allow $C(K)$ to depend on the distribution.

Step 1. Proposition 7.4 implies that Lemmas 4.3 and 4.6 hold for Steinhaus and Gaussian coefficients with the same constants, for Gaussian coefficients on the envelope event $\max_{k \leq 2N} |\xi_k| \leq N$. Hence ρ , a , M_2 and d_0 and the bounds (29)–(33) are unchanged. For bounded coefficients, the proof of Proposition 7.4 yields the tail amplitude Ba , with $K+2$ in place of K and $m_{\max} = m_j$, and the second-derivative bound BM_2 . Since (29) does not involve the coefficients, (30)–(33) become

$$\begin{aligned} Ba &\leq 45Be^{2(K+2)}N^{7/16}\sqrt{\log N}, \\ \frac{2Ba}{d_0} &\leq 90Be^{2(K+2)}N^{-67/64}\sqrt{\log N}, \\ \frac{2NBa}{d_0} &\leq 90Be^{2(K+2)}N^{-3/64}\sqrt{\log N}, \\ \frac{4(BM_2)(Ba)}{d_0^2} &\leq 180 \cdot 100B^2e^{K+4}e^{2(K+2)}N^{-1/32}\log N, \end{aligned}$$

which are those bounds multiplied by B , B , B and B^2 . The right-hand sides of the last two lines tend to zero, so that $2Ba/d_0 \leq 1/N$ and $4(BM_2)(Ba) \leq d_0^2$ for all N large enough, depending on K and B .

Step 2. For Steinhaus and Gaussian coefficients this step is the same as before, since a , M_2 and d_0 are unchanged. For bounded coefficients, we apply Lemma 5.4 to each good annular root α , with $P = f_N$, the amplitude Ba , $|f'_N(\alpha)| \geq d_0$ and $|f''_N| \leq BM_2$ on the disk of radius $2Ba/d_0$ about α . This disk lies in $|z| \leq 1 + (K+1)/N$, since $2Ba/d_0 \leq 1/N$, and we have $4(BM_2)(Ba) \leq d_0^2$ by Step 1. The lemma then shows that the disk contains exactly one zero and that $|f_N| \geq 3Ba/2$ on its boundary, and we choose the common regular level in $(Ba, 3Ba/2)$.

Step 3. The proof of Corollary 3.3 relies on the deterministic Lemmas 2.4 and 2.5, which require a nonzero polynomial of degree at most N . For Steinhaus coefficients $f_N(0) = \xi_0 \neq 0$, and for Gaussian coefficients $f_N(0) = \xi_0 \neq 0$ almost surely. For bounded coefficients, $f_N \equiv 0$ implies $V = 0$ at every radius, so that the zero polynomial lies in the energy exception E^c . It follows that, with the concentration bound of Proposition 7.4 instead of Theorem 3.1, the corollary holds for each distribution, with its constants, outside four events at the width K and five events at the width 2, of probabilities at most $\Pi_K(N) + \mathbb{P}(E^c)$ and $\Pi_2(N) + \mathbb{P}(E^c)$, with $\mathbb{P}(E^c)$ at the widths K and 2. Using the bound (49) or (50) on $\mathbb{P}\{B_{K,N} > N^{31/32}\}$ from the proof of Proposition 7.4 instead of Lemma 4.5 (for Steinhaus coefficients (27) still holds), we obtain (34) with the constant $C(K)$ of the distribution. For bounded coefficients, the crossing bound (36) needs $2Ba/d_0 < N^{-65/64}$ instead of $2a/d_0 < N^{-65/64}$, and by Step 1 we have

$$N^{65/64} \frac{2Ba}{d_0} \leq 90Be^{2(K+2)}N^{-67/64+1+1/64}\sqrt{\log N} = 90Be^{2(K+2)}N^{-1/32}\sqrt{\log N}.$$

This tends to zero, as (35) does. The bound on c_N/N and the case $q = 0$ of (37) then hold with the constant $C(2)$ of the distribution.

Step 4. As in Step 4 of Section 6.1, outside one maximal tail event we have $|f_n - f_N| < a$, or $|f_n - f_N| < Ba$ for bounded coefficients, on every selected boundary. Theorem 5.1 needs nonzero polynomials, and outside the events of Step 3 we have $f_N \neq 0$, which implies $f_n \neq 0$, since its first $N + 1$ coefficients are those of f_N . If $\mathbb{P}(\xi_0 = 0) = 0$, every coefficient is nonzero almost surely, so that $\deg f_n = n$ and $\deg f_N = N$, and Step 4 holds as before, with (39).

Suppose now that $\mathbb{P}(\xi_0 = 0) > 0$. Then the zero-polynomial event is contained in the amplitude exception. Since the probabilities of the trailing-run events below are summable along $N = j^8$ once $8D|\log \mathbb{P}(\xi_0 = 0)| > 1$, any $D > 1/(8|\log \mathbb{P}(\xi_0 = 0)|)$ would do. We choose $D > 4/|\log \mathbb{P}(\xi_0 = 0)|$, which keeps a margin. Let the trailing-run event be the event that the trailing run of zero coefficients at $N = j^8$ exceeds $\lceil D \log N \rceil$. Since $0 < \mathbb{P}(\xi_0 = 0) < 1$, this event has probability at most $\mathbb{P}(\xi_0 = 0)^{\lceil D \log N \rceil} \leq \mathbb{P}(\xi_0 = 0)^{D \log N} = N^{-D|\log \mathbb{P}(\xi_0 = 0)|}$, and, since $8D|\log \mathbb{P}(\xi_0 = 0)| > 32$, we also have $\sum_{j=1}^{\infty} j^{-8D|\log \mathbb{P}(\xi_0 = 0)|} < \infty$. Outside the trailing-run event we have $\deg f_N \geq N - \lceil D \log N \rceil$ and $\deg f_n \leq n$, so that, for every $N \leq n \leq N + m_j$,

$$\begin{aligned} \deg f_n - \deg f_N &\leq n - N + \lceil D \log N \rceil, \\ 0 &\leq \frac{\deg f_n - \deg f_N}{N} \leq \frac{m_j}{N} + \frac{\lceil D \log N \rceil}{N} \longrightarrow 0, \end{aligned}$$

where $0 \leq$ holds since the prefix degrees are nondecreasing, and the limit follows from $n - N \leq m_j$ and (29). Thus the actual-degree term in Theorem 5.1, divided by N , tends to zero. Since each selected domain contains exactly one root of f_N and U_N counts the other roots of f_N , we still have $\deg f_N - s = U_N$, although $\deg f_N$ may now be smaller than N . Hence, by Theorem 5.1 and the degree bound above, for $N \leq n \leq N + m_j$ we have

$$\begin{aligned} |\nu_n(1) - \nu_N(1)| &\leq c_N + (\deg f_N - s) + \max(0, \deg f_n - \deg f_N) \\ &\leq c_N + U_N + m_j + \lceil D \log N \rceil, \\ \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N + U_N}{N} + \frac{3m_j}{2N} + \frac{\lceil D \log N \rceil}{N}, \end{aligned}$$

where the third line follows from the second as in the chain of (39), with the second line in place of the matching bound there and $\lceil D \log N \rceil/n \leq \lceil D \log N \rceil/N$. The initial zero multiplicity, the index of the first nonzero coefficient, is finite almost surely, since the coefficients are independent and $\mathbb{P}(\xi_0 \neq 0) > 0$, and so, divided by N , it tends to zero, and the bound on U_N/N already counts these zeros at the origin,

since they are unselected roots. There are only finitely many zero polynomials, and their counts do not affect the limit.

Step 5. We now also include in the event $E_{K,j}$ the energy exceptions at the four radii of the width K and at the five radii of the width 2, the additional local-count exception and, for Gaussian coefficients, the complement of the envelope event, which the derivative and tail bounds share. For bounded distributions with an atom at zero we also include the trailing-run event. For Steinhaus coefficients (40) still holds. For Gaussian coefficients we have, for all large j ,

$$\begin{aligned} \mathbb{P}(E_{K,j}) \leq & 4\Pi_K(N) + 5\Pi_2(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4 \\ & + 4e^{-3N/(32e^{6K})} + 5e^{-3N/(32e^{12})} + e^{-3(N+1)/32} + 2(2N+1)e^{-N^2/2}. \end{aligned}$$

Similarly, for bounded coefficients we have, for all large j ,

$$\begin{aligned} \mathbb{P}(E_{K,j}) \leq & 4\Pi_K(N) + 5\Pi_2(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4 \\ & + 4e^{-N/(2B^4e^{6K})} + 5e^{-N/(2B^4e^{12})} + e^{-(N+1)/(2B^4)} + N^{-D|\log \mathbb{P}(\xi_0=0)|}, \end{aligned}$$

where the last term is present only if $\mathbb{P}(\xi_0 = 0) > 0$. In both bounds, the first two exponential terms are the energy exceptions of Proposition 7.4 at the widths K and 2, and the third is the additional exception of Lemma 7.3 or 7.2. At $N = j^8$ the terms of (40) are summable as in Step 5 of Section 6.1, the exponential terms by (52), and the trailing-run term by the sum above. Hence the Borel–Cantelli argument of Step 5 applies, with the additional term $\lceil D \log N \rceil / N \rightarrow 0$ in the final bound when $\mathbb{P}(\xi_0 = 0) > 0$. $\square$

Proof of Theorem 1.1 for the other distributions. We follow the proof in Section 6.2 for Rademacher coefficients step by step, with the normalization $\mathbb{E}|\xi_0|^2 = 1$ and the changes written out in the proof of Corollary 1.2 above. Let $\Pi_K(N)$ be as in that proof, and let $\Pi_R(N)$ be $\Pi_K(N)$ with R in place of K . Fix K, R, x . We use the logarithmic, mesh, local-count and tail estimates from Section 7, and for Gaussian and bounded distributions the probabilities of the additional exceptional events are exponential in N . As in that proof, the variance profile is unchanged. The concentration bound of Proposition 7.4 at the five radii r_q of width R shows that (41), (42) and (43) hold with the constant $C(R)$ of the distribution, outside five events of probabilities at most $\Pi_R(N) + \mathbb{P}(E^c)$, with $\mathbb{P}(E^c)$ at the width R . The selected domains are the same as in Steps 1 and 2 of that proof. For bounded coefficients, the band condition involves $2Ba/d_0$ instead of $2a/d_0$, and, as in Section 6.2, we have

$$\frac{N}{h} \left(\frac{2Ba}{d_0} + \frac{|x|m_j}{N^2} \right) \leq 90Be^{2(K+2)} N^{-67/64+65/64} \sqrt{\log N} + 9|x|N^{7/8-2+65/64}$$

$$= 90Be^{2(K+2)}N^{-1/32}\sqrt{\log N} + 9|x|N^{-7/64},$$

which tends to zero.

If $\mathbb{P}(\xi_0 = 0) \in (0, 1)$, we take D and the trailing-run event as in that proof. Outside that event, the degree correction $\max(0, \deg Q - \deg P)$ in Corollary 5.3 is bounded by $m_j + \lceil D \log N \rceil$, and the extra term divided by N tends to zero. Indeed, the three-term bound of Section 6.2 becomes

$$\begin{aligned} \left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| &\leq \varepsilon_N + \frac{2\varepsilon_N N + U_N + m_j + \lceil D \log N \rceil}{N} + \frac{m_j}{N} \\ &= 3\varepsilon_N + \frac{U_N}{N} + \frac{2m_j}{N} + \frac{\lceil D \log N \rceil}{N}. \end{aligned}$$

For Gaussian coefficients we have, for all large j ,

$$\begin{aligned} \mathbb{P}(E_{K,R,x,j}) &\leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4 \\ &\quad + 4e^{-3N/(32e^{6K})} + 5e^{-3N/(32e^{6R})} + e^{-3(N+1)/32} + 2(2N+1)e^{-N^2/2}, \end{aligned}$$

and for bounded coefficients,

$$\begin{aligned} \mathbb{P}(E_{K,R,x,j}) &\leq 4\Pi_K(N) + 5\Pi_R(N) + N^{-3} + 2N^{-10} + C(K)N^{-3/8}(\log N)^4 \\ &\quad + 4e^{-N/(2B^4e^{6K})} + 5e^{-N/(2B^4e^{6R})} + e^{-(N+1)/(2B^4)} + N^{-D|\log \mathbb{P}(\xi_0=0)|}, \end{aligned}$$

where the last term is present only if $\mathbb{P}(\xi_0 = 0) > 0$. These bounds are obtained from those of Step 5 in that proof by replacing the five thin-band terms at the width 2 with the five logarithmic terms at the width R , as (44) replaces (40). For Steinhaus coefficients (44) still holds. As in that proof, these bounds are summable along $N = j^8$. The conclusion now follows from the fixed- K intersection and the rational- x argument of Section 6.2. $\square$

8 Quantitative refinements and weighted profiles

Throughout this section the coefficients form one nested Rademacher sequence, except where another coefficient distribution is specified. The estimates of the preceding sections allow the annular width to vary with the degree, and their constants and thresholds are given in Appendix A.

8.1 A logarithmic almost-sure rate

We obtain the constant in Theorem 1.3 from the explicit bounds below. The deterministic analytic thresholds depend on C_*, c_{HW} , while the last exceptional block is random.

Proof of Theorem 1.3. The idea is to run the block argument of Section 6.1 at a width $K = K_N$ that grows with N , while C_* and c_{HW} stay fixed. By (34) the unselected roots contribute $2 \log 2/K$ to the error in a block, so an error of order $1/\log n$ requires K_N to grow at least like a multiple of $\log N$, while the other constants grow exponentially in K , which bounds $K_N/\log N$ from above. We derive these upper bounds, choose K_N , and then follow Steps 1–5 of Section 6.1 at the width K_N .

Table 6 lists the constants of the argument that depend on K , and $T = (\log N)/32$ is the clipping level of Section A.2. It follows that each bound in Table 7

Quantity	Dependence on K	Source
H_K	$10^8 e^{20(K+1)} = H_0 e^{20K}$	Section A.2
$\sqrt{2 + H_K}$	$\leq \sqrt{2 + H_0} e^{10K}$	H_K and $2 \leq 2e^{20K}$
D_K	$10(K+1)^2 e^{4K}$	Section A.1
C_K	$\leq C_{\text{bad}}(K+1)^4 e^{13K}$	Section A.12
S_K	$S_0 e^K$	Section A.12
Mesh mean threshold, 1st term	number times $(K+1)e^{8K}$	Section A.12
Mesh mean threshold, 2nd term	number times $(K+1)e^{11K}$	Section A.12
$C_1(K+2)$	$\leq C_{10}(K+1)$	Section A.12
Tail amplitude a	factor $e^{2(K+2)} = e^4 e^{2K}$	Step 1, Section 6.1

Table 6: The constants of the proof in Section 8.1 that depend on the width K . Here $H_0 = 10^8 e^{20}$, and S_0 , C_{10} , C_{bad} and the numbers in the mesh mean threshold do not depend on K . The constant C_{bad} depends only on c_{HW} .

is at most of the form $C(K+1)^p (\log N)^q e^{bK} N^{-\beta}$, with C independent of K , real p, q , and $b, \beta > 0$. Since $e^{bK} N^{-\beta} = N^{bK/\log N - \beta}$, and since powers of $K+1 \leq \log N$ and of $\log N$ are dominated by any power of N , if $K \leq c \log N$ then

$$C(K+1)^p (\log N)^q e^{bK} N^{-\beta} \begin{cases} \text{tends to zero} & \text{if } c < \beta/b, \\ \text{is summable along } N = j^8 & \text{if } c < (\beta - 1/8)/b, \end{cases}$$

where in the second case the exponent of $j = N^{1/8}$ stays below -1 . In the first part of Table 7, the two terms of $\varepsilon_K^*(N)$ that grow with K , and D_K/N , enter the bound on U_N/N below through $(8/K)(\varepsilon_K^*(N) + D_K/N)$, a term that is $o(1/\log N)$ when $K/\log N$ has a positive limit and $\varepsilon_K^*(N) + D_K/N$ tends to zero. Its other rows are ratios of Table 15, which must be at most one for the degree thresholds of Section A.12 to hold. The second part contains the exceptional probabilities in (64) that depend on K . The remaining ratios of Table 15, $2K/N$ and the excluded-sector

Bound	e^{bK}	$N^{-\beta}$	$K/\log N$ below
<i>Must tend to zero</i>			
Lower logarithmic tail in $\varepsilon_K^*(N)$	e^{10K}	$N^{-1/32}$	1/320
Mesh mean threshold, first term	e^{8K}	$N^{-1/32}$	1/256
Taylor ratio $4M_2a/d_0^2$	e^{3K}	$N^{-1/32}$	1/96
H_K^4/N	e^{80K}	N^{-1}	1/80
$N^{65/64} \cdot 2a/d_0$	e^{2K}	$N^{-1/32}$	1/64
Truncation bias $2H_KTN^{-1/2}$ in $\varepsilon_K^*(N)$	e^{20K}	$N^{-1/2}$	1/40
Mesh mean threshold, second term	e^{11K}	$N^{-7/16}$	7/176
$(6400e^{8K})^2/N$	e^{16K}	N^{-1}	1/16
D_K/N and $2(1+2K)e^{4K}/N$	e^{4K}	N^{-1}	1/4
<i>Must be summable along $N = j^8$</i>			
Occupation failure $H_KN^{-3/8}$	e^{20K}	$N^{-3/8}$	1/80
Clipped-integral failure $\frac{H_K}{256}N^{-7/16}(\log N)^{-12}$	e^{20K}	$N^{-7/16}$	1/64
Small-derivative failure $C_KN^{-3/8}(\log N)^4$	e^{13K}	$N^{-3/8}$	1/52

Table 7: The bounds of the proof in Section 8.1 that grow exponentially with the width K , and the upper bound on $K/\log N$ that each requires: β/b in the first part and $(\beta - 1/8)/b$ in the second.

ratio $80C_{10}(K+1)(\log N)N^{-15/32}$, have no exponential factor and tend to zero whenever $K \leq \log N$, and its condition $\varepsilon_K^*(N) + D_K/N \leq K/16$ holds for large N once $\varepsilon_K^*(N)$ and D_K/N tend to zero. The strongest condition is $K/\log N < 1/320$, which comes from the lower logarithmic tail. We choose

$$K_N = \left\lfloor \frac{\log N}{1000} \right\rfloor,$$

where $N = j^8$ as in Section 6.1. Then $K_N/\log N \leq 1/1000 < 1/320$, so that every condition in Table 7 holds. At $K = K_N$ the error bounds and deterministic thresholds of Section A.12 become (66), and every threshold ratio tends to zero, so that the degree thresholds of that section hold for all large j . The block argument of Section 6.1 applies at this width, and its exceptional probability is bounded by (64) with $K = K_N$ and $R = 2$, that is, by

$$4p_{K_N}^*(N) + 5p_2^*(N) + N^{-3} + 2N^{-10} + C_{K_N}N^{-3/8}(\log N)^4,$$

in the notation of Appendix A. Here the term N^{-3} is the local-count exception, with $K_N + 2$ in place of K . Since this parameter grows with N , we take it from

Lemma 4.1 with $a_k = 1$, $B = 1$, $\eta = 1$ and $\mathcal{E} = \Omega$. Then A and $t_N = A \log N/N$ are as in the proof of Lemma 4.2, the thresholds $N \geq e$ and $t_N \leq 1$ do not depend on K , and the constant is $C_1(K_N + 2) = (4A + 4K_N + 12)/\log 2$. Since $K_N \leq (\log N)/1000$, we have $e^{bK_N} \leq e^{(b/1000)\log N} = N^{b/1000}$ for $b \geq 0$, so that the factors e^{20K} of H_K and e^{13K} of C_K give, with $N = j^8$,

$$\begin{aligned} N^{-3/8} e^{20K_N} &\leq N^{-3/8+20/1000} = N^{-71/200} = j^{-71/25}, \\ N^{-3/8} e^{13K_N} &\leq N^{-3/8+13/1000} = N^{-181/500} = j^{-362/125}. \end{aligned}$$

These are the two slowly decaying powers in (66) that come from the variable width. By the third and fourth lines of (66), with $K_N + 1 \leq \log N = 8 \log j$, the terms of (64) that depend on K_N satisfy

$$\begin{aligned} 4p_{K_N}^*(N) + C_{K_N} N^{-3/8} (\log N)^4 &\leq 4H_0 N^{-71/200} + 4N^{-12} + \frac{H_0}{64} N^{-167/400} (\log N)^{-12} \\ &\quad + C_{\text{bad}} (K_N + 1)^4 N^{-181/500} (\log N)^4 \\ &\leq 4H_0 j^{-71/25} + 4j^{-96} + \frac{H_0}{64} j^{-167/50} (8 \log j)^{-12} \\ &\quad + C_{\text{bad}} (8 \log j)^8 j^{-362/125}. \end{aligned}$$

Their exponents in j are less than -1 , and their remaining factors are polynomial in $\log j$. The other terms, $5p_2^*(N) + N^{-3} + 2N^{-10}$, do not depend on K_N , and by Table 14 at the width 2 they are summable along $N = j^8$. Thus every term in (64) is summable.

The bounds (29)–(33) of Step 1 in Section 6.1 are deterministic for $j \geq 60$. At the width K_N the constant of Lemma 4.3 is $100e^{K_N+4}$, and (31), (33) and $e^{bK_N} \leq N^{b/1000}$ give

$$\begin{aligned} \frac{2a}{d_0} &\leq 90e^{2(K_N+2)} N^{-67/64} \sqrt{\log N} \leq 90e^4 N^{-67/64+1/500} \sqrt{\log N}, \\ N^{65/64} \frac{2a}{d_0} &\leq 90e^4 N^{-1/32+1/500} \sqrt{\log N} = 90e^4 \sqrt{\log N} N^{-117/4000}, \\ \frac{4M_2 a}{d_0^2} &\leq 180 \cdot 100e^{K_N+4} e^{2(K_N+2)} N^{-1/32} \log N \\ &= 18000e^8 e^{3K_N} N^{-1/32} \log N \leq 18000e^8 (\log N) N^{-113/4000}. \end{aligned}$$

Since the last two right-hand sides tend to zero, for all large j we have $2a/d_0 < N^{-65/64}$, so that $2a/d_0 \leq 1/N$, and $4M_2 a \leq d_0^2$. These are the last two rows of Table 15, in which $180S_0 e^4 = 18000e^8$.

Let

$$\delta_N = \max_{q=-1,0,1} \left| \frac{\nu_N(1 + qN^{-65/64})}{N} - \frac{1}{2} \right|.$$

By the sparse estimate (37), whose five events are taken at the fixed width 2, we have

$$\delta_N \leq 2N^{-1/64} + C(2)N^{-1/64}(\log N)^7 = O(N^{-1/64}(\log N)^7).$$

Since $2a/d_0 < N^{-65/64}$, the bound (38) on c_N/N holds at the width K_N . Hence $|\nu_N(1)/N - 1/2| \leq \delta_N$ by the case $q = 0$, and $c_N/N \leq 2\delta_N$ by (38) and the cases $q = \pm 1$.

By the first line of (34), the number U_N of unselected roots is bounded by the number of roots with $||\alpha| - 1| > K_N/N$ plus $B_{K_N,N}$. Outside the four radial events at the width K_N , Lemma 2.4 applies with $\varepsilon = \varepsilon_{K_N}^*(N)$, by (54), and with $D = D_{K_N}$, by Lemma 2.3 and Section A.1, since $\varepsilon_{K_N}^*(N) + D_{K_N}/N \leq K_N/16 \leq K_N/8$ for large N . Outside the event of (61) we have $B_{K_N,N} \leq N^{31/32}$. This is (62) at $K = K_N$. Dividing it by N and using (66), we get

$$\begin{aligned} \frac{U_N}{N} &\leq N^{-1/32} + \frac{2 \log 2}{K_N} + \frac{8}{K_N} \left(\varepsilon_{K_N}^*(N) + \frac{D_{K_N}}{N} \right) \\ &\leq N^{-1/32} + \frac{2 \log 2}{K_N} + \frac{8}{K_N} \left(E_0 N^{-17/800} (\log N)^7 + 10(K_N + 1)^2 N^{-249/250} \right) \\ &\leq \frac{2 \log 2}{K_N} + o(1/\log N), \end{aligned}$$

since $K_N/\log N \rightarrow 1/1000$, so that $\log N$ times each term other than $2 \log 2/K_N$ tends to zero. By the normalized matching inequality (39) of Step 4 in Section 6.1, we therefore have, uniformly for $N \leq n \leq N + m_j$,

$$\begin{aligned} \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N + U_N}{N} + \frac{3m_j}{2N} \leq 3\delta_N + \frac{U_N}{N} + \frac{3m_j}{2N} \\ &\leq 3\delta_N + \frac{2 \log 2}{K_N} + o(1/\log N) + \frac{27}{2} N^{-1/8} = \frac{2 \log 2}{K_N} + o(1/\log N), \end{aligned}$$

where $3m_j/(2N) \leq \frac{27}{2} N^{-1/8}$ by (29), and $(\log N)\delta_N \rightarrow 0$. With every error term written out, this inequality is (67). Since the exceptional probabilities are summable, it follows from the Borel–Cantelli lemma that, almost surely, only finitely many exceptional events occur. Since $m_j \leq N$, we have $\log N \leq \log n \leq \log(2N)$, so that $\log n/\log N \rightarrow 1$ uniformly within the block. Since $(\log N)/1000 - 1 < K_N \leq (\log N)/1000$, we also have $K_N/\log N \rightarrow 1/1000$ and $(\log N) 2 \log 2/K_N \rightarrow 2000 \log 2$. Almost surely, every large n lies in a block $N \leq n \leq N + m_j$ whose exceptional event does not occur, and therefore, by the bound above in the block

of n ,

$$\begin{aligned} \limsup_{n \rightarrow \infty} (\log n) \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \limsup_{j \rightarrow \infty} \log(2N) \left(\frac{2 \log 2}{K_N} + o(1/\log N) \right) \\ &= \limsup_{j \rightarrow \infty} \frac{\log(2N)}{\log N} \left((\log N) \frac{2 \log 2}{K_N} + o(1) \right) = 2000 \log 2. \end{aligned}$$

□

8.2 Choice of the sparse degree sequence

Proposition 8.1 (Sparse-degree exponents). *With logarithmic cutoff $(\log N)/32$, deviation $N^{-1/32}$, derivative parameter $L = N^{1/64}$, exceptional root count $N^{31/32}$, and thin-band width $N^{-1-1/64}$, one may replace j^8 by $\lfloor j^q \rfloor$ for every*

$$\frac{8}{3} < q < 16.$$

In particular j^3 works. After retuning these powers, every real $q > 2$ works for the fixed- K strong law and radial-profile proof. For this construction, the second-moment conditions require $q > 2$.

Proof. The idea is to rerun the block argument of Section 6.1, and its form at moving radii in Section 6.2, along the degrees $\lfloor j^q \rfloor$, with the powers of the statement left as parameters, and to collect the inequalities that each step imposes on these powers and on q . We solve them for every $q > 2$, and then for the fixed powers of the statement.

In this proof we use $N = \lfloor j^q \rfloor$ and $m = \lfloor (j+1)^q \rfloor - \lfloor j^q \rfloor$ in place of j^8 and m_j . We claim that if $q > 1$, then eventually $m \leq (2 + q2^q)N^{1-1/q}$ and $m \leq N$. Indeed, for every $j \geq 1$,

$$\begin{aligned} m &\leq (j+1)^q - j^q + 1 \leq q(j+1)^{q-1} + 1 \leq q2^{q-1}j^{q-1} + 1 \\ &\leq q2^{q-1}(2N)^{1-1/q} + 1 \leq q2^q N^{1-1/q} + 1 \leq (2 + q2^q)N^{1-1/q}, \end{aligned}$$

where we use $\lfloor (j+1)^q \rfloor \leq (j+1)^q$ and $\lfloor j^q \rfloor > j^q - 1$, the mean value theorem for the q -th power on $[j, j+1]$, where its derivative is at most $q(j+1)^{q-1}$ since $q > 1$, and $j^{q-1} = (j^q)^{1-1/q}$ with $j^q < N+1 \leq 2N$. The second bound follows once $N^{1/q} \geq 2 + q2^q$. Also $m \geq 1$, since $(j+1)^q - j^q \geq qj^{q-1} \geq 1$. Since $\lfloor j^q \rfloor \geq j^q/2$, a bound on an exceptional probability that is a power of N times a power of $\log N$ is summable over j when q times the exponent of N is less than -1 .

We choose a tail disk of radius $1 + (K+2)/N$. The maximal-tail estimate applies with $a = 15e^{2(K+2)}\sqrt{m \log N}$, and its exceptional event still has probability

at most N^{-10} . This is Lemma 4.6 with $K + 2$ in place of K and $m_{\max} = m$, which applies since $1 \leq m \leq N$ and $N \geq \max\{4, K + 2\}$ for large j . By the bound on m , we have

$$a \leq 15e^{2(K+2)}(2 + q2^q)^{1/2}N^{1/2-1/(2q)}\sqrt{\log N}.$$

We take the derivative parameter N^ℓ in place of $L = N^{1/64}$, so that $d_0 = N^{3/2-\ell}$. For the deterministic localization bounds we have

$$\begin{aligned} \frac{2Na}{d_0} &= 2N^{\ell-1/2}a = O_K(N^{\ell-1/(2q)}\sqrt{\log N}), \\ \frac{4aN^{2\ell}}{N^3} \sup |f_N''| &\leq \frac{4aN^{2\ell}}{N^3} \cdot 100e^{K+4}N^{5/2}\sqrt{\log N} = O_K(N^{2\ell-1/(2q)}\log N), \end{aligned}$$

where the supremum is over $|z| \leq 1 + (K + 1)/N$ and is bounded by Lemma 4.3 outside its exceptional event, the two O_K bounds use the bound on a , and the implied constants also depend on the fixed q . It follows that Taylor isolation holds when $\ell < 1/(4q)$. Indeed, then $\ell - 1/(2q) < 2\ell - 1/(2q) < 0$, so both bounds tend to zero, and for large N they imply that $2a/d_0 \leq 1/N$ and $4M_2a \leq d_0^2$, where $M_2 = 100e^{K+4}N^{5/2}\sqrt{\log N}$ and $4aN^{2\ell}/N^3 = 4a/d_0^2$. A band of width $N^{-1-\kappa}$ contains the localization disks when $\kappa < 1/(2q) - \ell$, since $N^{1+\kappa} \cdot 2a/d_0 = N^\kappa \cdot 2Na/d_0 = O_K(N^{\kappa+\ell-1/(2q)}\sqrt{\log N})$. The moving target in Theorem 1.1 has scaled drift $O_x(N^{-1/q})$, which the same band absorbs. Indeed, for $N \leq n \leq N + m$ the target radius moves by $|x/n - x/N| \leq |x|m/N^2$, as in Section 6.2, and by the bound on m we have

$$N \cdot \frac{|x|m}{N^2} \leq (2 + q2^q)|x|N^{-1/q}, \quad N^{1+\kappa} \cdot \frac{|x|m}{N^2} \leq (2 + q2^q)|x|N^{\kappa-1/q}.$$

Since $\kappa < 1/(2q) - \ell < 1/q$, the second right-hand side tends to zero.

For the logarithmic clipping we set $T = t \log N$ and $u = N^{-w}$, and we apply Remark B.2 with $t = w > 0$, moment parameter $6 \log N$, and threshold N^{-2t} . By (13), the one- and two-point errors of Theorem B.1 are at most $C(K)N^{-1/2}$, and the removed angles and pairs there have measures at most $2N^{-1/2}$ and $10N^{-1/2}$. The quantities d_1 and v of the remark are the one-point error plus the removed measure of angles, and the two-point error plus the removed measure of pairs plus twice d_1 . Both are therefore at most $C(K)N^{-1/2}$, with a larger $C(K)$, and the remark applies with its exponent equal to $1/2$ and with $C(K)$ for both of its constants. Here the moment bound is (15), whose constant is C_* and whose exponent is 6, and the energy identity is (16). Therefore the logarithmic error satisfies

$$\begin{aligned} \eta_{\log} &\leq N^{-t} + 2tC(K)(\log N)N^{-1/2} \\ &\quad + e(12C_* \log N)^6 \sqrt{2N^{-2t} + C(K)N^{-1/2}} + \frac{3}{2}N^{-2t} = O_K(N^{-t}(\log N)^6) \end{aligned}$$

for $0 < t \leq 1/4$, since then $N^{-1/2} \leq N^{-2t} \leq N^{-t}$, and $1 \leq \log N \leq (\log N)^6$ for $N \geq e$. The first condition of (S) below implies that $t < 1/8$. The probabilities of the three exceptional events are bounded by constants times $N^{-1/2+4t}$, N^{-12} and $N^{-1/2+2t}(\log N)^2$, the three terms of the probability bound of Remark B.2 with $w = t$ and moment parameter $6 \log N$.

The mesh geometry and the normal approximation apply to a general derivative parameter N^ℓ . Indeed, in Section A.6 the bound $D'_K N L^2 \log N$ on the number of mesh points, the one-point probability (59) and the covariance bound $(3\mathcal{B}_K + \mathcal{T}_K)N^{-1/2}$ for a pair are written with L as a parameter, and the normal-approximation errors in them, $\mathcal{B}_K N^{-1/2}$ and $\mathcal{T}_K N^{-1/2}$, do not involve L . For a root-count threshold N^{1-d} , the mean and variance bounds in Corollary B.5, with both of its exponents equal to $1/2$ and its sector count $b_0 = O_K(\sqrt{N} \log N)$, lead to conditions (iii)–(vi) of the table below. We use $d < 1/2$ for the excluded-sector bound $O_K(\sqrt{N} \log N)$, which counts at most $C(K+2) \log N$ roots in each of $O(\sqrt{N})$ disks by Lemma 4.2, with $K+2$ in place of K . For logarithmic concentration and the thin band, we impose

$$(S) \quad q \left(\frac{1}{2} - 4t \right) > 1, \quad q \left(\frac{1}{2} - 2t \right) > 1, \quad 0 < \kappa < \min \left(t, \frac{1}{2q} - \ell \right).$$

At the moving radii of Theorem 1.1, the probes of Section 6.2 have spacing $h = N^{-\kappa}$. Fix an integer $R \geq |x| + 3$, as in Section 6.2, let $N \geq 2R$, and let y be one of the probes $x - h$, x and $x + h$. We apply Lemma B.3 at y , with R in place of K , which applies since $|y| + h \leq |x| + 2h < R < N$. We take the centering $\frac{1}{2} \log N$, the profile F of Proposition 2.2, $M_F = 1/2$ and $B_F = 1$, since $0 \leq \Phi' \leq 1/2$ and $0 \leq \Phi \leq 1$ (Section 6.2). In place of ε we take the variance-profile error $(R^2 + e^{4R}/2)/N$ of Lemma 2.3, and we use $\eta_{\log} = O_x(N^{-t}(\log N)^6)$, which is the bound on $\eta_{\log}$ above with R in place of K , since $|y| < R$, so that its constant depends on x through R . Combined with $|\Phi(y) - \Phi(x)| \leq |y - x|/2$, the lemma yields

$$\begin{aligned} \left| \frac{\nu_N(1 + y/N)}{N} - \Phi(x) \right| &\leq \frac{|y - x|}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2(\eta_{\log} + (R^2 + e^{4R}/2)/N)(1 + R/N)}{h} \\ &\leq \frac{7}{4} N^{-\kappa} + 4N^\kappa (\eta_{\log} + (R^2 + e^{4R}/2)N^{-1}) \\ &= O_x(N^{-\kappa} + N^{\kappa-t}(\log N)^6), \end{aligned}$$

where we use $|y - x| \leq h = N^{-\kappa}$, $R/N \leq N^{-\kappa}$ and $1 + R/N \leq 2$ for large N . This tends to zero since $\kappa < t$. The strict power inequalities dominate the above logarithmic factors and all fixed- K constants. We therefore obtain the nine conditions below, of which rows (ii), (vii), (viii) and (ix) together form (S).

	Condition	Where it arises
(i)	$\ell < 1/(4q)$	Taylor isolation
(ii)	$\kappa < 1/(2q) - \ell$	band containing the localization disks
(iii)	$0 < d < 4\ell$	mean threshold of Corollary B.5
(iv)	$d < \frac{1}{2} - 2\ell$	mean threshold of Corollary B.5
(v)	$d < \frac{1}{2}$	excluded sectors
(vi)	$q(\frac{1}{2} - 4\ell - 2d) > 1$	root-count failure, summable
(vii)	$q(\frac{1}{2} - 4t) > 1$	occupation failure, summable
(viii)	$q(\frac{1}{2} - 2t) > 1$	clipped-integral failure, summable
(ix)	$0 < \kappa < t$	thin band, probes at the moving radii

For every $q > 2$ we choose

$$\Delta = \frac{1}{2} - \frac{1}{q}, \quad t = w = \frac{\Delta}{16},$$

$$\ell = \min\left(\frac{\Delta}{32}, \frac{1}{16q}\right), \quad d = \ell, \quad \kappa = \min\left(\frac{t}{2}, \frac{1}{8q}\right).$$

Then $0 < \Delta < 1/2$, all four powers are positive, and the nine conditions follow from

$$\begin{aligned} \text{(iii)} \quad d = \ell < 4\ell, & \quad \text{(v)} \quad d \leq \frac{\Delta}{32} < \frac{1}{2}, & \quad \text{(iv)} \quad d + 2\ell = 3\ell \leq \frac{3\Delta}{32} < \frac{1}{2}, \\ \text{(vii)} \quad 4t = \frac{\Delta}{4} < \Delta, & \quad \text{(viii)} \quad 2t = \frac{\Delta}{8} < \Delta, & \quad \text{(vi)} \quad 4\ell + 2d = 6\ell \leq \frac{3\Delta}{16} < \Delta, \\ \text{(i)} \quad 2\ell \leq \frac{1}{8q} < \frac{1}{2q}, & \quad \text{(ix)} \quad \kappa \leq \frac{t}{2} < t, & \quad \text{(ii)} \quad \ell + \kappa \leq \frac{1}{16q} + \frac{1}{8q} = \frac{3}{16q} < \frac{1}{2q}, \end{aligned}$$

where the middle row implies the summability conditions (vii), (viii) and (vi), since $q(\frac{1}{2} - \Delta) = 1$, and subtracting from $\frac{1}{2}$ a number smaller than Δ leaves more than $\frac{1}{2} - \Delta$. The remaining exceptional probabilities N^{-3}, N^{-10}, N^{-12} are summable as well. The conclusion then follows from the fixed- K argument and the countable intersection over K .

For the fixed powers of the statement, $t = w = 1/32$, $\ell = 1/64$, $d = 1/32$ and $\kappa = 1/64$, the conditions read

$$\begin{aligned} \text{(vii)} \quad q\left(\frac{1}{2} - 4t\right) &= \frac{3q}{8}, & \text{(vi)} \quad q\left(\frac{1}{2} - 4\ell - 2d\right) &= \frac{3q}{8}, & \text{(viii)} \quad q\left(\frac{1}{2} - 2t\right) &= \frac{7q}{16}, \\ \text{(i)} \quad \frac{1}{4q} - \ell &= \frac{16 - q}{64q}, & \text{(ii)} \quad \frac{1}{2q} - \ell - \kappa &= \frac{16 - q}{32q}, \end{aligned}$$

where the three quantities of the first row must exceed one and the two of the second row must be positive, while rows (ix), (iii) and (iv), $\kappa < t$, $d < 4\ell$ and $d < \frac{1}{2} - 2\ell$, hold since $1/64 < 1/32 < 1/16 < 15/32$, as does (v). For the fixed

powers above, the slow exponent of the exceptional probabilities is $3/8$, so that summability requires $q > 8/3$. The Taylor condition requires $q < 16$. At $q = 8/3$ the mesh bound is of order $j^{-1}(\log j)^4$, and at $q = 16$ the Taylor ratio grows logarithmically. For $q \leq 2$ and $t > 0$, the inequality $q(\frac{1}{2} - 4t) > 1$ cannot hold, since $q(\frac{1}{2} - 4t) < q/2 \leq 1$. These endpoint restrictions belong to the present second-moment estimates. $\square$

8.3 Sublevel components meeting nearby circles

Corollary 8.2 (Sublevel components meeting a nearby circle). *For a fixed tail bound a_0 , and $r_{\pm} = 1 \pm N^{-1-\kappa}$ with fixed $0 < \kappa \leq 1/12$, the number of roots in components of $\{|f_N| < a_0\}$ meeting those circles is $o(N)$ outside a single exceptional sequence summable at $N = j^8$.*

Proof. The idea is to bound, for each fixed K , the roots in question by the roots of f_N in a thin band about the unit circle, which (20) controls, and by the unselected roots U_N of Section 6.1, and then to let K tend to infinity by Lemma B.7. With $N = j^8$, let Y_j denote the number of roots of f_N in components of $\{|f_N| < a_0\}$ that meet at least one of the circles $|z| = r_+$ and $|z| = r_-$. We show that there exist deterministic numbers $\varepsilon_j \rightarrow 0$ and events F_j with $\sum_j \mathbb{P}(F_j) < \infty$ such that $Y_j \leq \varepsilon_j N$ off F_j .

Fix an integer $K \geq 2$, and let $a = 15e^{2(K+2)}\sqrt{m_j \log N}$ be the maximal-tail level of Step 1 of Section 6.1. Since $a \geq 15\sqrt{\log N}$, we have $a_0 \leq a$ for all sufficiently large N . Every component of $\{|f_N| < a_0\}$ lies in a component of $\{|f_N| < a\}$, and if the first meets a circle, then so does the second. Thus, for each circle, the number of roots in components of $\{|f_N| < a_0\}$ meeting it is at most the number of roots in components of $\{|f_N| < a\}$ meeting it. A component containing a good annular root lies in the selected domain of that root, which is a component of the sublevel set of Step 2 of Section 6.1, whose level exceeds a . By that step, it therefore has exactly one root and lies in the open disk of radius $2a/d_0$ about it. Every other root is included in U_N . If this component meets either nearby circle at a point z , then $||\alpha| - 1| \leq |\alpha - z| + ||z| - 1| < N^{-1-\kappa} + 2a/d_0$. We enlarge the band to half-width $N^{-1-\kappa_*}$, which must contain this range for large N and to which (20) must apply. This requires, first, $0 < \kappa_* < \kappa$, so that $N^{-1-\kappa}$ is small compared with $N^{-1-\kappa_*}$, second, $\kappa_* < 3/64$, so that $2a/d_0$ is small compared with $N^{-1-\kappa_*}$ by (31), since $1 - 67/64 = -3/64$, and third, $\kappa_* \leq 1/64$, the range of (20). We choose $\kappa_* = \min(\kappa/2, 1/128)$. Since $\kappa_* \leq \kappa/2$ and $\kappa_* \leq 1/128$, (31) gives

$$N^{1+\kappa_*} (N^{-1-\kappa} + 2a/d_0) \leq N^{\kappa_*-\kappa} + 90e^{2(K+2)} N^{1+\kappa_*-67/64} \sqrt{\log N}$$

$$\leq N^{-\kappa/2} + 90e^{2(K+2)}N^{-5/128}\sqrt{\log N},$$

and the right-hand side tends to zero, so that $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$ for all N large enough. Since $0 < \kappa_* \leq 1/128 \leq 1/64$, (20) applies with 2 in place of K and κ_* in place of κ . Outside five events, one at each radius $1 + qN^{-1-\kappa_*}$ with $q = -2, \dots, 2$, it yields

$$\left| \frac{\nu_N(1 \pm N^{-1-\kappa_*})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa_*} + C(2)N^{\kappa_*-1/32}(\log N)^7.$$

This five-radius estimate bounds the number of roots in this band by $o(N)$:

$$\begin{aligned} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| < N^{-1-\kappa_*}\} &\leq \nu_N(1 + N^{-1-\kappa_*}) - \nu_N(1 - N^{-1-\kappa_*}) \\ &= N \left(\frac{\nu_N(1 + N^{-1-\kappa_*})}{N} - \frac{1}{2} \right) - N \left(\frac{\nu_N(1 - N^{-1-\kappa_*})}{N} - \frac{1}{2} \right) \\ &\leq 2N(2N^{-\kappa_*} + C(2)N^{\kappa_*-1/32}(\log N)^7) = o(N), \end{aligned}$$

where the first inequality holds as in (36), and the last step since $\kappa_* > 0$ and $\kappa_* - 1/32 \leq 1/128 - 4/128 = -3/128$. Each of the five events has probability at most the right-hand side of (8) at $K = 2$, which does not depend on the radius. For fixed K , the exceptional probabilities are therefore bounded by the right-hand side of (40), whose five thin-band terms now bound the events at the exponent κ_* , and they are summable. The explicit form of this bound is (64), with thin-band parameter κ_* . Let $E_{K,j}(\kappa_*)$ denote the event $E_{K,j}$ of Step 5 of Section 6.1 with its five thin-band events taken at the exponent κ_* . Let $j_0(K) \geq 60$ be a deterministic index beyond which the thresholds of Section 6.1 at the width K , the bound $a_0 \leq a$, the strict inequality $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$ and the five-radius estimate hold. Each root counted by Y_j lies in a component of $\{|f_N| < a\}$ meeting one of the circles. Hence it is either a good annular root, which lies in the band $||\alpha| - 1| < N^{-1-\kappa_*}$ by the strict inequality, or it is counted by U_N . Therefore, for $j \geq j_0(K)$ and outside $E_{K,j}(\kappa_*)$, by the band count and (34) we have

$$\begin{aligned} \frac{Y_j}{N} &\leq \frac{1}{N} \#\{\alpha : f_N(\alpha) = 0, \ ||\alpha| - 1| < N^{-1-\kappa_*}\} + \frac{U_N}{N} \\ &\leq 4N^{-\kappa_*} + 2C(2)N^{\kappa_*-1/32}(\log N)^7 \\ &\quad + \frac{2\log 2}{K} + C(K)N^{-1/32}(\log N)^7 + N^{-1/32}. \end{aligned}$$

To apply Lemma B.7, which concerns normalized block increments, with the exponent 8, so that $n_j = j^8$, we take $X_n = 0$ when n is an eighth power and $X_n = Y_j$ when $j^8 < n < (j+1)^8$. Since $X_{n_j} = 0$ and each block contains more than

one degree, we have $\max_{n_j \leq n < n_{j+1}} |X_n - X_{n_j}|/n_j = Y_j/N$. For each integer $K \geq 2$ we take the index of the lemma to be $K - 1$, which runs over the positive integers, and put

$$b_{K-1}(j) = 4N^{-\kappa_*} + 2C(2)N^{\kappa_*-1/32}(\log N)^7 + C(K)N^{-1/32}(\log N)^7 + N^{-1/32},$$

and let $p_{K-1}(j)$ be the bound (40) for $\mathbb{P}(E_{K,j}(\kappa_*))$ when $j \geq j_0(K)$, and $p_{K-1}(j) = 1$ when $j < j_0(K)$. By the last display, $\mathbb{P}\{Y_j/N > 2 \log 2/K + b_{K-1}(j)\} \leq p_{K-1}(j)$ for every j . Moreover $2 \log 2/K \rightarrow 0$ as $K \rightarrow \infty$, and, for fixed K , $b_{K-1}(j) \rightarrow 0$ as $j \rightarrow \infty$, since $\kappa_* > 0$ and $\kappa_* - 1/32 \leq -3/128$. Also $\sum_j p_{K-1}(j) < \infty$, since only finitely many terms equal one and the others are summable by Step 5 of Section 6.1. These are the hypotheses of Lemma B.7 at the index $K - 1$, with $2 \log 2/K$ as the term that tends to zero as the index grows, and the lemma gives deterministic numbers $\varepsilon_j \rightarrow 0$ and events F_j , the events E_j of the lemma, with $\sum_j \mathbb{P}(F_j) < \infty$ such that $Y_j \leq \varepsilon_j N$ off F_j . This proves the conclusion for both signs simultaneously. $\square$

8.4 Polynomially varying deterministic coefficients

Let $(\xi_k)_{k \geq 0}$ be random variables on one probability space, let (c_k) be deterministic complex weights, and consider the prefixes

$$f_N(z) = \sum_{k=0}^N \xi_k c_k z^k.$$

Note that every degree uses the same coefficient sequence. For independent centered coefficients with unit variance, the normalization is the square root of $\sum_{k=0}^N |c_k|^2 r^{2k}$.

Proposition 8.3 (Polynomially weighted logarithmic variance). *Fix $\tau > -1/2$, put $c_k = k^\tau$ for $k \geq 1$, and choose a fixed finite $c_0 \neq 0$. For the normalization*

$$\sigma_{N,\tau}(r)^2 = |c_0|^2 + \sum_{k=1}^N k^{2\tau} r^{2k},$$

we have

$$\log \sigma_{N,\tau}(1 + x/N) - (\tau + 1/2) \log N \longrightarrow F_\tau(x) := \frac{1}{2} \log \int_0^1 t^{2\tau} e^{2xt} dt$$

uniformly on compact intervals of x . Moreover,

$$F'_\tau(x) = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt}, \quad F'_\tau(0) = \frac{2\tau + 1}{2\tau + 2}.$$

The four-secant formula in Lemma 2.1 applies, where $F_{N,\tau}(x) = \log \sigma_{N,\tau}(1 + x/N) - (\tau + 1/2) \log N$. With the centering $\frac{1}{2} \log N$, we add $\tau \log N$ to $F_{N,\tau}$, a term that cancels in every secant. Both the convention at $k = 0$ and the restriction $\tau > -1/2$ are needed for the integral profile.

Proof. Since every compact interval of x lies in some $[-K, K]$, we fix $K > 0$ and work with $|x| \leq K$. The idea is to show that $\sigma_{N,\tau}(1 + x/N)^2/N^{2\tau+1}$ converges to $\int_0^1 t^{2\tau} e^{2xt} dt$ uniformly for $|x| \leq K$, and then to take logarithms. For $|x| \leq K$ and $0 < t \leq 1$, let

$$g_x(t) = t^{2\tau} e^{2xt}, \quad I_\tau(x) = \int_0^1 g_x(t) dt = \int_0^1 t^{2\tau} e^{2xt} dt.$$

For the weighted sum, we have

$$S_{N,\tau}(x) := \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} e^{2xk/N} = \frac{1}{N} \sum_{k=1}^N (k/N)^{2\tau} e^{2xk/N} = \frac{1}{N} \sum_{k=1}^N g_x(k/N).$$

Step 1: the radial factor. Let $|x| \leq K$ and $N \geq 2K$. Multiplying the comparison (5) by $k^{2\tau} e^{2xk/N}/N^{2\tau+1} > 0$ and summing over $1 \leq k \leq N$, we get, by the definition of $S_{N,\tau}$,

$$e^{-2K^2/N} S_{N,\tau}(x) \leq \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1 + x/N)^{2k} \leq e^{2K^2/N} S_{N,\tau}(x).$$

Step 2: the Riemann sums for $\tau \geq 0$. When $\tau \geq 0$, the integrand is continuous, since the map $(x, t) \mapsto g_x(t)$ extends continuously to the compact set $[-K, K] \times [0, 1]$, and hence it is uniformly continuous there. Let $\eta(\delta)$ be the supremum of $|g_x(t) - g_x(t')|$ over $|x| \leq K$ and $t, t' \in [0, 1]$ with $|t - t'| \leq \delta$, so that $\eta(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. For $|x| \leq K$ we have

$$\begin{aligned} |S_{N,\tau}(x) - I_\tau(x)| &= \left| \sum_{k=1}^N \frac{1}{N} g_x(k/N) - \sum_{k=1}^N \int_{(k-1)/N}^{k/N} g_x(t) dt \right| \\ &= \left| \sum_{k=1}^N \int_{(k-1)/N}^{k/N} (g_x(k/N) - g_x(t)) dt \right| \\ &\leq \sum_{k=1}^N \int_{(k-1)/N}^{k/N} |g_x(k/N) - g_x(t)| dt \leq \eta(1/N), \end{aligned}$$

where the last step uses $|k/N - t| \leq 1/N$ on the k -th interval, whose lengths add up to one.

Step 3: the Riemann sums for $-1/2 < \tau < 0$. When $-1/2 < \tau < 0$, we fix $0 < \varepsilon < 1$ and split the sum at $k_\varepsilon = \lfloor \varepsilon N \rfloor$, where $N \geq 2/\varepsilon$, so that $2 \leq k_\varepsilon < N$.

Then

$$\begin{aligned} S_{N,\tau}(x) - I_\tau(x) &= \left[\frac{1}{N} \sum_{k=1}^{k_\varepsilon} g_x(k/N) - \int_0^{k_\varepsilon/N} g_x(t) dt \right] \\ &\quad + \left[\frac{1}{N} \sum_{k=k_\varepsilon+1}^N g_x(k/N) - \int_{k_\varepsilon/N}^1 g_x(t) dt \right]. \end{aligned}$$

Since $t \mapsto t^{2\tau}$ is decreasing, we have $\frac{1}{N}(k/N)^{2\tau} \leq \int_{(k-1)/N}^{k/N} t^{2\tau} dt$ for $k \geq 2$, and the term $k = 1$ is $\frac{1}{N}(1/N)^{2\tau} = N^{-(2\tau+1)}$. Also $e^{2xt} \leq e^{2K}$ for $|x| \leq K$ and $0 \leq t \leq 1$, and $k_\varepsilon/N \leq \varepsilon$. Hence the normalized initial part satisfies

$$\begin{aligned} 0 \leq \frac{1}{N} \sum_{k=1}^{k_\varepsilon} g_x(k/N) &\leq \frac{e^{2K}}{N} \sum_{k=1}^{k_\varepsilon} (k/N)^{2\tau} = e^{2K} \left(N^{-(2\tau+1)} + \sum_{k=2}^{k_\varepsilon} \frac{1}{N} (k/N)^{2\tau} \right) \\ &\leq e^{2K} \left(N^{-(2\tau+1)} + \int_{1/N}^{k_\varepsilon/N} t^{2\tau} dt \right) \leq e^{2K} \left(\frac{\varepsilon^{2\tau+1}}{2\tau+1} + N^{-(2\tau+1)} \right), \end{aligned}$$

where the intervals $[(k-1)/N, k/N]$ with $2 \leq k \leq k_\varepsilon$ make up $[1/N, k_\varepsilon/N]$. By the same two facts, the corresponding integral satisfies

$$0 \leq \int_0^{k_\varepsilon/N} g_x(t) dt \leq e^{2K} \int_0^\varepsilon t^{2\tau} dt = e^{2K} \frac{\varepsilon^{2\tau+1}}{2\tau+1}.$$

Since both initial parts are nonnegative, the first bracket is at most $e^{2K}(\varepsilon^{2\tau+1}/(2\tau+1) + N^{-(2\tau+1)})$ in absolute value. For the remaining Riemann sum, every interval $[(k-1)/N, k/N]$ with $k \geq k_\varepsilon + 1$ lies in $[\varepsilon/2, 1]$, since $k_\varepsilon/N \geq \varepsilon - 1/N \geq \varepsilon/2$. On the compact set $[-K, K] \times [\varepsilon/2, 1]$ the map $(x, t) \mapsto g_x(t)$ is uniformly continuous. Let $\eta_\varepsilon(\delta)$ be the supremum of $|g_x(t) - g_x(t')|$ over $|x| \leq K$ and $t, t' \in [\varepsilon/2, 1]$ with $|t - t'| \leq \delta$, so that $\eta_\varepsilon(\delta) \rightarrow 0$ as $\delta \rightarrow 0$. Then the second bracket satisfies

$$\begin{aligned} \left| \frac{1}{N} \sum_{k=k_\varepsilon+1}^N g_x(k/N) - \int_{k_\varepsilon/N}^1 g_x(t) dt \right| &= \left| \sum_{k=k_\varepsilon+1}^N \int_{(k-1)/N}^{k/N} (g_x(k/N) - g_x(t)) dt \right| \\ &\leq \sum_{k=k_\varepsilon+1}^N \int_{(k-1)/N}^{k/N} |g_x(k/N) - g_x(t)| dt \leq \eta_\varepsilon(1/N), \end{aligned}$$

where the last step holds as in Step 2, since k/N and t lie in $[\varepsilon/2, 1]$. Therefore

$$\sup_{|x| \leq K} |S_{N,\tau}(x) - I_\tau(x)| \leq e^{2K} \left(\frac{\varepsilon^{2\tau+1}}{2\tau+1} + N^{-(2\tau+1)} \right) + \eta_\varepsilon(1/N).$$

Since $2\tau + 1 > 0$, the right-hand side tends to $e^{2K}\varepsilon^{2\tau+1}/(2\tau+1)$ as $N \rightarrow \infty$, so that the limit superior of the left side as $N \rightarrow \infty$ is at most this value for every $0 < \varepsilon < 1$, and it is therefore zero.

We conclude that in both cases $S_{N,\tau}(x) \rightarrow I_\tau(x)$ uniformly for $|x| \leq K$.

Step 4: the variance. Let $|x| \leq K$ and $N \geq 2K$. By the definition of $\sigma_{N,\tau}$ and Step 1, we have

$$\begin{aligned} \left| \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} - I_\tau(x) \right| &\leq \frac{|c_0|^2}{N^{2\tau+1}} + \left| \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1+x/N)^{2k} - S_{N,\tau}(x) \right| \\ &\quad + |S_{N,\tau}(x) - I_\tau(x)| \\ &\leq \frac{|c_0|^2}{N^{2\tau+1}} + (e^{2K^2/N} - 1) S_{N,\tau}(x) + |S_{N,\tau}(x) - I_\tau(x)| \\ &\leq \frac{|c_0|^2}{N^{2\tau+1}} + (e^{2K^2/N} - 1) \left(\frac{e^{2K}}{2\tau+1} + |S_{N,\tau}(x) - I_\tau(x)| \right) \\ &\quad + |S_{N,\tau}(x) - I_\tau(x)|, \end{aligned}$$

where the second inequality also uses $1 - e^{-2K^2/N} \leq e^{2K^2/N} - 1$, which holds since $e^u + e^{-u} \geq 2$, and the last uses $I_\tau(x) \leq e^{2K} \int_0^1 t^{2\tau} dt = e^{2K}/(2\tau+1)$. Since $2\tau+1 > 0$, the term with the fixed c_0 also vanishes after normalization, and by Steps 2 and 3 the other terms tend to zero uniformly for $|x| \leq K$, since $e^{2K^2/N} - 1 \rightarrow 0$. Thus $\sigma_{N,\tau}(1+x/N)^2/N^{2\tau+1} \rightarrow \int_0^1 t^{2\tau} e^{2xt} dt$ uniformly for $|x| \leq K$.

Step 5: the logarithm. For $|x| \leq K$, the integral is at least $e^{-2K}/(2\tau+1) > 0$, since $I_\tau(x) \geq e^{-2K} \int_0^1 t^{2\tau} dt$. By Step 1 we have

$$\begin{aligned} \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} &\geq \frac{1}{N^{2\tau+1}} \sum_{k=1}^N k^{2\tau} (1+x/N)^{2k} \geq e^{-2K^2/N} S_{N,\tau}(x) \\ &\geq e^{-2K^2/N} (I_\tau(x) - |S_{N,\tau}(x) - I_\tau(x)|) \\ &\geq e^{-2K^2/N} \left(\frac{e^{-2K}}{2\tau+1} - |S_{N,\tau}(x) - I_\tau(x)| \right), \end{aligned}$$

where the first inequality drops $|c_0|^2/N^{2\tau+1} \geq 0$. Also $I_\tau(x)$ is at least the same lower bound, since $e^{-2K^2/N} \leq 1$. Since $|\log a - \log b| \leq |a-b|/\min(a,b)$ for $a, b > 0$, we get, for all N so large that this lower bound is positive for $|x| \leq K$,

$$\begin{aligned} &|\log \sigma_{N,\tau}(1+x/N) - (\tau+1/2) \log N - F_\tau(x)| \\ &= \frac{1}{2} \left| \log \frac{\sigma_{N,\tau}(1+x/N)^2}{N^{2\tau+1}} - \log I_\tau(x) \right| \\ &\leq \frac{| \sigma_{N,\tau}(1+x/N)^2/N^{2\tau+1} - I_\tau(x) |}{2e^{-2K^2/N} (e^{-2K}/(2\tau+1) - |S_{N,\tau}(x) - I_\tau(x)|)}, \end{aligned}$$

since $F_\tau = \frac{1}{2} \log I_\tau$ and both arguments of the logarithms are at least the lower bound above. By Steps 2 to 4, the numerator tends to zero and the denominator

tends to $2e^{-2K}/(2\tau + 1)$, both uniformly for $|x| \leq K$. Hence the left side tends to zero uniformly for $|x| \leq K$, which proves the profile limit of the proposition.

By differentiating the positive integral, we get

$$\frac{d}{dx} \left(\frac{1}{2} \log \int_0^1 t^{2\tau} e^{2xt} dt \right) = \frac{1}{2} \cdot \frac{\int_0^1 2t \cdot t^{2\tau} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt} = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt},$$

where differentiation under the integral sign is allowed since, for $|x| \leq K$, the x -derivative $2t^{2\tau+1}e^{2xt}$ of the integrand is at most $2e^{2K}t^{2\tau}$, which is integrable on $[0, 1]$ since $2\tau > -1$. At zero this quotient is $\int_0^1 t^{2\tau+1} dt / \int_0^1 t^{2\tau} dt = (2\tau + 1)/(2\tau + 2)$. $\square$

Remark 8.4. For $\tau \leq -1/2$, the integral defining F_τ diverges at zero. The finite-degree radial secant estimate still applies with the exact normalization. To deduce an almost-sure root law from a variance profile, one must also establish the occupation and logarithmic hypotheses and the degree-block estimates.

The integral diverges since its integrand is at least $e^{-2|x|}t^{2\tau}$ and $2\tau \leq -1$. The finite-degree radial secant estimate of the remark is (2) of Lemma 2.1, which allows any normalization $\sigma_N(r) > 0$, and in particular $\sigma_{N,\tau}$.

Proposition 8.5 (Weighted polynomials with random signs). *Take $c_0 = 1$ and $c_k = k^\tau$ for $k \geq 1$, where $\tau > -1/2$. For independent signs put*

$$f_N^{(\tau)}(z) = \sum_{k=0}^N \epsilon_k c_k z^k, \quad X_r(\theta) = \frac{f_N^{(\tau)}(re^{i\theta})}{\sigma_{N,\tau}(r)}.$$

The uniform-on-compact variance limit in Proposition 8.3 holds for these polynomials. Its candidate radial profile is

$$\Phi_\tau(x) = F'_\tau(x) = \frac{\int_0^1 t^{2\tau+1} e^{2xt} dt}{\int_0^1 t^{2\tau} e^{2xt} dt}, \quad \Phi_\tau(0) = \frac{2\tau + 1}{2\tau + 2}.$$

The normalized logarithmic moments and angular energy satisfy

$$\mathbb{E} \int |\log |X_r(\theta)||^p d\mu(\theta) \leq (C_* p)^{6p}, \quad p > 1, \quad \int |X_r(\theta)|^2 d\mu(\theta) = 1.$$

The energy identity holds for every sign vector.

Proof. The idea is that X_r is a Rademacher Fourier series whose deterministic coefficients $c_k r^k / \sigma_{N,\tau}(r)$ have squared moduli summing to one, which is all that the logarithmic bound and the energy identity of Section 3.2 use. The variance convergence follows from the preceding proposition. By the definition of $\sigma_{N,\tau}$ with $c_0 = 1$, the deterministic Fourier coefficients satisfy

$$\sum_{k=0}^N \left| \frac{c_k r^k}{\sigma_{N,\tau}(r)} \right|^2 = \frac{1 + \sum_{k=1}^N k^{2\tau} r^{2k}}{\sigma_{N,\tau}(r)^2} = 1.$$

By Corollary 1.2 of Nazarov–Nishry–Sodin, the above logarithmic bound holds with the same absolute constant C_* . Since that corollary holds for every $p \geq 1$, the logarithmic bound also holds for $p = 1$, as in (15). By Parseval’s identity, $\epsilon_k^2 = 1$ and the preceding identity, we obtain for every sign vector the energy identity

$$\int |X_r(\theta)|^2 d\mu(\theta) = \frac{1}{\sigma_{N,\tau}(r)^2} \sum_{k=0}^N \epsilon_k^2 c_k^2 r^{2k} = 1.$$

□

The conclusions of Proposition 8.5 hold for any fixed finite nonzero c_0 , with $|c_0|^2$ in place of 1 in the variance.

9 Open problems

The almost-sure radial law and its proof lead us to pose three problems.

Problem 9.1 (The moment threshold). *Let $(\xi_k)_{k \geq 0}$ be independent and identically distributed complex random variables, not almost surely constant, and let f_n and ν_n be as in Theorem 1.1. Does the conclusion of Theorem 1.1 hold whenever $\mathbb{E}|\xi_0|^2 < \infty$?*

Theorem 1.1 establishes the law for five families of coefficient distributions, and our proof uses the distribution only through the estimates of Table 5 and Proposition 7.4. The second moment cannot be weakened. If $\xi_n \neq 0$, Jensen’s formula for $w^n f_n(1/w)$ on the circle $|w| = e^{1/n}$ and the concavity of the logarithm give

$$n - \nu_n(1) \leq \frac{n}{2} \log \left(1 + e^2 \sum_{k < n} \frac{|\xi_k|^2}{|\xi_n|^2} \right).$$

When $\mathbb{E}|\xi_0|^2 = \infty$, the ratio $|\xi_n|^2 / \sum_{k < n} |\xi_k|^2$ is unbounded almost surely, by a theorem of Kesten [14], and hence $\limsup_n \nu_n(1)/n = 1$. At a fixed degree, with mean zero and a finite second moment, the zeros near each point of the unit circle converge in distribution at the scale $1/n$ [11], and equidistribution of the zeros on the unit circle needs only a logarithmic moment [10].

Problem 9.2 (Fluctuations). *For Rademacher coefficients, is $(\nu_n(1) - n/2)/\sqrt{n}$ asymptotically normal with a positive variance? Is*

$$\limsup_{n \rightarrow \infty} \frac{\nu_n(1) - n/2}{\sqrt{n \log \log n}}$$

a positive and finite constant almost surely?

Theorem 1.3 gives $\nu_n(1) - n/2 = O(n/\log n)$ almost surely. Positive answers would identify the exact almost-sure order $\sqrt{n \log \log n}$, as in the law of the iterated logarithm for sums of independent random variables. Along the nested sequence, the increments $\nu_{n+m}(1) - \nu_n(1)$ measure the flow of zeros through the unit circle as coefficients are appended, and their size is part of the question.

Problem 9.3 (Almost-sure local universality). *Let the coefficients be as in Theorem 1.1, with $\mathbb{E}|\xi_0|^2 = 1$, and let $G(\zeta) = \int_0^1 e^{t\zeta} dW(t)$, where W is a complex Brownian motion with $\mathbb{E}|W(t)|^2 = t$. Almost surely, do the correlation functions of the zeros of $\zeta \mapsto f_n(e^{i\theta}(1 + \zeta/n))$, averaged over θ , converge to those of the zeros of G ?*

The zeros of G have intensity $\Phi'(\operatorname{Re} \zeta)/(2\pi)$, and Theorem 1.1 is the one-point case of the problem. At a fixed degree, local universality of the zeros near the unit circle was proved by Tao and Vu [20], and in distribution under a finite second moment by Kabluchko, Khoruzhenko and Marynych [11]. The problem asks for it almost surely, at every degree of one nested sequence. In the proof of Theorem 1.1, root matching (Theorem 5.1) keeps each selected zero of f_N near the unit circle within $N^{-65/64} = o(1/N)$ of a zero of f_n , at every degree n of its block, so that local configurations at the sparse degrees persist at every degree.

10 Numerical observations

In this section we consider radial counts, logarithmic integrals, derivative sizes, and perturbations of nested polynomials. The limits that hold with probability one follow from the proofs in the previous sections, and these proofs do not use the computations below. The figures describe measurements at finite degrees, and the fitted slopes describe only the measured range. The residuals and grid comparisons quantify numerical sensitivity rather than interval certification.

We use 100 independent sequences of random signs at the degrees $N = 1000, 3000, 10000, 30000$ and 100000 , and the block length is $\lfloor N^{7/8} \rfloor$. Table 8 contains the numbers of root samples that were completed at the collection cutoff, and we retained every completed base sample, independently of its root counts. For 78 sequences at $N = 1000$ we computed all of the 421 intermediate polynomials, while at the larger degrees there are 3, 3, 2, and 0 completed three-point blocks, respectively. Two further midpoint measurements belong to blocks that were not completed.

For the base sample and the retained block measurements, we apply three Newton refinements to the output of MPSolve's secular isolation. By independent

N	Base samples	Endpoint samples	Maximum base residual
1000	92	78	1.79711×10^{-15}
3000	92	58	3.46309×10^{-15}
10000	92	57	6.89178×10^{-15}
30000	91	53	1.20336×10^{-14}
100000	77	27	2.11195×10^{-14}

Table 8: Completed base and endpoint samples. The endpoint is $N + \lfloor N^{7/8} \rfloor$. The $N = 1000$ samples include every intermediate degree. The last column gives the maximum normalized residual at each base degree.

Horner evaluation we get the residual $|f_N(z)| / \sum_{k=0}^N |z|^k$. For $|z| > 1$ we evaluate the reciprocal polynomial, which yields the same quotient with controlled numerical magnitude. Over all base samples the maximum normalized residual is below 3×10^{-14} , and Table 8 gives this maximum at each base degree.

For Figure 9 we use the degree-1000 coordinates unchanged, and we compute the roots of the two additional prefixes, at degrees 1210 and 1421, with a balanced real companion-matrix eigensolver. We then refine these roots by Newton steps that reduce the residual. With an independent 60-digit evaluation at every stored coordinate we obtain the maximum normalized residuals 3.170×10^{-15} , 3.824×10^{-15} and 4.317×10^{-15} at degrees 1000, 1210, and 1421, respectively. In all three cases the coefficient vector is the same saved sequence.

We determine the roots at 1 and -1 , with their multiplicities, by exact integer division. For the remaining roots, the band $||z| - 1| \leq 10^{-9}$ yields a sensitivity interval, whose two endpoints count every root in that band as outside or inside the disk. We check the counting procedure by direct evaluation and on polynomials with known boundary roots. Over all base samples, the numbers of the remaining roots in this band are 0, 0, 2, 28, 284, in increasing degree order, and they belong to 0, 0, 1, 12, 61 polynomials, respectively. Thus the largest unnormalized sensitivity intervals have widths 0, 0, 2, 4, 14.

We compute the angular occupations and the logarithmic integral on a shifted FFT grid. A doubled grid yields 5500 comparisons over 100 seeds, five degrees, and eleven radii, and the largest changes in absolute value are 3.599×10^{-4} for the logarithmic average and 1.840×10^{-3} for occupation. Percentile bootstrap intervals resample whole sequences, and in particular each resample retains the nesting across degrees. If a logarithmic fit omits zero observations, then its interval is conditional on the positive observations that are retained.

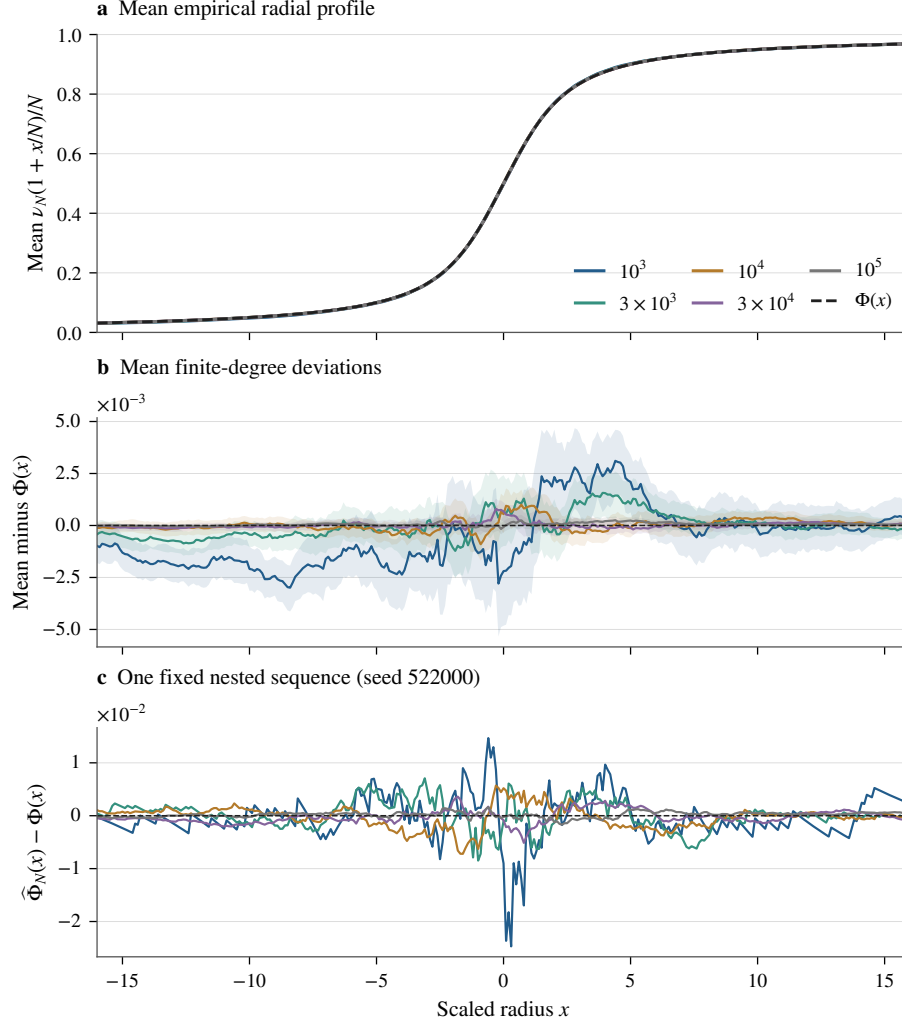


Figure 10: Mean empirical radial profile over 77 sequences at five degrees, which cannot be distinguished from Φ (a), and its deviations (b, c).

Radial mass and logarithmic concentration. We compare the empirical profile $\hat{\Phi}_N(x) = \nu_N(1 + x/N)/N$ of Figure 10 with $\Phi(x)$, and the outside-annulus count with $2 \log 2/K$ and with

$$1 - \{\Phi(K) - \Phi(-K)\} = \frac{1}{K} - \frac{2}{e^{2K} - 1}, \quad K = 2, 4, 8, 16, 32.$$

We also compare the occupation means with $1 - e^{-t^2}$, and we fit the variance of occupation and the centered logarithmic mean square over the available degrees (Table 9).

For the profile and root-count comparisons we use the 77 sequences that were completed at all five base degrees, while the angular comparisons use all 100 sequences. The bootstrap uses 2000 whole-sequence draws, with seed 522987, and

the bands and intervals are pointwise 95% percentile intervals. Representative log-log fits are given in Table 9.

Statistic	Slope	95% interval
Var A_1 at $r = 1$	-1.053	$[-1.130, -0.980]$
Centered log-integral mean square at $r = 1$	-0.975	$[-1.052, -0.903]$
Annular small-derivative fraction versus L	-2.293	$[-2.301, -2.285]$
Global small-derivative fraction versus L	-0.687	$[-0.688, -0.686]$
Larger-degree endpoint change	0.878	$[0.873, 0.883]$
Endpoint change after removing degree drift	0.460	$[0.364, 0.559]$

Table 9: Log-log fits across the measured degrees or thresholds. The first two rows use the doubled angular grids and five base degrees. The derivative rows use $N = 10^5$ and $L = 2, 4, 8, 16$. The endpoint rows use 27 common sequences at the four larger base degrees. Every bootstrap draw is retained for these six fits.

Small derivatives. In the threshold comparison we use $L = 2, 4, 8, 16$ in the $K = 2$ annulus and include the two thin sectors around the real axis, while larger thresholds and retained-angle counts provide supplementary comparisons. We compare the finite-range slope with the small-threshold power -4 , and the global counts show the effect of the annular restriction. At $N = 10^5$, the supplementary retained-angle fit over $L = 16, 32, 64, 128$ has slope -3.852 and interval $[-4.476, -3.450]$, which omits 96 of the 2000 resamples with a zero statistic. Over the primary threshold range the slope is close to -2.3 .

For the Gaussian local comparison, we set

$$I(x) = \int_0^1 e^{2xt} dt, \quad B(x) = \int_0^1 t e^{2xt} dt, \quad V(x) = \int_0^1 t^2 e^{2xt} dt - \frac{B(x)^2}{I(x)}.$$

The Gaussian calculation then integrates

$$\frac{2V(x)}{I(x)} \left\{ 1 - \left(1 + \frac{L^{-2}}{V(x)} \right) e^{-L^{-2}/V(x)} \right\}$$

over the annulus, for comparison with the data for polynomials with random signs.

Localization. The measured amplitude is $a = 15e^4 \sqrt{[N^{7/8}] \log N}$, which is that of Lemma 4.6 with 2 in place of K . We use the two adaptive radii $2a/|f'_N(\alpha)|$ and $2aL/|f'_N(\alpha)|$, and the uniform radius $2aL/N^{3/2}$. For containment in the auxiliary $K = 2$ annulus of the proof we use that lemma with 4 in place of K , and at the same block length its amplitude and radii are then e^4 times larger.

Degree blocks. At $N = 1000$ the maxima use every intermediate polynomial in each of the 78 complete blocks, while at the larger degrees we obtain endpoint changes from the endpoint samples. From the midpoint samples we also obtain three-point maxima. Within the count sensitivity above, the endpoint and three-point maxima are lower bounds for the full-block maximum. We measure both the uncentered changes and the changes after subtracting the degree drift.

The centered logarithmic mean square is that of $\hat{I}_N(1 + x/N) + \gamma/2$, where $\hat{I}_N(r)$ is the average of $\log |f_N/\sigma_N(r)|$ over the shifted FFT grid on $|z| = r$, which approximates $J_N(r) - \log \sigma_N(r)$. For the endpoint changes we use 27 common sequences at the larger degrees and 78 complete blocks at $N = 1000$. For the radius $2a/|f'_N(\alpha)|$ above and each L , we take its median over the roots with $||\alpha| - 1| \leq 2/N$ and $|f'_N(\alpha)| \geq N^{3/2}/L$, and we average these medians over 77 common samples. These averages lie between $3.8 \times 10^3/N$ and $7.9 \times 10^3/N$, and every finite-constant disk that we recorded contains more than one computed root.

A Finite constants

In this appendix we collect the constants and the sufficient degree thresholds used above. Let C_* be the positive universal constant in Nazarov–Nishry–Sodin, Corollary 1.2, and let $c_{\text{HW}} > 0$ be the constant in Rudelson–Vershynin, Theorem 1.1. We keep the dependence on both constants in all the bounds below. We choose, for fixed parameters, the least index beyond which every displayed inequality holds, so that all degree cutoffs are deterministic. Each subsection below corresponds to one result of the main text, and Table 10 lists its degree thresholds and its exceptional probability.

A.1 The variance profile

We begin with the constant of Lemma 2.3, and set, for an integer $K \geq 2$,

$$D_K = 10(K+1)^2 e^{4K}.$$

For $N \geq 2K$ and $|x| \leq K$, Lemma 2.3 and $1 \leq 9(K+1)^2$ give

$$\begin{aligned} \left| \log \sigma_N(1 + x/N) - \frac{1}{2} \log N - F(x) \right| &\leq \frac{K^2 + e^{4K}/2}{N} \leq \frac{(K+1)^2 e^{4K} + e^{4K}}{N} \\ &\leq \frac{10(K+1)^2 e^{4K}}{N} = \frac{D_K}{N}. \end{aligned}$$

Subsection	Degree thresholds	Exceptional probability
A.1, Lemma 2.3	$N \geq 2K$	none
A.2, Theorem 3.1	$K \geq 2$ and $N \geq H_K^4$, uniformly for $r \in [1 - K/N, 1 + K/N]$	$p_K^*(N)$
A.3, Corollary 3.3	those of A.2, $N \geq 2K$, $N \geq 2(1 + 2K)e^{4K}$ and $\varepsilon_K^*(N) + D_K/N \leq K/16$	$4p_K^*(N)$ for (55), $5p_K^*(N)$ for (56), $5p_2^*(N)$ for its form at $K = 2$
A.4, Lemma 4.2	$t_N \leq 1$ and $N \geq e$	N^{-3}
A.5, Lemma 4.3	$N \geq \max\{2, (K + 1)/3\}$	N^{-10}
A.6, Lemma 4.5	$N \geq \lceil e^{X_0} \rceil$, and those of A.4 and A.8	the right-hand side of (61)
A.7, Lemma 4.6	$j \geq 60$ and $N \geq \max\{4, K\}$	N^{-10}
A.8, Lemma 5.4	$2aL/N^{3/2} \leq 1/N$ and $4S_K aL^2 N^{-1/2} \sqrt{\log N} \leq 1$	none
A.9, Section 6.1	those of A.2–A.8 at the width K , where Lemma 4.6 enters with $K + 2$ in place of K (so that $N \geq \max\{4, K + 2\}$), those of A.3 at $K = 2$, and $j \geq 60$	(64), with $R = 2$
A.10, Section 6.2	$R \geq x + 3$, $K \geq R$, $N \geq 2R$ and $N \geq H_R^4$, which implies $N > R$, and those of A.9	$5p_R^*(N)$ for (65), and (64)
A.12, Section 8.1	the conditions of Table 15, $j \geq 60$, $K \geq 2$, $A \log N/N \leq 1$ and $N \geq K + 2$	(64), with $K = K_N$ and $R = 2$
A.13, Corollary 8.2	$0 < \kappa_* < \min(\kappa, 1/64)$ and $N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*}$	(64), with $R = 2$

Table 10: The degree thresholds and exceptional probabilities of the subsections of Appendix A, with the result of the main text that each one treats. For other coefficient distributions, Section A.11 enlarges the constants and adds exceptional probabilities.

A.2 Concentration of the angular logarithm

We now turn to the constants of Theorem 3.1. The one- and two-point bounds of its proof lead to the occupation constant H_K , and the clipped integral and its tails lead to the logarithmic error $\varepsilon_K^*(N)$ and the exceptional probability $p_K^*(N)$. Let $K \geq 2$ be an integer, set

$$H_K = 10^8 e^{20(K+1)},$$

and recall that $T = (\log N)/32$ is the clipping level of Lemma 3.2. In the proof of Theorem 3.1, N is taken large in the three places of Table 11. Each condition there

is a lower bound on N that holds at $N = H_K^4 = 10^{32}e^{80(K+1)}$, so that the degree threshold $N \geq H_K^4$ covers all three.

Use in the proof of Theorem 3.1	Condition	At $N = H_K^4$
The summation by parts (9) and the radial bounds (10)	$N \geq 2K$	$2K \leq e^{2K} < H_K^4$
The eigenvalues of the covariance matrices Σ of Section 3.1 and of the matrices between Σ and $\frac{1}{2}I_d$ lie in $[1/4, 3/4]$ once $\ \Sigma - \frac{1}{2}I_d\ \leq 1/4$, and this norm is at most $40e^{8K}N^{-1/2}$	$N \geq (160e^{8K})^2$	$(160e^{8K})^2 = 25600e^{16K} < H_K^4$
$q \leq 2 \log N$ for $q = \lceil \log N \rceil$, $TN^{-1/2} \leq N^{-1/32}(\log N)^7$ and $N^{-1/16} \leq N^{-1/32}(\log N)^7$, in Lemma 3.2	$N \geq e$	$e < 10^{32} < H_K^4$

Table 11: The three places where N is taken large in the proof of Theorem 3.1. The second condition is $40e^{8K}N^{-1/2} \leq 1/4$, and the last column uses $y \leq e^y$, $2K < 80(K+1)$, $25600 < 10^{32}$ and $16K < 80(K+1)$.

Let $K \geq 2$, $N \geq H_K^4$ and $r \in [1 - K/N, 1 + K/N]$. In (13), the line before the last is $N^{-1/2}$ times $(42d^{1/4} + 16) \cdot 46e^{9K} + 320e^{8K}$, where the dimension is $d \in \{2, 4\}$, so that $d^{1/4} \leq 4^{1/4}$. Since $42 \cdot 4^{1/4} + 16 = 42\sqrt{2} + 16 < 75.4$, $e^{8K} = e^{-K}e^{9K} \leq e^{-2}e^{9K}$, $75.4 \cdot 46 = 3468.4$ and $320e^{-2} < 44$, we have

$$\begin{aligned} (42 \cdot 4^{1/4} + 16) \cdot 46e^{9K} + 320e^{8K} &< 75.4 \cdot 46e^{9K} + 320e^{-2}e^{9K} \\ &< 3469e^{9K} + 44e^{9K} < 4000e^{9K}. \end{aligned}$$

Hence the one- and two-point differences in (13) are at most $4000e^{9K}N^{-1/2}$ on the retained angles and pairs. On the removed sets, whose measures are at most $2N^{-1/2}$ and $10N^{-1/2}$, the integrands of $\mathbb{E}A_t$ and $\mathbb{E}A_t^2$ differ from $F_G(t)$ and $F_G(t)^2$ by at most one. As in the first and third lines of (14), we get, uniformly for $t \geq 0$,

$$|\mathbb{E}A_t - F_G(t)| \leq (2 + 4000e^{9K})N^{-1/2}, \quad |\mathbb{E}A_t^2 - F_G(t)^2| \leq (10 + 4000e^{9K})N^{-1/2}.$$

Since $12000e^{9K} + 14 \leq 12014e^{9K} < 10^8e^{20(K+1)} = H_K$ and $F_G(t) = 1 - e^{-t^2}$, it follows from these bounds, the identity $\text{Var}(A_t) = \mathbb{E}A_t^2 - F_G(t)^2 - (\mathbb{E}A_t - F_G(t))(\mathbb{E}A_t + F_G(t))$ in (14) and $0 \leq \mathbb{E}A_t + F_G(t) \leq 2$ that

$$\begin{aligned} |\mathbb{E}A_t - (1 - e^{-t^2})| &\leq (2 + 4000e^{9K})N^{-1/2} \leq H_K N^{-1/2}, \\ \text{Var}(A_t) &\leq |\mathbb{E}A_t^2 - F_G(t)^2| + 2|\mathbb{E}A_t - F_G(t)| \end{aligned}$$

$$\begin{aligned}
&\leq (10 + 4000e^{9K})N^{-1/2} + 2(2 + 4000e^{9K})N^{-1/2} \\
(53) \quad &= (14 + 12000e^{9K})N^{-1/2} \leq H_K N^{-1/2}.
\end{aligned}$$

This two-point estimate also permits different radii and thresholds. We take the deviation for the truncated logarithmic integral, that is, the parameter u of Theorem B.1, to be $N^{-1/32}(\log N)^7$, and we set

$$\begin{aligned}
\varepsilon_K^*(N) &= N^{-1/32}(\log N)^7 + 2H_K T N^{-1/2} \\
&\quad + (4eC_*)^6 \sqrt{2 + H_K} N^{-1/32}(\log N)^7 + \frac{3}{2}N^{-1/16}, \\
p_K^*(N) &= H_K N^{-3/8} + N^{-12} + 4H_K T^2 N^{-7/16}(\log N)^{-14}.
\end{aligned}$$

Then

$$\begin{aligned}
(54) \quad &\mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon_K^*(N)\} \leq p_K^*(N), \\
&\varepsilon_K^*(N) = O_K(N^{-1/32}(\log N)^7).
\end{aligned}$$

Table 12 lists the source of each term of $\varepsilon_K^*(N)$ and of $p_K^*(N)$.

Source	Term
<i>Logarithmic error $\varepsilon_K^*(N)$</i>	
Deviation of the truncated logarithmic integral	$N^{-1/32}(\log N)^7$
Bias in the truncated expectation	$2H_K T N^{-1/2}$
Lower logarithmic tail	$(4eC_*)^6 \sqrt{2 + H_K} N^{-1/32}(\log N)^7$
Upper logarithmic tail and the two Gaussian tails	$\frac{3}{2}N^{-1/16}$
<i>Exceptional probability $p_K^*(N)$</i>	
Small-value occupation	$H_K N^{-3/8}$
Angular logarithmic L^2 norm	N^{-12}
Deviation of the truncated logarithmic integral	$4H_K T^2 N^{-7/16}(\log N)^{-14}$

Table 12: The terms of the logarithmic error and of the exceptional probability in (54).

The first line of (54) is Lemma 3.2 with $H = H_K$, $C = C_*$, $E = \Omega$ and $B_E = 1$, which applies for $N \geq H_K^4$ by (53), (15) and (16). Its error is $\varepsilon_K^*(N)$, since $B_E/2 + 1 = \frac{3}{2}$, and its probability bound is $p_K^*(N)$, since $4T^2 = (\log N)^2/256$. In its proof, $\mathbb{P}\{U > (4eC_*)^{12}(\log N)^{14}\} \leq N^{-12}$ for $U = \int |\log |X_r||^2 d\mu$, and outside this event $\sqrt{U} \leq (4eC_*)^6(\log N)^7$, which is the square-root threshold in Theorem B.1. For the second line of (54), we have $2H_K T N^{-1/2} = (H_K/16)N^{-1/2} \log N$, and, for $N \geq e$, $N^{-1/2} \log N \leq N^{-1/32}(\log N)^7$ and $N^{-1/16} \leq N^{-1/32}(\log N)^7$, so that

$$\varepsilon_K^*(N) \leq \left(1 + \frac{H_K}{16} + (4eC_*)^6 \sqrt{2 + H_K} + \frac{3}{2}\right) N^{-1/32}(\log N)^7,$$

whose constant depends only on K , since C_* is an absolute constant.

A.3 Annular mass and thin bands

Next we compute the constants of Corollary 3.3. The four radii $1 - K/N$, $1 - K/(2N)$, $1 + K/(2N)$ and $1 + K/N$ are $1 + x/N$ with $|x| \leq K$, and they lie in $[1 - K/N, 1 + K/N]$. Hence their variance errors are at most D_K/N for $N \geq 2K$, by Section A.1, and each logarithmic error is at most $\varepsilon_K^*(N)$, outside probability $p_K^*(N)$, for $N \geq H_K^4$, by (54). For the fixed-width radial tail we require

$$N \geq 2K, \quad N \geq 2(1 + 2K)e^{4K}, \quad \varepsilon_K^*(N) + D_K/N \leq K/16.$$

The last condition implies the hypothesis $\varepsilon + D/N \leq K/8$ of Lemma 2.4, with $\varepsilon = \varepsilon_K^*(N)$ and $D = D_K$. Since $f_N(0) = \varepsilon_0 \neq 0$, this lemma applies outside the four logarithmic exceptional events, whose union has probability at most $4p_K^*(N)$, and gives

$$(55) \quad \frac{\#\{\alpha : ||\alpha| - 1| > K/N\}}{N} \leq \frac{2 \log 2}{K} + \frac{8}{K}(\varepsilon_K^*(N) + D_K/N).$$

We use all five radii $1 + qN^{-1-\kappa}$, $q = -2, \dots, 2$, for $0 < \kappa \leq 1/64$. They lie in the annulus of Theorem 3.1, since $|qN^{-1-\kappa}| \leq 2N^{-1-\kappa} \leq 2/N \leq K/N$, and $N \geq H_K^4$ implies the condition $N \geq 4$ of Lemma 2.5, since $H_K^4 > 10^{32}$. By that lemma with $\varepsilon = \varepsilon_K^*(N)$, whose hypothesis holds outside the five events of (54) at these radii, for $q = -1, 0, 1$ the same sample satisfies

$$(56) \quad \left| \frac{\nu_N(1 + qN^{-1-\kappa})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa} + 4\varepsilon_K^*(N)N^\kappa$$

outside a union of probability at most $5p_K^*(N)$. These errors tend to zero, since $\kappa > 0$ and $4\varepsilon_K^*(N)N^\kappa = O_K(N^{\kappa-1/32}(\log N)^7)$ by (54), with $\kappa \leq 1/64 < 1/32$. The probabilities of the exceptional events are summable along $N = j^8$, since $4T^2(\log N)^{-14} = (\log N)^{-12}/256 \leq 1$ for $N \geq e$, so that the three terms of $p_K^*(j^8)$ are at most $H_K j^{-3}$, j^{-96} and $H_K j^{-7/2}$.

At $K = 2$ we have $H_2 = 10^8 e^{20(2+1)} = 10^8 e^{60}$, and $\varepsilon_2^*(N)$ and $p_2^*(N)$ are $\varepsilon_K^*(N)$ and $p_K^*(N)$ with $10^8 e^{60}$ in place of H_K , with the degree threshold $N \geq H_2^4 = 10^{32} e^{240}$. Since the five radii lie in $[1 - 2/N, 1 + 2/N]$, (56) holds at $K = 2$, with $\varepsilon_2^*(N)$, outside probability $5p_2^*(N)$. This form, with $\kappa = 1/64$, is used in Step 3 of Section 6.1.

A.4 Local zero counts

In this subsection we compute the constant of Lemma 4.2. For the local root counts we set

$$A = \max(1, 1024\pi/c_{\text{HW}}), \quad t_N = A \log N/N, \quad C_1(K) = \frac{4A + 4K + 4}{\log 2},$$

and we require $t_N \leq 1$ and $N \geq e$. The proof of Lemma 4.2 applies Lemma 4.1 with $a_k = 1$, $B = 1$ and $\eta = 1$, at which its constants $A(1, 1)$, $t_N(1, 1)$ and $C_1(K, 1, 1)$ are these. In the proof of Lemma 4.1 at these values, the $\lceil 2\pi/t_N \rceil$ arcs have radian length $2t_N$ and normalized measure t_N/π , since $t_N \leq 1 < \pi$. By the trace identity of that proof and $Nt_N = A \log N$, the Gram matrix of each arc satisfies $\text{tr } A_I = (N+1)t_N/\pi \geq (A/\pi) \log N$, and since $\|A_I\|_{\text{HS}}^2 \leq \text{tr } A_I$ and $\|A_I\| \leq 1$, the minimum in the Hanson–Wright inequality at half the trace, with sub-Gaussian norm bound two, is at least $\text{tr } A_I/64$. Since $A \geq 1024\pi/c_{\text{HW}}$, we have $c_{\text{HW}} \text{tr } A_I/64 \geq c_{\text{HW}} A \log N/(64\pi) \geq 16 \log N$, which gives the bound $2N^{-16}$ for each arc and, over at most $8N$ arcs, $16N^{-15} \leq N^{-3}$ for $N \geq 2$. In Jensen’s formula, the inner and outer radii are $R = (2K+1)/N + 2t_N$ and $2R$, so that $2NR = 4K + 2 + 4A \log N$, and the Jensen bound of that proof is $C_1(K) \log N$, since $1 + (4K+3) + 4A = 4A + 4K + 4$. We use $\log N \geq 1$ in the bound on the number of arcs and in the last step of the Jensen bound. With $K+2$ in place of K ,

$$C_1(K+2) = \frac{4A + 4K + 12}{\log 2}.$$

We conclude that the simultaneous local bound is $C_1(K) \log N$ in every closed disk $\{z : |z - z_0| \leq 2K/N\}$ with $||z_0| - 1| \leq 1/N$.

A.5 The second derivative

Next we consider the constant of Lemma 4.3. We set $S_K = 100e^{K+4}$, which is the constant of that lemma, so that, for $N \geq \max\{2, (K+1)/3\}$, the second derivative satisfies

$$(57) \quad \sup_{|z| \leq 1+(K+1)/N} |f_N''(z)| \leq S_K N^{5/2} \sqrt{\log N}$$

outside probability N^{-10} . This degree threshold is the range of Lemma 4.3: in its proof, $N \geq (K+1)/3$ keeps the sample circle of radius $1 + (K+1)/N$ inside $|z| \leq 4$, and $N \geq 2$ ensures the interpolation bound and the union bound $(2N^3)(4N^{-100}) \leq N^{-10}$.

A.6 Few small derivatives

We now compute the constants of Lemma 4.5, following its proof at the annular width K , with the local-count and tail lemmas at the width $K+2$. With S_K as in Section A.5 and $C_1(K+2)$ as in Section A.4, we set

$$\lambda_K = e^{-4K}/80, \quad D'_K = 2^{18}(K+1)S_K^2,$$

$$\begin{aligned}
\mathcal{B}_K &= 2000e^{3K}(2/\lambda_K)^{3/2}, & \mathcal{T}_K &= 80e^{4K}/\lambda_K, \\
V_K &= 50 + 3\mathcal{B}_K + \mathcal{T}_K, & c_K &= \frac{9}{1024S_K^2\lambda_K^2}, \\
C_K &= 16C_1(K+2)^2V_K(D'_K)^2.
\end{aligned}$$

Here $\mathcal{B}_K$ is the Raič bound in the proof of Lemma 4.5, which bounds the normal-approximation error and the one-point error, and $\mathcal{T}_K$ bounds the Gaussian comparison error.

In that proof, the mesh constant must be at most $1/(2S_K)$ for the Taylor bounds and at most one for the mesh count. Both conditions hold for the value $1/(64S_K)$, since $S_K \cdot 1/(64S_K) = 1/64 \leq 1/2$ and $S_K \geq 1$. With $L = N^{1/64}$, let

$$h = \frac{1}{64S_KNL\sqrt{\log N}}.$$

We use exactly $\lceil 2K/(Nh) \rceil + 1$ equally spaced radial rows, including both endpoints, and $\lceil 8\pi/h \rceil$ angles on each row. This mesh has spacing h , and by the count of mesh points in the proof of Lemma 4.5, with $1/(64S_K)$ as its mesh constant, and $53 \cdot 64^2 = 217088 < 262144 = 2^{18}$, its total size satisfies

$$\begin{aligned}
m &\leq 53(K+1)(64S_K)^2NL^2\log N = 53 \cdot 64^2(K+1)S_K^2NL^2\log N \\
&\leq 2^{18}(K+1)S_K^2NL^2\log N = D'_KNL^2\log N.
\end{aligned}$$

Since $Nh = 1/(64S_KL\sqrt{\log N})$, $N^2h^2\sqrt{\log N} = 1/(4096S_K^2L^2\sqrt{\log N})$ and $1/64 + 1/8192 < 1/16$, in the two Taylor chains of that proof, with $C(K) = S_K$ and the mesh constant $1/(64S_K)$, we have

$$\frac{Nh}{L} + \frac{S_K}{2}N^2h^2\sqrt{\log N} = \left(\frac{1}{64} + \frac{1}{8192}\right)\frac{1}{S_KL^2\sqrt{\log N}} \leq \frac{1}{16S_KL^2\sqrt{\log N}},$$

and $2(1 + S_K/(64S_K))/L = 2(1 + 1/64)/L = 65/(32L) < 3/L$. It follows that a retained mesh point detecting a small-derivative zero satisfies

$$(58) \quad |f_N(z)|/\sqrt{N} \leq u = (16S_K)^{-1}L^{-2}(\log N)^{-1/2}, \quad |zf'_N(z)|/N^{3/2} \leq v = 3/L.$$

Table 13 lists the degree conditions of that proof. By Section A.2 and $H_K^4 > 10^{32}$, the first four hold for $N \geq H_K^4$, and the norm bounds of the last two, from which the covariance lower bounds follow, each hold if and only if $N^{1/2} \geq 6400e^{8K}$, as shown in the proof of Lemma 4.5. We therefore require

$$N \geq H_K^4, \quad N \geq (6400e^{8K})^2.$$

Use in the proof of Lemma 4.5	Condition
(9) and the radial weights	$N \geq 2K$
The bound $ z \leq 2$	$N \geq K$
The retained angles on each row	$N \geq 4$
$h \leq 1/N$	$N \geq e$
Covariance lower bound λ_K in dimension four (24), from $20e^{4K}N^{-1/2} \leq e^{-4K}/320$	$N^{1/2} \geq 6400e^{8K}$
Covariance lower bound $\lambda_K/2$ in dimension eight, from $40e^{4K}N^{-1/2} \leq \lambda_K/2$	$N^{1/2} \geq 6400e^{8K}$

Table 13: The degree conditions in the proof of Lemma 4.5.

Since $u^2 = (16S_K)^{-2}L^{-4}(\log N)^{-1}$ and $v^2 = 9L^{-2}$ by (58), the bound on the Gaussian probability of the product of two disks in that proof is

$$\frac{u^2v^2}{4\lambda_K^2} = \frac{1}{4\lambda_K^2} \cdot \frac{9}{256S_K^2} \cdot \frac{L^{-6}}{\log N} = \frac{9}{4 \cdot 256S_K^2\lambda_K^2} \cdot \frac{L^{-6}}{\log N} = \frac{c_K L^{-6}}{\log N}.$$

Hence the line $u^2v^2/(4\lambda_K^2) + 2000e^{3K}(2/\lambda)^{3/2}N^{-1/2}$ of (25), which follows from the Gaussian volume and Raič bounds, gives at $\lambda = \lambda_K$ the one-point probability

$$(59) \quad c_K L^{-6}/\log N + \mathcal{B}_K N^{-1/2}.$$

For a retained separated pair, write I_z and I_w for the indicators of the test (58) at the two points. By the covariance chain in the proof of Lemma 4.5, whose three terms are the eight-dimensional Raič bound, the total variation bound (26) and twice the four-dimensional Raič bound,

$$|\text{Cov}(I_z, I_w)| \leq \mathcal{B}_K N^{-1/2} + \frac{80e^{4K}}{\lambda_K \sqrt{N}} + 2\mathcal{B}_K N^{-1/2} = (3\mathcal{B}_K + \mathcal{T}_K)N^{-1/2}.$$

By that proof, at most $10mN^{-1/2}$ partners of each mesh point belong to the removed bands. Let Z be the number of successful tests. Summing (59) over the at most m retained mesh points, and bounding each of the at most $m \cdot 10mN^{-1/2}$ ordered pairs in the removed bands by one and each other pair by the covariance bound, we get, by the bound on m ,

$$\begin{aligned} \mathbb{E}Z &\leq m \left\{ \frac{c_K L^{-6}}{\log N} + \mathcal{B}_K N^{-1/2} \right\} \leq D'_K \{c_K N L^{-4} + \mathcal{B}_K N^{1/2} L^2 \log N\}, \\ \text{Var } Z &\leq 10m^2 N^{-1/2} + (3\mathcal{B}_K + \mathcal{T}_K)m^2 N^{-1/2} \leq V_K m^2 N^{-1/2} \\ (60) \quad &\leq V_K (D'_K)^2 N^{3/2} L^4 (\log N)^2. \end{aligned}$$

In the proof of Lemma 4.5, the number $B_{K,N}$ of roots in $||z| - 1| \leq K/N$ whose derivatives have modulus less than $N^{3/2}/L$ satisfies $B_{K,N} \leq C_1(K+2)(\log N)Z + 40C_1(K+2)\sqrt{N}\log N$, where the second term bounds the roots in the at most $40\sqrt{N}$ disks of the real-sector cover, and Z is compared with the level $Q = N^{31/32}/(2C_1(K+2)\log N)$ by Chebyshev's inequality at half this level, which requires $\mathbb{E}Z$ to be at most half of it. We therefore impose the mean and excluded-sector thresholds

$$\begin{aligned} D'_K\{c_K N L^{-4} + \mathcal{B}_K N^{1/2} L^2 \log N\} &\leq \frac{N^{31/32}}{4C_1(K+2)\log N}, \\ 40C_1(K+2)\sqrt{N}\log N &\leq N^{31/32}/2. \end{aligned}$$

Then $\mathbb{E}Z \leq Q/2$ by (60), and the disk cover assigns at most $N^{31/32}/2$ roots to the sectors, so that $B_{K,N} > N^{31/32}$ forces $Z > Q$. Since $L^4 = N^{1/16}$ and $Q^2 = N^{31/16}/(4C_1(K+2)^2(\log N)^2)$, (60) and $3/2 + 1/16 - 31/16 = -3/8$ give

$$\begin{aligned} \mathbb{P}\{Z > Q\} &\leq \mathbb{P}\{|Z - \mathbb{E}Z| \geq Q/2\} \leq \frac{4 \operatorname{Var} Z}{Q^2} \\ &\leq \frac{4V_K(D'_K)^2 N^{3/2} L^4 (\log N)^2 \cdot 4C_1(K+2)^2 (\log N)^2}{N^{31/16}} \\ &= 16C_1(K+2)^2 V_K(D'_K)^2 N^{3/2+1/16-31/16} (\log N)^4 = C_K N^{-3/8} (\log N)^4. \end{aligned}$$

Since $B_{K,N} > N^{31/32}$ implies $Z > Q$ on the local-count event of Section A.4 and the derivative event of Section A.5, it follows that

$$(61) \quad \mathbb{P}\{B_{K,N} > N^{31/32}\} \leq N^{-3} + N^{-10} + C_K N^{-3/8} (\log N)^4.$$

In this union, the local and derivative events are counted once.

A closed-form mesh threshold. Let $A, \beta > 0$ and $p \in \mathbb{N}$, and define

$$\Psi(A, \beta, p) = \max\{0, (2/\beta) \log[A(2p/\beta)^p]\}.$$

Then, for $x \geq \Psi(A, \beta, p)$,

$$x^p \leq (2p/\beta)^p e^{\beta x/2}, \quad Ax^p \leq A(2p/\beta)^p e^{\beta x/2} \leq e^{\beta x}.$$

The first inequality holds for every $x \geq 0$, since with $y = \beta x/2$ it reads $y^p \leq p^p e^y$, which follows by raising $y/p \leq e^{y/p}$ to the power p , and the last holds since $x \geq \Psi(A, \beta, p)$ implies $A(2p/\beta)^p \leq e^{\beta x/2}$. With $x = \log N$, the two mean terms, each compared with half of the mean bound, and the sector count satisfy, by

$L = N^{1/64}$ and $1 - 1/16 - 31/32 = -1/32$, $1/2 + 1/32 - 31/32 = -7/16$ and $1/2 - 31/32 = -15/32$,

$$\begin{aligned}\frac{D'_K c_K N L^{-4}}{N^{31/32}/(8C_1(K+2)\log N)} &= 8C_1(K+2)D'_K c_K x e^{-x/32}, \\ \frac{D'_K \mathcal{B}_K N^{1/2} L^2 \log N}{N^{31/32}/(8C_1(K+2)\log N)} &= 8C_1(K+2)D'_K \mathcal{B}_K x^2 e^{-7x/16}, \\ \frac{40C_1(K+2)\sqrt{N} \log N}{N^{31/32}/2} &= 80C_1(K+2)x e^{-15x/32}.\end{aligned}$$

By the bound $Ax^p \leq e^{\beta x}$, each right-hand side is at most one when x is at least the corresponding value of Ψ in the last three entries of

$$\begin{aligned}X_0 = \max\{ &\log 16, \log(2(K+1)), 4\log H_K, 2\log(6400e^{8K}), \\ &\Psi(8C_1(K+2)D'_K c_K, 1/32, 1), \Psi(8C_1(K+2)D'_K \mathcal{B}_K, 7/16, 2), \\ &\Psi(80C_1(K+2), 15/32, 1)\}.\end{aligned}$$

Hence $N \geq \lceil e^{X_0} \rceil$ implies the mesh mean and sector thresholds, and the first four entries of X_0 ensure that $N \geq 16$, $N \geq 2(K+1)$, $N \geq H_K^4$ and $N \geq (6400e^{8K})^2$. We also impose the local-count thresholds $t_N \leq 1$ and $N \geq e$ of Section A.4 and the localization thresholds $2a/d_0 \leq 1/N$ and $4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1$ of Section A.8.

A.7 Appended tails

Next we turn to the constants of Lemma 4.6, for its parameter K , at the degrees $N = j^8$ of Section 6, with $j \geq 60$ and $m_{\max} = m_j$, where $m_j = (j+1)^8 - j^8$ as there. In that lemma, the radius and the amplitude are

$$\rho = 1 + K/N, \quad a(K) = 15e^{2K} \sqrt{m_j \log N},$$

where we write $a(K)$ for the amplitude at the width K , since the later subsections use it at the width $K+2$. By (29), $m_j \leq 9N^{7/8} = 9N/j \leq N$, so that $1 \leq m_{\max} \leq N$, as Lemma 4.6 requires, and $a(K) \leq 15e^{2K} \sqrt{9N^{7/8} \log N} = 45e^{2K} N^{7/16} \sqrt{\log N}$. That lemma also requires $N \geq \max\{4, K\}$, and its single exceptional event, of probability at most N^{-10} , controls every degree in the block.

A.8 Localization

We next consider the constants of Lemma 5.4 at the parameters of Step 1 of Section 6.1, at the width K . Recall that there $a = a(K+2) = 15e^{2(K+2)} \sqrt{m_j \log N}$

is the amplitude of Section A.7 with $K + 2$ in place of K , $M_2 = S_K N^{5/2} \sqrt{\log N}$ by (57), and $d_0 = N^{3/2}/L$. For a good annular zero, it is sufficient that

$$2aL/N^{3/2} \leq 1/N, \quad 4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1.$$

These are the conditions $2a/d_0 \leq 1/N$ and $4M_2 a \leq d_0^2$ of Step 1, since $2a/d_0 = 2aL/N^{3/2}$ and $4M_2 a/d_0^2 = 4S_K N^{5/2} \sqrt{\log N} a L^2/N^3 = 4S_K a L^2 N^{-1/2} \sqrt{\log N}$. As in the proof of Lemma 5.4, on the localization circle $|z - \alpha| = 2a/d_0$ the Taylor remainder is at most $\frac{1}{2}M_2(2a/d_0)^2 = 2M_2 a^2/d_0^2 \leq a/2$, since $4M_2 a \leq d_0^2$, and the linear term has modulus $|f'_N(\alpha)| \cdot 2a/d_0 \geq 2a$, since $|f'_N(\alpha)| \geq d_0$.

A.9 The unit circle

In this subsection we write out in finite form the bounds of the proof in Section 6.1. Recall that $N = j^8$, $m_j = (j+1)^8 - j^8$ and $L = N^{1/64}$. Fix integers $K, R \geq 2$, and let $H_R, D_R, \varepsilon_R^*(N)$ and $p_R^*(N)$ be $H_K, D_K, \varepsilon_K^*(N)$ and $p_K^*(N)$ at the width R , with the degree threshold $N \geq H_R^4$. At $R = 2$ these are $\varepsilon_2^*(N)$ and $p_2^*(N)$ of Section A.3. By the first line of the bound on U_N in Step 3 of Section 6.1, by (55), multiplied by N , and since $B_{K,N} \leq N^{31/32}$ outside the event of (61), on the radial and derivative events the number U_N of unselected roots in the matching construction satisfies

$$(62) \quad \begin{aligned} U_N &\leq \#\{\alpha : f_N(\alpha) = 0, ||\alpha| - 1| > K/N\} + B_{K,N} \\ &\leq N^{31/32} + N \left[\frac{2 \log 2}{K} + \frac{8}{K} \{\varepsilon_K^*(N) + D_K/N\} \right]. \end{aligned}$$

Every selected domain meeting the unit circle has its root in the strict band $1 - N^{-65/64} < |\alpha| < 1 + N^{-65/64}$. Indeed, as in Step 3 of Section 6.1, such a root lies within $2a/d_0$ of the unit circle, and $2a/d_0 < N^{-65/64}$, since $d_0 = N^{3/2}/L$, $L = N^{1/64}$ and $65/64 + 1/64 - 3/2 = -15/32$, and since $a \leq N^{15/32}/(4S_K \sqrt{\log N})$ by the condition $4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1$ of Section A.8 with $L^2 = N^{1/32}$, so that

$$N^{65/64} \cdot \frac{2a}{d_0} = 2aN^{-15/32} \leq \frac{1}{2S_K \sqrt{\log N}} < 1.$$

By the five-radius estimate of Section A.3 at $K = 2$, with $\kappa = 1/64$, in the cases $q = 0$ and $q = \pm 1$, and by the crossing bound (38), which relies on the strict band above, we have

$$(63) \quad \begin{aligned} \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| &\leq 2N^{-1/64} + 4\varepsilon_2^*(N)N^{1/64}, \\ \frac{c_N}{N} &\leq 2(2N^{-1/64} + 4\varepsilon_2^*(N)N^{1/64}) = 4N^{-1/64} + 8\varepsilon_2^*(N)N^{1/64}. \end{aligned}$$

Collecting the exceptional probabilities, we find that the common exceptional event has probability at most

$$(64) \quad 4p_K^*(N) + 5p_R^*(N) + N^{-3} + 2N^{-10} + C_K N^{-3/8}(\log N)^4.$$

This union consists of the four radial logarithmic events, used by the radial secants, the five local logarithmic events, used by the sparse estimate, and the local-count, derivative, maximal-tail and mesh-concentration events. The shared local-count and derivative events are counted once each, and a single maximal-tail event controls every degree in the block. For the unit-disk argument we take $R = 2$. Table 14 lists the individual exceptional probabilities in (64) and their size at $N = j^8$. For every fixed K, R , the series of (64) over $N = j^8$ is finite, since every exponent of j in the last column of the table is less than -1 , with H_R in place of H_K for the terms of $p_R^*(N)$.

Event	Failure at N	At $N = j^8$, ignoring constants
Radial occupation	$H_K N^{-3/8}$	j^{-3}
Logarithmic norm	N^{-12}	j^{-96}
Clipped integral	$4H_K T^2 N^{-7/16}(\log N)^{-14}$	$j^{-7/2}(\log j)^{-12}$
Annular bad roots	$C_K N^{-3/8}(\log N)^4$	$j^{-3}(\log j)^4$
Local zero count	N^{-3}	j^{-24}
Derivative supremum	N^{-10}	j^{-80}
Block tail	N^{-10}	j^{-80}

Table 14: Exceptional probabilities at degree N for fixed K , and their size at $N = j^8$, ignoring constants.

A.10 Moving radii

We now do the same for the bounds of the proof in Section 6.2, with H_R , D_R , $\varepsilon_R^*(N)$ and $p_R^*(N)$ as in Section A.9. For the moving radius, fix x and an integer $R \geq |x| + 3$, and recall from Section 6.2 that $h = N^{-1/64}$ and $r_N = 1 + x/N$. We apply the logarithmic estimates at $x + qh$, $q = -2, \dots, 2$, and Lemma B.3 at $x + qh$, with R in place of K , logarithmic error $\eta_{\log} = \varepsilon_R^*(N)$, profile error $\varepsilon = D_R/N$, $F' = \Phi$, $M_F = 1/2$ and $B_F = 1$, since $0 \leq \Phi \leq 1$ and $0 \leq \Phi' = 2\text{Var}_x(t) \leq 1/2$ by Section 6.2, which also implies that $|\Phi(x + qh) - \Phi(x)| \leq |q|h/2$. For $N \geq 2R$ we get

$$\left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| \leq \frac{|q|h}{2} + \frac{h}{4} + \frac{R}{N} + \frac{2(\varepsilon_R^*(N) + D_R/N)(1 + R/N)}{h}.$$

Since $|q|h/2 + h/4 \leq 2h$ for $|q| \leq 1$ and $2(1 + R/N) \leq 4$, we obtain

$$(65) \quad \left| \frac{\nu_N(r_N + qh/N)}{N} - \Phi(x) \right| \leq \delta_R^*(N), \quad q = -1, 0, 1,$$

$$\delta_R^*(N) := 2h + \frac{R}{N} + \frac{4(\varepsilon_R^*(N) + D_R/N)}{h}.$$

Here the exceptional probability is at most $5p_R^*(N)$, and $\delta_R^*(N) \rightarrow 0$. By (65) with $q = \pm 1$, as in (43), the strict band $||\alpha| - r_N| < h/N$ contains at most $2\delta_R^*(N)N$ roots. This band contains the root of every selected domain whose closure meets the closed annulus between the circles of radii r_N and $1 + x/n$, for $N \leq n \leq N + m_j$, once $2a/d_0 + |x|m_j/N^2 < h/N$, as in Section 6.2. At the width $K \geq R$ this follows from the thresholds. Indeed, by $N/h = N^{65/64}$, the bound on $N^{65/64} \cdot 2a/d_0$ in Section A.9 and (29),

$$\begin{aligned} \frac{N}{h} \left(\frac{2a}{d_0} + \frac{|x|m_j}{N^2} \right) &= N^{65/64} \cdot \frac{2a}{d_0} + |x|m_j N^{-63/64} \\ &\leq \frac{1}{2S_K \sqrt{\log N}} + 9|x|N^{-7/64} < \frac{1}{2} + \frac{1}{2} = 1, \end{aligned}$$

since $S_K \sqrt{\log N} > 1$ and $18|x| < 18R < 10^{3.5} e^{8.75(R+1)} = H_R^{7/16} \leq N^{7/64}$, by $|x| < R$, $18R < e^{8.75(R+1)}$, $(10^8)^{7/16} = 10^{3.5}$, $20 \cdot 7/16 = 8.75$ and $N \geq H_R^4$. Therefore the number $c(r_N, 1 + x/n)$ of these domains is at most $2\delta_R^*(N)N$, and the matching bound of Section 6.2 becomes $|\nu_n(1 + x/n) - \nu_N(r_N)| \leq 2\delta_R^*(N)N + U_N + m_j$. The chain of that proof that splits $\nu_n(1 + x/n)/n - \Phi(x)$ into three terms, with $\delta_R^*(N)$ in place of ε_N , then gives

$$\left| \frac{\nu_n(1 + x/n)}{n} - \Phi(x) \right| \leq 3\delta_R^*(N) + \frac{U_N}{N} + \frac{2m_j}{N}, \quad N \leq n \leq N + m_j.$$

Here (62) and (64) apply with $K \geq R$.

A.11 Other coefficient distributions

This subsection contains the constants of Proposition 7.4, which are used in the proofs of Theorem 1.1 and Corollary 1.2 for the other distributions. We recompute, for each coefficient distribution, the constants of the sign case from the inputs that depend on the distribution: its logarithmic-moment constant, its energy bound, its local-count constant and, for unbounded coefficients, an envelope on their size. For Steinhaus coefficients, we use the same constants as for Rademacher coefficients, since the amplitudes have modulus one, so that conditional signs yield the same C_* and $C_1(K + 2)$.

Real Gaussian coefficients. We use

$$(C, \beta) = (16, 1), \quad B_E = 2, \quad p_E = \exp[-3N/(32e^{6K})].$$

These choices are valid, since $(16p)^p \leq (16p)^{6p}$ for $p \geq 1$, so that, by (47), the logarithmic moments satisfy (15) with 16 in place of C_* . In the proof of Proposition 7.4, the Gaussian occupation constants $2 + 320e^{8K}$ and $14 + 960e^{8K}$ are at most H_K , since $2 + 320e^{8K} < 14 + 960e^{8K} \leq 974e^{9K} < 10^8 e^{20(K+1)} = H_K$, so that (53) holds for these coefficients. Hence Lemma 3.2 applies with $H = H_K$, $C = 16$, $E = \{V \leq B_E\}$ and $B_E = 2$, where $V = \int |X_r|^2 d\mu$ as in Proposition 7.4, and $\mathbb{P}(E^c) \leq p_E$ by (48). Since $4e \cdot 16 = 64e$, $B_E/2 + 1 = 2$ and $4T^2 = (\log N)^2/256$, it gives, for $N \geq H_K^4$ and $r \in [1 - K/N, 1 + K/N]$,

$$\begin{aligned} & \mathbb{P}\left\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > N^{-1/32}(\log N)^7 + 2H_K T N^{-1/2}\right. \\ & \quad \left.+ (64e)^6 \sqrt{2 + H_K} N^{-1/32}(\log N)^7 + 2N^{-1/16}\right\} \\ & \leq H_K N^{-3/8} + N^{-12} + 4H_K T^2 N^{-7/16}(\log N)^{-14} + p_E. \end{aligned}$$

For these coefficients, the error and the probability on the two sides of this display replace $\varepsilon_K^*(N)$ and $p_K^*(N)$ at each radius in (55), (56), (62), (63), (64) and (65), and in the radial thresholds of Section A.3.

We use the same derivative and tail amplitudes, with $\lceil N^4 \rceil \leq 2N^4$ boundary samples. On the coefficient envelope $\max_{k \leq 2N} |g_k| \leq N$, the third-derivative bound in the proof of Lemma 4.3 gains a factor N , and the distance to the nearest sample loses one:

$$\sup_{|z| \leq 1 + (K+1)/N} |f_N'''(z)| \leq e^{K+1} N^5, \quad \frac{4\pi}{N^4} \cdot e^{K+1} N^5 = 4\pi e^{K+1} N.$$

Here the first bound is the bound $e^{K+1} N^4$ of that proof with the extra factor $|g_k| \leq N$ in each term, and $4\pi/N^4$ bounds the arc length from a point of the circle to its nearest sample, since the circle has length at most 8π and carries $\lceil N^4 \rceil$ equally spaced samples. Thus the interpolation error is again $4\pi e^{K+1} N$. The scalar Gaussian tails bound the exceptional probability of each sample at half the threshold by $4N^{-100}$, and the union bound over the samples yields $(2N^4)(4N^{-100}) = 8N^{-96} \leq N^{-10}$. The common coefficient-envelope event costs at most $\mathbb{P}\{\max_{k \leq 2N} |g_k| > N\} \leq 2(2N + 1)e^{-N^2/2}$, by the union bound over the $2N + 1$ coefficients and $\mathbb{P}\{|g_k| > N\} \leq 2e^{-N^2/2}$. For the local counts we use the constant of Lemma 7.3 with $K + 2$ in place of K ,

$$C_{1,G}(K + 2) = C_1(K + 2) + 2/\log 2 = \frac{4A + 4(K + 2) + 6}{\log 2} = \frac{4A + 4K + 14}{\log 2},$$

and the local-count exceptional probability is $N^{-3} + \exp[-3(N+1)/32]$. We replace $C_1(K+2)$ by $C_{1,G}(K+2)$ in the mean and sector conditions of Section A.6, in X_0 and in the Chebyshev bound there, whose constants D'_K , c_K , $\mathcal{B}_K$ and V_K are unchanged. On the common coefficient-envelope event, the small-derivative exceptional probability, which replaces the right-hand side of (61), is then

$$N^{-3} + \exp[-3(N+1)/32] + N^{-10} + 16C_{1,G}(K+2)^2 V_K (D'_K)^2 N^{-3/8} (\log N)^4.$$

For interpolation we impose $N \geq e^2$ and $\rho \leq 2$. These imply the conditions $N \geq 4$ and $\rho \leq 2$ of the interpolation step in the proof of Lemma 4.6, and that step has the same error here, since the tail derivatives gain the factor N and the sample spacing loses it. We use at most $2N^5$ prefix/sample pairs, and under the tail threshold the union bound over these pairs yields $(2N^5)(4N^{-25}) = 8N^{-20} \leq N^{-10}$. All other exceptional probabilities are those of the sign case.

Standard complex Gaussian coefficients. By the proof of Proposition 7.4, with Lemma 7.3 for the local count, these coefficients have the constants of the real Gaussian coefficients above.

Centrally symmetric bounded coefficients. Assume that the coefficient distribution is nondegenerate and centrally symmetric, as defined in Section 7. We normalize $\mathbb{E}|\xi_0|^2 = 1$ and assume $|\xi_0| \leq B$, so that $B \geq 1$. We use V , $E = \{V \geq 1/2\}$, J_N and $C_B = 2\{C_* + \max(\frac{1}{2} \log 2, \log B) + 1\}$ as in Proposition 7.4 and its proof. Set

$$\begin{aligned} H_{K,B} &= (1 + B^3)H_K, & S_{K,B} &= BS_K, & D'_{K,B} &= B^2 D'_K, \\ \mathcal{B}_{K,B} &= B^3 \mathcal{B}_K, & c_{K,B} &= c_K/B^2, & B_E &= B^2. \end{aligned}$$

For these coefficients the tail amplitude is Ba , where $a = 15e^{2(K+2)}\sqrt{m_j \log N}$ is the tail amplitude of Section A.7 with $K+2$ in place of K , and we use the same λ_K and $\mathcal{T}_K$. The local-count constants are those of Lemma 7.2 with $K+2$ in place of K ,

$$A_B = \max(1, 2048\pi B^2/c_{\text{HW}}), \quad C_{1,B}(K+2) = \frac{4A_B + 4(K+2) + 4 + \log B}{\log 2},$$

and since the conditional trace event adds $\exp[-(N+1)/(2B^4)]$, the local-count exceptional probability is $N^{-3} + \exp[-(N+1)/(2B^4)]$. We define $V_{K,B}$ and $C_{K,B}$ by the formulas for V_K and C_K , with these constants and with the local-count constant $C_{1,B}(K+2)$. Since $\mathcal{B}_{K,B} = B^3 \mathcal{B}_K$, $2/\lambda_K = 160e^{4K}$, $(160e^{4K})^{3/2} = 160^{3/2}e^{6K}$,

$80e^{4K}/\lambda_K = 6400e^{8K}$, $D'_{K,B} = B^2 D'_K$, $(D'_K)^2 = 2^{36}(K+1)^2 S_K^4$, $S_K^4 = 10^8 e^{4K+16}$ and $16 \cdot 2^{36} = 2^{40}$,

$$\begin{aligned} V_{K,B} &= 50 + 3\mathcal{B}_{K,B} + \mathcal{T}_K = 50 + 6000B^3 e^{3K} (2/\lambda_K)^{3/2} + \frac{80e^{4K}}{\lambda_K} \\ &= 50 + 6000 \cdot 160^{3/2} B^3 e^{9K} + 6400e^{8K}, \\ C_{K,B} &= 16C_{1,B}(K+2)^2 V_{K,B} (D'_{K,B})^2 = 16B^4 C_{1,B}(K+2)^2 V_{K,B} (D'_K)^2 \\ &= 2^{40} \cdot 10^8 B^4 (K+1)^2 e^{4K+16} C_{1,B}(K+2)^2 V_{K,B}. \end{aligned}$$

For the restricted logarithmic hypothesis we use

$$(C, \beta) = (C_B, 6), \quad p_H = \exp[-N/(2B^4 e^{6K})].$$

For the occupation bounds, the constant c_B of the proof of Proposition 7.4 satisfies, as in Section A.2 and with $B \geq 1$ in the last step,

$$c_B = (42 \cdot 4^{1/4} + 16) \cdot 46B^3 e^{9K} + 320e^{8K} < 3469B^3 e^{9K} + 44e^{9K} < 4000B^3 e^{9K}.$$

Hence, by $B^3 e^{9K} \geq 1$, $12014 < 10^8$ and $9K < 20(K+1)$, the occupation constants $2 + c_B < 14 + 3c_B$ of that proof satisfy $14 + 3c_B < 14 + 12000B^3 e^{9K} \leq 12014B^3 e^{9K} < B^3 H_K < H_{K,B}$, so that (53) holds with $H_{K,B}$ in place of H_K . Hence Lemma 3.2 applies with $H = H_{K,B}$, $C = C_B$, $E = \{V \geq 1/2\}$ and $B_E = B^2$, by the moment bound on E in Proposition 7.4 and since $V \leq B^2$. With $B_E/2 + 1 = B^2/2 + 1$ and $4T^2 = (\log N)^2/256$, its error is

$$\begin{aligned} \varepsilon_{K,B}^*(N) &= N^{-1/32} (\log N)^7 + 2H_{K,B} T N^{-1/2} \\ &\quad + (4eC_B)^6 \sqrt{2 + H_{K,B}} N^{-1/32} (\log N)^7 + (B^2/2 + 1) N^{-1/16}, \end{aligned}$$

and it gives

$$\begin{aligned} \mathbb{P}\{|J_N(r) - \log \sigma_N(r) + \gamma/2| > \varepsilon_{K,B}^*(N)\} &\leq H_{K,B} N^{-3/8} + N^{-12} \\ &\quad + 4H_{K,B} T^2 N^{-7/16} (\log N)^{-14} + \exp[-N/(2B^4 e^{6K})], \end{aligned}$$

where the last term is the probability of E^c from (51). As in the real Gaussian case, $\varepsilon_{K,B}^*(N)$ and the right-hand side of the last display replace $\varepsilon_K^*(N)$ and $p_K^*(N)$ at each radius. Similarly, the small-derivative exceptional probability, which replaces the right-hand side of (61), is

$$N^{-3} + \exp[-(N+1)/(2B^4)] + N^{-10} + C_{K,B} N^{-3/8} (\log N)^4.$$

We require $A_B \log N/N \leq 1$ and $N \geq e^2$, and all previous thresholds with the enlarged constants. Since $D'_{K,B} C_{K,B} = D'_K C_K$, $D'_{K,B} \mathcal{B}_{K,B} = B^5 D'_K \mathcal{B}_K$ and $S_{K,B} \cdot Ba = B^2 S_K a$, these are

$$N \geq H_{K,B}^4 = (1 + B^3)^4 H_K^4, \quad N \geq (6400e^{8K})^2,$$

$$\begin{aligned}
N &\geq 2K, & N &\geq 2(1+2K)e^{4K}, & \varepsilon_{K,B}^*(N) + D_K/N &\leq K/16, \\
D'_K\{c_K N L^{-4} + B^5 \mathcal{B}_K N^{1/2} L^2 \log N\} &\leq \frac{N^{31/32}}{4C_{1,B}(K+2) \log N}, \\
40C_{1,B}(K+2)\sqrt{N} \log N &\leq N^{31/32}/2, \\
2BaL/N^{3/2} &\leq 1/N, & 4B^2 S_K a L^2 N^{-1/2} \sqrt{\log N} &\leq 1,
\end{aligned}$$

together with the thresholds $N \geq \max\{2, (K+1)/3\}$, $j \geq 60$ and $N \geq \max\{4, K+2\}$, which do not change. In X_0 , H_K and $C_1(K+2)$ become $H_{K,B}$ and $C_{1,B}(K+2)$, and $D'_K \mathcal{B}_K$ becomes $B^5 D'_K \mathcal{B}_K$. These inequalities hold eventually for each fixed distribution, with cutoffs that depend on B .

If $\mathbb{P}(\xi_0 = 0) > 0$, the trailing-zero allowance $\lceil D \log N \rceil$, with $D > 4/|\log \mathbb{P}(\xi_0 = 0)|$, and the bound $N^{-D|\log \mathbb{P}(\xi_0 = 0)|}$ on the probability of its exceptional event at each degree are those of the proof of Corollary 1.2 for the other distributions in Section 7, which also treats the initial zero multiplicity and the actual degrees in Theorem 5.1.

A.12 Bounds for a growing annulus

In this subsection we compute the constants of the proof in Section 8.1. We write each constant as a number independent of K , times powers of $K+1$ and a factor e^{bK} , and then bound e^{bK} by $N^{b/1000}$ at the width $K = K_N$. For the logarithmic rate, we set $K = K_N$, where $K_N = \lfloor (\log N)/1000 \rfloor$ is the width of Section 8.1, with $N = j^8$ and $L = N^{1/64}$ as before, and use the local-count and tail lemmas with $K+2$ in place of K . The following bounds hold whenever the displayed deterministic conditions hold. Set

$$\begin{aligned}
H_0 &= 10^8 e^{20}, & S_0 &= 100 e^4, & D_0 &= 2^{18} S_0^2, \\
B_0 &= 2000 \cdot 160^{3/2}, & V_0 &= 50 + 3B_0 + 6400, & Q_0 &= \frac{9 \cdot 80^2}{1024 S_0^2}, \\
A &= \max(1, 1024\pi/c_{\text{HW}}), & C_{10} &= (4A + 12)/\log 2, & C_{\text{bad}} &= 16C_{10}^2 V_0 D_0^2.
\end{aligned}$$

These constants are independent of K, N . By the definitions we have, for $K \geq 0$,

$$\begin{aligned}
H_K &= H_0 e^{20K}, & S_K &= S_0 e^K, & D'_K &= D_0 (K+1) e^{2K}, \\
\mathcal{B}_K &= B_0 e^{9K}, & \mathcal{T}_K &= 6400 e^{8K}, & V_K &\leq V_0 e^{9K}, \\
c_K &= Q_0 e^{6K}, & C_1(K+2) &\leq C_{10}(K+1), & C_K &\leq C_{\text{bad}}(K+1)^4 e^{13K}.
\end{aligned}$$

Indeed, the first three hold since $H_K = 10^8 e^{20} e^{20K}$, $S_K = 100 e^4 e^K$ and $D'_K = 2^{18} (K+1) S_K^2 = 2^{18} S_0^2 (K+1) e^{2K}$. For the others we use $1/\lambda_K = 80 e^{4K}$, so that

$2/\lambda_K = 160e^{4K}$ and $\mathcal{T}_K = 80e^{4K} \cdot 80e^{4K} = 6400e^{8K}$, and, for the bounds, $e^{8K} \leq e^{9K}$, $1 \leq e^{9K}$, $4K \leq (4A + 12)K$ and $C_1(K + 2) = (4A + 4(K + 2) + 4)/\log 2$:

$$\begin{aligned}\mathcal{B}_K &= 2000e^{3K}(160e^{4K})^{3/2} = 2000 \cdot 160^{3/2}e^{9K} = B_0e^{9K}, \\ V_K &= 50 + 3B_0e^{9K} + 6400e^{8K} \leq (50 + 3B_0 + 6400)e^{9K} = V_0e^{9K}, \\ c_K &= \frac{9 \cdot 80^2e^{8K}}{1024S_0^2e^{2K}} = Q_0e^{6K}, \\ C_1(K + 2) &= \frac{4A + 4K + 12}{\log 2} \leq \frac{(4A + 12)(K + 1)}{\log 2} = C_{10}(K + 1), \\ C_K &= 16C_1(K + 2)^2V_K(D'_K)^2 \leq 16C_{10}^2(K + 1)^2 \cdot V_0e^{9K} \cdot D_0^2(K + 1)^2e^{4K} \\ &= C_{\text{bad}}(K + 1)^4e^{13K}.\end{aligned}$$

Set

$$E_0 = 1 + (4eC_*)^6\sqrt{2 + H_0 + H_0/16} + 3/2.$$

At $K = K_N$ we have $K \leq (\log N)/1000$, and hence $e^{bK} \leq e^{(b/1000)\log N} = N^{b/1000}$ for $b \geq 0$. Using this at $K = K_N$, we get for $N \geq e$

$$\begin{aligned}\varepsilon_K^*(N) &\leq E_0N^{-17/800}(\log N)^7, \\ D_K/N &\leq 10(K + 1)^2N^{-249/250}, \\ p_K^*(N) &\leq H_0N^{-71/200} + N^{-12} + \frac{H_0}{256}N^{-167/400}(\log N)^{-12}, \\ (66) \quad C_KN^{-3/8}(\log N)^4 &\leq C_{\text{bad}}(K + 1)^4N^{-181/500}(\log N)^4.\end{aligned}$$

By the bound on e^{bK} with $b = 20$ and $b = 10$, and $2 \leq 2e^{20K}$,

$$\begin{aligned}H_K &= H_0e^{20K} \leq H_0N^{1/50}, \\ \sqrt{2 + H_K} &\leq \sqrt{(2 + H_0)e^{20K}} = \sqrt{2 + H_0}e^{10K} \leq \sqrt{2 + H_0}N^{1/100}.\end{aligned}$$

We substitute these bounds into $\varepsilon_K^*(N)$, whose second term is $(H_K/16)N^{-1/2}\log N$, with $1/50 - 1/2 = -12/25$ and $1/100 - 1/32 = -17/800$. Then the bounds $N^{-1/32}, N^{-12/25}, N^{-1/16} \leq N^{-17/800}$ and $1 \leq \log N \leq (\log N)^7$ for $N \geq e$ give the first line of (66):

$$\begin{aligned}\varepsilon_K^*(N) &\leq N^{-1/32}(\log N)^7 + \frac{H_0}{16}N^{-12/25}\log N \\ &\quad + (4eC_*)^6\sqrt{2 + H_0}N^{-17/800}(\log N)^7 + \frac{3}{2}N^{-1/16} \leq E_0N^{-17/800}(\log N)^7.\end{aligned}$$

The second line follows from $D_K/N = 10(K + 1)^2e^{4K}N^{-1} \leq 10(K + 1)^2N^{1/250-1} = 10(K + 1)^2N^{-249/250}$. Since $4T^2 = (\log N)^2/256$ and $H_K \leq H_0N^{1/50}$, with $1/50 - 3/8 = -71/200$ and $1/50 - 7/16 = -167/400$, the third line follows from

$$p_K^*(N) = H_KN^{-3/8} + N^{-12} + \frac{H_K}{256}N^{-7/16}(\log N)^{-12}$$

$$\begin{aligned}
&\leq H_0 N^{1/50-3/8} + N^{-12} + \frac{H_0}{256} N^{1/50-7/16} (\log N)^{-12} \\
&= H_0 N^{-71/200} + N^{-12} + \frac{H_0}{256} N^{-167/400} (\log N)^{-12}.
\end{aligned}$$

Finally, by the bound on C_K above, the bound on e^{bK} with $b = 13$ and $13/1000 - 3/8 = -181/500$, the fourth line follows from

$$\begin{aligned}
C_K N^{-3/8} (\log N)^4 &\leq C_{\text{bad}} (K+1)^4 e^{13K} N^{-3/8} (\log N)^4 \\
&\leq C_{\text{bad}} (K+1)^4 N^{13/1000-3/8} (\log N)^4 = C_{\text{bad}} (K+1)^4 N^{-181/500} (\log N)^4.
\end{aligned}$$

Each ratio in Table 15 tends to zero, and the indicated condition holds once the ratio is at most one, where for the two mean terms we require that their sum be at most one. Indeed, at $K = K_N$ every ratio other than $2K/N$ and the fourth one is a constant times powers of $K_N + 1 \leq \log N$ and of $\log N$, times a negative power of N . Also $2K_N/N \leq 2(\log N)/N$, and, once $K_N \geq 2$, the fourth ratio is at most 8 times its bracket, which tends to zero for the same reason. These rates are compared in Section 8.1. We require the last ratio to be strictly below one, which

Condition	Sufficient ratio bound
$H_K^4 \leq N$	$H_0^4 N^{-23/25}$
$(6400e^{8K})^2 \leq N$	$6400^2 N^{-123/125}$
$2K \leq N$,	$2K/N$, $2(1+2K)N^{-249/250}$
$2(1+2K)e^{4K} \leq N$	
$\varepsilon_K^*(N) + D_K/N \leq K/16$	$16[E_0 N^{-17/800} (\log N)^7 + 10(K+1)^2 N^{-249/250}]/K$
First term of the mesh mean threshold	$4C_{10}D_0Q_0(K+1)^2(\log N)N^{-93/4000}$
Second term of the mesh mean threshold	$4C_{10}D_0B_0(K+1)^2(\log N)^2N^{-853/2000}$
Excluded-sector root threshold	$80C_{10}(K+1)(\log N)N^{-15/32}$
Taylor localization inequality	$180S_0e^4(\log N)N^{-113/4000}$
$2a/d_0 < N^{-65/64}$	$90e^4\sqrt{\log N}N^{-117/4000}$

Table 15: Sufficient ratio bounds for the degree conditions at the width $K = K_N$.

also implies $2a/d_0 \leq 1/N$, since then $2a/d_0 < N^{-65/64} \leq N^{-1}$.

We obtain the ratios of the table by dividing each condition by its right-hand side and substituting the bounds above. For the first three rows, by the bound on

e^{bK} with $b = 80, 16$ and 4 ,

$$\begin{aligned} \frac{H_K^4}{N} &= H_0^4 e^{80K} N^{-1} \leq H_0^4 N^{2/25-1} = H_0^4 N^{-23/25}, \\ \frac{(6400e^{8K})^2}{N} &= 6400^2 e^{16K} N^{-1} \leq 6400^2 N^{2/125-1} = 6400^2 N^{-123/125}, \\ \frac{2(1+2K)e^{4K}}{N} &\leq 2(1+2K)N^{1/250-1} = 2(1+2K)N^{-249/250}. \end{aligned}$$

For the fourth row we divide $\varepsilon_K^*(N) + D_K/N$ by $K/16$ and use the first two lines of (66). We multiply the mesh mean threshold of Section A.6 by $4C_1(K+2)(\log N)N^{-31/32}$, and use $L^{-4} = N^{-1/16}$, $L^2 = N^{1/32}$, $C_1(K+2) \leq C_{10}(K+1)$, $D'_K c_K = D_0 Q_0(K+1)e^{8K}$, $D'_K \mathcal{B}_K = D_0 B_0(K+1)e^{11K}$, $1 - 1/16 - 31/32 = -1/32$ and $1/2 + 1/32 - 31/32 = -7/16$. Its two terms then satisfy

$$\begin{aligned} &4C_1(K+2)D'_K c_K (\log N) N^{1-1/16-31/32} \\ &\leq 4C_{10}D_0 Q_0(K+1)^2 e^{8K} (\log N) N^{-1/32}, \\ &4C_1(K+2)D'_K \mathcal{B}_K (\log N)^2 N^{1/2+1/32-31/32} \\ &\leq 4C_{10}D_0 B_0(K+1)^2 e^{11K} (\log N)^2 N^{-7/16}, \end{aligned}$$

and the bound on e^{bK} with $b = 8$ and $b = 11$, with $1/125 - 1/32 = -93/4000$ and $11/1000 - 7/16 = -853/2000$, gives the entries of the table. For the excluded-sector root threshold, the ratio $80C_1(K+2)x e^{-15x/32}$ of Section A.6, with $x = \log N$, is at most $80C_{10}(K+1)(\log N)N^{-15/32}$. The last two rows, for the Taylor localization inequality $4S_K a L^2 N^{-1/2} \sqrt{\log N} \leq 1$ of Section A.8, which is $4M_2 a \leq d_0^2$, and for the strict band $N^{65/64} \cdot 2a/d_0 < 1$, are computed in the proof of Theorem 1.3 from (31) and (33), with $180S_0 e^4 = 18000e^8$, $7/16 + 1/32 - 1/2 = -1/32$, $3/1000 - 1/32 = -113/4000$, $7/16 - 15/32 = -1/32$ and $2/1000 - 1/32 = -117/4000$. They use the bounds $m_j \leq 9N^{7/8}$ and $a \leq 45e^{2(K+2)} N^{7/16} \sqrt{\log N} = 45e^4 e^{2K} N^{7/16} \sqrt{\log N}$ of (29) and (30).

We also impose

$$j \geq 60, \quad K \geq 2, \quad A \log N / N \leq 1, \quad N \geq K + 2.$$

These conditions are eventual. Beyond a fixed cutoff, the profile and thin-band radii belong to their required annuli. In this way we obtain all local-count, covariance, mean, sector, tail, localization and radial cutoffs used in the proof.

For a growing annulus, the sparse unit-circle error is at most $2N^{-1/64} + 4\varepsilon_2^*(N)N^{1/64}$, and the crossing count is at most $2N$ times this bound, by (63), whose five radii are taken at the fixed width 2. By the block bound (39) of Step 4

of Section 6.1, the two bounds of (63), and (62) at $K = K_N$ divided by N , the bound that we use for the logarithmic rate is, for $N \leq n \leq N + m_j$,

$$\begin{aligned}
 \left| \frac{\nu_n(1)}{n} - \frac{1}{2} \right| &\leq \left| \frac{\nu_N(1)}{N} - \frac{1}{2} \right| + \frac{c_N + U_N}{N} + \frac{3m_j}{2N} \\
 &\leq 3(2N^{-1/64} + 4\varepsilon_2^*(N)N^{1/64}) + \frac{U_N}{N} + \frac{3m_j}{2N} \\
 &\leq 3(2N^{-1/64} + 4\varepsilon_2^*(N)N^{1/64}) + N^{-1/32} + \frac{2\log 2}{K_N} \\
 (67) \quad &+ \frac{8}{K_N}(\varepsilon_{K_N}^*(N) + D_{K_N}/N) + \frac{3m_j}{2N}.
 \end{aligned}$$

Here $\varepsilon_{K_N}^*(N)$, $H_{K_N} = 10^8 e^{20(K_N+1)}$ and $D_{K_N} = 10(K_N + 1)^2 e^{4K_N}$ are $\varepsilon_K^*(N)$, H_K and D_K at $K = K_N$. For this bound, the exceptional probability is (64) with $K = K_N$ and $R = 2$, where $C_{K_N} = 16C_1(K_N + 2)^2 V_{K_N}(D'_{K_N})^2$ and $p_{K_N}^*(N)$ is the function $p_K^*(N)$ at $K = K_N$. By the third and fourth lines of (66), with $4 \cdot H_0/256 = H_0/64$, its terms $4p_{K_N}^*(N) + C_{K_N}N^{-3/8}(\log N)^4$ are bounded as in the proof of Theorem 1.3. It follows from the majorants in (66) that every term in (67) other than $2\log 2/K_N$ is $o(1/\log N)$, as in Section 8.1, since $K_N/\log N \rightarrow 1/1000$ and $m_j/N \leq 9N^{-1/8}$.

A.13 Nearby circles

Finally, we compute the constants of the proof of Corollary 8.2 in Section 8.3. The good annular roots counted there lie in a thin band about the unit circle that contains both circles, and the five-radius estimate at an exponent $\kappa_* < \kappa$ counts the roots of that band. For a target exponent $\kappa > 0$, we choose a wider band with

$$0 < \kappa_* < \min(\kappa, 1/64),$$

as does the choice $\kappa_* = \min(\kappa/2, 1/128)$ in the proof of Corollary 8.2. Then the band $||z| - 1| < N^{-1-\kappa_*}$ contains the circles $|z| = 1 \pm N^{-1-\kappa}$, and κ_* lies in the range $0 < \kappa \leq 1/64$ of the thin-band estimate of Section A.3. We impose

$$N^{-1-\kappa} + 2a/d_0 < N^{-1-\kappa_*},$$

where $2a/d_0$ is the localization radius of Section A.8. A good annular root whose localization disk meets one of the circles $|z| = 1 \pm N^{-1-\kappa}$ satisfies $||\alpha| - 1| \leq N^{-1-\kappa} + 2a/d_0$, so this condition places it in the band $||\alpha| - 1| < N^{-1-\kappa_*}$. By the form of (56) at $K = 2$ in Section A.3, with κ_* in place of κ ,

$$\left| \frac{\nu_N(1 + qN^{-1-\kappa_*})}{N} - \frac{1}{2} \right| \leq 2N^{-\kappa_*} + 4\varepsilon_2^*(N)N^{\kappa_*}, \quad q = -1, 0, 1,$$

outside probability $5p_2^*(N)$. Since $p_2^*(N)$ does not depend on the thin-band exponent, the exceptional probability of the block is (64) with $R = 2$.

B General concentration and counting estimates

Let $(a_k)_{k \geq 0}$ be a sequence of random complex coefficients on one probability space, and put

$$f_N(z) = \sum_{k=0}^N a_k z^k.$$

The same coefficient sequence is used for all degrees, and in particular $\deg f_N \leq N$ when f_N is nonzero. Let $\sigma_N(r) > 0$ be deterministic, and write

$$J_N(r) = \int_{\mathbb{T}} \log |f_N(re^{i\theta})| d\mu(\theta),$$

where μ is the normalized angular measure. Deterministic counting statements concern nonzero polynomials, and in probabilistic applications we include the case of an identically zero polynomial in an explicit exceptional event.

B.1 Angular occupation and logarithmic integrals

One- and two-point occupation bounds. For fixed N, r , we set

$$X(\theta) = \frac{f_N(re^{i\theta})}{\sigma_N(r)}.$$

Let G have density $\pi^{-1}e^{-|z|^2}$ on $\mathbb{C}$. We set

$$F_G(t) = \mathbb{P}(|G| \leq t) = 1 - e^{-t^2}.$$

There are measurable exceptional sets $E_1 \subset \mathbb{T}$, $E_2 \subset \mathbb{T}^2$ with $\mu(E_1) \leq \eta_1$, $(\mu \otimes \mu)(E_2) \leq \eta_2$, such that, for all $s, t \geq 0$,

$$|\mathbb{P}(|X(\theta)| \leq t) - F_G(t)| \leq \delta_1 \quad (\theta \notin E_1),$$

$$|\mathbb{P}(|X(\theta)| \leq s, |X(\phi)| \leq t) - F_G(s)F_G(t)| \leq \delta_2 \quad ((\theta, \phi) \notin E_2).$$

In the two-point bound we compare with a pair of independent standard complex Gaussians, and if the normal approximation yields a correlated Gaussian pair, then δ_2 includes the error of the comparison with this independent pair.

Logarithmic moments and energy. For fixed constants $C > 0$, $\beta > 0$, and for all real $p > 1$,

$$(68) \quad \mathbb{E} \int_{\mathbb{T}} |\log |X(\theta)||^p d\mu(\theta) \leq (Cp)^{\beta p}.$$

We will also use the following eventwise bound on an event E_H with $\mathbb{P}(E_H^c) \leq p_H$:

$$(69) \quad \mathbb{E} \left[\mathbf{1}_{E_H} \int_{\mathbb{T}} |\log |X(\theta)||^p d\mu(\theta) \right] \leq (Cp)^{\beta p} \quad (p > 1).$$

Here the expectation is the integral over E_H . We evaluate the integrand only on that event. This also defines the eventwise bound when the angular integral is infinite on E_H^c . The exact angular energy identity is

$$(70) \quad \int_{\mathbb{T}} |X(\theta)|^2 d\mu(\theta) = 1 \quad \text{almost surely.}$$

The upper-energy probability bound is

$$(71) \quad \mathbb{P} \left(\int_{\mathbb{T}} |X(\theta)|^2 d\mu(\theta) > B_E \right) \leq p_E \quad (B_E > 0).$$

The full bound (68) is the case $E_H = \Omega$ and $p_H = 0$ of (69). We use the tail form (71) when the coefficient moduli are random, and under (70) we can take $B_E = 1$, $p_E = 0$. For the coefficient distributions of Sections 3 and 7, these logarithmic bounds hold for every $p \geq 1$ (see (15), (47) and Proposition 7.4), but in the proof of Theorem B.1 we use them only for $p = 2q \geq 2$.

Theorem B.1 (Logarithmic integral concentration). *Assume the one- and two-point occupation bounds, the eventwise logarithmic bound (69), and the energy bound (71). We set*

$$d_1 = \delta_1 + \eta_1, \quad d_2 = \delta_2 + \eta_2, \quad v = d_2 + 2d_1.$$

For all $T > 0$, $u > 0$, $b > 0$, $q \geq 1$, and $\lambda > 1$, we set

$$(72) \quad \eta_{\log} = u + 2Td_1 + \lambda(2Cq)^{\beta} \sqrt{e^{-2T} + d_1 + b} + (B_E/2 + 1)e^{-2T},$$

$$(73) \quad p_{\text{fail}} = v/b^2 + \lambda^{-2q} + 4T^2 v/u^2 + p_E + p_H.$$

Then

$$\mathbb{P}(|J_N(r) - \log \sigma_N(r) - \mathbb{E} \log |G|| > \eta_{\log}) \leq p_{\text{fail}}, \quad \mathbb{E} \log |G| = -\gamma/2.$$

The same conclusion holds if, in place of the occupation bounds, $|\mathbb{E} A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$ for all $t \geq 0$ and some $d_1, v \geq 0$, where $A_t = \mu\{\theta : |X(\theta)| \leq t\}$. If moreover $\int_{\mathbb{T}} |X|^2 d\mu \leq B_E$ on E_H , then $p_E + p_H$ in (73) may be replaced by $\mathbb{P}(E_H^c)$.

Proof. The idea is to clip the logarithm at $\pm T$, to estimate the mean and the variance of the clipped integral by the occupation bounds, and to control the lower tail that the clipping removes by the logarithmic moments and the occupation bound at e^{-T} , and the upper tail by the energy bound. Table 16 lists the five exceptional events, one for each term of (73).

We set $A_t = \int_{\mathbb{T}} \mathbf{1}_{\{|X(\theta)| \leq t\}} d\mu(\theta)$. By Tonelli's theorem, we have

$$\begin{aligned} \mathbb{E}A_t &= \int_{\mathbb{T}} \mathbb{P}(|X(\theta)| \leq t) d\mu(\theta), \\ \mathbb{E}A_t^2 &= \mathbb{E} \int_{\mathbb{T}^2} \mathbf{1}_{\{|X(\theta)| \leq t\}} \mathbf{1}_{\{|X(\phi)| \leq t\}} d\mu(\theta) d\mu(\phi) \\ &= \int_{\mathbb{T}^2} \mathbb{P}(|X(\theta)| \leq t, |X(\phi)| \leq t) d\mu(\theta) d\mu(\phi). \end{aligned}$$

By these identities and the [one- and two-point occupation bounds](#), with $s = t$ in the second, and since the probabilities and $F_G(t)$ lie in $[0, 1]$, we have, for every $t \geq 0$,

$$\begin{aligned} |\mathbb{E}A_t - F_G(t)| &= \left| \int_{\mathbb{T}} (\mathbb{P}(|X(\theta)| \leq t) - F_G(t)) d\mu(\theta) \right| \\ &\leq \int_{\mathbb{T} \setminus E_1} \delta_1 d\mu + \int_{E_1} 1 d\mu \leq \delta_1 + \mu(E_1) \leq d_1, \\ |\mathbb{E}A_t^2 - F_G(t)^2| &= \left| \int_{\mathbb{T}^2} (\mathbb{P}(|X(\theta)| \leq t, |X(\phi)| \leq t) - F_G(t)^2) d(\mu \otimes \mu) \right| \\ &\leq \int_{\mathbb{T}^2 \setminus E_2} \delta_2 d(\mu \otimes \mu) + \int_{E_2} 1 d(\mu \otimes \mu) \leq \delta_2 + (\mu \otimes \mu)(E_2) \leq d_2. \end{aligned}$$

Since both $\mathbb{E}A_t$ and $F_G(t)$ belong to $[0, 1]$, we have

$$\begin{aligned} |(\mathbb{E}A_t)^2 - F_G(t)^2| &= |\mathbb{E}A_t - F_G(t)| |\mathbb{E}A_t + F_G(t)| \leq 2d_1, \\ \text{Var}(A_t) = \mathbb{E}A_t^2 - (\mathbb{E}A_t)^2 &\leq |\mathbb{E}A_t^2 - F_G(t)^2| + |(\mathbb{E}A_t)^2 - F_G(t)^2| \leq d_2 + 2d_1 = v. \end{aligned}$$

From here on, the occupation bounds enter only through the bounds $|\mathbb{E}A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$, which proves the first variant in the statement.

Since $F_G(e^{-T}) = 1 - \exp(-e^{-2T}) \leq e^{-2T}$, taking $t = e^{-T}$ in these bounds and applying Chebyshev's inequality, we obtain

$$\begin{aligned} \mathbb{E}A_{e^{-T}} &\leq F_G(e^{-T}) + d_1 \leq e^{-2T} + d_1, \\ \mathbb{P}(A_{e^{-T}} > e^{-2T} + d_1 + b) &\leq \mathbb{P}(A_{e^{-T}} - \mathbb{E}A_{e^{-T}} > b) \leq \frac{\text{Var}(A_{e^{-T}})}{b^2} \leq \frac{v}{b^2}. \end{aligned}$$

We call the event $\{A_{e^{-T}} > e^{-2T} + d_1 + b\}$ the Chebyshev event for $A_{e^{-T}}$.

On E_H , we set $U = \int_{\mathbb{T}} |\log |X(\theta)||^2 d\mu(\theta)$. By Jensen's inequality and (69) with $p = 2q$, we have

$$U^q = \left(\int_{\mathbb{T}} |\log |X(\theta)||^2 d\mu(\theta) \right)^q \leq \int_{\mathbb{T}} |\log |X(\theta)||^{2q} d\mu(\theta),$$

$$\mathbb{E}[\mathbf{1}_{E_H} U^q] \leq \mathbb{E} \left[\mathbf{1}_{E_H} \int_{\mathbb{T}} |\log |X(\theta)||^{2q} d\mu(\theta) \right] \leq (2Cq)^{2\beta q}.$$

Applying Markov's inequality to U^q on E_H , we get

$$\mathbb{P}\{\omega \in E_H : U > \lambda^2(2Cq)^{2\beta}\} \leq \frac{\mathbb{E}[\mathbf{1}_{E_H} U^q]}{\lambda^{2q}(2Cq)^{2\beta q}} \leq \frac{(2Cq)^{2\beta q}}{\lambda^{2q}(2Cq)^{2\beta q}} = \lambda^{-2q}.$$

We call the event on the left the Markov event for U .

We now truncate the logarithm and set $\ell_T(z) = \max(-T, \min(T, \log |z|))$ and $L_T = \int_{\mathbb{T}} \ell_T(X) d\mu$. For every $z \in \mathbb{C}$, including $z = 0$, we have the layer-cake identity

$$\ell_T(z) = T - \int_{-T}^T \mathbf{1}_{\{|z| \leq e^\tau\}} d\tau,$$

since the set of $\tau \in [-T, T]$ with $|z| \leq e^\tau$ is $[\max(-T, \log |z|), T]$ if $\log |z| \leq T$ (with $\log 0 = -\infty$), and is empty otherwise. By the layer-cake identity at $z = X(\theta)$ and at $z = G$, Tonelli's theorem and $\mathbb{P}(|G| \leq e^\tau) = F_G(e^\tau)$, we have

$$\begin{aligned} L_T &= \int_{\mathbb{T}} \left(T - \int_{-T}^T \mathbf{1}_{\{|X(\theta)| \leq e^\tau\}} d\tau \right) d\mu(\theta) = T - \int_{-T}^T A_{e^\tau} d\tau, \\ \mathbb{E}L_T - \mathbb{E}\ell_T(G) &= \int_{-T}^T (F_G(e^\tau) - \mathbb{E}A_{e^\tau}) d\tau, \\ L_T - \mathbb{E}L_T &= - \int_{-T}^T (A_{e^\tau} - \mathbb{E}A_{e^\tau}) d\tau. \end{aligned}$$

By these identities, the bounds $|\mathbb{E}A_t - F_G(t)| \leq d_1$ and $\text{Var}(A_t) \leq v$, and the Cauchy-Schwarz inequality, we have

$$\begin{aligned} |\mathbb{E}L_T - \mathbb{E}\ell_T(G)| &\leq \int_{-T}^T |\mathbb{E}A_{e^\tau} - F_G(e^\tau)| d\tau \leq 2Td_1, \\ |\text{Cov}(A_{e^\tau}, A_{e^{\tau'}})| &\leq \sqrt{\text{Var}(A_{e^\tau}) \text{Var}(A_{e^{\tau'}})} \leq v, \end{aligned}$$

and hence, by Fubini's theorem, which applies since $|A_{e^\tau} - \mathbb{E}A_{e^\tau}| \leq 1$,

$$\begin{aligned} \text{Var}(L_T) &= \mathbb{E} \left(\int_{-T}^T (A_{e^\tau} - \mathbb{E}A_{e^\tau}) d\tau \right)^2 \\ &= \int_{-T}^T \int_{-T}^T \mathbb{E}[(A_{e^\tau} - \mathbb{E}A_{e^\tau})(A_{e^{\tau'}} - \mathbb{E}A_{e^{\tau'}})] d\tau d\tau' \\ &= \int_{-T}^T \int_{-T}^T \text{Cov}(A_{e^\tau}, A_{e^{\tau'}}) d\tau d\tau' \leq \int_{-T}^T \int_{-T}^T |\text{Cov}(A_{e^\tau}, A_{e^{\tau'}})| d\tau d\tau' \leq 4T^2v. \end{aligned}$$

Applying Chebyshev's inequality again, we get

$$\mathbb{P}(|L_T - \mathbb{E}L_T| > u) \leq \frac{\text{Var}(L_T)}{u^2} \leq \frac{4T^2v}{u^2}.$$

We call the event on the left the Chebyshev event for L_T .

On E_H , outside the Chebyshev event for $A_{e^{-T}}$ and the Markov event for U , by the Cauchy–Schwarz inequality we bound the negative truncation error by

$$\begin{aligned} \int_{\{|X| < e^{-T}\}} (-\log |X| - T) d\mu &\leq \int_{\{|X| < e^{-T}\}} |\log |X|| d\mu \\ &\leq \left(\int_{\mathbb{T}} |\log |X||^2 d\mu \right)^{1/2} \left(\int_{\mathbb{T}} \mathbf{1}_{\{|X| < e^{-T}\}} d\mu \right)^{1/2} \\ &\leq \sqrt{U A_{e^{-T}}} \leq \lambda(2Cq)^\beta \sqrt{e^{-2T} + d_1 + b}. \end{aligned}$$

On the energy event $\{\int_{\mathbb{T}} |X|^2 d\mu \leq B_E\}$, by Tonelli's theorem applied to $(y - T)_+ = \int_T^\infty \mathbf{1}_{\{y > \tau\}} d\tau$ at $y = \log |X|$ and by Markov's inequality for μ , we bound the positive truncation error by

$$\begin{aligned} \int_{\{|X| > e^T\}} (\log |X| - T) d\mu &= \int_T^\infty \mu\{|X| > e^\tau\} d\tau \leq \int_T^\infty e^{-2\tau} \left(\int_{\mathbb{T}} |X|^2 d\mu \right) d\tau \\ &\leq \int_T^\infty B_E e^{-2\tau} d\tau = \frac{B_E}{2} e^{-2T}. \end{aligned}$$

For the Gaussian tails, $\int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau$ and $\int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau$ are at most $\frac{1}{2}e^{-2T}$ by (17). On E_H , outside the Markov event for U , the integral $\int_{\mathbb{T}} \log |X| d\mu$ is finite, since $\int_{\mathbb{T}} |\log |X||^2 d\mu = U \leq \lambda^2(2Cq)^{2\beta}$. Since

$$|\ell_T(z) - \log |z|| = (-\log |z| - T)_+ + (\log |z| - T)_+,$$

on E_H and on the energy event, outside the Chebyshev event for $A_{e^{-T}}$ and the Markov event for U , it follows from the layer-cake identity, the two truncation bounds and the two Gaussian tail bounds that

$$\begin{aligned} \left| \int_{\mathbb{T}} \log |X| d\mu - L_T \right| &\leq \int_{\mathbb{T}} |\log |X| - \ell_T(X)| d\mu \\ &= \int_{\{|X| < e^{-T}\}} (-\log |X| - T) d\mu + \int_{\{|X| > e^T\}} (\log |X| - T) d\mu \\ &\leq \lambda(2Cq)^\beta \sqrt{e^{-2T} + d_1 + b} + \frac{B_E}{2} e^{-2T}, \\ |\mathbb{E}\ell_T(G) - \mathbb{E}\log |G|| &\leq \mathbb{E}|\ell_T(G) - \log |G|| = \mathbb{E}(-\log |G| - T)_+ + \mathbb{E}(\log |G| - T)_+ \\ &= \int_T^\infty \mathbb{P}(|G| < e^{-\tau}) d\tau + \int_T^\infty \mathbb{P}(|G| > e^\tau) d\tau \leq e^{-2T}. \end{aligned}$$

Finally, $J_N(r) - \log \sigma_N(r) = \int \log |X| d\mu$, and outside the five exceptional events of Table 16, by the previous display and the bounds on $|L_T - \mathbb{E}L_T|$ and $|\mathbb{E}L_T - \mathbb{E}\ell_T(G)|$, we have

$$\begin{aligned} \left| \int \log |X| d\mu - \mathbb{E}\log |G| \right| &\leq \left| \int \log |X| d\mu - L_T \right| + |L_T - \mathbb{E}L_T| \\ &\quad + |\mathbb{E}L_T - \mathbb{E}\ell_T(G)| + |\mathbb{E}\ell_T(G) - \mathbb{E}\log |G|| \end{aligned}$$

Exceptional event	Defined by	Probability at most
Chebyshev event for $A_{e^{-T}}$	$A_{e^{-T}} > e^{-2T} + d_1 + b$	v/b^2
Markov event for U	$\omega \in E_H, U > \lambda^2(2Cq)^{2\beta}$	λ^{-2q}
Chebyshev event for L_T	$ L_T - \mathbb{E}L_T > u$	$4T^2v/u^2$
Complement of the energy event	$\int_{\mathbb{T}} X ^2 d\mu > B_E$	p_E
Complement of E_H		p_H

Table 16: The exceptional events in the proof of Theorem B.1. The sum of the bounds on their probabilities is (73).

$$\leq \lambda(2Cq)^\beta \sqrt{e^{-2T} + d_1 + b} + \frac{B_E}{2} e^{-2T} + u + 2Td_1 + e^{-2T} = \eta_{\log}.$$

Under (70) and (68) the last two events are absent. If $\int_{\mathbb{T}} |X|^2 d\mu \leq B_E$ on E_H , then the complement of the energy event is contained in E_H^c , so that the last two events have probability at most $\mathbb{P}(E_H^c)$ together, which proves the second variant in the statement. By (69), the logarithmic integral is finite almost surely on E_H , even for polynomials with roots on the integration circle. Since $|G|^2$ has the exponential distribution with mean one, (18) gives $\mathbb{E} \log |G| = -\gamma/2$. $\square$

Remark B.2 (Polynomial choices of parameters). If $d_1 \leq DN^{-a}$ and $v \leq VN^{-a}$, choose $T = t \log N$, $b = N^{-2t}$, $u = N^{-w}$, and $q = c \log N \geq 1$. Under (68) and (70), we may take $p_H = 0$, $B_E = 1$ and $p_E = 0$. With $\lambda = e$, so that $\lambda(2Cq)^\beta = e(2Cc \log N)^\beta$, the terms of (72) and (73), other than $u = N^{-w}$ and $p_E = p_H = 0$, satisfy

$$\begin{aligned} 2Td_1 &\leq 2t(\log N)DN^{-a}, & (B_E/2 + 1)e^{-2T} &= \tfrac{3}{2}N^{-2t}, \\ e^{-2T} + d_1 + b &\leq N^{-2t} + DN^{-a} + N^{-2t} = 2N^{-2t} + DN^{-a}, \\ v/b^2 &\leq VN^{-a}/N^{-4t} = VN^{4t-a}, & \lambda^{-2q} &= e^{-2c \log N} = N^{-2c}, \\ 4T^2v/u^2 &\leq 4t^2(\log N)^2VN^{-a}/N^{-2w} = 4t^2V(\log N)^2N^{2w-a}. \end{aligned}$$

By substituting these bounds, we get

$$\begin{aligned} \eta_{\log} &\leq N^{-w} + 2tD(\log N)N^{-a} + e(2Cc \log N)^\beta \sqrt{2N^{-2t} + DN^{-a}} + \tfrac{3}{2}N^{-2t}, \\ p_{\text{fail}} &\leq VN^{4t-a} + N^{-2c} + 4t^2V(\log N)^2N^{2w-a}. \end{aligned}$$

The failure terms tend to zero when $4t < a$ and $2w < a$. For $N = j^s$, the three failure terms are

$$Vj^{-s(a-4t)}, \quad j^{-2cs}, \quad 4t^2Vs^2(\log j)^2j^{-s(a-2w)},$$

and they are summable if

$$s(a - 4t) > 1, \quad 2cs > 1, \quad s(a - 2w) > 1.$$

In particular, take $a = 1/2$, $t = w = 1/32$, $c = 6$, $\beta = 6$ and $s = 8$. Then $s(a - 4t) = 8(\frac{1}{2} - \frac{1}{8}) = 3$, $2cs = 2 \cdot 6 \cdot 8 = 96$ and $s(a - 2w) = 8(\frac{1}{2} - \frac{1}{16}) = \frac{7}{2}$ all exceed one, and for $N \geq e$,

$$\begin{aligned} \eta_{\log} &\leq N^{-1/32} + \frac{D}{16}(\log N)N^{-1/2} + e(12C \log N)^6 \sqrt{2N^{-1/16} + DN^{-1/2}} + \frac{3}{2}N^{-1/16} \\ &\leq \left(\frac{5}{2} + \frac{D}{16} + e(12C)^6 \sqrt{2 + D} \right) N^{-1/32} (\log N)^6, \end{aligned}$$

by the bound on $\eta_{\log}$ above with $2tD = D/16$, $2Cc = 12C$, $2t = 1/16$ and $w = 1/32$, and by $\log N \geq 1$ and $N^{-1/2} \leq N^{-1/16} \leq N^{-1/32}$. So we get an error bounded by the above constant times $N^{-1/32}(\log N)^6$ and a summable failure on j^8 . For large enough degree the error is at most $N^{-1/32}(\log N)^7$, with a deterministic threshold depending on the fixed constant: it suffices that $\log N$ is at least the above constant. We use the moments of the angular logarithmic norm in (68). The proof of Lemma 3.2 takes instead $u = N^{-1/32}(\log N)^7$, $q = \lceil \log N \rceil$ and $\lambda = (2e \log N/q)^6 \log N \geq e^6 \log N$, for which $\lambda(2Cq)^6 = (4eC)^6(\log N)^7$ and $\lambda^{-2q} \leq N^{-12}$. The choice $q = 6 \log N$, $\lambda = e$ above gives the same failure bound $N^{-2c} = N^{-12}$ with $\lambda(2Cq)^6 = e(12C \log N)^6$. The extra factor $\log N$ in λ is the difference between the exponents 7 and 6.

B.2 A local radial secant estimate

Lemma B.3 (Local approximation of radial counts). *Let f_N be a nonzero polynomial of degree at most N , and let $\sigma_N(r) > 0$ be deterministic. Suppose that the function $F_N(x)$, equal to $\log \sigma_N(1 + x/N)$ minus a centering independent of x , satisfies $|F_N(x) - F(x)| \leq \varepsilon$ on $[-K, K]$. Suppose also $|F'| \leq B_F$ and $|F'(x) - F'(y)| \leq M_F|x - y|$ there. Suppose the three logarithmic deviations*

$$|J_N(1 + y/N) - \log \sigma_N(1 + y/N) - \mathbb{E} \log |G||, \quad y = x - h, x, x + h,$$

are at most $\eta_{\log}$ outside their respective exceptional events. If $h > 0$ and $|x| + h \leq K < N$, then outside the union of those three events,

$$(74) \quad \left| \frac{\nu_N(1 + x/N)}{N} - F'(x) \right| \leq \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h}.$$

In particular an error $\eta_{\log} + \varepsilon \leq N^{-1/32}(\log N)^7$, with $h = N^{-\kappa}$, $0 < \kappa \leq 1/64$, gives convergence to $F'(0)$ uniformly for deterministic $|x| \leq h$.

Proof. By Jensen's formula, $\nu_N(r)/N$ lies between two secant slopes of J_N in the variable $N \log \rho$, one on each side of r , and the idea is to replace J_N by F at the three probe points, up to a constant which cancels, and then to compare the secant slopes of F with $F'(x)$. This leads to three errors, which come from the curvature of F , the change of the denominators, and the deviations of J_N from F at the probe points.

Let $r = 1 + x/N$ and set

$$d_- = N \log \frac{r}{r - h/N}, \quad d_+ = N \log \frac{r + h/N}{r}.$$

Since $|x| + h \leq K < N$, we have $r - h/N = 1 + (x - h)/N \geq 1 - K/N > 0$. By the integral form (4) of Jensen's formula, in which ν_N counts the roots in the closed disk, and since ν_N is nondecreasing, we have

$$\begin{aligned} J_N(r) - J_N(r - h/N) &= \int_{r-h/N}^r \frac{\nu_N(\rho)}{\rho} d\rho \leq \nu_N(r) \log \frac{r}{r - h/N} = \frac{\nu_N(r)d_-}{N}, \\ J_N(r + h/N) - J_N(r) &= \int_r^{r+h/N} \frac{\nu_N(\rho)}{\rho} d\rho \geq \nu_N(r) \log \frac{r + h/N}{r} = \frac{\nu_N(r)d_+}{N}. \end{aligned}$$

Dividing by $d_{\pm} > 0$, we obtain the Jensen bounds

$$\frac{J_N(r) - J_N(r - h/N)}{d_-} \leq \frac{\nu_N(r)}{N} \leq \frac{J_N(r + h/N) - J_N(r)}{d_+}.$$

Since $N \log(1 + u/N)$ has derivative $1/(1 + u/N)$ in u , we have

$$\begin{aligned} d_- &= N \log(1 + x/N) - N \log(1 + (x - h)/N) = \int_{x-h}^x \frac{du}{1 + u/N}, \\ d_+ &= N \log(1 + (x + h)/N) - N \log(1 + x/N) = \int_x^{x+h} \frac{du}{1 + u/N}. \end{aligned}$$

Since $|x| + h \leq K < N$, every $u \in [x - h, x + h]$ satisfies $0 < 1 - K/N \leq 1 + u/N \leq 1 + K/N$, and taking reciprocals and integrating over an interval of length h , we obtain

$$\begin{aligned} \frac{1}{1 + K/N} &\leq \frac{1}{1 + u/N} \leq \frac{1}{1 - K/N}, & \frac{h}{1 + K/N} &\leq d_{\pm} \leq \frac{h}{1 - K/N}, \\ 1 - \frac{K}{N} &\leq \frac{h}{d_{\pm}} \leq 1 + \frac{K}{N}, & \left| \frac{h}{d_{\pm}} - 1 \right| &\leq \frac{K}{N}. \end{aligned}$$

Since F' is Lipschitz on $[-K, K]$, which contains $x \pm t$ for $0 \leq t \leq h$, for the ordinary secants we have

$$\left| \frac{F(x + h) - F(x)}{h} - F'(x) \right| = \left| \frac{1}{h} \int_0^h (F'(x + t) - F'(x)) dt \right|$$

$$\begin{aligned}
&\leq \frac{1}{h} \int_0^h M_F t \, dt = \frac{M_F h}{2}, \\
\left| \frac{F(x) - F(x-h)}{h} - F'(x) \right| &= \left| \frac{1}{h} \int_0^h (F'(x-t) - F'(x)) \, dt \right| \\
&\leq \frac{1}{h} \int_0^h M_F t \, dt = \frac{M_F h}{2}.
\end{aligned}$$

Each ordinary secant has absolute value at most B_F , since it is the mean of F' over an interval in $[-K, K]$. Replacing its denominator h by $d_{\pm}$ multiplies it by $h/d_{\pm}$, and

$$\begin{aligned}
\left| \frac{F(x+h) - F(x)}{d_+} - \frac{F(x+h) - F(x)}{h} \right| &= \left| \frac{F(x+h) - F(x)}{h} \right| \left| \frac{h}{d_+} - 1 \right| \\
&\leq B_F \left| \frac{h}{d_+} - 1 \right| \leq \frac{B_F K}{N}, \\
\left| \frac{F(x) - F(x-h)}{d_-} - \frac{F(x) - F(x-h)}{h} \right| &= \left| \frac{F(x) - F(x-h)}{h} \right| \left| \frac{h}{d_-} - 1 \right| \\
&\leq B_F \left| \frac{h}{d_-} - 1 \right| \leq \frac{B_F K}{N}.
\end{aligned}$$

Let c be the centering, so that $F_N(x) = \log \sigma_N(1+x/N) - c$. For $y = x-h, x, x+h$, which lie in $[-K, K]$, we have

$$\begin{aligned}
&J_N(1+y/N) - \mathbb{E} \log |G| - c - F(y) \\
&= \left\{ J_N(1+y/N) - \log \sigma_N(1+y/N) - \mathbb{E} \log |G| \right\} + \left\{ F_N(y) - F(y) \right\}.
\end{aligned}$$

The constant $\mathbb{E} \log |G| + c$ does not depend on y , so that it cancels from each secant, and outside the exceptional event at y the first brace is at most $\eta_{\log}$ in absolute value and the second is at most ε . Since $r \pm h/N = 1 + (x \pm h)/N$, by subtracting the identity at two neighboring probe points we obtain

$$\begin{aligned}
\left| J_N(r+h/N) - J_N(r) - (F(x+h) - F(x)) \right| &\leq 2(\eta_{\log} + \varepsilon), \\
\left| J_N(r) - J_N(r-h/N) - (F(x) - F(x-h)) \right| &\leq 2(\eta_{\log} + \varepsilon).
\end{aligned}$$

By the lower bound $d_{\pm} \geq h/(1+K/N)$, the secant error is therefore at most $2(\eta_{\log} + \varepsilon)/d_{\pm} \leq 2(\eta_{\log} + \varepsilon)(1+K/N)/h$. Adding the three errors to the Jensen bounds, we obtain

$$\begin{aligned}
\frac{\nu_N(r)}{N} &\leq \frac{J_N(r+h/N) - J_N(r)}{d_+} \leq \frac{F(x+h) - F(x)}{d_+} + \frac{2(\eta_{\log} + \varepsilon)(1+K/N)}{h} \\
&\leq \frac{F(x+h) - F(x)}{h} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1+K/N)}{h}
\end{aligned}$$

$$\begin{aligned}
&\leq F'(x) + \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h}, \\
\frac{\nu_N(r)}{N} &\geq \frac{J_N(r) - J_N(r - h/N)}{d_-} \geq \frac{F(x) - F(x - h)}{d_-} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} \\
&\geq \frac{F(x) - F(x - h)}{h} - \frac{B_F K}{N} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} \\
&\geq F'(x) - \frac{M_F h}{2} - \frac{B_F K}{N} - \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h},
\end{aligned}$$

which together give (74). The Jensen bounds include roots on the counting circle. Since the three exceptional events may be arbitrarily dependent, we add their three probabilities. To prove the last assertion, let $|x| \leq h = N^{-\kappa}$ and let N be so large that $2h \leq K$, so that $|x| + h \leq K$. Since $|F'(x) - F'(0)| \leq M_F |x| \leq M_F h$, it follows from (74) and $\eta_{\log} + \varepsilon \leq N^{-1/32}(\log N)^7$ that

$$\begin{aligned}
\left| \frac{\nu_N(1 + x/N)}{N} - F'(0) \right| &\leq \left| \frac{\nu_N(1 + x/N)}{N} - F'(x) \right| + |F'(x) - F'(0)| \\
&\leq \frac{M_F h}{2} + \frac{B_F K}{N} + \frac{2(\eta_{\log} + \varepsilon)(1 + K/N)}{h} + M_F h \\
&= \frac{3M_F}{2} N^{-\kappa} + \frac{B_F K}{N} + 2 \left(1 + \frac{K}{N} \right) (\eta_{\log} + \varepsilon) N^\kappa \\
&\leq \frac{3M_F}{2} N^{-\kappa} + \frac{B_F K}{N} + 2 \left(1 + \frac{K}{N} \right) N^{\kappa-1/32} (\log N)^7,
\end{aligned}$$

which tends to zero since $\kappa \leq 1/64 < 1/32$. $\square$

B.3 A second-moment bound for small-derivative roots

Mesh and local multiplicity. Let $\mathcal{A}$ be an annulus contained in $\{|z| \leq R\}$ and let $z_1, \dots, z_m$ be retained mesh points. Apart from at most b_0 roots in explicitly excluded sectors, every root of f_N in $\mathcal{A}$ is within distance τ of a retained point. The segment between them lies in a specified domain D , and $|z_i| \leq R$. Except on events of probabilities at most p_2 and p_0 , respectively,

$$\sup_D |f_N''| \leq M_2, \quad \text{each mesh cell contains at most } q_0 \text{ roots.}$$

The probability p_0 also includes failure of the sector bound.

For the cell bound we can use a disk count, provided that each cell lies in one of the disks of a local count such as Lemma 4.1.

Jet approximation. We write the pair

$$W(z_i) = (W_0(z_i), W_1(z_i)) = (f_N(z_i)/\sqrt{N}, z_i f'_N(z_i)/N^{3/2})$$

as a real four-vector. For every retained i there is a mean-zero Gaussian four-vector $H_i = ((H_i)_0, (H_i)_1)$, written as a pair like $W(z_i)$, with covariance least eigenvalue at least $\lambda > 0$. For the product-of-disks event $E_i = \{|W_0(z_i)| \leq u, |W_1(z_i)| \leq v_0\}$, put $g_i = \mathbb{P}(|(H_i)_0| \leq u, |(H_i)_1| \leq v_0)$ and assume

$$|\mathbb{P}(E_i) - g_i| \leq \delta.$$

All but at most $b_* m^2$ ordered pairs (i, j) additionally obey

$$|\mathbb{P}(E_i \cap E_j) - g_i g_j| \leq \delta_* + \chi.$$

For example, the pair estimate can be obtained by an eight-dimensional Gaussian approximation with error δ_* , followed by a comparison with the independent marginal Gaussians with total variation error χ . We include in the exceptional count the ordinary and conjugate angular diagonals, the radial duplicates, and $i = j$, unless they satisfy the above pair estimate.

Theorem B.4 (Annular small-derivative count). *Assume the [mesh and local multiplicity](#) bounds and the [jet approximation](#), and denote by B_N the number of roots in $\mathcal{A}$ satisfying $|f'_N(\alpha)| < N^{3/2}/L$. In the jet hypothesis use*

$$u = \frac{N^{3/2}\tau/L + M_2\tau^2/2}{\sqrt{N}}, \quad v_0 = \frac{R(N^{3/2}/L + M_2\tau)}{N^{3/2}}.$$

We set

$$\mu_* = m \left(\frac{u^2 v_0^2}{4\lambda^2} + \delta \right), \quad V_* = b_* + \delta_* + \chi + 2\delta.$$

For every $t > 0$,

$$(75) \quad \mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \frac{m^2 V_*}{t^2}.$$

Proof. The idea is to detect each counted root outside the excluded sectors at its assigned mesh point, where both components of W are small, and to bound the number of mesh points with small W by its mean and its variance, which the jet approximation controls. We then bound B_N by Chebyshev's inequality and the cell bound.

Let α be a root in B_N outside the excluded sectors, and let z_i be its assigned mesh point. On the derivative event $\{\sup_D |f''_N| \leq M_2\}$, since $f_N(\alpha) = 0$, $|f'_N(\alpha)| < N^{3/2}/L$, $|z_i - \alpha| \leq \tau$, and the segment $[\alpha, z_i]$ lies in D by the [mesh and local multiplicity](#) bounds, it follows from Taylor's theorem with integral remainder along the segment that

$$|f_N(z_i)| = \left| f'_N(\alpha)(z_i - \alpha) + \int_{[\alpha, z_i]} (z_i - \zeta) f''_N(\zeta) d\zeta \right|$$

$$\begin{aligned} &\leq |f'_N(\alpha)| |z_i - \alpha| + \frac{M_2}{2} |z_i - \alpha|^2 \leq \frac{N^{3/2}\tau}{L} + \frac{M_2\tau^2}{2}, \\ |f'_N(z_i)| &= \left| f'_N(\alpha) + \int_{[\alpha, z_i]} f''_N(\zeta) d\zeta \right| \leq |f'_N(\alpha)| + M_2 |z_i - \alpha| \leq \frac{N^{3/2}}{L} + M_2\tau. \end{aligned}$$

Since $|z_i| \leq R$, dividing by the normalizations in $W(z_i)$ we obtain

$$\frac{|f_N(z_i)|}{\sqrt{N}} \leq \frac{N^{3/2}\tau/L + M_2\tau^2/2}{\sqrt{N}} = u, \quad \frac{|z_i f'_N(z_i)|}{N^{3/2}} \leq \frac{R(N^{3/2}/L + M_2\tau)}{N^{3/2}} = v_0,$$

so that the event E_i occurs at every assigned point.

We set $Z = \sum_{i=1}^m \mathbf{1}_{E_i}$. Let Σ_i be the covariance matrix of H_i , whose eigenvalues are at least λ , so that it has determinant at least λ^4 . Hence the density of H_i is bounded by $(2\pi)^{-2}\lambda^{-2}$, since at every $y \in \mathbb{R}^4$ it satisfies

$$(2\pi)^{-2}(\det \Sigma_i)^{-1/2} \exp\left(-\frac{1}{2}\langle \Sigma_i^{-1}y, y \rangle\right) \leq (2\pi)^{-2}(\det \Sigma_i)^{-1/2} \leq (2\pi)^{-2}\lambda^{-2}.$$

Since the product of the two complex disks has four-dimensional volume $\pi^2 u^2 v_0^2$, by the [jet approximation](#) we have

$$\begin{aligned} g_i &\leq (2\pi)^{-2}\lambda^{-2}\pi^2 u^2 v_0^2 = \frac{u^2 v_0^2}{4\lambda^2}, \\ \mathbb{E}Z &= \sum_{i=1}^m \mathbb{P}(E_i) \leq \sum_{i=1}^m (g_i + \delta) \leq m \left(\frac{u^2 v_0^2}{4\lambda^2} + \delta \right) = \mu_*. \end{aligned}$$

For an admissible ordered pair (i, j) , that is, one that obeys the pair estimate of the jet approximation, since $\mathbb{P}(E_i), g_j \leq 1$ we have

$$\begin{aligned} |\mathbb{P}(E_i)\mathbb{P}(E_j) - g_i g_j| &= \left| \mathbb{P}(E_i)(\mathbb{P}(E_j) - g_j) + g_j(\mathbb{P}(E_i) - g_i) \right| \\ &\leq \mathbb{P}(E_i)|\mathbb{P}(E_j) - g_j| + g_j|\mathbb{P}(E_i) - g_i| \leq 2\delta, \\ |\text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j})| &= |\mathbb{P}(E_i \cap E_j) - \mathbb{P}(E_i)\mathbb{P}(E_j)| \\ &\leq |\mathbb{P}(E_i \cap E_j) - g_i g_j| + |g_i g_j - \mathbb{P}(E_i)\mathbb{P}(E_j)| \leq \delta_* + \chi + 2\delta. \end{aligned}$$

On each of the at most $b_* m^2$ exceptional pairs we use the bound one, and summing over all m^2 ordered pairs, including the diagonal, we obtain

$$\begin{aligned} \text{Var } Z &= \sum_{i,j=1}^m \text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j}) \leq \sum_{i,j=1}^m |\text{Cov}(\mathbf{1}_{E_i}, \mathbf{1}_{E_j})| \\ &\leq b_* m^2 + m^2(\delta_* + \chi + 2\delta) = m^2 V_*. \end{aligned}$$

Since $\mathbb{E}Z \leq \mu_*$, Chebyshev's inequality yields

$$\mathbb{P}(Z > \mu_* + t) \leq \mathbb{P}(Z - \mathbb{E}Z > t) \leq \frac{\text{Var } Z}{t^2} \leq \frac{m^2 V_*}{t^2}.$$

On the two geometric good events, each cell receives at most q_0 roots, and the excluded sectors contain at most b_0 roots. On these events and on the derivative event, by the first part of the proof every counted root outside the excluded sectors is assigned to a retained point z_i at which E_i occurs, and the cell bound of the [mesh hypothesis](#), applied to the cell of z_i , which contains the roots assigned to z_i , shows that at most q_0 roots are assigned to each such point. Hence $B_N \leq b_0 + q_0 Z$, so that $B_N > b_0 + q_0(\mu_* + t)$ implies $Z > \mu_* + t$ outside the geometric and derivative exceptional events, and

$$\mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \mathbb{P}(Z > \mu_* + t) \leq p_0 + p_2 + \frac{m^2 V_*}{t^2}.$$

□

Corollary B.5 (Polynomial parameters for the mesh bound). *Suppose the following bounds hold with fixed positive constants:*

$$\begin{aligned} R &\leq 2, & A &\geq S, & \tau &= (ANL\sqrt{\log N})^{-1}, \\ M_2 &= SN^{5/2}\sqrt{\log N}, & m &\leq DNL^2 \log N, & q_0 &\leq C_0 \log N. \\ \delta &\leq D_0 N^{-\beta_1}, & V_* &\leq VN^{-\beta}, & L &= N^\ell. \end{aligned}$$

Then

$$(76) \quad \mu_* \leq \frac{9D}{A^2\lambda^2} NL^{-4} + DD_0 N^{1-\beta_1} L^2 \log N.$$

For a chosen $d > 0$, impose the explicit deterministic thresholds

$$b_0 \leq \frac{1}{2} N^{1-d}, \quad C_0 \log N \left[\frac{9D}{A^2\lambda^2} NL^{-4} + DD_0 N^{1-\beta_1} L^2 \log N \right] \leq \frac{1}{4} N^{1-d}.$$

Then

$$(77) \quad \mathbb{P}(B_N > N^{1-d}) \leq p_0 + p_2 + 16C_0^2 D^2 V N^{-\beta+4\ell+2d} (\log N)^4.$$

The mean thresholds are eventual if $d < 4\ell$, $d < \beta_1 - 2\ell$, and $b_0 = o(N^{1-d})$. The last failure term is summable on $N = j^s$ whenever

$$s(\beta - 4\ell - 2d) > 1.$$

In particular $\beta = \beta_1 = 1/2$, $\ell = 1/64$, $d = 1/32$ give $B_N \leq N^{31/32}$. The probability of the exceptional event, in addition to $p_0 + p_2$, is at most

$$16C_0^2 D^2 V N^{-3/8} (\log N)^4.$$

Proof. We first substitute the parameters into the definitions of u , v_0 and μ_* in Theorem B.4, and then apply this theorem with a deviation t such that $b_0 + q_0(\mu_* + t) \leq N^{1-d}$.

With the chosen parameters, and since $S \leq A$ and $R \leq 2$, we have

$$\begin{aligned} \frac{N^{3/2}\tau}{L} &= \frac{N^{1/2}}{AL^2\sqrt{\log N}}, & \frac{M_2\tau^2}{2} &= \frac{SN^{1/2}}{2A^2L^2\sqrt{\log N}}, \\ u &= \left(\frac{1}{A} + \frac{S}{2A^2}\right) L^{-2}(\log N)^{-1/2} \leq \frac{3}{2A} L^{-2}(\log N)^{-1/2}, \\ M_2\tau &= \frac{S}{A} \cdot \frac{N^{3/2}}{L}, & v_0 &= \frac{R}{L} \left(1 + \frac{S}{A}\right) \leq \frac{4}{L}, \\ \frac{u^2v_0^2}{4\lambda^2} &\leq \frac{1}{4\lambda^2} \frac{9}{4A^2L^4\log N} \frac{16}{L^2} = \frac{9}{A^2\lambda^2\log N} L^{-6}. \end{aligned}$$

Using also $m \leq DNL^2\log N$ and $\delta \leq D_0N^{-\beta_1}$, we have

$$\mu_* \leq DNL^2\log N \left(\frac{9}{A^2\lambda^2\log N} L^{-6} + D_0N^{-\beta_1} \right) = \frac{9D}{A^2\lambda^2} NL^{-4} + DD_0N^{1-\beta_1} L^2\log N.$$

This proves (76).

We now choose the deviation t in Theorem B.4. The event $\{B_N > N^{1-d}\}$ is contained in the event of (75) when $b_0 + q_0(\mu_* + t) \leq N^{1-d}$. Since the probability bound m^2V_*/t^2 decreases in t , we take the largest t with $C_0t\log N \leq \frac{1}{4}N^{1-d}$,

$$t = \frac{N^{1-d}}{4C_0\log N}.$$

By the two deterministic thresholds, the mean bound (76) with $q_0 \leq C_0\log N$, and this choice of t , we have

$$\begin{aligned} b_0 + q_0(\mu_* + t) &= b_0 + q_0\mu_* + q_0t \\ &\leq \frac{1}{2}N^{1-d} + C_0\log N \left[\frac{9D}{A^2\lambda^2} NL^{-4} + DD_0N^{1-\beta_1} L^2\log N \right] \\ &\quad + C_0t\log N \leq \frac{1}{2}N^{1-d} + \frac{1}{4}N^{1-d} + \frac{1}{4}N^{1-d} = N^{1-d}. \end{aligned}$$

Hence, by (75), $m \leq DNL^2\log N$, $V_* \leq VN^{-\beta}$ and $L = N^\ell$, we have

$$\begin{aligned} \mathbb{P}(B_N > N^{1-d}) &\leq \mathbb{P}\{B_N > b_0 + q_0(\mu_* + t)\} \leq p_0 + p_2 + \frac{m^2V_*}{t^2}, \\ \frac{m^2V_*}{t^2} &\leq D^2N^2L^4(\log N)^2 VN^{-\beta} \frac{16C_0^2(\log N)^2}{N^{2-2d}} \\ &= 16C_0^2D^2VN^{-\beta}L^4N^{2d}(\log N)^4 = 16C_0^2D^2VN^{-\beta+4\ell+2d}(\log N)^4. \end{aligned}$$

We have thus proved (77). With $L = N^\ell$, the left-hand side of the second threshold equals

$$\begin{aligned} & \frac{9C_0D}{A^2\lambda^2} N^{1-4\ell} \log N + C_0DD_0N^{1-\beta_1+2\ell}(\log N)^2 \\ &= \frac{1}{4} N^{1-d} \left(\frac{36C_0D}{A^2\lambda^2} N^{d-4\ell} \log N + 4C_0DD_0N^{d-\beta_1+2\ell}(\log N)^2 \right). \end{aligned}$$

When $d < 4\ell$ and $d < \beta_1 - 2\ell$, the factor in parentheses tends to zero, and so the second threshold holds for all large N . The first threshold holds for all large N when $b_0 = o(N^{1-d})$. With $N = j^s$, the last failure term equals $16C_0^2D^2Vj^{-s(\beta-4\ell-2d)}(s \log j)^4$, which is summable when $s(\beta - 4\ell - 2d) > 1$. For the particular powers of the corollary, $4\ell = 2d = 1/16$, and the three exponents, the conditions $d < 4\ell$ and $d < \beta_1 - 2\ell$, and the summability at $N = j^8$ read

$$\begin{aligned} 1 - 4\ell &= \frac{15}{16}, & 1 - \beta_1 + 2\ell &= \frac{17}{32}, & -\beta + 4\ell + 2d &= -\frac{3}{8}, \\ 4\ell = \frac{1}{16} &> \frac{1}{32} = d, & \beta_1 - 2\ell &= \frac{15}{32} > d, & 8(\beta - 4\ell - 2d) &= 3 > 1. \end{aligned}$$

Since $1 - d = 31/32$, once the two thresholds hold, we obtain $B_N \leq N^{31/32}$ outside an event of probability at most $p_0 + p_2 + 16C_0^2D^2VN^{-3/8}(\log N)^4$. $\square$

B.4 From a sparse subsequence of degrees to the full sequence

Theorem B.6 (Sparse-to-dense convergence). *Let $s \geq 1$ be fixed, $n_j = \lfloor j^s \rfloor$, and let (X_n) be real random variables on a common probability space. Suppose $X_{n_j}/n_j \rightarrow c$ almost surely. If there are deterministic $\varepsilon_j \rightarrow 0$ and events E_j such that*

$$\sum_j \mathbb{P}(E_j) < \infty, \quad \max_{n_j \leq n < n_{j+1}} |X_n - X_{n_j}| \leq \varepsilon_j n_j \quad \text{on } E_j^c,$$

then $X_n/n \rightarrow c$ almost surely.

The numbers ε_j need not be monotone. In particular, the case $c = 1/2$ yields the strong law needed for the unweighted Rademacher model.

Proof. We compare each degree n of the block $n_j \leq n < n_{j+1}$ with n_j , and use the Borel–Cantelli lemma to discard the events E_j . For every J , by the union bound we have

$$\mathbb{P}\left(\bigcup_{j \geq J} E_j\right) \leq \sum_{j \geq J} \mathbb{P}(E_j) \rightarrow 0$$

as $J \rightarrow \infty$, since the series converges. It follows that almost surely only finitely many E_j occur, which is the Borel–Cantelli lemma. For $n_j \leq n < n_{j+1}$ we write

$$\frac{X_n}{n} - c = \frac{n_j}{n} \left(\frac{X_{n_j}}{n_j} - c \right) + \frac{X_n - X_{n_j}}{n} + c \left(\frac{n_j}{n} - 1 \right).$$

Outside E_j , by the block bound $|X_n - X_{n_j}| \leq \varepsilon_j n_j$, it follows that

$$\begin{aligned} \left| \frac{X_n}{n} - c \right| &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \frac{|X_n - X_{n_j}|}{n} + |c| \left| \frac{n_j}{n} - 1 \right| \\ &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \frac{\varepsilon_j n_j}{n} + |c| \frac{n - n_j}{n} \\ &\leq \frac{n_j}{n} \left| \frac{X_{n_j}}{n_j} - c \right| + \varepsilon_j + |c| \frac{n_{j+1} - n_j}{n_j}, \end{aligned}$$

where the last step uses $\varepsilon_j \geq 0$, which holds on E_j^c since there $0 \leq |X_n - X_{n_j}| \leq \varepsilon_j n_j$. Since $j^s - 1 < n_j \leq j^s$ for every j , we have, for $j \geq 2$,

$$\frac{n_{j+1}}{n_j} < \frac{(j+1)^s}{j^s - 1} = \frac{(1 + 1/j)^s}{1 - j^{-s}} \rightarrow 1.$$

Since also $n_{j+1} \geq n_j$, it follows that $n_{j+1}/n_j \rightarrow 1$, and hence the last ratio, which equals $n_{j+1}/n_j - 1$, tends to zero. As $n \rightarrow \infty$, the index j with $n_j \leq n < n_{j+1}$ tends to infinity. Almost surely, $X_{n_j}/n_j \rightarrow c$ and E_j occurs for only finitely many j , and then the three terms of the last bound tend to zero, the first since $n_j/n \leq 1$ and the second since $\varepsilon_j \rightarrow 0$. This proves the assertion. $\square$

B.5 Deterministic choice of a vanishing block envelope

Lemma B.7 (Deterministic diagonalization). *Let $s \geq 1$, $n_j = \lfloor j^s \rfloor$, and let (X_n) be real random variables on a common probability space. Suppose that for each positive integer k ,*

$$\mathbb{P} \left\{ \max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} > a_k + b_k(j) \right\} \leq p_k(j),$$

where $a_k \rightarrow 0$, $b_k(j) \rightarrow 0$ for fixed k , and $\sum_j p_k(j) < \infty$ for fixed k . Then there are deterministic numbers $\varepsilon_j \rightarrow 0$ and events E_j with $\sum_j \mathbb{P}(E_j) < \infty$ such that

$$\max_{n_j \leq n < n_{j+1}} \frac{|X_n - X_{n_j}|}{n_j} \leq \varepsilon_j \quad \text{on } E_j^c.$$

In particular, if $X_{n_j}/n_j \rightarrow c$ almost surely, then $X_n/n \rightarrow c$ almost surely.

Proof. The idea is to let the index k grow slowly with j , along a deterministic diagonal $k(j) \rightarrow \infty$ chosen so that the probabilities of the exceptional events along the diagonal are still summable, and then to apply Theorem B.6. Choose increasing deterministic integers J_k , with $J_k \geq k$ and $J_k > J_{k-1}$, such that for all $j \geq J_k$ we have $b_k(j) \leq 1/k$ and $\sum_{j \geq J_k} p_k(j) \leq 2^{-k}$, which is possible since $b_k(j) \rightarrow 0$ and $\sum_j p_k(j) < \infty$. Choosing the least admissible integer makes each J_k deterministic, and this definition does not require a numerical bound for J_k . For $j \geq J_1$, we set

$$k(j) = \max\{k : J_k \leq j\}, \quad \varepsilon_j = a_{k(j)} + \frac{1}{k(j)}.$$

Since $J_k \geq k$ and the J_k increase strictly, $k(j)$ is finite and tends to infinity, and therefore $\varepsilon_j \rightarrow 0$. Let E_j be the event that the normalized block increment exceeds ε_j , and for $j < J_1$ take $E_j = \Omega$ and choose ε_j arbitrarily. For $j \geq J_1$ we have $J_{k(j)} \leq j$, hence $b_{k(j)}(j) \leq 1/k(j)$ and $\varepsilon_j \geq a_{k(j)} + b_{k(j)}(j)$. By the hypothesis with $k = k(j)$ we therefore have $\mathbb{P}(E_j) \leq p_{k(j)}(j)$, and these bounds satisfy

$$\sum_{j \geq J_1} p_{k(j)}(j) \leq \sum_{k \geq 1} \sum_{j \geq J_k} p_k(j) \leq \sum_{k \geq 1} 2^{-k} = 1,$$

where the first step holds since each $j \geq J_1$ contributes the single nonnegative term $p_{k(j)}(j)$, with $j \geq J_{k(j)}$, which appears in the inner sum for $k = k(j)$. Since only finitely many $j < J_1$ remain, it follows that $\sum_j \mathbb{P}(E_j) < \infty$. Finitely many initial degrees do not affect the limit, and the final assertion follows from Theorem B.6. The conclusion holds for arbitrary dependence between degrees, blocks, and exceptional events. $\square$

B.6 The mean for coefficients in $\{0, 1\}$

Remark B.8 (The mean for coefficients in $\{0, 1\}$). Let $K \geq 2$, $N \geq 2K$, $|r - 1| \leq K/N$ and $\text{dist}(\theta, 2\pi\mathbb{Z}) > 0$, let $\sigma_N(r)^2 = \sum_{k=0}^N r^{2k}$, as in Section 2, and let $D_N(z) = \sum_{k=0}^N z^k$. Then

$$\frac{|D_N(re^{i\theta})|}{\sigma_N(r)} \leq \frac{10e^{4K}}{\sqrt{N} \text{dist}(\theta, 2\pi\mathbb{Z})}.$$

Indeed, $D_N(re^{i\theta}) = \{1 - (re^{i\theta})^{N+1}\}/(1 - re^{i\theta})$. For the numerator, $r \leq 1 + K/N$ gives $|1 - (re^{i\theta})^{N+1}| \leq 1 + (1 + K/N)^{N+1} \leq 1 + e^{2K} \leq 2e^{2K}$. For the denominator, $|1 - re^{i\theta}|^2 = (1 - r)^2 + 4r \sin^2(\theta/2) \geq 4r \sin^2(\theta/2)$, and $r \geq 1 - K/N \geq 1/2$ together with $|\sin(\theta/2)| \geq \text{dist}(\theta, 2\pi\mathbb{Z})/\pi$, by the concavity of the sine on $[0, \pi/2]$, gives $|1 - re^{i\theta}| \geq (\sqrt{2}/\pi) \text{dist}(\theta, 2\pi\mathbb{Z})$. For the normalization, every term of $\sigma_N(r)^2$ is at least $(1 - K/N)^{2N} \geq e^{-4K}$, by $\log(1 - u) \geq -2u$ for $0 \leq u \leq 1/2$, so that $\sigma_N(r) \geq \sqrt{N}e^{-2K}$. Combining the three bounds and $\sqrt{2}\pi < 10$ gives the claim.

C The Lean formalization

Theorem 1.1 and Corollary 1.2 for all coefficient distributions considered, Theorem 1.3, and the results of Section 8 are formalized in Lean 4 [4] with Mathlib [21] (Lean v4.34.0, Mathlib commit 5ed2965), in release 1.0.0 of [13] (Lean sources `swb:1:dir:383ccdcc9544c05b9f977e4d6e61a026014648a2`). The formalization also uses the Hanson–Wright inequality from SLT [26] and Bentkus’s normal approximation [2] from ProbabilityApproximation [25]. The declarations are checked by the Lean kernel and depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The formal proofs follow the architecture of this paper, with different intermediate estimates and constants in places.

The principal declarations, in the namespace `Erdos522`, are

```
Corollary 1.2 (Erdős #522)   erdos_522
Theorem 1.1 (Rademacher)    radial_profile_compact_uniform_prefixes
Theorem 1.3                  ae_rademacher_logarithmic_root_rate_eventually
```

Theorem 1.3 is formalized in the equivalent form “for every $c > 2000 \log 2$, eventually $(\log n) |\nu_n(1)/n - 1/2| \leq c$ ”, and the radial laws for all degrees, obtained through the degrees $\lfloor j^q \rfloor$ of Proposition 8.1, are formalized at each fixed x . The other formal statements coincide with, or are stronger than, those of the paper. The file `lean/CORRESPONDENCE.md` in [13] lists the declaration of each result, including those for the other coefficient distributions and Section 8, and compares the formal and informal proofs.

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